

GMES Initial Operations / Copernicus Land monitoring services – Validation of products - (Second Specific Contract)

Validation Services for the geospatial products of the
Copernicus land Continental and local components
including in-situ data (lot 1)

Open Call for Tenders - EEA/MDI/14/010/SC2

CORINE LAND COVER 2012 FINAL VALIDATION REPORT



Document: GIO_CLC2012_Validation_Report_SC2_1_4

Issue: 1.4

Date: 10/02/2017

Consortium Partners

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Main Partners	GAF AG, Germany Joanneum Research, Austria e-GEOS, Italy
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Document Issue Record

Issue	Date	Author(s)	Description of Change
1.0	13/07/2015	Christophe Sannier	Document creation
1.1	14/11/2016	Gabriel Jaffrain, Michel Bossard, Jan Feranec	First draft
1.2	23/12/2016	Gabriel Jaffrain	Second draft – insertion of plausibility analysis
1.3	12/01/2017	Alexandre Pennec	Integration of 95% CI results
1.4	10/02/2017	Alexandre Pennec, Annalaura Di Federico	Final review and revision

Executive Summary

This report provides the evaluation results of the CORINE Land Cover (CLC) data layers, specifically the Land Cover 2012 status and the Land Cover 2006-2012 change layers. This analysis was performed over 100% of coverage for each of the EEA39 countries during the two Specific Contracts (SC) of the validation project. No semantic checks were envisaged for this product and documentation on previous semantic checks was not made available to the validation team.

The thematic accuracy assessment was conducted in a two-stage process:

1. An initial blind interpretation in which the validation team did not have knowledge of the product's thematic classes. However, the product polygon was provided to the validation team together with the point sample unit to consider boundary effects and geometric differences between the validation and production data (without access to the different attributes)
2. A plausibility analysis was performed on all sample units in disagreement with the production data to consider the following cases:
 - 1: Uncertain code, both producer and operator codes are plausible. Final validation code used is producer code
 - 2: Error from first validation interpretation. Final validation used is producer code
 - 3: Error from producer. Final validation code used is from first validation interpretation
 - 4: Producer and operator are both wrong. Final Validation code used is a new code from this second interpretation.

Resulting to this two-stage approach, it should be noticed that the plausibility analysis shows better results than the blind analysis (for each level of reporting: biogeographical or country level).

The reporting for validation was done at 3 levels: pan-European, country (or group of countries) and by biogeographical regions. A dedicated sampling design has been developed and applied using two levels of stratification according to (1) countries and group of countries greater than 90,000km² and (2) the CLC2012 or CLCC2006-2012 classes. The thematic accuracy assessment (in a blind approach) was carried out over 25,182 point (including sample units for SC1 and SC2) covering the EEA39.

The final results show that the CLC2012 products meet the thematic validation requirement, i.e. 85 % in a plausibility analysis, even if the blind interpretation results are slightly above the threshold. Indeed, the overall accuracies obtained for CLC2012 are **83.6 %** for the blind analysis and **89.7 %** for the plausibility analysis. The overall accuracy for the CLCCH2006-2012 data layer at CLC level 3 is **81.8%** for the blind analysis and **88.6 %** for the plausibility analysis. Those fully acceptable results are related to the fact that the 2012 dataset was the third update of CORINE Land Cover data, and that each time the database has undergone corrections performed by the national teams.

The lower results can be observed in the Anatolian and Black Sea region as well as in the Mediterranean area. The most consistent results can be observed within the Eastern temperate zones.

For each country or biogeographical region, the overall accuracy for the CLCCH2006-2012 change layer is systematically lower than the CLC2012 overall accuracy. However it is quite surprising to note that good results for the CLC2012 status layer does not guarantee good results for the changes CLCCH2006-2012 layer, and vice versa. But a deeper analysis shows that there is generally a link between landscape complexity, diversity of actual changes and the overall accuracy of change detection. The more various are the changes (at level 3), the weaker overall accuracy will be.

Quality checks of the validation points have been made by European experts, partners in the consortium. It should be noted that for the blind analysis, the 2006 and 2012 source images were used as reference data for validation, with the only help of Google Earth as very high resolution ancillary data. Unlike the national photo-interpretation teams, we did not have any additional thematically source of information such as forest stand maps, peatland maps, etc. Control has thus been performed with a lack of source information.

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List of Abbreviations

CLC	CORINE Land Cover
CLCCH2006-2012	Corine land cover change between 2006 and 2012
EEA	European Environment Agency
EU-DEM	Digital Elevation Model over Europe
ESA	European Spatial Agency
FAO	Food and Agriculture Organization of the United Nations
FTY	Forest Type
GIO	GMES Initial Operations
GMES	Global Monitoring for Environment and Security
HRL	High Resolution Layer
IGN FI	IGN France International
JRC	Joint Research Centre
LAEA	Lambert Azimuthal Equal-Area
LUCAS	Land Use/Cover Area frame Survey
MMU	Minimum Mapping Unit
MMW	Minimum Mapping Width
PSU	Primary Sample Unit
SP	Service Provider
TCD	Tree Cover Density
UA	Urban Atlas

1. Validation Framework

The validation framework is defined by a comprehensive analysis of the product specifications to determine the criteria to be used for the validation exercise.

1.1. Products to be validated

The CORINE Land Cover (CLC) dataset was initiated for the reference year 1990 in the late 80s. Updates were then regularly produced for 2000, 2006 and the latest 2012. The CLC applies a pragmatic nomenclature of 44 land cover/land use classes with a Minimum Mapping Unit (MMU) of 25 hectares (ha) and a Minimum Mapping Width (MMW) of 100 m.

Updates are created by generating a change layer. Interpreted changes that have occurred between the two reference years considered are mapped with a 5 ha MMU. It means that the change layer has a higher level of spatial detail than the status layer. Changes are then aggregated to unchanged areas to produce the updated status layer with a 25 ha MMU.

Therefore, the products to be validated as part of the CLC2012 exercise are the following:

- CORINE Land Cover 2012 map (CLC2012) nomenclature with 44 classes
- CORINE Land Cover 2006-2012 change map (CLCCH2006-2012)

Table 1: Attribute name and attribute type of CLC2012 database

ATTRIBUTE NAME	ATTRIBUTE TYPE
OBJECTID	Object ID
Shape	Geometry
ID	String, Length 18
Remark	String, Length 20
Area_ha	Double
Code_12	String, Length 3
Shape_Length	Double
Shape_Area	Double

Table 2: Attribute name and attribute type of CLCCH2006-2012 database

ATTRIBUTE NAME	ATTRIBUTE TYPE
OBJECTID	Object ID
Shape	Geometry
ID	String, Precision 18
Change	String, Precision 7
Code_06	String, Precision 3
Code_12	String, Precision 3
Chtype	String, Precision 1
Remark	String, Precision 20
Area_ha	Double
Shape_Length	Double
Shape_Area	Double

1.2. Validation Criteria

1.2.1. CLC2012

The validation exercise focuses on thematic accuracy. The CLC expected overall accuracy is greater than 85% for both the status and change layers. Former validation exercises have been undertaken for previous version of CLC; the most recent available by Büttner *et al.* (2012).

1.2.2. CLCCH2006-2012

The validation exercise should contribute to determine a suitable accuracy threshold for CLC changes between 2006 and 2012. An overall accuracy will be calculated and another will be specified by country or group of small countries and by biogeographical region. The European countries are divided into 23 areas.

2. Validation approach

The validation approach will provide guidance on how the products will be validated by defining suitable indicators or metrics. Detailed completeness and Logical consistency checks are performed as part of the semantic checks undertaken by ETC ULS for most products. Therefore, the aim of this validation exercise is not to repeat these, but to review the existing documentation and perform additional checks if deemed necessary.

Thematic accuracy will represent the bulk of the work undertaken as part of this validation exercise.

The quality assessment is performed according to INSPIRE Data Specifications. The data quality elements considered are: (i) Completeness, (ii) Logical Consistency, (iii) Positional Accuracy, (iv) Thematic Accuracy, (v) Temporal quality and (vi) Usability. Each of them forms a section in the Validation Check list.

Logical consistency checks do not consist in a duplication of Semantic checks, but are performed to identify missing information if relevant.

2.1. Completeness

Completeness should only be performed when the product is complete

Description: For non-thematic raster products (Image mosaics & EU-DEM), Completeness provides an indication for missing data or omission within the intended area. For land cover and land use products (both raster & vector), the notion of Completeness in INSPIRE provides an indication of omission and commission errors.

Indicators: the rate of excess items is used for commission errors and the rate of missing items is used to verify omission errors.

2.2. Logical consistency

Logical consistency evaluates the degree of adherence to logical rules of data structure, attribution and relationships. In INSPIRE Data Specifications; Logical Consistency comprises four sub-elements

described hereafter: conceptual consistency, domain consistency, format consistency and topological consistency.

2.2.1. Conceptual consistency

Description: indicates that the data structure follows the data specifications in terms of data model and relationships.

Indicators:

- Type of feature used
- Minimum Mapping Unit (MMU)
- Coordinate Reference System
- The presence of a unique identifier for each feature
- Nomenclature used

2.2.2. Domain consistency

Description: involves the detection of attribute values that are outside the pre-defined range of values. For vector data each attribute has a pre-defined set of range of values. For raster data, the correct encoding of data is checked.

Indicator: Value domain non-conformance: number of items not in conformance with their expected value domain.

2.2.3. Format consistency

Description: includes detection of file format, file or attribute names or attribute types which do not correspond to the specifications. In addition, for raster data the pixel depth is also considered here.

Indicators:

- File format conformance
- File name conformance
- Attribute names conformance
- Attribute types conformance

2.2.4. Topological consistency

Description: topological consistency is applicable to vector data and describes the degree of correctness of the topological characteristics described in the product specification of the dataset.

Indicators:

- | | |
|-----------------------------------|--------------------------------|
| - Number of overlaps | - Number of null geometries |
| - Number of gaps | - Number of unclosed rings |
| - Number of multipart features | - Number of duplicate vertex |
| - Number of neighbouring features | - Number of pseudo nodes |
| - Number of self-intersections | - Number of non-matching nodes |

2.2.5. Additional logical consistency checks

- **Labelling or symbology:** the conformity of a layer with the symbology or style given in the product specifications is checked;
- **Map projection:** the conformity of the map projection parameters is also checked.

2.3. Positional Accuracy

Detailed positional accuracy as described below is only required for the validation of image mosaics. CLC2012 is dependent on the underlying imagery, but some scattered systematic shifts still remain due to historical reasons. Assessments of these systematic shifts are envisaged based on the comparison of CLC2012 with HRL PWB which provides a basis for comparison considering the distinct nature of the Permanent Water Bodies features. Any discrepancies detected by overlaying CLC2012 with HRL PWB will be assessed visually to ensure that the differences are effectively due to geometric shifts.

Positional accuracy of HR and VHR mosaics will be evaluated by RMSE of planimetry as an indication of the closeness of the position values of gridded data (for raster) or vertices along lines (for vector) to values accepted as being true. However, it should be noted that RMSE does represent an “average” error more or less representing 68% of the total errors (assuming errors are normally distributed). The Root Mean Square Error of Planimetry (RMSEP) is given by:

$$RMSEP = \sqrt{\frac{1}{n} \sum_{i=1}^n (x_{Ref_i} - x_{Map_i})^2 + (y_{Ref_i} - y_{Map_i})^2}$$

There is a close relationship between positional accuracy and the scale of the data according to INSPIRE Technical guidelines document on land cover.

Table 3: relationship between positional accuracy and the scale of the data according to INSPIRE Technical guidelines document on land cover

Scale, 1:	Estimated Positional accuracy
1 000 000	500 m
500 000	250 m
250 000	125 m
100 000	50 m
50 000	25 m
20 000	10 m
10 000	5 m
5 000	2,5 m
2 000	1 m
1 000	0,5 m

For raster elevation data, positional accuracy metrics are generally related to the vertical height and are expressed as an absolute Linear Error (LE) expressed in metre at 90% or 95% confidence interval meaning that an LE95 of 8m indicates that 95% of the height measurements are within 8m. LE90 or

LE95 can only be estimated and require the knowledge of how errors are distributed. It is usually assumed that errors are normally distributed. In which case, Root Mean Square Error (RMSE) can be used to estimate a DEM's absolute accuracy with 68% of values within the RMSE value and $LE95 \approx 1.96 \cdot RMSE$.

RMSE is calculated based on the comparison between reference and DEM elevation values according to the following formula: $RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (Z_{Ref_i} - Z_{DEM_i})^2}$, where n is the number of observations.

Concerning the positional accuracy of CLC2012, only areas where the problem of systematic shift occurs are identified (as most of the shifts were corrected in previous updates), and therefore no comprehensive assessment of the whole dataset is performed. A pragmatic approach is to compare CLC2012 with distinct thematic features such as water bodies. Therefore, the detection of systematic geometric shifts in CLC2012 is performed based on the comparison with HRL PWB. Areas where discrepancies are detected through this automated comparison are further investigated by visual comparison.

2.4. Thematic Accuracy

2.4.1. Level of reporting

The level of reporting for the validation results is at pan-European level. However, results are also provided at different levels of aggregation:

- Countries or group of countries with an area greater than 90,000km² level (Figure 1)
- European Biogeographical regions 2016 level (Figure 2). The French overseas departments are aggregated into one specific biogeographical region named DOM. In fact, this region corresponds to the tropical climate and includes French Guiana, Martinique, Guadeloupe, La Reunion and Mayotte.

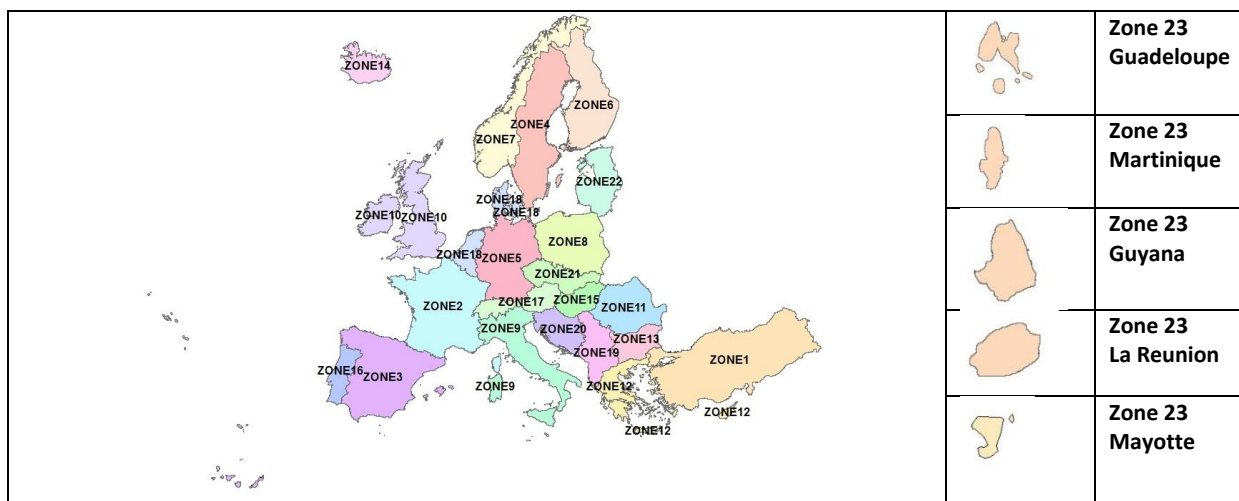


Figure 1: Stratification according to countries and group of countries greater than 90,000km² (total of 23 zones including French overseas department)

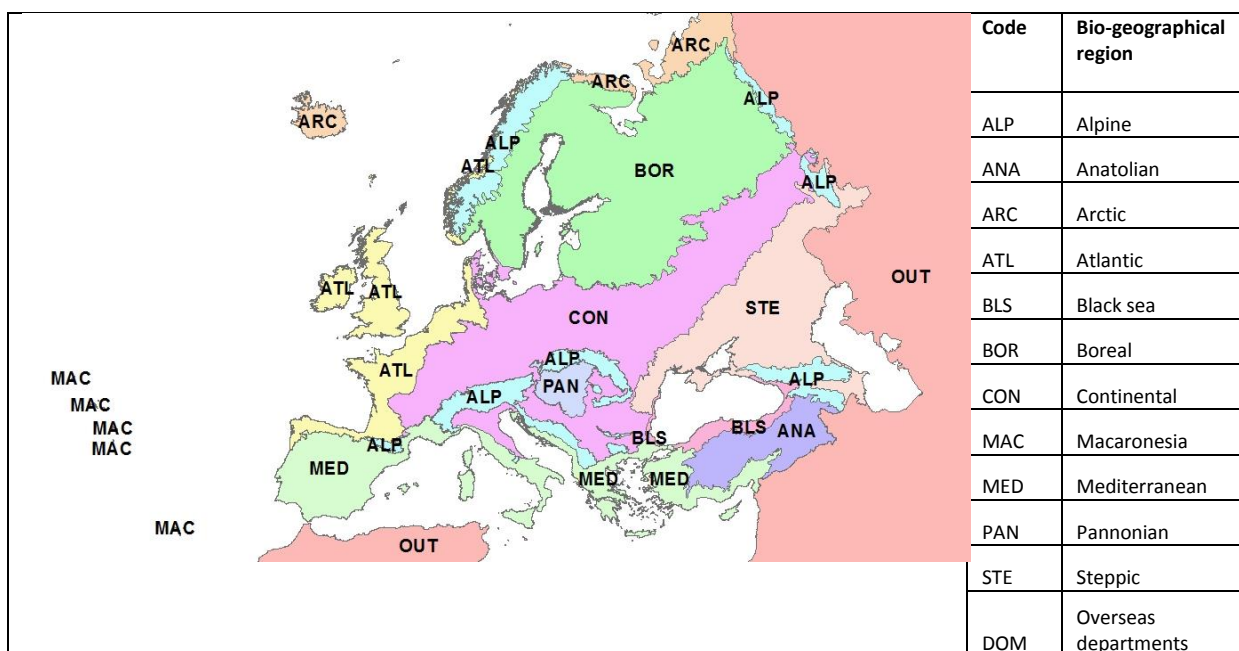


Figure 2: The European Biogeographical region 2016

Analysis of the results at a more disaggregated scale will contribute to assess regional differences if any and the causes of these differences.

2.4.2. Stratification and sample design

A stratified systematic sampling approach based on the LUCAS sampling frame is used for all thematic layers adapting the number of replicates to each stratum. The LUCAS sampling is densified for small strata based on a 200m grid. Using LUCAS sampling ensures coherence between the different layers and traceability.

A set of 81 points located on an 18x18 km square constitutes a group (red points shown in Figure 3) in which every point is associated with a number comprised between 0 and 81 (the numbers do not follow each other spatially). The same pattern with the same numbers allocation is repeated all over the grid. A replicate refers to the points with the same number selected on the whole LUCAS grid.

At first, the number of samples to allocate to each stratum (or land cover class) was calculated as a function of their area. In this manner the sampling design is not only systematic but also stratified. A minimum number of sample units per stratum was defined to ensure that even small strata are represented in the sample.

The number of replicates to be selected for a stratum depends on its area and the number of LUCAS points intersecting the stratum.

For land cover classes covering a large proportion of the study area, 1 replicate may already exceed the defined number of samples for this class. To solve this problem, replicates are split into four sub-replicates, as illustrated by the blue numbers in Figure 3.

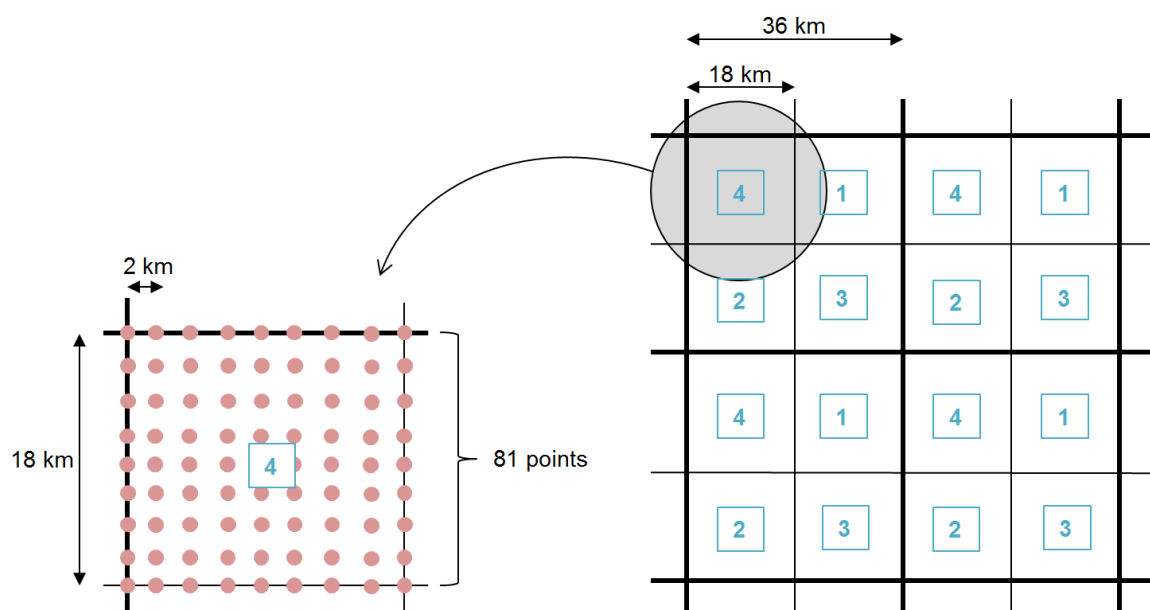


Figure 3: Replicates and sub-replicates used on LUCAS grid

The opposite problem is encountered for land cover classes covering a small proportion of the study area: even by selecting 81 replicates (the maximum number), the intersecting area between the stratum and LUCAS points is too small to reach the required number of samples. Therefore LUCAS grid was densified by creating one point every 200 m.

For the CLC2012 status and change layers, stratification is applied at two levels:

1. Stratification according to countries and group of countries greater than 90,000km²
2. Stratification based on the CLC2012 classes and CLCCH2006-2012 classes based on the map products

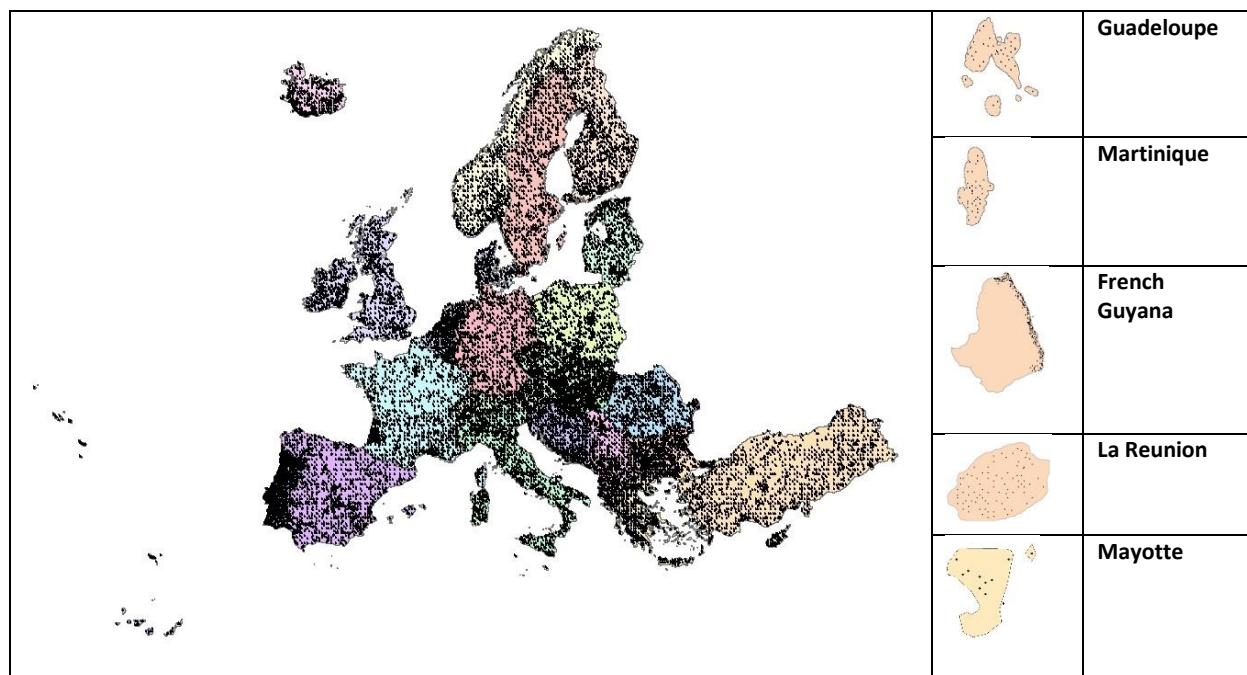


Figure 4: Distribution of the sample units according to the stratification into countries and group of countries

The number of sample units per stratum should be such to ensure a sufficient level of precision at reporting level. The minimum number of sample units per stratum should be set at 5 if possible. Priority is given to strata which are known to be difficult to map: i.e. changes and difficult classes.

The validation exercise covers the whole study area in order to be valid.

There were a total of 25,182 sample units selected and covering the entire of EEA39 area. The overall share of sample units per main strata and per country (or aggregated countries) is presented in Table 4.

Table 4: Distribution of sample units per main strata and substrata for the CLC012 and CLCCH2006-12 products

		TR	FR	ES	SE	DE	FI	NO	PL	IT	IE+UK	RO	EL	BG	IS	HU	PT	AT+CH+LI	BE+LU+NL+DK	AL+ME+MK+RS+XK	BA+HR+SI	CZ+SK	EE+LT+LV	DOM	
	CLASS/ ZONE	Z1	Z2	Z3	Z4	Z5	Z6	Z7	Z8	Z9	Z10	Z11	Z12	Z13	Z14	Z15	Z16	Z17	Z18	Z19	Z20	Z21	Z22	Z23	Total général
111	Continuous urban fabric	25	10	20	5	10	0	15	10	20	20	2	15	1	0	2	20	20	20	2	5	10	2	1	235
112	Discontinuous urban fabric	29	50	20	20	50	20	20	20	50	30	50	25	20	20	20	20	20	20	30	20	30	20	20	624
121	Industrial or commercial units	30	20	20	15	30	20	20	20	20	20	20	20	20	15	20	20	20	20	20	20	20	20	1	451
122	Road and rail networks and associated structures	8	10	10	10	10	1	15	20	20	20	15	15	15	1	20	20	20	20	4	20	20	20	1	315
123	Port areas	2	10	5	5	10	5	15	5	20	20	5	5	2	5	1	10	2	20	2	2	1	10	1	163
124	Airports	8	10	10	10	10	10	15	15	20	20	10	10	5	3	5	10	10	15	3	3	10	10	2	224
131	Mineral extraction sites	25	10	10	10	20	20	20	20	20	20	20	15	20	5	20	20	20	20	23	20	20	20	1	399
132	Dump sites	3	5	10	10	10	15	3	20	10	20	10	1	5	0	15	10	1	15	4	3	20	10	1	201
133	Construction sites	10	5	10	5	5	5	5	20	20	20	10	10	2	2	10	15	5	50	3	10	10	10	1	243
141	Green urban areas	6	10	10	10	20	20	20	20	20	20	15	5	10	5	20	10	20	20	11	10	20	20	1	323
142	Sport and leisure facilities	8	20	10	15	20	20	20	20	20	20	20	15	20	20	20	20	20	20	16	20	20	20	1	405
211	Non-irrigated arable land	70	50	50	50	60	50	20	20	50	50	50	35	50	5	50	50	50	50	35	20	50	20	20	955
212	Permanently irrigated land	60	0	40	0	0	0	0		5	0	3	30	0	0	0	50	0	0	2	2	0	0	0	192
213	Rice fields	30	10	20	1	0	0	0	0	20	1	5	15	5	0	2	20	0	0	1	0	0	0	2	132
221	Vineyards	30	50	40	0	20	0	0	0	20	0	20	20	20	0	20	20	20	2	20	20	20	0	0	342
222	Fruit trees and berry plantations	40	20	40	5	20	0	0	20	20	15	20	25	20	0	50	20	10	20	25	20	30	20	10	450
223	Olive groves	21	10	40	0	0	0	0	0	20	0	0	30	0	0	0	20	0	0	11	20	0	0	0	172
231	Pastures	40	50	20	20	50	20	20	20	20	50	50	25	50	30	50	20	20	20	30	20	50	20	10	705
241	Annual crops associated with pastures	1	5	10	0	0	0	0	0	20	0	0	15	0	0	0	20	0	1	2	1	0	0	0	75
242	Complex cultivation patterns	60	50	40	20	20	0	20	20	50	20	20	30	20	20	20	20	20	50	30	20	20	20	10	600
243	Land principally occupied by agriculture in other areas	60	50	40	30	20	50	20	20	50	50	50	35	20	0	20	20	20	20	53	50	20	20	20	738
244	Agro-forestry areas		1	40	0	0	0	0	0	20	0	0	0	0	0	0	20	0	0	1	0	0	0	1	83
311	Broad-leaved forest	60	50	40	50	50	50	50	20	50	20	50	35	50	20	50	50	50	20	53	50	50	20	40	978
312	Coniferous forest	60	50	40	50	50	50	50	20	50	50	50	30	50	10	50	50	50	50	30	50	50	20	1	961
313	Mixed forest	40	50	40	50	50	50	50	20	20	50	50	30	50	15	50	50	50	50	30	20	30	20	1	866
321	Natural grasslands	80	50	40	20	20	15	50	20	20	20	20	35	20	20	20	20	20	20	30	20	20	20	10	610
322	Moors and heathland	1	20	40	50	20	20	50	5	20	20	20	20	15	20	0	20	20	20	20	20	15	20	10	466
323	Sclerophyllous vegetation	21	20	40	0	0	0	0	0	50	0	0	40	0	0	0	20	0	0	30	20	0	0	10	251
324	Transitional woodland-shrub	60	50	40	50	30	50	15	20	50	50	50	35	50	20	50	50	50	20	53	50	50	20	20	933
331	Beaches, dunes, sands	20	10	10	5	10	3	15	10	20	20	20	15	10	50	0	15	5	20	13	2	0	15	1	289
332	Bare rocks	40	20	20	20	10	20	20	5	20	20	15	15	20	20	0	9	20	0	20	20	5	0	1	340
333	Sparsely vegetated areas	80	20	20	30	10	20	50	10	20	20	10	25	20	30	5	20	20	0	30	20	10	5	10	485
334	Burnt areas	2	1	10	4	0	0	1	0	20	5	0	15	1	0	0	30	0	0	8	3	1	0	0	101
335	Glaciers and perpetual snow	1	10	1	10	0	0	20	0	20	0	0	0	0	30	0	0	50	0	0	1	0	0	0	143
411	Inland marshes	30	10	10	15	10	20	1	20	20	20	20	15	20	20	20	5	20	20	18	20	20	20	30	404
412	Peat bogs		5	1	50	20	50	50	15	1	50	0	0	1	20	10	3	5	20	1	1	10	20	0	333
421	Salt marshes	21	10	10	5	10	20	0	0	20	20	1	15	0	10	0	20	0	20	2	1	0	1	1	187
422	Salines	5	10	10	0	0	0	0	0	5	0	0	5	1	0	0	10	0	0	2	1	0	0	0	49
423	Intertidal flats		5	2	0	10	0	20	0	0	20	0	15	0	20	0	5	0	15	0	1	0	0	1	114
511	Water courses	20	20	10	20	20	20	20	10	20	10	20	0	5	20	10	5	5	5	10	10	10	10	10	290
512	Water bodies	30	20	20	50	30	50	50	20	20	20	20	20	20	20	20	20	20	20	25	20	20	20	1	556
521	Coastal lagoons	11	10	5	5	3	0	0	1	20	5	1	5	1	15	0	5	0	10	4	1	0	2	1	105
522	Estuaries	1	10	5	5	3	0	0	0	1	20	0	3	0	5	0	2	0	1	0	0	0	0	10	66
523	Sea and ocean	12	1	10	10	3	15	15	1	2	3	1	5	1	1	0	1	0	1	4	1	0	1	10	98
	TOTAL CLC2012 classes	1161	908	939	740	744	709	775	487	1004	859	743	774	640	497	650	865	683	715	711	638	662	476	272	16652
	CLCCH2006-2012	535	520	530	360	355	590	195	470	570	450	171	326	220	135	620	525	250	355	223	195	455	340	140	8530
Total général		1696	1428	1469	1100	1099	1299	970	957	1574	1309	914	1100	860	632	1270	1390	933	1070	934	833	1117	816	412	25182

The sample units were provided to the bulk interpretation team as one shapefile along with the polygons from the original dataset in which all the information on strata or thematic classes were removed to ensure the independence of the interpretation and to ensure that actual thematic interpretation errors are separated from thematic errors due to potential geometric shifts.

2.4.3. Response Design

LUCAS points are re-interpreted based on available in situ data. LUCAS thematic information is not used directly.

Response design for most data set are based on the interpretation of thematic class at the point level taking into account product specifications (MMU, MMW, class definitions,...) based on combination of available in situ data. For CLC2012, the VHR dataset can be used alongside the imagery used in production (CORE01 imagery with a 20 meters spatial resolution) and virtual globes data (Bing or Google Earth images) as illustrated in Figure 5. These images are used in case there is a discrepancy between the independent imagery acquisition date and the imagery used in production. It is also used to confirm geometry in case of a geometric shift between the thematic data and the reference data. In detail, hereafter the reference data used during the validation are listed, covering the EEA39 area at a wall-to-wall basis:

- CORE3 VHR image Mosaic 2012, RGB with spatial resolution of 2.5m (Web map service provided by JRC) during SC1
- VHR Image 2012 NIR with spatial resolution of 2.5m for the SC2

A double blind approach guarantees complete independence from the map products and may underestimate their accuracy for complicated and difficult classes when sometimes several LC/LU classes are possible. This is resolved by the plausibility approach for which the interpreter checks the map value to assess whether it can be considered correct or not, within the frame of accepted product specifications. However, the plausibility analysis should be combined with the double-blind approach to ensure full traceability and transparency of the validation process.

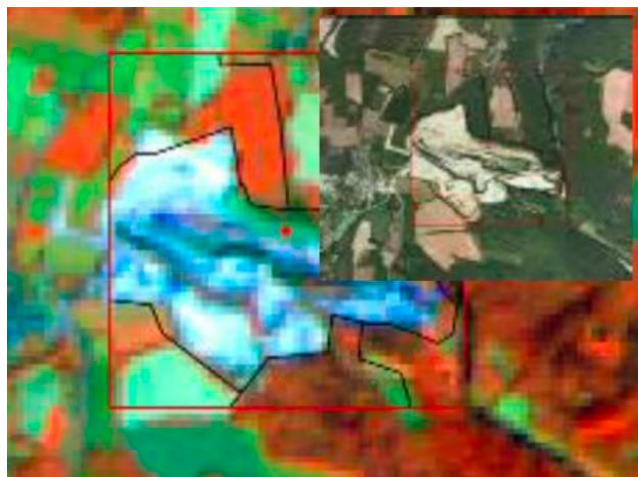


Figure 5: Example of a sample unit used for the validation of CLC2012

A double-blind approach was first applied. This consists in constructing the validation data set without any knowledge about the corresponding map layer information, i.e. the validation team did not have knowledge of the product's thematic classes for the selected sample units. However, the product polygon was provided as guidance to the validation team together with the sample unit to consider

boundary effects and geometric differences between the validation and production data. Two parameters were considered to interpret the sample unit along with the polygon:

- the area to be checked
- the size of the polygon

The maximum area to be verified was 100ha (=1 sq.km), e.g. 4 times the size of MMU (25ha).

The expert only interpreted the polygon which contains the sample unit. For each sample unit, two thematic fields were filled:

- code2006: the land-cover class value for 2006 (one of the 44 CLC codes)
- code2012: the land-cover class value for 2012 (one of the 44 CLC codes)

Additional fields were filled if necessary:

- uncertain: drop-down menu (2006/2012/2006&2012/none) in case of uncertainty of the interpretation
- comment: free text if any comment is needed

The sample unit was first blindly interpreted by the validation expert, i.e. only the limit of the polygon is visible without knowing the CLC thematic attribute of the area; but interpretation have to be made at sample unit level.

A second interpretation, carried out by a second interpreter as part of the plausibility analysis, was done only on the sample units in disagreement with product codes. The operator assigned a QC code modifying the validation code if necessary:

- 1: Uncertain code, both producer and operator codes are plausible. Final validation code used is producer code
- 2: Error from first validation interpretation. Final validation used is producer code
- 3: Error from producer. Final validation code used is from first validation interpretation
- 4: Error of both producer and operator. The controller should inform the new CLC field by a new land cover classes. Final validation code used is the new CLC code.

The Plausibility Analysis resulted in revised accuracy metrics.

2.4.4. Estimation and analyses procedures

Thematic accuracy should be presented in the form of an error matrix. Unequal sampling intensity resulting from the stratified systematic sampling approach should be accounted for by applying a weight factor (p) to each sample unit based on the ration between the number of samples and the size of the stratum considered:

$$\hat{p}_{ij} = \left(\frac{1}{N}\right) \sum_{x \in (i,j)} \frac{1}{\pi_{uh}^*}$$

Where i and j are the columns and rows in the matrix, N is the total number of possible units (population) and π is the sampling intensity for a given stratum.

Overall accuracy and User and producer accuracy should be computed for all thematic classes and 95% confidence intervals should be calculated for each overall accuracy.

The standard error of the error rate can be calculated as follows: $\sigma_h = \sqrt{\frac{p_h(1-p_h)}{n_h}}$ where n_h is the sample size for stratum h and p_h is the expected error rate. The standard error is calculated for each stratum and an overall standard error is calculated based on the following formula:

$$\sigma = \sqrt{\sum w_h^2 \cdot \sigma_h^2}$$

In which is the proportion of the total area covered by each stratum. The 95% confidence interval is +/- 1.96.

2.5. Temporal Quality

Temporal quality is evaluated by providing an indication of the closeness of the acquired image data to the reference year, e.g. the percentage area covered outside the accepted reference period as defined in the tender/product specification i.e. 2012 +/- 1-2 year(s).

2.6. Usability

Usability relates to the appropriateness of the metadata description and accompanying documentation to describe the processes and workflows involved in the production of the data. Although it is difficult to describe usability in quantitative terms, it provides a clear evaluation based on objective criteria of any limitation in the intended use of the data.

2.7. INSPIRE compliant metadata

Presence of INSPIRE compliant metadata should be verified.

3. Validation check list

Table 5: Validation check list

PRODUCT: CLC2012						
VALIDATION LEVEL: EEA39						
SERVICE PROVIDER: IGN FI				SERVICE USER:	ISSUE/REVISION:	
VALIDATION DATE: 28/10/2015				REVIEW DATE:		
CONDUCTED BY: SIRS				REVIEWED BY:	APPROVED BY:	
No.	DATA QUALITY SUB-	DATA QUALITY MEASURE	DATA QUALITY RESULT	COMMENTS BY AUDIT TEAM	DRAFT AUDIT CONCLUSION	FINAL AUDIT CONCLUSION
1	COMPLETENESS					
1.1	Commission	Rate of excess items	n/a	Please refer to the chapter 4		
1.2	Omission	Rate of missing items	n/a	Please refer to the chapter 4		
2	LOGICAL CONSISTENCY					
2.1	Format consistency	File format	This was not assessed			
2.2		File name	X	File name expected is CLC12_FR		
2.3		Attributes names	X	Attribute names expected: "Remark" and "Code_12"		
2.4		Attributes types	Conform with criteria			

PRODUCT: CLC2012						
VALIDATION LEVEL: EEA39						
SERVICE PROVIDER: IGN FI				SERVICE USER:	ISSUE/REVISION:	
VALIDATION DATE: 28/10/2015				REVIEW DATE:		
CONDUCTED BY: SIRS				REVIEWED BY:	APPROVED BY:	
No.	DATA QUALITY SUB-	DATA QUALITY MEASURE	DATA QUALITY RESULT	COMMENTS BY AUDIT TEAM	DRAFT AUDIT CONCLUSION	FINAL AUDIT CONCLUSION
2.5	Conceptual consistency	Feature type	Conform with the criteria			
2.6		MMU	X	Polygons < 25ha are present: 5019 features		
2.7		Coordinate reference system	Conform with the criteria			
2.8		Unique identifier	Conform with the criteria			
2.9		Nomenclature	Conform with the criteria			
2.10	Domaine consistency	Value domain non-conformance	Conform with the criteria			
2.11	Topological consistency	Overlaps	Conform with the criteria	No overlapping polygons		
2.12		Gaps	This was not assessed	Request for clarification or completion		

PRODUCT: CLC2012						
VALIDATION LEVEL: EEA39						
SERVICE PROVIDER: IGN FI				SERVICE USER:	ISSUE/REVISION:	
VALIDATION DATE: 28/10/2015				REVIEW DATE:		
CONDUCTED BY: SIRS				REVIEWED BY:	APPROVED BY:	
No.	DATA QUALITY SUB-	DATA QUALITY MEASURE	DATA QUALITY RESULT	COMMENTS BY AUDIT TEAM	DRAFT AUDIT CONCLUSION	FINAL AUDIT CONCLUSION
2.13		Multipart features	Conform with the criteria	No multipart polygons		
2.14		Neighbouring features	X	Neighbouring polygons with the same code are present: 142 features (maximum).		
2.15		Self-intersections	X	Polygons with self-intersections are present: 817 features		
2.16		Null geometry	Conform with the criteria			
2.17		Unclosed rings	Conform with the criteria			
2.18		Duplicate vertex	Conform with the criteria			
2.19		Pseudo nodes	This was not assessed			
2.20		Non matching nodes	Conform with the criteria			

PRODUCT: CLC2012						
VALIDATION LEVEL: EEA39						
SERVICE PROVIDER: IGN FI				SERVICE USER:	ISSUE/REVISION:	
VALIDATION DATE: 28/10/2015				REVIEW DATE:		
CONDUCTED BY: SIRS				REVIEWED BY:	APPROVED BY:	
No.	DATA QUALITY SUB-	DATA QUALITY MEASURE	DATA QUALITY RESULT	COMMENTS BY AUDIT TEAM	DRAFT AUDIT CONCLUSION	FINAL AUDIT CONCLUSION
3	POSITIONAL ACCURACY					
3.1	Absolute or external accuracy	RMSEP	n/a	This is dependent on the assessment of the CORE001 mosaic		
3.2	Relative or internal accuracy	RMSEP	n/a	This is dependent on the assessment of the CORE001 mosaic		
4	THEMATIC ACCURACY					
4.1	Classification correctness	Overall accuracy	83.3%	Including the overall area	See & 4.4	
4.2		Min. producer's accuracy	34.0%	Class 133 – Construction site	See & 4.4	
4.3		Min. user's accuracy	29.0%	Class 241 – Annual crops associated with permanent crops	See & 4.4	
4.4		Kappa				
5	TEMPORAL QUALITY					

PRODUCT: CLC2012						
VALIDATION LEVEL: EEA39						
SERVICE PROVIDER: IGN FI				SERVICE USER:	ISSUE/REVISION:	
VALIDATION DATE: 28/10/2015				REVIEW DATE:		
CONDUCTED BY: SIRS				REVIEWED BY:	APPROVED BY:	
No.	DATA QUALITY SUB-	DATA QUALITY MEASURE	DATA QUALITY RESULT	COMMENTS BY AUDIT TEAM	DRAFT AUDIT CONCLUSION	FINAL AUDIT CONCLUSION
5.1	Temporal quality	Closeness of the acquired image data to the reference		Acquisition windows from 2011 to 2013 'More detailed is not assessed'		
6	USABILITY					
6.1	Usability	Usability description		N/A		
7	METADATA					
7.1	INSPIRE compliant metadata	Presence		N/A		
7.2		File format		N/A		
7.3		File name		N/A		
7.4		INSPIRE compliance		N/A		

4. Additional tables

4.1. Coordinate reference system description

Table 6: Coordinate reference system description

NAME OF COORDINATE REFERENCE SYSTEM	ETRS_1989_LAEA
WKID	EPSG:3035
PROJECTION	Lambert_Azimuthal_Equal_Area
FALSE EASTING	4321000.0
FALSE NORTHING	3210000.0
CENTRAL MERIDIAN	10.0
LATITUDE OF ORIGIN	52,0
LINEAR UNIT	(1,0)
GEOGRAPHIC COORDINATE SYSTEM	GCS_ETRS_1989
ANGULAR UNIT	Degree (0.0174532925199433)
PRIME MERIDIAN	Greenwich (0,0)
DATUM	D_ETRS_1989
SPHEROID	GRS_1980
SEMIMAJOR AXIS	6378137.0
SEMIMINOR AXIS	6356752.314140356
INVERSE FLATTENING	298.257222101

4.2. Nomenclature

Table 7: Nomenclature table

CODE	CLASS NAME	CODE	CLASS NAME
111	Continuous urban fabric	311	Broad-leaved forests
112	Discontinuous urban fabric	312	Coniferous forests
121	Industrial and commercial units	313	Mixed forests
122	Road and rail network and associated land	321	Natural grassland
123	Port areas	322	Moors and heathland
124	Airports	323	Sclerophyllous vegetation
131	Mineral extraction sites	324	Transitional woodland/shrub
132	Dump sites	331	Beaches, dunes, and sand plains
133	Construction sites	332	Bare rocks
141	Green urban areas	333	Sparsely vegetated areas
142	Sport and leisure facilities	334	Burnt area
211	non-irrigated arable land	335	Glaciers and perpetual snow
212	Permanently irrigated land	411	Inland marshes
213	Rice field	412	Peatbogs
221	Vineyard	421	Salt marshes
222	Fruit trees and berry plantation	422	Salines
223	Olive grove	423	Intertidal flats
231	Pastures	511	Water courses
241	Annual crops associated with permanent crops	512	Water bodies
242	Complex cultivation patterns	521	Coastal lagoons
243	Land principally occupied by agriculture, with significant areas of natural vegetation	522	Estuaries
244	Agro-forestry areas	523	Sea and Ocean

4.3. Thematic accuracy

This section provides the evaluation results of the CORINE Land Cover layers, specifically the Land Cover 2012 status and the change layer for 2006 and 2012. This analysis was performed over 100% of the coverage. The thematic accuracy assessment was conducted in a two-stage process: an initial blind interpretation in which the validation team did not have knowledge of the product's thematic classes and a plausibility analysis performed on all sample units in disagreement with the production. This chapter provides both results. It should be noticed that among the 25,182 validation points, 442 sample units (around 1.7 %) could not be checked due to the lack of suitable data source imagery (either satellite images 2006 or satellite image 2012 or both).

4.3.1. CORINE Land Cover 2012 layer

4.3.1.1. Results

The overall accuracies obtained in the frame of the Validation exercise are **83.6%** for the blind interpretation and **89.7%** for the plausibility analysis, so the results are, respectively, under and above the 85% threshold expected. The confusion matrices at the European level including the blind and plausibility analysis are provided in Annex 1.

Table 8 summarizes overall accuracies obtained by zone (country or group of countries) for the blind and plausibility interpretations including the 95% confidence interval. For the detailed analysis, for both blind and plausibility analysis, accuracy assessments and metrics (Confusion matrices, Producer, User and overall accuracies per zones) are provided in Annex 1, Annex 3 and Annex 6.

Table 8 Overall accuracy for blind and plausibility accuracy and confidence interval (95%) for CLC2012 per country or group of countries.

		Overall accuracy	Confidence interval of 95%	Overall accuracy	Confidence interval of 95%
Z1	Turkey	78,61%	2,80%	85,17%	2,44%
Z2	France	86,35%	3,33%	89,33%	3,11%
Z3	Spain	85,10%	2,68%	89,42%	2,27%
Z4	Sweden	87,91%	3,92%	93,08%	3,10%
Z5	Germany	82,83%	4,16%	89,05%	3,43%
Z6	Finland	74,32%	6,11%	81,66%	5,73%
Z7	Norway	93,22%	2,66%	96,61%	1,99%
Z8	Poland	92,06%	2,37%	94,15%	2,21%
Z9	Italy - Sardinia	76,01%	4,15%	85,01%	3,53%
Z10	United Kingdom - Ireland	87,07%	3,75%	92,67%	2,78%
Z11	Roumania	78,90%	4,11%	91,21%	2,61%
Z12	Greece	80,21%	3,63%	87,28%	3,01%
Z13	Bulgaria	86,90%	3,54%	94,53%	2,03%
Z14	Island	85,13%	7,41%	93,68%	4,91%
Z15	Hungaria	86,08%	3,97%	91,28%	3,28%
Z16	Portugal	75,91%	3,68%	84,80%	3,17%
Z17	Switzerland - Austria	76,07%	4,74%	86,77%	3,72%
Z18	Belgium - Danmark - Luxembourg - Netherlands	84,75%	4,21%	88,18%	4,05%
Z19	Albania - Serbia - Macedonia - Montenegro	82,08%	3,87%	93,46%	2,18%
Z20	Bosnia Herzegovina - Croatia - Slovenia	77,38%	5,20%	89,27%	3,81%
Z21	Czech Republic - Slovakia	85,68%	3,77%	93,31%	2,40%
Z22	Estonia - Latvia - Lithuania	88,63%	3,68%	91,69%	3,22%
Z23	French overseas departments	96,41%	1,03%	97,61%	0,90%
	GLOBAL ACCURACY	83,60%	0,91%	89,66%	0,77%

	Accuracy better than 85%
	Accuracy between 80% and 85%
	Accuracy less than 80%

The overall accuracies at the biogeographical regions level illustrated in Table 9 show good to very good final results except for Black Sea regions with slightly lower accuracies. Annex 2 provides the producer and user accuracies for each land cover classes and the Annex 4 accuracies for each biogeographical region. Confusion matrices per biogeographical regions are presented in Annex 5.

Table 9: Overall accuracy for blind and plausibility analysis per bio-geographical region

	Overall accuracy CLC2012	
	Blind analysis	Plausibility analysis
Alpine	84,29%	92,49%
Anatolian	77,15%	86,44%
Arctic	85,29%	93,75%
Atlantic	87,75%	91,30%
BlackSea	81,70%	84,45%
Boreal	84,35%	89,12%
Continental	83,91%	90,74%
Macaronesia	80,32%	90,33%
Mediterranean	80,69%	86,54%
Pannonian	86,87%	93,48%
Steppic	87,03%	96,40%
Oversea department	96,41%	97,61%

	Accuracy better than 85%
	Accuracy between 80% and 85%
	Accuracy less than 80%

Table 10 summarizes Producer and User accuracies obtained by biogeographical region, for the blind and plausibility interpretations.

Table 10: Producer and user accuracy of CLC2012 given by land cover classes– Blind and plausibility analysis

	Code CLC	111	112	121	122	123	124	131	132	133	141	142	211	212	213	221	222	223	231	241	242	243	244
BLIND ANALYSIS	User accuracy	0,773	0,903	0,895	0,892	0,954	0,960	0,840	0,777	0,763	0,857	0,881	0,899	0,942	0,952	0,674	0,679	0,854	0,769	0,312	0,720	0,785	0,928
	Producer accuracy	0,831	0,891	0,743	0,663	0,878	0,965	0,806	0,804	0,633	0,663	0,719	0,937	0,966	0,941	0,792	0,768	0,886	0,802	0,207	0,625	0,652	0,751
PLAUSIBILITY ANALYSIS	User accuracy	0,791	0,948	0,938	0,919	0,954	0,981	0,868	0,860	0,793	0,887	0,929	0,932	0,979	0,984	0,766	0,740	0,898	0,846	0,412	0,798	0,844	0,930
	Producer accuracy	0,837	0,929	0,838	0,858	0,884	0,966	0,976	0,929	0,624	0,686	0,765	0,954	0,981	0,973	0,938	0,896	0,918	0,870	0,279	0,736	0,747	0,789
	Code CLC	311	312	313	321	322	323	324	331	332	333	334	335	411	412	421	422	423	511	512	521	522	523
BLIND ANALYSIS	User accuracy	0,873	0,847	0,856	0,690	0,851	0,780	0,779	0,861	0,780	0,744	0,800	0,985	0,858	0,854	0,570	0,849	0,728	0,902	0,984	0,968	0,860	0,989
	Producer accuracy	0,945	0,945	0,753	0,684	0,837	0,802	0,607	0,768	0,929	0,841	0,626	0,817	0,791	0,921	0,865	0,939	0,882	0,917	0,985	0,969	0,639	0,933
PLAUSIBILITY ANALYSIS	User accuracy	0,918	0,902	0,943	0,806	0,912	0,875	0,884	0,924	0,862	0,842	0,790	0,992	0,913	0,965	0,582	0,932	0,855	0,930	0,989	0,978	0,938	0,993
	Producer accuracy	0,976	0,976	0,817	0,813	0,939	0,847	0,734	0,867	0,963	0,955	0,784	0,868	0,804	0,958	0,928	0,951	0,958	0,986	0,987	0,982	0,676	0,998

	Accuracy better than 85%
	Accuracy between 80% and 85%
	Accuracy less than 80%

4.3.1.2. Main findings and discussion

The CLC2012 status product exceeds the accuracy requirements for the plausibility analysis. In addition, these results are relatively homogenous across the European zones or bio-geographical regions from an overall accuracy of 84.5% for the Black Sea region to 97% for the French overseas departments. For the blind analysis, the requirement is not met but results show that the product is close to the thematic threshold, i.e. 85 % (the overall accuracy at the pan-European scale assessed for CLC2012 is 83.6%).

The differences between the blind interpretation and plausibility results (for some particular biogeographical regions) highlights the complexity of the nomenclature in liaison with the complexity of the landscape and suggests that several interpretations are sometimes possible for a given land unit despite the detailed documentation available

The CLC2012 product shows both low producer and user accuracies for blind and plausibility analysis for the following land cover classes: 133, 221, 222, 241, 242, 243, 324 and 334. Reasons for main sources of errors can be summarised as follows:

- **Main sources of both omission and commission errors (producer and user accuracy):**
 - Class 111 "Continuous urban fabric": some difficulties in the visual interpretation of the urban density. Mainly the confusion concerns the class 112 "discontinuous urban fabric"
 - Class 132 "Dump sites": some confusion with the classes 131 "Mineral extraction sites" and 133 "Construction sites".
 - Class 133 "Construction sites": some confusion with artificial classes 112, 121, 122 due to temporal issues (progress of construction development)
 - Class 221 "Vineyards" and 222 "Fruit trees and berry plantations": confusion with the classes 242 "Complex cultivation patterns" when the density of plots planted with permanent crops is not enough sufficient and not representative (Density of permanent crops must occupy more than 75 %). Other possible confusion with the class 223 "Olive groves" in the Mediterranean countries. According to the type of plantation possible confusion between 221 and 222.
 - Class 231 "pasture". Possible confusion with 321 "Natural grassland"
 - Class 241 "annual crops associated with permanent crops". The use of the production imagery alone is not sufficient to characterise some of the complexity of the nomenclature (which leads to misunderstanding of the land-cover classes' content by the interpreters) and the use of very high resolution data such as google earth or aerial photography is necessary in order to identify the land-cover strata.
 - Class 242 and 243: "heterogeneous agricultural areas" including the complex cultivation pattern and the land principally occupied by agriculture with significant areas of natural vegetation. Classes in general underestimated or overestimated in the CLC production. Bad assessment of mosaic composed of permanent & annual crops / grassland or mixed with natural area. Classes easily generalizable where it is possible to identify other homogeneous land cover classes more than 25 ha inside the polygon such as 211, 231...
 - Class 321 "Natural grassland" possible confusion according to the country with 231 "pasture", 322 "Moors and heathland", 333 "sparsely vegetation", 324 "transitional woodland".
 - Class 323 "sclerophyllous vegetation" possible confusion with 321 "Natural grassland", 324 "transitional woodland", 333 "sparsely vegetation", in Mediterranean countries*

- Class 324 “transitional woodland” which characterizes both a dynamic aspect of the vegetation and a physiognomic state of the vegetation everywhere in Europe (from Boreal to Mediterranean region). Possible confusion with forest according to the forest growing 311 “ broad-leaved forest”/ 312 “coniferous forest” / 313” mixed forest” or 321 “Natural grassland” /323 “sclerophyllous vegetation”
- Class 332 “Bare rock” , 333 “sparsely vegetation” possible confusion between them.
- Class 334: burnt area” Class difficult to assess from the satellite images (acquisition dates of satellite images, and vegetation growing). Confusion with 324
- Class 421 “salt marshes” Possible confusion with humid grass and water 411 “ inland marshes”, and sometime with 423 “intertidal flats”.
- Class 423 “intertidal flats”. Possible confusion with 421 “salt marshes” , 522 “ estuaries”
- Main sources of omission errors (producer accuracy):
- Class 121 “industrial area”/122 “Road”. Confusion with 133 “construction sites” and vice-versa: the reason is the precision to identify the building or houses. Need to respect the date of the google earth regarding to the date of satellite image 2012.
- Class 132 “Dump site”/133 “construction site”. Internal confusion between these classes. Need to have ancillary data in order to identify 131 from 132. Class 133 could be classify into 112/ 121/ 122 according to progress of the construction site.
- Class 133 “construction site”: In direct relation with the progress of construction development (possible confusion with 112, 121, 122)
- Class 141 “green urban area”: Confused class when it’s localized around cities!
- Class 142 “sport and leisure facilities”- Need to analyses per zone (cottage communities confusion with 112)
- Class 241 “annual crops associated with permanent crops”. The use of the production imagery alone is not sufficient to characterise some of the complexity of the nomenclature (which leads to misunderstanding of the land-cover classes’ content by the interpreters) and the use of very high resolution data such as google earth or aerial photography is necessary in order to identify the land-cover strata.
- Class 242 and 243: “heterogeneous agricultural areas” including the complex cultivation pattern and the land principally occupied by agriculture with significant areas of natural vegetation. Classes in general underestimated or overestimated in the CLC production. Bad assessment of mosaic composed of permanent & annual crops / grassland or mixed with natural area. Classes easily generalizable where is it possible to identify other homogeneous land cover classes more than 25 ha inside the polygon such as 211, 231...
- Class 313 “mixed forest” possible confusion with 312 / 311 or 324
- Class 321 “Natural grassland” possible confusion according the country with 231, 322, 333, 324
- Class 324 “transitional woodland” Transitional woodland” which characterizes both a dynamic aspect of the vegetation and a physiognomic state of the vegetation everywhere in Europe (from Boreal to Mediterranean region). - Possible confusion with 31x or 321 /323
- Class 334”burnt area” Class difficult to assess from the satellite images (acquisition dates of satellite images, and vegetation growing). Confusion with 324
- Class 411 “inland marshes”. Possible confusion with humid grass and water
- Class 423 “intertidal flats”. Need topographic maps. Possible confusion with 421, 521, 522, 523
- Class 522 “estuaries” is difficult to delineate and need to have bathymetry data and other marine maps. Possible confusion with 511 or 523

The stratification and sampling design proved efficient ensuring a good representation of all Land-Cover classes whilst providing precise overall thematic accuracy results at EEA39, zones and biogeographical region level.

4.3.2. CLCCH2006-2012 change layer

4.3.2.1. Main observation at the European and bio-geographical region

The first observation is **an overestimation of changes** during the interpretation phase. This is because the satellite image used during the validation step allowed identifying a land cover change but in some case it's not clear which kind of change has occurred. Changes are identified and considered when the difference in the spectral radiometry of satellite images acquired between 2006 and 2012 corresponds to a real land cover or land use change.

For the blind analysis, the overall accuracy of CLCCH2006-2012 at the third level of nomenclature is 81.8%. The overall accuracy of CLCCH2006-2012 at the level 2 is 85.9%.

For the plausibility analysis, the overall accuracy of CLCCH2006-2012 at the third level is 88.6%. The overall accuracy of CLCCH2006-2012 at the second level is 91.2%.

The level 3 of the nomenclature permits a high number of changes combinations (44 x 43 classes = 1892 possible changes), which makes the analysis very complex. Since the goal of the validation is to provide an overall picture about the accuracy of CLC-Change, the analyses were focused essentially at CLC level 2. Main findings are only provided where significant differences are found between the validation.

The matrix of change at the European level (CLC level 2) is provided in Annex 6 and at the biogeographical region in Annex 8.

Table 11 shows the overall accuracy of CLCCH2006-2012 for each bio-geographical region at the second and third level both for the blind and the plausibility analysis. The plausibility analysis shows a very good overall accuracy both at the second and at the third level.

Confusion matrix of CLCCH2006-2012 per Bio-geographical region is presented in Annex 8 of the document.

Table 11: assessment of CLCCH2006-2012 for blind and plausibility analysis per bio-geographical region at the second and third CLC level

	Overall accuracy CLCCH2006-2012 level 3		Overall accuracy CLCCH2006-2012 level 2	
	Blind analysis CLC level 3	Plausibility analysis CLC level 3	Blind analysis CLC level 2	Plausibility analysis CLC level 2
Alpine	83,47%	91,94%	88,55%	94,62%
Anatolian	75,62%	85,93%	79,28%	88,01%
Arctic	83,12%	93,38%	93,16%	99,47%
Atlantic	85,89%	89,81%	87,97%	91,36%
BlackSea	81,54%	84,31%	86,32%	87,89%
Boreal	80,21%	86,69%	84,78%	90,06%
Continental	82,98%	90,25%	85,31%	91,76%
Macaronesia	79,95%	89,95%	83,49%	93,23%
Mediterranean	79,06%	85,54%	85,80%	89,78%
Pannonian	85,29%	92,25%	85,83%	92,86%
Steppic	86,99%	96,38%	86,99%	96,38%
Oversea departments	96,38%	97,58%	97,58%	98,07%

	Accuracy better than 85%
	Accuracy between 80% and 85%
	Accuracy less than 80%

4.3.2.2. Assessment of CLCCH2006-2012 overall accuracy per country or group of countries at the second and third level

Table 12: assessment of CLCCH2006-2012 for blind and plausibility analysis and confidence interval (95%) per country or group of countries at the second or third CLC level

		Overall accuracy CLCCH2006-2012 level 3				Overall accuracy CLCCH2006-2012 level 2	
Country or group of countries		BLIND ANALYSIS	Confidence interval of 95%	PLAUSIBILITY ANALYSIS	Confidence interval of 95%	BLIND ANALYSIS	PLAUSIBILITY ANALYSIS
Z1	Turkey	77,30%	2,84%	84,50%	2,47%	83,16%	88,57%
Z2	France	84,80%	3,36%	88,11%	3,15%	88,37%	90,88%
Z3	Spain	84,23%	2,69%	88,71%	2,28%	89,58%	92,01%
Z4	Sweden	81,42%	4,77%	89,51%	3,59%	82,17%	90,02%
Z5	Germany	80,59%	4,37%	88,12%	3,62%	83,30%	89,96%
Z6	Finland	72,92%	6,15%	80,67%	5,76%	85,92%	89,87%
Z7	Norway	93,07%	2,67%	96,02%	1,99%	95,00%	97,59%
Z8	Poland	91,52%	2,37%	93,80%	2,21%	92,35%	94,81%
Z9	Italy - Sardinia	75,23%	4,16%	84,74%	3,53%	80,20%	87,37%
Z10	United Kingdom - Ireland	85,60%	3,77%	91,67%	2,79%	87,09%	92,72%
Z11	Roumania	77,65%	4,30%	90,94%	2,61%	79,74%	92,05%
Z12	Greece	78,64%	3,67%	86,28%	3,04%	85,08%	90,63%
Z13	Bulgaria	86,52%	3,55%	94,36%	2,03%	89,02%	95,62%
Z14	Island	82,94%	7,95%	93,31%	4,91%	93,09%	99,46%
Z15	Hungaria	84,01%	3,98%	89,67%	3,31%	84,73%	90,54%
Z16	Portugal	68,26%	3,88%	80,87%	3,43%	78,92%	85,63%
Z17	Switzerland - Austria	75,27%	4,75%	86,39%	3,72%	80,48%	88,72%
Z18	Belgium - Danmark - Luxembourg - Netherlands	84,45%	4,21%	87,77%	4,05%	86,81%	89,20%
Z19	Albania - Serbia - Macedonia - Montenegro	80,87%	4,02%	92,61%	2,21%	84,18%	94,18%
Z20	Bosnia Herzegovina - Croatia - Slovenia	76,89%	5,21%	88,89%	3,82%	84,83%	90,79%
Z21	Czech Republic - Slovakia	84,10%	3,80%	91,82%	2,45%	84,23%	92,96%
Z22	Estonia - Latvia - Lithuania	86,95%	3,81%	90,17%	3,40%	87,82%	90,72%
Z23	French overseas departments	96,38%	1,03%	97,58%	0,90%	97,58%	98,07%
	GLOBAL ACCURACY	81,80%	0,95%	88,60%	0,79%	85,90%	91,23%

Table 12 shows the overall accuracy of CLCCH2006-2012 for each zone (country or group of countries) at the second and third level for the blind and plausibility analysis. The European countries have been gathered into 23 areas. Overall, the land cover changes are very good at the second level and good at the third level

Confusion matrixes of CLCCH2006-2012 per zone/country, group of countries are presented in Annex 7.

Table per country are presented in Annex 9 of the document. These include observations and discussion for the main problematic countries (overall accuracy CLCCH2006-2012 less than 80%).

4.3.2.1. Main findings and discussion

Table 13 below summarizes and explains the main differences encountered between the validation and production interpretation at level 2 of the nomenclature for the 2006-2012 changes.

Table 13: Main problematic flow observed at the European level (commission and omission)

CLC Change Classes	Weight	User Accuracy	Comments
32-31	282.016	0.730	Overestimation of forest regeneration. It was still forest in 2006 (31-31) or still transitional woodland in 2012 (32-32).
21-23	17.832	0.292	Overestimation of agriculture intensification flow. Possible confusion with temporary grass which have to be classified into 211. Confusion with 21-21 or 23-23
23-21	15.865	0.635	Overestimation of intensification of agriculture area It was still grass in 2012 (23-23) or already annual crops in 2012 (21-21)
21-13	7.260	0.818	Construction site area slightly overestimated.
33-32	7.236	0.771	Overestimation of change – it was already 32 in 2006 (32-32)
21-22	6.927	0.813	Difficult change to assess. Need VHR data both in 2006 and in 2012 in order to better analyse this change. Change overestimated. Often there were still annual crops (21-21) in the satellite images 2012 (confirmed by historical images in comparison with google earth).
21-12	5.836	0.802	Commission, with other land cover change and mainly with grassland (23) in 2006. Classic confusion and not important confusion.
32-33	5.483	0.645	Classical confusion (321, 322 /333, Mountain region, due to the density of vegetation and the weakness resolution of satellite images to appreciate the density of the vegetation Not a real change
22-21	5.252	0.499	Difficult change to assess. Need VHR data both in 2006 and in 2012 in order to better analyse this change. Change overestimated. Often there was still

			permanent crops (22-22) in the satellite images 2012 (Confirmed by historical images in google earth)
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CLC Change classes	Weight	Producer Accuracy	Comments
31-32	322.54	0.545	Underestimation of change. It should be analysed at country level because it's a significant change
32-31	249.77	0.824	Slight underestimation of forest regeneration
24-21	15.99	0.035	Intensification of agriculture areas – Missing of change
21-12	15.21	0.308	Problem of industrial building density or localisation of the point or the MMU.
23-21	14.69	0.686	Under-estimation of intensification of agriculture area.
13-12	13.13	0.309	Industrial areas into the construction site are more or less clearly identified.
23-24	10.90	0.025	
21-13	10.19	0.583	Some missing change in 2006- Building and houses into the construction site are more or less clearly identified. Omission from the classes 13-13. We have to consider the date of the satellite images used for the production and quality assessment. ,

5. Conclusions and recommendations

This validation exercise shows anyway satisfactory overall accuracies, both for CLC2012 and for CLCCH2006-2012 changes products, for the blind and the plausibility analysis. Indeed, the CLC2012 and CLCCH2006-2012 products show an overall accuracy which meets the product specification in the plausibility analysis (and nearly meets the threshold of 85% for the blind approach). The plausibility analysis metrics are significantly better than the blind interpretation because the plausibility analysis results in revised accuracy metrics based on the results of the blind analysis.

However, disparities exist from a country to another, from 74% (blind analysis) of overall accuracy for Finland to 96% for French overseas departments regarding the CLC2012 product.

The CLCCH2006-2012 change product also contains a high disparity in overall accuracy, from 68% (blind analysis) for Portugal to 96% for French overseas departments. There is an obvious link between landscape complexity and weaker overall accuracy.

	Overall accuracy CLC 2012		Overall accuracy CLCCH2006-2012 level 3		Overall accuracy CLCCH2006-2012 level 2	
	Blind analysis	Plausibility analysis	Blind analysis	Plausibility analysis	Blind analysis	Plausibility analysis
Alpine	84,29%	92,49%	83,47%	91,94%	88,55%	94,62%
Anatolian	77,15%	86,44%	75,62%	85,93%	79,28%	88,01%
Arctic	85,29%	93,75%	83,12%	93,38%	93,16%	99,47%
Atlantic	87,75%	91,30%	85,89%	89,81%	87,97%	91,36%
BlackSea	81,70%	84,45%	81,54%	84,31%	86,32%	87,89%
Boreal	84,35%	89,12%	80,21%	86,69%	84,78%	90,06%
Continental	83,91%	90,74%	82,98%	90,25%	85,31%	91,76%
Macaronesia	80,32%	90,33%	79,95%	89,95%	83,49%	93,23%
Mediterranean	80,69%	86,54%	79,06%	85,54%	85,80%	89,78%
Pannonian	86,87%	93,48%	85,29%	92,25%	85,83%	92,86%
Steppic	87,03%	96,40%	86,99%	96,38%	86,99%	96,38%
Oversea departments	96,41%	97,61%	96,38%	97,58%	97,58%	98,07%

Considering the plausibility approach the CLC2012 product show that the most problematic biogeographical regions are the ones located in the south of Europe including the Black Sea, Anatolian and Mediterranean region with respectively 84%, 86% and 87% overall accuracies. The results from the plausibility analysis of the CLCCH2006-2012 product show the same problematic biogeographical regions with respectively 84%, 86% and 86% overall accuracies.

It should be noted that lower level of changes were found by the validation team with respect to the ones in the production data (conducting to an overestimations of the changes). For the plausibility analysis, the overall accuracy of CLCCH2006-2012 at the second level is very good and higher than 90% while at the third level the overall accuracy is around 88%.

Although the overall accuracy obtained by the plausibility analysis improves the one from the blind analysis and exceeds the expected value, better results in terms of overall accuracy might be obtained by introducing an error typology.

A typology of mistake or confusion would allow classifying the relevance of the errors without affecting the final accuracy. Indeed “minor confusion” might be filtered such as:

- Natural grassland (321), heathland (322) and sparsely vegetation (333) in Anatolian region or Mountain region are sometime difficult to determine precisely.
- In Mediterranean region sclerophyllous vegetation (323) are also confused with transitional woodland (324) according to the density of trees.

- In Arctic and boreal region peatbogs (412) heathland and moor land (322) and natural grassland (321) are confused in some part and needed to have thematic data or to organize some field control.

There are **“minor error or normal confusion”** met regarding data source resolution and difficult to discriminate without field true or other very precise thematic data. They happen in transitions between neighbouring classes with uncertain thresholds such as broadleaves (or coniferous) forests vs. mixed forests, shrubs vs. low density forests, altitude grassland vs. sparsely vegetated areas. Ecotone and other transitional natural formation constitute part of these difficulties.

If we integrate this typology of “minor error or normal confusion” the overall accuracy will be still better than 90 %.

Some other errors can be explained by various reasons:

- A correct identification of certain CLC classes requires the use of topical ancillary data. These were missing during our control work (Forest, peat, sclerophyllous vegetation maps...)
- Errors might be related to the interpretation of CLC classes by the national team interpreters (333/421 for inland salt marshes with halophyte communities must be classified into 333 and not 421...).
- The location of the validation points could sometimes be confusing inside the selected polygon (e.g. on the boundary between two land cover types bigger than 25 ha).
- The spatial resolution of satellite images might not be sufficient to identify specific classes (e.g. young vineyards and orchards)
- Generalisation of some polygons including different land cover type.
- The temporal resolution of satellite images is not enough to distinguish grasslands from arable land with vegetation.
- Different acquisition period of satellite images (1 or two years) for the same area could induce some interpretation difference (notably for land cover changes) in forest area.
- Problem of the MMU for the land cover change assessment. Indeed in some cases the photo-interpreter could generalize the change while the check point itself falls in a part without change.

These errors and difficulties can only be overcome/resolved if:

- Very High Resolution satellite imagery is used for the production
- Multi-temporal HR satellite imagery is available (Landsat or Sentinel-2)
- Auxiliary data sources are available.
- Organisation of some field control

The analysis of the **validation results at class and disaggregated zone or biogeographical level** should provide insights on where the product could be improved thematically focusing on weaker classes and geographical regions.

Another recommendation would be to use the user and producer accuracy of the change classes (or regrouped change classes at Level 2 of the nomenclature) as an indicator of the quality of the change layer rather than overall accuracy which is always likely to be dominated by unchanged areas.

ANNEX

Annex 1. Confusion matrix of CLC2012 European level – Blind and plausibility analysis

A total of 25,182 sample units located according to the sampling design have been interpreted by an independent expert's team. Table 14 and Table 15 shows the confusion matrix between the validation results in column and the production map in line for both blind and plausibility analysis.

Table 14: Confusion matrix of CLC 2012 – blind analysis

CLC Classe	111	112	121	122	123	124	131	132	133	141	142	211	212	213	221	222	223	231	241	242	243	244	311	312	313	321	322	323	324	331	332	333	334	335	411	412	421	422	423	511	512	521	522	523 Total	User Acc
111	22,783	4,347	0,578						0,657	0,074																0,509		0,509																29,457	0,773
112	4,523	632,927	5,443	2,238	0,014		0,038		3,039	4,065	0,022	4,526					0,444	0,021		26,246	12,102		2,238	1,306	0,528	1,088			0,068															700,876	0,903
121		3,924	119,153	0,640	0,493		0,095	0,777	2,475	0,023	0,110	0,453			0,007		0,035	1,520		0,559	0,177			0,709	0,701	0,086	0,034		0,925			0,018				0,010				0,160				133,084	0,895
122		0,140	0,445	16,239	0,019		0,058		0,348			0,143						0,234		0,225	0,056		0,216	0,019		0,034			0,029															18,204	0,892
123	0,021	0,021	0,002	0,050	3,981						0,029							0,034			0,034									0,029														4,172	0,954
124		0,020	0,141		14,377				0,015		0,082	0,044					0,025	0,221									0,048																	14,973	0,960
131		0,261	0,280	0,068			33,663	0,161	0,775	0,103	0,114	0,280			0,020			0,836		0,068	0,327		0,087	0,179	0,203	0,109			0,892			0,230		0,034	0,016			0,059	1,289				40,055	0,840	
132		0,024	0,173	0,019			0,392	4,246	0,066	0,042	0,043	0,090						0,146						0,037				0,149			0,015									0,020			5,461	0,777	
133	0,050	1,241	1,278	0,437	0,008		0,226	0,052	15,930		0,597	0,355				0,040		0,096		0,141	0,091		0,010	0,004	0,062	0,009	0,042		0,056			0,136					0,005				0,025			20,891	0,763
141		0,366	0,293							12,980	0,399	0,198						0,566		0,187	0,064		0,026				0,014		0,014					0,019						0,017				15,145	0,857
142		1,375	0,058			0,277			0,029	1,213	46,897	0,625						1,170		0,156	0,059		0,017	0,475	0,030	0,103	0,028		0,148	0,012		0,152		0,341					0,042				53,204	0,881	
211		10,889	13,759	4,023			0,036	0,015			10,539	4594,710	2,360	0,049	13,046	16,776	0,080	117,047	6,805	208,774	90,644	6,755	0,121	0,061	1,304	0,007		0,012	13,684							0,023								5111,519	0,899
212		4,522	0,036								12,007	415,669	0,066					5,048		3,596									0,009															441,049	0,942
213											0,413	0,062	31,515					0,096		0,290			0,617							0,006								0,099						33,098	0,952
221		0,384	0,168								10,005	0,037		118,644	1,835	1,332	0,772	1,126	33,731	6,799		0,435				0,384		0,007	0,273															175,934	0,674
222		1,223	1,199								8,267	1,241		0,693	119,788	1,652	0,229	5,869	19,744	13,055		0,955	0,015		1,199			1,409																176,538	0,679
223		0,045									3,300				3,680	184,778		5,042	5,850	8,615		1,110	0,045	0,045			1,027	2,813															216,351	0,854	
231		19,393	3,614	0,082			0,334	0,023	0,153		4,035	85,863	1,576		1,701	0,161	0,330	1371,727		101,762	109,313		9,589	9,773	5,106	29,469	1,261	1,547	20,907			0,066	0,019		5,856	0,012				0,593				1784,264	0,769
241		1,376	0,096								0,888	0,022				0,028		1,973	7,838	6,185	4,590	1,802					0,096		0,216															25,111	0,312
242		8,472	6,546			0,250	0,016		0,030		0,542	82,342	6,022	1,872	14,756	9,087	9,681	69,095	4,918	763,987	71,319	3,011	4,376			0,250	0,214	0,007	3,500						0,193								1060,485	0,720	
243		5,242							0,003			53,843				1,594	1,485	56,804	6,296	45,721	901,866	1,485	14,975	6,017	6,431	3,385	0,503	5,905	26,540		6,277			2,391	0,503			1,784					1149,049	0,785	
244											1,311									1,311	2,980	127,138						2,588	3,036															137,053	0,928
311		0,038	1,275								7,160	0,013				0,891		4,205		0,023	40,247	11,584	2178,595	2,678	105,284	2,069	0,201	13,977	126,619							0,021					0,013	2494,893	0,873		
312		0,800	0,400	0,076			6,632				1,063	11,916						12,335		17,981			2,923	3062,486	202,111		0,034	9,662	257,417				2,812			23,662			0,030		4,956	3617,294	0,847		
313		2,084							1,428		0,542	2,064						3,939		1,216	6,878	1,470	32,688	52,745	1080,394	1,027	0,063		74,512					0,760	0,006								1261,818	0,856	
321		4,122	0,018						0,084	1,032		6,209					8,626	46,466		2,388	16,824	12,903	4,091	5,347	4,285	656,051	39,212	27,569	65,614	3,715	6,203	39,603	0,067	0,564			0,054					951,048	0,690		
322		4,024					0,010											1,057			2,693			4,194		30,204	614,609		32,247	0,203	3,937	28,784		0,028									721,988	0,851	
323			5,211									0,852					0,025	3,462		16,633		0,865		3,037	12,964		350,843	55,833														449,725	0,780		
324		2,448	0,014	0,615			0,057		0,046	0,008	0,081	8,294	2,789		0,950	1,970	0,041	5,393		1,784	42,760	3,120	49,863	91,835	24,292	31,886	23,835	13,743	1129,929	0,206	0,111	7,013	0,262		0,009	6,063		0,313					1449,733	0,779	
331	0,027						0,096																																						

CLC Class	111	112	121	122	123	124	131	132	133	141	142	211	212	213	221	222	223	231	241	242	243	244	311	312	313	321	322	323	324	331	332	333	334	335	411	412	421	422	423	511	512	521	522	523	Total	User Acc		
111	23,310	4,042	0,503						0,509	0,074																0,509		0,509																29,457	0,791			
112	4,515	664,721	3,157	2,242					2,348	2,085	2,015	0,117					0,006	2,238		10,013	2,208				3,537	0,533	0,070			1,069															700,876	0,948		
121		1,795	124,767	0,160	0,493		0,012	0,201		0,014	0,087	0,160	0,409					0,729		0,417	0,157				0,024	0,691	0,065	0,016		0,930		0,018			0,461										133,084	0,938		
122		0,068	0,364	16,724			0,058		0,150			0,091						0,202		0,225	0,024		0,216		0,036				0,046																	18,204	0,919	
123	0,021	0,021	0,036	0,050	3,981					0,029											0,034																									4,172	0,954	
124		0,020	0,082			14,683					0,082							0,057								0,048																				14,973	0,981	
131		0,244	0,204			34,743	0,054	0,625	0,103	0,114	0,082							0,706		0,454		0,054	0,104	0,055	0,109			0,968		0,107		0,034	0,016					1,270						40,046	0,868			
132		0,024	0,099	0,019			0,215	4,695	0,084		0,037	0,090						0,030											0,149										0,020						5,461	0,860		
133		0,937	1,039	0,202	0,008		0,315	0,098	16,541		0,585	0,714						0,107	0,063	0,034		0,007	0,004	0,039	0,009	0,017	0,013	0,054		0,025			0,005			0,014	0,025							20,857	0,793			
141		0,288	0,307							13,429	0,331	0,030						0,527	0,164	0,012		0,026					0,014				0,019															15,145	0,887	
142		0,764	0,027		0,262				0,005	0,288	49,321	0,625						0,923	0,060	0,059			0,006	0,319	0,030	0,170		0,139		0,067							0,030									53,094	0,929	
211		3,537	6,755	0,004			0,009	0,007		3,537	6,763	4763,782	0,195	0,049	1,751	1,065	0,056	86,277	6,755	135,673	71,162	6,755	0,102	8,415				0,036	8,801					0,005													5111,490	0,932
212		4,522	0,012							0,381	431,856	0,019			0,223			0,028		3,824																												

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Annex 2. Producer and user accuracy of CLC2012 per country or group of countries

Table 16: Producer, user and overall accuracy per country (from zone 1 to zone 6) – BLIND and PLAUSIBILITY analysis

CLASSES	Z1				Z2				Z3				Z4				Z5				Z6			
	Turkey				France				Spain				Sweden				Germany				Finland			
	Blind analysis		Plausibility analysis		Blind analysis		Plausibility analysis		Blind analysis		Plausibility analysis		Blind analysis		Plausibility analysis		Blind analysis		Plausibility analysis		Blind analysis		Plausibility analysis	
	Producer	User	Producer	User	Producer	User	Producer	User	Producer	User	Producer	User	Producer	User	Producer	User	Producer	User	Producer	User	Producer	User	Producer	User
111	0,656	1,000	0,665	1,000	0,700	1,000	0,700	1,000	0,800	0,989	0,800	0,995	0,800	1,000	0,800	1,000	0,800	1,000	0,900	1,000				
112	0,980	0,765	0,994	0,771	0,999	0,989	0,999	0,992	0,950	0,760	0,948	0,758	0,997	0,998	0,998	0,998	0,940	0,915	0,960	0,987	0,790	0,963	0,948	0,973
121	0,893	0,919	0,900	0,933	0,992	0,982	0,992	0,985	0,952	0,633	0,952	0,677	0,870	0,984	0,875	0,984	0,776	0,988	0,875	0,990	0,898	0,647	0,893	0,779
122	0,948	1,000	0,966	1,000	0,915	1,000	0,915	1,000	0,934	0,979	0,934	1,000	1,000	0,982	1,000	0,990	0,900	0,975	0,900	0,975	1,000	1,000	1,000	1,000
123	1,000	0,209	1,000	0,209	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	0,800	1,000	0,800	1,000	0,988	0,988	0,988	0,988
124	1,000	1,000	1,000	1,000	1,000	0,890	1,000	0,890	0,800	1,000	0,800	1,000	1,000	0,713	1,000	0,713	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000
131	0,862	0,546	0,881	1,000	0,740	0,994	0,740	0,994	0,987	1,000	0,987	1,000	0,923	0,947	0,923	0,947	0,919	0,947	0,963	0,955	0,855	0,835	0,861	0,905
132	0,513	1,000	0,513	1,000	0,600	1,000	0,800	0,863	0,934	0,940	0,934	0,940	0,900	0,658	0,900	0,658	0,900	0,954	0,900	0,954	0,475	1,000	0,766	1,000
133	0,807	0,624	0,857	0,908	0,696	0,806	0,758	0,830	0,820	0,802	0,767	0,786	0,976	0,969	0,976	0,090	0,738	0,159	0,766	0,162	0,860	0,163	0,916	0,884
141	0,802	1,000	0,802	1,000	1,000	0,504	1,000	1,000	0,752	1,000	0,752	1,000	0,987	1,000	0,987	1,000	0,850	0,553	0,850	0,553	0,611	0,866	0,722	1,000
142	0,809	1,000	0,809	1,000	0,806	0,996	0,855	0,989	1,000	0,773	1,000	0,773	1,000	1,000	1,000	1,000	0,852	0,975	0,951	0,978	0,908	0,977	0,952	0,977
211	0,841	0,882	0,873	0,924	0,920	0,966	0,920	0,976	0,960	0,969	0,980	0,969	0,700	0,991	0,720	0,991	0,916	0,948	0,950	0,953	0,917	0,706	0,979	0,750
212	0,930	0,971	0,984	0,973					0,974	0,987	0,974	0,999												
213	0,941	0,993	0,946	0,993	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000												
221	0,414	1,000	0,486	1,000	0,900	0,894	0,920	0,896	0,679	0,996	0,754	0,997					0,850	0,970	0,850	0,970				
222	0,707	0,863	0,739	0,874	0,501	0,982	0,501	0,992	0,708	0,936	0,781	0,989	0,600	1,000	0,600	1,000	0,500	0,959	0,550	1,000				
223	0,854	0,580	0,854	0,692	0,400	1,000	0,400	1,000	0,973	0,932	0,998	0,972												
231	0,762	0,584	0,843	0,667	0,860	0,849	0,880	0,890	0,743	0,510	0,743	0,552	0,501	0,181	0,601	0,251	0,636	0,858	0,717	0,986	0,220	0,058	0,220	0,059
241							0,200	1,000	0,700	0,074	0,900	0,103												
242	0,666	0,718	0,682	0,797	0,760	0,775	0,820	0,791	0,848	0,699	0,873	0,816	0,300	0,129	0,350	0,149	0,800	0,023	0,850	0,038				
243	0,769	0,716	0,823	0,766	0,940	0,480	0,960	0,530	0,872	0,583	0,872	0,661	0,867	0,686	0,867	0,642	0,550	0,063	0,750	0,107	0,646	0,848	0,708	0,901
244									0,925	0,877	0,925	0,889												
311	0,822	0,918	0,884	0,943	0,856	0,986	0,899	0,992	0,950	0,972	0,975	0,989	0,860	1,000	0,900	1,000	0,780	0,938	0,840	0,956	0,635	0,768	0,727	0,999
312	0,905	0,910	0,961	0,937	0,880	0,947	0,940	0,984	0,900	0,953	0,925	0,933	0,874	0,976	0,935	0,991	0,840	0,987	0,940	1,000	0,686	0,873	0,702	0,970
313	0,744	0,774	0,843	0,859	0,914	0,624	0,973	0,652	0,824	0,900	0,950	0,963	0,911	0,896	0,979	0,902	0,840	0,596	0,979	0,690	0,815	0,565	0,926	0,650
321	0,774	0,610	0,883	0,729	0,440	0,801	0,560	0,866	0,599	0,734	0,675	0,871	0,900	0,337	0,950	0,653	0,351	0,861	0,551	0,925	0,533	0,195	0,733	0,941
322					0,550	0,323	0,600	0,436	0,727	0,999	0,777	0,999	0,920	0,996	0,980	1,000	0,600	0,981	0,600	0,981	0,500	0,574	0,611	0,822
323	0,700	0,638	0,750	0,847	1,000	1,000	1,000	0,959	0,725	0,835	0,849	0,895												
324	0,869	0,792	0,904	0,836	0,859	0,708	0,930	0,865	0,784	0,484	0,832	0,585	0,888	0,548	0,954	0,727	0,752	0,413	0,862	0,454	0,609	0,598	0,861	0,633
331	0,900	0,944	0,927	0,896	0,900	0,828	1,000	0,842	0,888	1,000	0,986	1,000	0,600	1,000	0,600	1,000	0,477	0,492	0,618	0,604				
332	0,669	0,927	0,716	0,932	0,750	0,936	0,800	0,940	0,740	1,000	0,740	1,000	0,750	0,918	0,900	0,930	0,700	1,000	0,800	1,000	0,367	1,000	0,577	1,000
333	0,571	0,797	0,729	0,917	0,400	0,673	0,500	0,837	0,548	0,894	0,794	0,929	0,767	0,818	0,933	0,963	0,585	0,678	0,708	1,000	0,950	0,639	0,950	0,466
334			0,500	0,494	1,000	0,102	1,000	1,000	0,931	0,951	0,861	1,000	0,500	0,008	0,750	1,000								
335	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	0,700	1,000	0,700	1,000								
411	0,927	0,917	0,965	0,917	0,997	0,997	0,997	0,997	1,000	0,961	1,000	1,000	0,867	0,998	0,867	0,998	0,900	0,678	0,900	0,816	0,850	0,969	0,900	0,969
412					1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	0,975	1,000	0,975	0,895	0,994	0,994	1,000	0,805	0,694	0,957	0,843
421	0,101	1,000	0,101	1,000	0,900	1,000	0,900	1,000	0,900	0,942	0,900	0,942	0,800	1,000	0,800	1,000	0,900	0,798	0,900	0,812	0,632	0,498	0,833	0,452
422	0,800	1,000	0,800	1,000	0,900	1,000	1,000	1,000	0,800	1,000	0,900	1,000												
423					1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000					0,700	0,749	0,900	0,794				
511	0,792	1,000	0,803	1,000	0,950	1,000	0,950	1,000	0,900	1,000	0,900	1,000	0,950	1,000	0,950	1,000	1,000	0,908	1,000	0,908	0,895	1,000	0,895	1,000
512	0,999	0,964	0,999	0,964	0,949	0,922	0,949	0,922	1,000	0,981	1,000	0,983	0,980	0,998	0,980	0,998	0,965	1,000	0,999	1,000	1,000	0,985	1,000	0,986
521	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	0,333	1,000	0,667	1,000				
522	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	0,947	1,000	0,947	0,800	1,000	0,800	1,000	1,000	1,000	1,000	1,000				
523	0,910	1,000	1,000	1,000	1,000	1,000	1,000	1,000	0,900	1,000	0,900	1,000	1,000	1,000	1,000	1,000	0,667	0,022	0,667	1,000	0,667	1,000	0,733	1,000
Overall Acc	0,786		0,852		0,863		0,893		0,851		0,894		0,879		0,931		0,828		0,890		0,743		0,817	

Table 17: Producer, user and overall accuracy per country (from zone 07 to zone 12) - BLIND and PLAUSIBILITY analysis

CLASSES	Z7				Z8				Z9				Z10				Z11				Z12			
	Norway				Poland				Italy Sardinia				United Kingdom				Roumania				Greece			
	Blind analysis		Plausibility analysis		Blind analysis		Plausibility analysis		Blind analysis		Plausibility analysis		Blind analysis		Plausibility analysis		Blind analysis		Plausibility analysis		Blind analysis		Plausibility analysis	
	Producer	User	Producer	User	Producer	User	Producer	User	Producer	User	Producer	User	Producer	User	Producer	User	Producer	User	Producer	User	Producer	User	Producer	User
111	0,933	1,000	0,933	1,000	0,900	1,000	0,900	1,000	0,898	0,753	0,900	0,754	0,600	1,000	0,650	1,000			0,500	1,000	0,933	1,000	0,933	1,000
112	0,996	0,994	0,997	0,996	0,849	0,990	0,900	0,996	0,917	0,844	0,937	0,894	0,933	0,803	0,966	0,812	0,560	0,861	0,900	0,995	0,960	0,953	1,000	0,969
121	0,882	0,964	0,887	0,964	0,900	0,992	0,951	0,987	0,949	0,493	0,997	0,566	0,843	0,930	0,894	0,946	0,934	0,660	0,984	0,995	0,883	0,680	0,891	0,655
122	0,643	1,000	0,597	1,000	0,838	0,893	0,984	0,953	0,807	1,000	0,845	0,990	0,800	0,970	0,800	0,970	0,602	0,695	0,763	0,959	0,895	0,863	0,895	0,854
123	1,000	0,320	1,000	0,310	1,000	1,000	1,000	1,000	0,907	1,000	0,907	1,000	0,950	1,000	0,950	1,000	1,000	0,874	1,000	1,000	1,000	1,000	1,000	1,000
124	0,933	1,000	0,933	1,000	0,852	1,000	0,934	1,000	0,950	1,000	0,950	1,000	0,950	1,000	1,000	1,000	0,900	1,000	1,000	1,000	1,000	1,000	1,000	1,000
131	0,842	0,986	0,927	0,987	0,905	0,972	0,910	0,974	0,718	0,989	0,764	0,985	0,857	0,974	0,859	0,984	0,718	0,988	0,873	0,990	0,876	0,923	0,938	0,897
132	0,342	1,000	0,342	1,000	0,850	1,000	0,950	1,000	0,800	0,473	0,900	1,000	0,650	1,000	0,750	0,943	1,000	0,404	1,000	0,968	1,000	1,000	1,000	1,000
133	0,725	0,410	0,725	0,380	0,679	0,735	0,774	0,989	0,936	0,976	0,941	0,978	0,588	0,225	0,614	0,387	0,387	0,246	0,721	0,064	0,798	0,831	0,760	0,953
141	0,950	1,000	0,950	1,000	0,850	0,622	0,900	0,840	0,850	1,000	0,900	1,000	0,795	0,461	0,844	0,960	0,933	0,947	0,933	1,000	0,800	1,000	0,800	1,000
142	0,901	0,998	0,938	1,000	0,850	0,973	0,950	0,991	1,000	0,158	1,000	0,158	0,898	0,996	0,948	0,828	0,633	0,128	0,962	1,000	0,856	1,000	0,856	1,000
211	0,899	0,994	0,949	0,999	1,000	0,972	1,000	0,983	0,738	0,902	0,818	0,931	0,958	0,960	0,958	0,961	0,880	0,944	0,960	0,978	0,828	0,971	0,828	0,969
212									0,800	0,991	1,000	1,000							0,333	1,000	0,998	1,000	0,999	1,000
213									0,943	0,866	0,996	1,000					1,000	1,000	1,000	0,619	0,873	0,954	0,937	0,987
221									0,451	0,857	0,700	1,000					0,263	0,311	0,368	0,997	0,300	1,000	0,650	1,000
222					0,896	1,000	0,945	0,672	0,751	0,766	0,801	0,952	0,600	1,000	0,733	1,000	0,399	0,361	0,548	0,912	0,800	0,679	0,840	0,690
223									0,750	0,912	0,850	0,887									0,733	0,972	0,767	0,946
231	0,800	1,000	0,900	0,995	0,699	0,931	0,749	0,998	0,700	0,243	0,800	0,376	0,897	0,949	0,958	0,959	0,660	0,793	0,900	0,899	0,679	0,226	0,798	0,429
241									0,350	0,257	0,450	0,362									0,400	0,210	0,400	0,210
242	0,800	0,809	0,900	1,000	0,800	0,542	0,800	0,577	0,540	0,496	0,700	0,706	0,550	0,188	0,550	0,976	0,700	0,394	0,950	0,552	0,633	0,650	0,700	0,670
243	0,950	0,935	0,950	0,950	0,683	0,579	0,842	0,684	0,600	0,613	0,700	0,736	0,673	0,366	0,755	0,561	0,571	0,370	0,694	0,822	0,886	0,677	0,885	0,759
244									0,900	0,443	0,950	0,471												
311	0,826	0,994	0,870	1,000	0,799	0,998	0,800	0,999	0,920	0,889	0,960	0,958	0,750	0,867	0,800	0,933	0,939	0,995	0,979	0,980	0,884	0,869	0,940	0,944
312	0,918	0,986	0,959	0,993	0,998	0,965	0,998	0,981	0,936	1,000	0,956	1,000	0,871	0,785	0,876	0,945	0,667	0,994	0,881	1,000	0,866	0,918	1,000	1,000
313	0,940	0,585	1,000	0,594	0,994	0,956	0,995	0,956	0,798	0,867	0,951	0,919	0,720	0,844	0,820	0,829	0,939	0,822	0,980	0,967	0,867	0,643	0,934	0,875
321					0,500	0,953	0,650	1,000	0,750	0,752	0,900	0,912	0,789	0,815	0,895	0,927			0,500	0,923	0,569	0,771	0,682	0,852
322	0,960	0,948	1,000	0,966	1,000	0,524	1,000	0,660	0,800	0,578	0,900	0,882	0,850	0,699	0,850	0,904	0,824	0,992	0,941	0,993	0,600	0,202	0,750	0,971
323									0,799	0,722	0,902	0,806									0,849	0,926	0,923	0,930
324	0,902	0,435	0,953	0,603	0,329	0,448	0,618	0,697	0,437	0,414	0,685	0,593	0,830	0,694	0,863	0,705	0,747	0,189	0,845	0,385	0,745	0,679	0,906	0,777
331	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	0,862	0,894	1,000	0,929	0,800	0,930	0,950	1,000	0,700	0,087	0,900	0,149	0,800	0,926	0,867	1,000
332	0,900	0,955	1,000	1,000	0,250	1,000	0,750	1,000	0,900	0,985	0,950	0,995	0,300	1,000	0,700	1,000	0,222	0,549	0,333	1,000	0,400	0,267	0,533	1,000
333	0,979	0,968	0,979	1,000	0,900	0,881	1,000	1,000	0,750	0,854	0,850	0,925	0,841	0,743	0,947	0,988	0,875	0,028	1,000	0,830	0,423	0,718	0,793	0,970
334			1,000	1,000					0,671	1,000	0,466	0,934	0,333	1,000	1,000	1,000					0,878	0,648	0,857	0,447
335	1,000	0,726	1,000	1,000					0,850	1,000	0,950	1,000												
411	1,000	1,000	1,000	1,000	0,700	0,330	0,950	0,400	0,892	0,977	0,940	1,000	0,695	1,000	0,851	1,000	0,835	0,920	0,885	0,844	0,867	0,521	0,867	0,978
412	0,939	1,000	1,000	1,000	0,714	0,545	0,786	1,000			1,000	1,000	0,720	0,987	0,920	0,986								
421									0,753	0,985	0,901	0,987	1,000	0,893	1,000	0,923					0,933	0,820	0,933	0,855
422									0,800	0,695	1,000	0,740									1,000	1,000	1,000	1,000
423	0,750	1,000	0,800	1,000									0,650	1,000	0,800	1,000								
511	1,000	1,000	1,000	1,000	0,900	1,000	0,900	1,000	0,752	0,444	0,901	1,000	0,900	0,957	0,900	0,957	0,950	0,984	0,950	1,000	0,933	0,934	1,000	1,000
512	1,000	1,000	1,000	1,000	0,999	0,981	0,999	0,994	0,999	0,981	1,000	0,981	1,000	0,988	1,000	0,988	0,901	0,900	0,901	0,902	1,000	1,000	1,000	1,000
521					1,000	1,000	1,000	1,000	0,900	0,974	0,900	0,974	0,800	1,000	0,800	1,000	1,000	1,000	1,000	1,000	1,000	0,930	1,000	1,000
522													0,750	0,709	0,800	0,958					0,667	1,000	0,667	1,000
523	0,933	1,000	0,933	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	0,892	1,000	0,908	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000
Overall Acc	0,932		0,966		0,921		0,942		0,760		0,850		0,871		0,927		0,789		0,912		0,802		0,873	

Table 18: Producer, user and overall accuracy per country (from zone 13 to zone 18) - BLIND and PLAUSIBILITY analysis

CLASSES	Z13				Z14				Z15				Z16				Z17				Z18			
	Bulgaria				Island				Hungaria				Portugal				Swiss Austria				Belgium-Danmark- Luxembourg- Netherlands			
	Blind analysis		Plausibility analysis		Blind analysis		Plausibility analysis		Blind analysis		Plausibility analysis		Blind analysis		Plausibility analysis		Blind analysis		Plausibility analysis		Blind analysis		Plausibility analysis	
	Producer	User	Producer	User	Producer	User	Producer	User	Producer	User	Producer	User	Producer	User	Producer	User	Producer	User	Producer	User	Producer	User	Producer	User
111	1,000	1,000	1,000	1,000					1,000	1,000	1,000	1,000	0,650	0,384	0,900	0,476	0,850	1,000	0,850	1,000	0,950	0,177	0,950	0,177
112	0,750	0,957	1,000	0,985	1,000	0,944	1,000	0,944	0,899	0,991	0,950	0,991	0,901	0,666	0,901	0,672	0,900	0,901	0,900	0,989	0,800	0,777	0,802	0,855
121	0,795	0,344	0,949	0,980	0,816	1,000	0,878	1,000	0,932	0,360	0,992	0,415	0,984	0,751	0,992	0,762	0,946	0,909	0,946	0,982	0,843	0,924	0,938	0,987
122	0,769	0,186	0,808	0,903	1,000	1,000	1,000	1,000	0,844	0,106	0,872	0,828	0,955	0,968	1,000	0,969	0,890	0,970	0,918	1,000	0,996	0,283	0,996	0,317
123	1,000	0,628	1,000	0,628	1,000	0,619	1,000	0,619	1,000	0,462	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	0,952	0,985	0,952	0,985
124	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	0,800	1,000	1,000	1,000	0,906	1,000	1,000	1,000	0,901	1,000	1,000	1,000	1,000	1,000	1,000	1,000
131	0,566	0,723	0,571	0,995	1,000	1,000	1,000	1,000	0,758	0,985	0,875	1,000	0,818	1,000	0,818	1,000	0,727	0,152	0,856	0,996	0,825	0,986	0,825	0,986
132	1,000	1,000	1,000	1,000					0,737	0,948	0,859	0,991	0,823	1,000	0,911	1,000			1,000	1,000	0,733	0,252	0,733	0,896
133	0,573	0,662	0,573	1,000	0,323	1,000	0,488	1,000	0,289	0,495	0,572	0,724	0,811	0,887	0,847	0,891	0,200	0,326	0,400	1,000	0,821	0,676	0,874	0,560
141	0,900	0,953	0,900	1,000	0,800	1,000	1,000	1,000	1,000	0,941	1,000	0,941	0,600	1,000	0,700	1,000	0,750	1,000	0,850	1,000	0,865	0,787	0,910	0,251
142	0,842	0,300	0,842	0,444	0,980	1,000	0,980	1,000	0,902	1,000	0,902	1,000	0,941	0,935	0,941	0,961	0,637	0,853	0,727	0,926	0,906	0,649	0,956	0,663
211	0,919	0,965	0,979	0,992	1,000	0,116	1,000	0,118	0,934	0,954	0,955	0,973	0,757	0,737	0,878	0,820	0,660	0,949	0,700	0,972	0,937	0,927	0,937	0,958
212													0,784	0,768	0,822	0,984								
213	1,000	1,000	1,000	1,000					0,500	0,778	1,000	0,875	0,953	1,000	1,000	1,000								
221	0,745	0,695	0,799	0,998					0,899	0,452	0,948	0,879	0,718	0,835	0,857	0,858	0,900	0,638	1,000	0,665	1,000	0,611	1,000	1,000
222	0,702	0,281	0,852	0,735					0,804	0,789	0,909	0,878	0,748	0,742	0,752	0,717	0,600	1,000	0,700	1,000	0,654	1,000	0,750	1,000
223													0,781	0,996	0,827	0,976								
231	0,757	0,701	0,917	0,691	0,897	0,991	0,934	0,992	0,784	0,817	0,840	0,919	0,659	0,494	0,903	0,689	0,473	0,584	0,842	0,740	0,850	0,804	0,850	0,846
241													0,250	0,773	0,350	0,878								
242	0,199	0,459	0,449	0,959	0,789	1,000	0,986	1,000	0,738	0,847	0,802	0,673	0,650	0,610	0,750	0,600	0,800	0,338	0,900	0,525	0,720	0,823	0,800	0,919
243	0,842	0,704	0,947	0,863					0,814	0,760	0,904	0,508	0,801	0,688	0,801	0,854	0,500	0,364	0,750	0,446	0,650	0,813	0,850	0,771
244													0,950	0,656	0,950	0,781								
311	0,979	0,991	1,000	0,992	0,611	1,000	0,778	1,000	0,793	0,929	0,878	0,968	0,734	0,959	0,816	0,973	0,842	0,879	0,899	0,991	0,991	0,809	0,997	0,971
312	0,860	0,998	0,980	0,998	0,500	1,000	0,700	1,000	0,808	0,981	0,953	0,914	0,959	0,853	0,959	0,840	0,734	0,949	0,857	0,974	0,796	0,933	0,819	0,850
313	0,898	0,926	0,939	1,000	0,667	0,845	0,667	0,891	0,768	0,730	0,833	0,789	0,827	0,981	0,941	0,995	0,920	0,797	1,000	0,908	0,821	0,880	0,894	0,897
321	0,750	0,925	0,800	0,975	0,890	0,969	0,951	0,972	0,650	0,975	0,700	0,999	0,598	0,281	0,801	0,577	0,840	0,835	0,997	0,920	0,635	0,843	0,660	0,880
322	0,800	0,475	0,929	0,595	0,850	0,962	0,950	0,991					0,802	0,454	1,000	0,779	0,842	0,823	0,842	1,000	0,701	0,977	0,843	1,000
323													0,947	0,947	0,947	0,884								
324	0,796	0,655	0,897	0,803	0,419	0,259	0,472	0,819	0,671	0,485	0,848	0,740	0,691	0,836	0,851	0,912	0,620	0,102	0,845	0,247	0,797	0,621	0,855	0,728
331	0,700	1,000	0,800	1,000	0,881	0,851	0,914	0,972					0,867	1,000	0,933	1,000	0,200	1,000	0,200	1,000	0,900	1,000	0,950	1,000
332	0,529	1,000	0,529	1,000	0,802	0,899	0,901	0,948					0,750	1,000	1,000	1,000	0,849	0,937	0,900	0,950				
333	0,632	0,717	0,684	0,852	0,832	0,593	0,931	1,000	1,000	0,837	1,000	1,000	0,649	1,000	0,900	1,000	0,783	0,763	0,836	0,925				
334													0,868	1,000	0,892	1,000								
335					1,000	0,814	1,000	0,815									0,940	0,991	0,980	0,999				
411	0,800	1,000	0,900	0,714	0,600	0,408	0,750	0,466	0,750	0,998	0,800	0,998	1,000	1,000	1,000	1,000	0,850	0,996	0,850	0,996	0,884	0,918	0,932	0,972
412	1,000	1,000	1,000	1,000	0,789	0,988	0,944	0,997	0,800	1,000	0,800	1,000	1,000	1,000	1,000	1,000	0,600	1,000	0,600	1,000	0,543	1,000	0,839	1,000
421					0,800	0,830	1,000	1,000					0,950	0,846	0,950	1,000					0,850	0,957	0,850	0,967
422	1,000	1,000	1,000	1,000									0,900	0,927	1,000	1,000								
423					0,750	1,000	0,800	1,000					0,800	1,000	1,000	1,000					0,667	0,747	0,867	1,000
511	0,600	1,000	1,000	1,000	0,931	0,918	0,931	1,000	1,000	1,000	1,000	1,000	1,000	0,752	1,000	0,752	1,000	1,000	1,000	1,000	0,793	0,540	0,793	0,908
512	0,901	0,939	0,901	1,000	1,000	0,985	1,000	0,985	0,950	0,897	0,998	0,906	0,856	1,000	0,904	1,000	1,000	1,000	1,000	1,000	0,802	0,985	0,901	0,987
521	1,000	1,000	1,000	1,000	1,000	0,921	1,000	0,921					0,800	1,000	1,000	1,000					1,000	0,674	1,000	0,674
522					0,400	0,727	1,000	0,870					0,500	0,774	0,500	0,774					1,000	0,101	1,000	0,101
523	1,000	0,827	1,000	0,827	1,000	1,000	1,000	1,000					1,000	1,000	1,000	1,000					1,000	0,889	1,000	0,960
Overall Acc	0,869		0,945		0,851		0,937		0,861		0,913		0,759		0,848		0,761		0,868		0,848		0,882	

Table 19: Producer, user and overall accuracy per country (from zone 19 to zone 23 - BLIND and PLAUSIBILITY analysis)

CLASSES	Z19				Z20				Z21				Z22				Z23			
	Albania - Serbia - Macedonia - Montenegro				Bosnia Herzegovina - Croatia -Slovenia				Czech Republic - Slovakia				Estonia - Latvia- Lithuania				French overseas departments			
	Blind analysis		Plausibility analysis		Blind analysis		Plausibility analysis		Blind analysis		Plausibility analysis		Blind analysis		Plausibility analysis		Blind analysis		Plausibility analysis	
	Producer	User	Producer	User	Producer	User	Producer	User	Producer	User	Producer	User	Producer	User	Producer	User	Producer	User	Producer	User
111	1,000	1,000	1,000	1,000	0,200	1,000	0,600	1,000	1,000	0,928	1,000	1,000	0,500	1,000	0,500	1,000	1,000	1,000	1,000	1,000
112	1,000	0,999	1,000	1,000	0,949	0,959	0,949	0,974	0,993	0,871	0,996	0,997	0,995	0,982	0,998	0,982	0,993	0,846	0,993	0,878
121	0,924	0,982	0,969	0,992	0,819	0,953	0,909	0,983	0,981	0,516	0,989	0,991	0,903	0,976	0,903	0,984	0,896	1,000	0,896	1,000
122	0,727	0,737	1,000	0,928	0,700	1,000	0,744	1,000	0,893	0,919	0,966	0,982	0,955	1,000	0,955	1,000	0,916	1,000	0,916	1,000
123	1,000	0,777	1,000	0,777	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000
124	1,000	0,882	1,000	0,937	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000
131	0,874	0,882	0,903	0,960	0,839	0,963	0,943	0,918	0,813	0,921	0,862	0,988	0,892	0,950	0,929	0,960	1,000	1,000	1,000	1,000
132	0,941	0,931	0,941	0,979	0,667	1,000	0,667	1,000	0,593	0,874	0,818	0,980	0,768	1,000	0,904	1,000	1,000	1,000	1,000	1,000
133	0,775	0,814	0,915	0,853	0,769	0,832	0,873	0,783	0,339	0,816	0,627	0,961	0,568	0,568	0,640	1,000	1,000	0,249	1,000	0,331
141	0,910	1,000	0,910	1,000	1,000	1,000	1,000	1,000	0,700	1,000	0,900	1,000	0,950	1,000	1,000	1,000	1,000	1,000	1,000	1,000
142	0,818	0,924	0,939	1,000	0,700	0,967	0,700	1,000	0,924	0,246	0,930	0,973	0,828	1,000	0,852	0,976	1,000	1,000	1,000	1,000
211	0,734	0,981	0,957	0,997	0,898	0,651	0,998	0,676	0,938	0,956	0,999	0,976	0,988	0,867	0,990	0,870	0,950	1,000	0,950	1,000
212	0,894	0,965	0,894	0,965	1,000	1,000	1,000	1,000												
213	1,000	1,000	1,000	1,000													1,000	1,000	1,000	1,000
221	0,734	0,226	0,838	0,990	0,502	0,393	0,509	1,000	0,830	0,930	0,830	0,930								
222	0,542	0,190	0,787	1,000	0,595	0,981	0,607	0,981	0,607	0,952	0,646	0,975	0,700	1,000	0,800	1,000	0,700	0,987	0,700	0,987
223	0,485	1,000	0,582	1,000	0,255	1,000	0,454	0,965												
231	0,566	0,520	0,733	0,745	0,749	0,769	0,899	0,860	0,767	0,931	0,835	0,943	0,496	0,851	0,598	0,881	0,691	0,994	0,789	0,995
241							1,000	1,000												
242	0,825	0,707	0,975	0,900	0,700	0,926	0,850	0,987	1,000	0,598	1,000	0,600	0,850	0,864	0,850	0,999	0,900	0,725	0,900	0,725
243	0,868	0,786	0,918	0,925	0,800	0,720	0,840	0,978	0,947	0,947	0,947	0,948	0,901	0,873	0,950	0,879	0,749	0,771	0,751	0,771
244																				
311	0,923	0,920	0,982	0,984	0,860	0,901	0,940	0,954	0,931	0,985	0,955	0,988	0,846	0,865	0,848	0,927	1,000	0,973	1,000	0,984
312	0,832	0,898	0,941	0,952	0,841	0,655	0,940	0,898	0,601	0,986	0,827	0,995	0,996	0,961	0,998	0,966	1,000	1,000	1,000	1,000
313	0,818	0,758	0,919	0,817	0,700	0,737	0,901	0,956	0,851	0,991	0,930	0,871	0,896	0,956	0,949	0,971			1,000	1,000
321	0,755	0,814	0,896	0,941	0,600	0,697	0,750	0,880	0,833	0,991	0,833	0,991	0,559	1,000	0,712	1,000	0,996	1,000	0,996	1,000
322	0,667	0,237	0,800	1,000	0,550	0,197	0,850	1,000	1,000	0,943	1,000	0,917	0,650	0,336	0,650	0,895	0,880	0,699	0,990	0,878
323	0,677	0,786	0,805	0,813	0,900	0,290	0,950	0,325									0,600	0,994	0,900	0,996
324	0,768	0,744	0,883	0,860	0,665	0,762	0,880	0,875	0,887	0,256	0,890	0,506	0,842	0,779	0,906	0,856	0,656	0,729	0,753	0,843
331	0,935	0,594	0,935	0,787			1,000	1,000					0,933	0,775	0,933	0,874	1,000	0,202	1,000	0,202
332	0,402	0,915	0,721	0,951	0,600	0,723	0,900	1,000	0,500	1,000	0,500	1,000					1,000	1,000	1,000	1,000
333	0,756	0,846	0,933	0,886	0,387	0,888	0,628	0,968	0,857	0,560	0,857	0,718					0,700	0,665	0,700	1,000
334	0,847	1,000	0,863	1,000	0,691	0,926	0,718	1,000												
335																				
411	0,785	0,977	1,000	0,982	0,756	0,935	0,756	1,000	0,750	0,802	0,750	1,000	0,900	0,601	0,900	0,609	0,965	0,993	0,999	0,993
412	1,000	1,000	1,000	1,000			1,000	1,000	0,400	0,915	0,500	0,931	0,849	0,958	0,999	0,964				
421	1,000	1,000	1,000	1,000			1,000	0,975					1,000	1,000	1,000	1,000				
422	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000												
423																	1,000	1,000	1,000	1,000
511	0,729	0,970	0,892	1,000	0,894	1,000	0,901	1,000	0,900	1,000	0,900	1,000	1,000	1,000	1,000	1,000	0,800	0,760	0,900	1,000
512	1,000	0,981	1,000	0,995	0,950	0,999	1,000	1,000	0,946	0,971	0,998	0,973	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000
521	1,000	1,000	1,000	1,000			1,000	1,000					1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000
522																	0,796	0,953	1,000	1,000
523	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000					1,000	1,000	1,000	1,000	1,000	0,998	1,000	1,000
Overall Acc	0,821		0,935		0,774		0,893		0,857		0,933		0,886		0,917		0,964		0,976	

Zone 1: part of Turkey Confusion matrix CLC2012 – Blind and plausibility analysis

Table 20: Confusion matrix CLC 2012 for zone 1 (Turkey) – BLIND ANALYSIS - In column the validation and line the production

Overall accuracy = 78.6%

Table 21: Confusion matrix CLC 2012 for zone 1 (Turkey) – PLAUSIBILITY ANALYSIS - In column the validation and line the production

CLC Clas	111	112	121	122	123	124	131	132	133	141	142	211	212	213	221	222	223	231	241	242	243	244	311	312	313	321	322	323	324	331	332	333	334	335	411	421	422	511	512	521	522	523	Total	User Acc				
111	1,83	0,92																																									2,757	0,665				
112		17,30							0,03											0,03							0,03																	17,400	0,994			
121		0,06	4,04		0,18															0,18							0,03																		4,489	0,900		
122			0,03	0,80																																									0,825	0,966		
123					0,05																																								0,048	1,000		
124						0,63																																							0,630	1,000		
131							2,64		0,11											0,11									0,14															3,000	0,881			
132								0,04				0,04																																0,077	0,513			
133		0,04	0,22						1,72																		0,01						0,01						0,01					2,005	0,857			
141			0,04							0,17																																		0,217	0,802			
142		0,15									0,67																0,01																		0,834	0,809		
211												213,88	0,02					7,80		3,35	20,05																								245,100	0,873		
212			2,26									0,03	147,39					0,02		0,04																									149,734	0,984		
213												0,02	0,02	3,14						0,14																									3,325	0,946		
221		0,19												2,18	0,77					0,96	0,19						0,19																		4,483	0,486		
222																19,14			0,81	2,42	3,54																								25,908	0,739		
223																0,48	8,43		0,48	0,48																									9,871	0,854		
231									0,02			1,62	0,79						36,72								4,38			0,01					0,02										43,551	0,843		
241																				0,00																								0,005	0,000			
242		1,51							0,02			6,22	3,01	0,02		1,52	1,50	3,20	63,05	10,87	1,50																								92,415	0,682		
243												9,63						7,19	2,40	8,40	128,43						0,01																		156,050	0,823		
311																							70,15			5,34				3,86															79,350	0,884		
312																								109,34	2,16				2,29																113,797	0,961		
313																							1,53	4,59	45,79				2,38																54,301	0,843		
321																2,25						2,25					173,68	2,25		9,07	2,81	4,50													196,791	0,883		
322																														0,00															0,004	0,000		
323																				1,79									13,40	2,68															17,867	0,750		
324											0,01							0,02		2,71	2,71		2,71	2,71		8,19			154,35			0,02													170,718	0,904		
331																	0,14														1,84														1,981	0,927		
332																										8,19																			53,521	0,716		
333																										43,28		2,42	9,66			148,86														204,219	0,729	
334																														0,02															0,041	0,500		
335																																	0,01												0,014	1,000		
411													0,01																						4,51			0,15						4,670	0,965			
421													0,19														0,19							1,95	0,39	0,39			0,78					3,894	0,101			
422																																			0,06			0,23							0,283	0,800		
511																													0,12	0,21								1,86	0,12						2,321	0,803		
512													0,03																									28,29							28,319	0,999		
521																																								0,16						0,156	1,000	
522																																									0,001					0,001	1,000	
523																																														1,026	1,000	
Total	1,83	22,43	4,33	0,80	0,23	0,63	2,64	0,04	1,89	0,17	0,67	231,46	151,44	3,16	2,18	21,90	12,18	55,09	3,69	79,07	167,69	3,75	74,39	116,65	53,29	238,19	2,25	15,82	184,60	2,05	41,11	162,40	0,04	0,01	4,91	0,39	0,23	1,86	29,36	0,16	0,001	1,026	1444,466					
Prod Acc	1,000	0,771	0,933	1,000	0,209	1,000	1,000	1,000	0,908	1,000	1,000	0,924	0,973	0,993	1,000	0,874	0,692	0,667	0,000	0,797	0,766	0,000	0,943	0,937	0,859	0,729	0,000	0,847	0,836	0,896	0,932	0,917	0,494	1,000	0,917	1,000	1,000	1,000	0,964	1,000	1,000	1,000					Global Acc	0,852

Overall accuracy = 85.2%

Table 22: Confusion matrix CLC 2012 for zone2 (France) – BLIND ANALYSIS- In column the validation and line the production

Overall accuracy = 86.3%

Table 23: Confusion matrix CLC 2012 for zone2 (France) – PLAUSIBILITY ANALYSIS- In column the validation and line the production

CLC Class	111	112	121	122	123	124	131	132	133	141	142	211	212	213	221	222	223	231	241	242	243	311	312	313	321	322	323	324	331	332	333	334	335	411	412	421	422	423	511	512	521	522	523	Total	User Acc		
111	0,79	0,34																																									1,127	0,700			
112		58,92										0,03								0,02								0,03															59,004	0,999			
121		0,02	10,95						0,01			0,04								0,01																								11,033	0,992		
122				1,42																		0,13																						1,552	0,915		
123					0,28																																							0,276	1,000		
124						1,21																																						1,211	1,000		
131							1,92		0,22																													0,45					2,592	0,740			
132								0,04		0,15																																		0,191	0,800		
133			0,07	0,13					0,02	1,16		0,03	0,12																																1,534	0,758	
141										0,64																																			0,637	1,000	
142						0,15					2,64	0,15						0,15																											3,092	0,855	
211							0,01					360,19	0,02					7,83		23,54																									391,592	0,920	
213														0,94																															0,940	1,000	
221															25,57					2,22																								27,793	0,920		
222												0,45				2,27				1,81																								4,528	0,501		
223			0,03														0,11				0,08		0,03	0,03																					0,273	0,400	
231												4,37						191,76		4,36	17,43							0,01																	217,928	0,880	
241												0,02				0,02			0,02	0,03																									0,086	0,200	
242												2,96			2,96			11,86		121,47	5,93			2,97																					148,153	0,820	
243												0,78									37,42	0,78																							38,975	0,960	
244																					0,01																							0,009	0,000		
311																				4,49	206,16		18,04					0,72																	229,414	0,899	
312																				1,71		84,25	3,41					0,30																	89,673	0,940	
313																					0,02	0,02	49,98					1,34																	51,349	0,973	
321																		3,13		1,88		0,63	17,54	5,64	0,63	1,88																			31,328	0,560	
322																		0,50		1,01		0,50				6,05		2,02																	10,079	0,600	
323																											14,66																		14,662	1,000	
324							0,01				0,04								0,01	0,68	0,45	0,82	1,55				47,28			0,01															50,854	0,930	
331																												0,80																	0,802	1,000	
332																								0,53	0,53			8,50	1,06															10,628	0,800		
333																								2,19	1,64			1,09		0,55	5,47													10,936	0,500		
334																															0,00														0,003	1,000	
335																																0,65													0,650	1,000	
411																		0,01																2,07										2,079	0,997		
412																																				0,12									0,118	1,000	
421																		0,17																											1,699	0,900	
422																																					0,32								0,322	1,000	
423																																						0,21							0,210	1,000	
511																													0,15																3,003	0,950	
512			0,01																			0,28																							5,575	0,949	
521																																														1,653	1,000
522																																														0,314	1,000
523																																														0,123	1,000
Total	0,789	59,382	11,117	1,420	0,276	1,361	1,929	0,177	1,401	0,637	2,673	369,151	0,023	0,940	28,533	2,287	0,109	215,414	0,017	153,475	70,633	207,816	85,623	76,605	20,262	13,858	15,289	54,674																			

Overall accuracy = 89.3%

Zone 3: Spain - confusion matrix CLC2012 - Blind and plausibility analysis

Table 24: Confusion matrix CLC 2012 for zone2 (SPAIN) – BLIND ANALYSIS- In column the validation and line the production

CLC Classe	111	112	121	122	123	124	131	132	133	141	142	211	212	213	221	222	223	231	241	242	243	244	311	312	313	321	322	323	324	331	332	333	334	335	411	412	421	422	423	511	512	521	522	523	Total	User Acc		
111	5,571	0,348							0,348																		0,348		0,348																6,964	0,800		
112	0,027	13,658																									0,697																		14,382	0,950		
121			7,769						0,372								0,024																												8,164	0,952		
122				1,571														0,112																											1,682	0,934		
123					0,128																																								0,128	1,000		
124						0,056					0,056																																		0,562	0,800		
131							2,491	0,034																																					2,525	0,987		
132								0,529	0,037																																				0,566	0,934		
133	0,034	0,244	0,054	0,033					3,247		0,346																																			3,958	0,820	
141		0,083								0,406	0,051																																		0,540	0,752		
142										1,546																																			1,546	1,000		
211											#####					0,032	0,032			5,826	5,738																								287,932	0,960		
212												68,830				0,065				1,753																										70,648	0,974	
213													3,907																																3,907	1,000		
221												0,032	0,026		21,471	0,026	0,770		0,770	8,506																										31,601	0,679	
222		0,820	0,820									1,640	0,820			23,927	0,024	2,460	2,460								0,820																				33,791	0,708
223																0,136	64,271		1,628																											66,035	0,973	
231																		18,639			2,468						3,986																			25,093	0,743	
241																			0,572	0,163									0,082																	0,817	0,700	
242										2,848			0,082	1,330		2,661	1,330	46,032																												54,283	0,848	
243															0,082	1,330	1,016	1,016	1,016	1,071	35,616	1,016					0,055			0,055																40,861	0,872	
244																				1,770	65,538								1,770	1,770																70,849	0,925	
311																					139,615				3,693					3,649																146,958	0,950	
312																					6,609					119,257			6,609											0,021						132,497	0,900	
313																				1,006	1,006	1,006				33,408			4,130																	40,555	0,824	
321		2,820							0,045			2,820					2,820	14,100		2,820	5,746						68,240		5,766	5,849																	113,846	0,599
322																										1,409		8,453	41,273		5,657																56,792	0,727
323			3,565																									3,565		103,892	28,776																143,362	0,725
324											1,463	0,045			0,045					1,463	1,463	3,055	1,605					0,027	4,388	49,303				0,048												62,904	0,784	
331																														0,877			0,097												0,988	0,888		
332																													0,406	0,406			6,094	1,325												8,232	0,740	
333																											6,682		1,158	2,113			12,044													21,996	0,548	
334																															0,069				0,929										0,998	0,931		
335																																			0,006											0,006	1,000	
411																																				1,293										1,293	1,000	
412																																					0,021									0,021	1,000	
421																											0,096												0,861								0,957	0,900
422																																					0,053		0,053	0,422							0,527	0,800
423																																							0,018								0,018	1,000
511																																																

Overall accuracy = 85.10%

Table 25: Confusion matrix CLC 2012 for zone2 (SPAIN) – PLAUSIBILITY ANALYSIS- In column the validation and line the production

CLC Class	111	112	121	122	123	124	131	132	133	141	142	211	212	213	221	222	223	231	241	242	243	244	311	312	313	321	322	323	324	331	332	333	334	335	411	412	421	422	423	511	512	521	522	523	Total	User Acc				
111	5,571	0,348							0,348																		0,348		0,348																6,964	0,800				
112	0,027	13,630	0,027																								0,348			0,348																14,382	0,948			
121			7,769						0,395																																					8,164	0,952			
122				1,571														0,112																												1,682	0,934			
123					0,128																																									0,128	1,000			
124						0,450					0,056																																			0,562	0,800			
131							2,491	0,034																																						2,525	0,987			
132								0,529	0,037																																					0,566	0,934			
133			0,278	0,054					3,037										0,346	0,244																										3,958	0,767			
141			0,083							0,406	0,051								0,051																											0,540	0,752			
142											1,546																																			1,546	1,000			
211												282,074					0,032	0,032	0,055						5,738																						287,932	0,980		
212													68,830					0,065			1,753																										70,648	0,974		
213														3,907																																	3,907	1,000		
221												0,770	0,026		23,815	0,026	0,770		0,770	5,424																												31,601	0,754	
222			0,820									0,820							3,280	2,460																												33,791	0,781	
223																0,136	65,899																														66,035	0,998		
231																		18,639		2,468							3,931				0,055																	25,093	0,743	
241																			0,735												0,082																0,817	0,900		
242											2,827				0,082			2,661	1,330	47,383																												54,283	0,873	
243																	1,016	1,016	1,016	1,071	35,616	1,016					0,055			0,055																		40,861	0,872	
244																				1,770	65,538								1,770	1,770																	70,849	0,925		
311																							143,309																									146,958	0,975	
312																				6,609										3,305																		132,497	0,925	
313																				1,006											1,006																40,555	0,950		
321			2,820					0,045				2,820						11,280			2,820	5,746						76,806		5,660	5,849																	113,846	0,675	
322																											5,635	44,113			7,044																		56,792	0,777
323				3,565																	3,565											121,739	14,494															143,362	0,849	
324											1,463	0,045				0,045						1,463	1,578	2,997	1,463			0,027	1,463	52,361																		62,904	0,832	
331																															0,975									0,014								0,988	0,986	
332																																0,406	0,406		6,094	1,325											8,232	0,740		
333																											1,268					1,158	2,113			17,458												21,996	0,794	
334																	0,069															0,069						0,860									0,998	0,861		
335																																			0,006												0,006	1,000		
411																																				1,293												1,293	1,000	
412																																					0,021											0,021	1,000	
421																										0,096													0,861									0,957	0,900	
422																																						0,053	0,474									0,527	0,900	
423																																									0,018								0,018	1,000
511																					</																													

Overall accuracy = 89.41%

Zone 4: Sweden confusion matrix CLC2012 - Blind and plausibility analysis

Table 26: Confusion matrix CLC 2012 for zone4 (Sweden – BLIND ANALYSIS - In column the validation and line the production

CLC Cl	111	112	121	122	123	124	131	132	133	141	142	211	222	231	242	243	311	312	313	321	322	324	331	332	333	334	335	411	412	421	511	512	521	522	523	Total	User A				
111	0,063	0,016																																		0,079	0,800				
112		8,973	0,014						0,003			0,005						0,002																		8,996	0,997				
121			1,375					0,099				0,009										0,099															1,581	0,870			
122				0,613																																	0,613	1,000			
123					0,058																																0,058	1,000			
124						0,315																															0,315	1,000			
131							0,378								0,032																						0,409	0,923			
132							0,021	0,190																													0,211	0,900			
133								0,085										0,002																			0,087	0,976			
141			0,009						0,670																												0,680	0,987			
142										1,199																											1,199	1,000			
211											44,596		11,471	5,100	2,547			0,003																		63,717	0,700				
213																						0,001															0,001	0,000			
222												0,023						0,008				0,008															0,038	0,600			
231												0,002		2,843		1,412		0,285	0,285									0,002									5,679	0,501			
242						0,127									0,762	0,762	0,762		0,127			0,003															2,544	0,300			
243												0,398		0,398		10,348			0,398			0,398															11,940	0,867			
311																0,013	36,877		1,724			4,290															42,905	0,860			
312				0,011								0,003				0,002		499,232	3,024			67,589				1,429											571,291	0,874			
313																		47,894				4,661															52,556	0,911			
321													0,205						3,687	0,205																	4,097	0,900			
322																	0,009	11,879	0,029							1,175											58,750	0,920			
324													0,029									94,578							0,011								106,536	0,888			
331														0,018								0,018	0,055														0,092	0,600			
332																			0,811					6,083	1,217													8,111	0,750		
333																			2,931						0,489	11,237												14,656	0,767		
334																	0,006					0,006					0,012										0,024	0,500			
335																								0,057	0,114		0,398										0,568	0,700			
411																												0,398										1,292	0,867		
412																												1,120					0,172					60,463	1,000		
421																	0,006													60,463		0,022						0,028	0,800		
511																		0,126														2,392						2,517	0,950		
512																														1,541				75,489					77,030	0,980	
521																																	0,057						0,057	1,000	
522																						0,013												0,054					0,067	0,800	
523																																			0,813					0,813	1,000
Total	0,063	8,989	1,398	0,624	0,058	0,442	0,399	0,288	0,088	0,670	1,199	45,013	0,023	15,726	5,894	15,085	36,892	511,543	53,483	10,954	54,255	172,513	0,055	6,629	13,742	1,441	0,398	1,122	62,014	0,022	2,392	75,661	0,057	0,054	0,813	967,003					
Prod A	1,000	0,998	0,984	0,982	1,000	0,713	0,947	0,658	0,969	1,000	1,000	0,991	1,000	0,181	0,129	0,686	1,000	0,976	0,896	0,337	0,996	0,548	1,000	0,918	0,818	0,008	1,000	0,998	0,975	1,000	1,000	0,998	1,000	1,000	1,000						
																																							Glob Acc 0,879		

Overall accuracy = 87.9%

Table 27: Confusion matrix CLC 2012 for zone4 (Sweden – PLAUSIBILITY ANALYSIS - In column the validation and line the production)

CLC Class	111	112	121	122	123	124	131	132	133	141	142	211	222	231	242	243	311	312	313	321	322	324	331	332	333	334	335	411	412	421	511	512	521	522	523	Total	User Acc	
111	0,063	0,016																																		0,079	0,800	
112		8,977	0,014						0,003										0,003																	8,996	0,998	
121			1,384					0,099														0,099														1,581	0,875	
122				0,613																																0,613	1,000	
123					0,058																															0,058	1,000	
124						0,315																														0,315	1,000	
131							0,378									0,032																				0,409	0,923	
132							0,021	0,190																												0,211	0,900	
133									0,085										0,002																	0,087	0,976	
141			0,009							0,670																										0,680	0,987	
142											1,199																									1,199	1,000	
211												45,870		8,923	5,100	3,821							0,003													63,717	0,720	
213																							0,001													0,001	0,000	
222													0,023						0,008				0,008													0,038	0,600	
231														3,413		1,130		0,003	0,282								0,002								5,679	0,601		
242						0,127									0,635	0,889	0,762		0,127				0,003													2,544	0,350	
243												0,398			0,398	10,348		0,398				0,398														11,940	0,867	
311									0,855							0,013	38,601		1,711				1,724													42,905	0,900	
312				0,006								0,003				0,002		534,373	3,024			33,882														571,291	0,935	
313																			51,459			1,097														52,556	0,979	
321													0,205							3,892																4,097	0,950	
322																			1,175	57,575																58,750	0,980	
324																0,002	0,009	4,856	0,059			101,609														106,536	0,954	
331														0,018								0,018	0,055													0,092	0,600	
332																				0,406				7,300	0,406											8,111	0,900	
333																				0,489				0,489	13,679											14,656	0,933	
334																		0,006								0,018										0,024	0,750	
335																								0,057	0,114		0,398									0,568	0,700	
411																												1,120				0,172					1,292	0,867
412																													60,463							60,463	1,000	
421																0,006														0,022						0,028	0,800	
511																	0,126														2,392					2,517	0,950	
512																													1,541				75,489				77,030	0,980
521																																0,057				0,057	1,000	
522																						0,013												0,054	0,067	0,800		
523																																			0,813	0,813	1,000	
Total	0,063	8,993	1,407	0,619	0,058	0,442	0,399	0,288	0,943	0,670	1,199	46,271	0,023	13,592	5,990	16,111	38,616	539,373	57,063	5,961	57,575	139,704	0,055	7,845	14,199	0,018	0,398	1,122	62,003	0,022	2,392	75,661	0,057	0,054	0,813	1023,844		
Prod Acc	1,000	0,998	0,984	0,990	1,000	0,713	0,947	0,658	0,090	1,000	1,000	0,991	1,000	0,251	0,149	0,642	1,000	0,991	0,902	0,653	1,000	0,727	1,000	0,930	0,963	1,000	1,000	0,998	0,975	1,000	1,000	0,998	1,000	1,000	1,000			
																																					Global Ac	0,931

Overall accuracy = 93.1%

Zone 5: Germany confusion matrix CLC2012 - Blind and plausibility analysis

Table 28: Confusion matrix CLC 2012 for zone5 (Germany) – BLIND ANALYSIS- In column the validation and line the production

CLC Cl	111	112	121	122	123	124	131	132	133	141	142	211	221	222	231	241	242	243	311	312	313	321	322	324	331	332	333	411	412	421	423	511	512	521	522	523	Total	User A		
111	0,307	0,038	0,038																																		0,383	0,800		
112		70,681							1,502	1,502		1,502																									75,186	0,940		
121		1,023	12,599						0,020			0,059			1,023	0,010				0,502	0,502			0,502													16,238	0,776		
122			0,062	0,557																																	0,619	0,900		
123					0,197										0,025		0,025																				0,246	0,800		
124						1,281																															1,281	1,000		
131		0,110					2,717					0,110			0,013					0,007																	2,957	0,919		
132								0,400																0,044													0,444	0,900		
133			0,049				0,037	0,019	0,295																												0,401	0,738		
141										1,858	0,109				0,109	0,109																						2,186	0,850	
142		0,247									4,260	0,247																0,247									5,002	0,852		
211												381,277			7,304	27,690								0,083													416,354	0,916		
221													3,194			0,564																						3,757	0,850	
222												0,460		2,300		0,230	1,150	0,230						0,230														4,599	0,500	
231		3,919		0,014					0,028			13,081			125,933	35,313	15,674	3,919	0,013		0,014			0,097													198,005	0,636		
242		0,098											0,098	0,098				1,574	0,098																			1,967	0,800	
243												0,135			0,135	0,539	1,481	0,269		0,135																	2,693	0,550		
311											4,223				2,112				82,672		16,907			0,115					0,013								106,043	0,780		
312															7,191			3,596		151,501	7,250			7,280											3,596	180,414	0,840			
313		0,883									0,883				0,883	0,883	0,883	0,883	0,892	0,910	37,269			0,901													44,385	0,840		
321															1,844			0,922				1,620		0,230													4,616	0,351		
322																		0,147				0,147	1,767	0,736	0,147												2,945	0,600		
324		0,227										0,237			0,069			0,227	0,352	0,508	0,513	0,014		7,199								0,227					9,574	0,752		
331							0,069	0,010															0,035		0,173		0,041					0,035					0,363	0,477		
332																				0,034		0,034					0,235	0,034									0,336	0,700		
333							0,026	0,007														0,053		0,026				0,158									0,270	0,585		
411																												1,019		0,113								1,132	0,900	
412															0,013													0,227	2,044									2,284	0,895	
421															0,064															0,575								0,639	0,900	
423															0,032										0,032					0,032	0,224							0,320	0,700	
511																																	2,235					2,235	1,000	
512							0,020											0,331									0,010						9,868					10,229	0,965	
521															0,040																		0,020					0,060	0,333	
522																																		0,715				0,715	1,000	
523																																0,040						0,081	0,121	0,667
Total	0,307	77,226	12,748	0,572	0,197	1,281	2,869	0,419	1,861	3,359	4,369	402,214	3,292	2,398	146,788	0,230	67,830	23,614	88,104	153,475	62,577	1,882	1,802	17,444	0,353	0,235	0,233	1,503	2,057	0,720	0,299	2,463	9,868	0,020	0,715	3,676	910,287			
Prod A	1,000	0,915	0,988	0,975	1,000	1,000	0,947	0,954	0,159	0,553	0,975	0,948	0,970	0,959	0,858	0,000	0,023	0,063	0,938	0,987	0,596	0,861	0,981	0,413	0,492	1,000	0,678	0,678	0,994	0,798	0,749	0,908	1,000	1,000	1,000	0,022				
																																							Global A	0,828

Overall accuracy = 82.8%

Table 29: Confusion matrix CLC 2012 for zone5 (Germany) – PLAUSIBILITY ANALYSIS- In column the validation and line the production

CLC Class	111	112	121	122	123	124	131	132	133	141	142	211	221	222	231	241	242	243	311	312	313	321	322	324	331	332	333	411	412	421	423	511	512	521	522	523	Total	User Acc
111	0,345		0,038																																	0,383	0,900	
112		72,183							1,502	1,502																										75,186	0,960	
121		0,502	14,203									0,020			0,502	0,010					0,502		0,502													16,238	0,875	
122			0,062	0,557																																0,619	0,900	
123			0,025		0,197												0,025																			0,246	0,800	
124						1,281																														1,281	1,000	
131		0,110					2,847																													2,957	0,963	
132				0,062				0,400																0,044												0,444	0,900	
133			0,019				0,037	0,019	0,307			0,019																								0,401	0,766	
141										1,858	0,109				0,109	0,109																				2,186	0,850	
142											4,755	0,247																								5,002	0,951	
211												395,562			0,083	20,625							0,083													416,354	0,950	
221													3,194			0,564																				3,757	0,850	
222												0,230	2,530		0,230	1,150	0,230						0,230													4,599	0,550	
231				0,014					0,013			16,816			141,902	19,639	15,674	3,919						0,028												198,005	0,717	
242		0,098											0,098				1,672	0,098																		1,967	0,850	
243															0,135	0,269	2,019	0,135		0,135															2,693	0,750		
311											2,112							89,094		14,782				0,055												106,043	0,840	
312																			169,598	3,596				7,220												180,414	0,940	
313																			0,018	43,466				0,901												44,385	0,979	
321															0,922		0,691				2,542		0,230					0,230								4,616	0,551	
322															0,147		0,147				0,147	1,767	0,589	0,147											2,945	0,600		
324		0,227							0,059			0,227							0,066	0,513				8,254								0,227				9,574	0,862	
331							0,069															0,035		0,225							0,035					0,363	0,618	
332																					0,034		0,034		0,269											0,336	0,800	
333							0,026														0,026		0,026				0,191									0,270	0,708	
411																												1,019		0,113							1,132	0,900
412																0,013													2,271							2,284	0,994	
421																0,064														0,575						0,639	0,900	
423															0,032																0,288					0,320	0,900	
511																																2,235					2,235	1,000
512									0,010																							0,020		10,219		10,229	0,999	
521																															0,020			0,040			0,060	0,667
522																																		0,715			0,715	1,000
523																																		0,081			0,121	0,667
Total	0,345	73,120	14,346	0,572	0,197	1,281	2,979	0,419	1,891	3,359	4,864	415,232	3,292	2,530	143,909	0,230	44,037	18,885	93,148	169,682	62,993	2,749	1,802	18,197	0,372	0,269	0,191	1,249	2,271	0,708	0,363	2,463	10,219	0,040	0,715	0,081	978,659	
Prod Acc	1,000	0,987	0,990	0,975	1,000	1,000	0,955	0,954	0,162	0,553	0,978	0,953	0,970	1,000	0,986	0,000	0,038	0,107	0,956	1,000	0,690	0,925	0,981	0,454	0,604	1,000	1,000	0,816	1,000	0,812	0,794	0,908	1,000	1,000	1,000	1,000		Global Ac 0,890

Overall accuracy = 89.0%

Zone 6: Finland confusion matrix CLC2012 - Blind and plausibility analysis

Table 30: Confusion matrix CLC 2012 for zone6 (Finland) – BLIND ANALYSIS- In column the validation and line the production

CLC Cl	112	121	122	123	124	131	132	133	141	142	211	212	231	242	243	311	312	313	321	322	323	324	332	333	411	412	421	511	512	523	Total	User A			
112	9,307	1,230						0,633							0,615																11,784	0,790			
121	0,247	2,296				0,009		0,002								0,002																2,555	0,898		
122			0,016																													0,016	1,000		
123		0,002		0,171																													0,173	0,988	
124					0,332																												0,332	1,000	
131	0,044	0,002				0,869		0,002								0,097						0,003											1,017	0,855	
132						0,161	0,165		0,021																								0,347	0,475	
133	0,012	0,004		0,002		0,002		0,125																									0,146	0,860	
141	0,050	0,013							0,138	0,013												0,013											0,225	0,611	
142										0,531												0,027							0,027				0,585	0,908	
211											52,538		0,014	1,166	2,341		0,021	1,175				0,057				0,011							57,322	0,917	
231													0,063		0,004	0,004	0,028	0,012	0,012			0,125		0,012	0,024				0,002				0,287	0,220	
243										11,111			1,010	1,010	31,312	0,002	3,030	1,012															48,488	0,646	
311																15,149		3,812				4,896				0,003							23,861	0,635	
312										10,655							353,662	117,289				12,272				21,318							515,196	0,686	
313																4,568	18,359	170,666				15,697				0,006							209,296	0,815	
321																			0,322					0,281									0,603	0,533	
322																			1,317	11,856		10,539											23,712	0,500	
324							0,003			0,024	2,454				2,513		30,027	7,773			2,454	78,207				4,914							128,370	0,609	
331																				0,031				0,015									0,046	0,000	
332																				1,414	0,566	0,870	1,980	0,566									5,396	0,367	
333																				0,081				1,546										1,627	0,950
411																									1,093	0,064			0,129					1,286	0,850
412							0,006									0,014	0,007		4,808		8,027				59,600			1,603						74,064	0,805
421															0,022	0,011	0,011			0,011		0,011			0,011		0,132						0,208	0,632	
511															0,115			0,115										1,951						2,180	0,895
512																													118,074					118,074	1,000
523										0,066								0,066				0,066					0,133				0,664		0,996	0,667	
Total	9,661	3,546	0,016	0,173	0,332	1,041	0,165	0,771	0,159	0,543	74,394	2,454	1,087	2,176	36,922	19,735	405,252	301,927	1,651	20,655	0,566	130,809	1,980	2,420	1,129	85,916	0,264	1,951	119,834	0,664	912,764				
Prod A	0,963	0,647	1,000	0,988	1,000	0,835	1,000	0,163	0,866	0,977	0,706	0,000	0,058	0,000	0,848	0,768	0,873	0,565	0,195	0,574	0,000	0,598	1,000	0,639	0,969	0,694	0,498	1,000	0,985	1,000					
																																		Glob Acc 0,743	

Overall accuracy = 74.3%

Table 31: Confusion matrix CLC 2012 for zone6 (Finland) – PLAUSIBILITY ANALYSIS- In column the validation and line the production

CLC Classe	112	121	122	123	124	131	132	133	141	142	211	231	242	243	311	312	313	321	322	324	332	333	411	412	421	511	512	523	Total	User Acc
112	11,170	0,615																											11,784	0,948
121	0,247	2,282				0,009		0,018																					2,555	0,893
122			0,016																										0,016	1,000
123		0,002		0,171																									0,173	0,988
124					0,332																								0,332	1,000
131	0,044	0,002				0,876									0,094					0,002									1,017	0,861
132						0,081	0,266																						0,347	0,766
133	0,006	0,002		0,002		0,002		0,134																					0,146	0,916
141	0,013	0,025							0,163	0,013										0,013									0,225	0,722
142										0,531																	0,027		0,558	0,952
211											56,101	0,002		1,166						0,038				0,004					57,311	0,979
231												0,063		0,004		0,028	0,008	0,012		0,133		0,012	0,027						0,287	0,220
243											8,081	1,010	1,010	34,342	0,002	3,030	1,012												48,488	0,708
311															17,319		2,167			4,342									23,828	0,727
312											10,655					353,931	95,891			33,222				10,663					504,362	0,702
313																	189,540			15,184				0,003					204,728	0,926
321																		0,442				0,161							0,603	0,733
322																			14,491	7,904		1,317							23,712	0,611
324										0,011			2,469		7,656	2,784			2,454	110,479				2,460					128,314	0,861
331																	0,015	0,031											0,046	0,000
332																			0,566	1,435	3,112	0,283							5,396	0,577
333																				0,081			1,546						1,627	0,950
411																							1,158	0,064		0,064			1,286	0,900
412															0,007					1,603				70,852		1,603			74,064	0,957
421												0,011	0,011										0,011		0,165				0,198	0,833
511													0,115			0,115										1,951			2,180	0,895
512																											118,074		118,074	1,000
523																				0,066					0,199		0,730		0,996	0,733
Total	11,479	2,927	0,016	0,173	0,332	0,968	0,266	0,151	0,163	0,543	74,847	1,075	1,010	38,096	17,332	364,757	291,517	0,470	17,623	174,421	3,112	3,319	1,195	84,047	0,364	1,951	119,768	0,730	990,235	
Prod Acc	0,973	0,779	1,000	0,988	1,000	0,905	1,000	0,884	1,000	0,977	0,750	0,059	0,000	0,901	0,999	0,970	0,650	0,941	0,822	0,633	1,000	0,466	0,969	0,843	0,452	1,000	0,986	1,000		
																													Global Acc	0,817

Overall accuracy = 81.7%

Zone 7: Norway confusion matrix CLC2012- Blind and plausibility analysis

Table 32: Confusion matrix CLC 2012 for zone 7 (Norway) – BLIND ANALYSIS- In column the validation and line the production

CLC Classe	111	112	121	122	123	124	131	132	133	141	142	211	231	242	243	311	312	313	321	322	324	331	332	333	335	411	412	423	511	512	523	Total	User Acc
111	0,055	0,004																													0,059	0,933	
112		5,481	0,006						0,009			0,002					0,005															5,503	0,996
121			0,596		0,063				0,017																							0,676	0,882
122				0,046					0,012								0,013															0,071	0,643
123					0,031																											0,031	1,000
124		0,014				0,199																										0,213	0,933
131			0,011				0,264										0,013							0,013			0,011					0,314	0,842
132			0,005					0,009									0,013															0,027	0,342
133					0,004				0,031							0,004												0,004				0,043	0,725
141										0,178		0,009																				0,187	0,950
142		0,017							0,003		1,423					0,012	0,026			0,013	0,071			0,013								1,579	0,901
211												15,446		0,858	0,858		0,012				0,008						0,008					17,189	0,899
231													0,599	0,037					0,037	0,037				0,037								0,748	0,800
242														3,792	0,948																	4,741	0,800
243															27,344						1,439											28,783	0,950
311																99,184	0,004	2,610			18,275											120,072	0,826
312																	163,301	7,124			7,424											177,849	0,918
313																0,585	0,004	13,747			0,292											14,628	0,940
322																			134,687				2,806	2,806								140,299	0,960
324						0,004					0,003	0,078				0,005	2,193	0,012			21,177											23,473	0,902
331																						0,047										0,047	1,000
332																							59,161	3,287	3,287							65,734	0,900
333																				4,848				227,852								232,700	0,979
334																					0,023											0,023	0,000
335																									8,692							8,692	1,000
411																										0,009						0,009	1,000
412																			2,543					1,271			58,482					62,296	0,939
423								0,003				0,003									0,003			0,003			0,003	0,052				0,069	0,750
511																													1,296			1,296	1,000
512																														39,812		39,812	1,000
523															0,087															1,215	1,302	0,933	
Total	0,055	5,516	0,618	0,046	0,098	0,199	0,268	0,009	0,077	0,178	1,426	15,539	0,599	4,688	29,236	99,790	165,585	23,493	0,037	142,128	48,712	0,047	61,967	235,283	11,979	0,009	58,508	0,052	1,296	39,812	1,215	884,206	
Prod Acc	1,000	0,994	0,964	1,000	0,320	1,000	0,986	1,000	0,410	1,000	0,998	0,994	1,000	0,809	0,935	0,994	0,986	0,585	0,000	0,948	0,435	1,000	0,955	0,968	0,726	1,000	1,000	1,000	1,000	1,000	1,000		

Overall accuracy = 93.2%

Table 33: Confusion matrix CLC 2012 for zone 7 (Norway) – PLAUSIBILITY ANALYSIS- In column the validation and line the production

CLC Class\	111	112	121	122	123	124	131	132	133	141	142	211	231	242	243	311	312	313	321	322	324	331	332	333	334	335	411	412	423	511	512	523	Total	User Acc	
111	0,055	0,004																														0,059	0,933		
112		5,485	0,006						0,009								0,003																5,503	0,997	
121			0,599		0,063				0,013																								0,676	0,887	
122				0,043					0,012						0,003						0,013												0,071	0,597	
123					0,031																												0,031	1,000	
124		0,014				0,199																											0,213	0,933	
131			0,011				0,291																					0,011					0,314	0,927	
132			0,005					0,009	0,013																								0,027	0,342	
133					0,004				0,031							0,004												0,004					0,043	0,725	
141										0,178		0,009																					0,187	0,950	
142		0,007							0,003		1,478					0,004	0,013				0,071												1,576	0,938	
211												16,319			0,862		0,008																17,189	0,949	
231													0,673						0,037	0,037													0,748	0,900	
242														4,266	0,474																		4,741	0,900	
243															27,344						1,439												28,783	0,950	
311																104,404	2,610				13,058												120,072	0,870	
312																	170,575	7,174			0,100												177,849	0,959	
313																		14,336															14,336	1,000	
322																			140,299															140,299	1,000
324							0,004										1,095	0,005			22,290												23,394	0,953	
331																						0,051											0,051	1,000	
332																							65,734										65,734	1,000	
333																			4,848					227,852									232,700	0,979	
334																								0,023									0,023	1,000	
335																									8,692								8,692	1,000	
411																										0,009							0,009	1,000	
412																											62,296						62,296	1,000	
423				0,003									0,003							0,003	0,003								0,055				0,069	0,800	
511																													1,296				1,296	1,000	
512																														39,812			39,812	1,000	
523															0,087																	1,215	1,302	0,933	
Total	0,055	5,509	0,622	0,043	0,102	0,199	0,295	0,009	0,083	0,178	1,478	16,329	0,677	4,266	28,769	104,412	171,694	24,125	0,037	145,188	36,975	0,051	65,734	227,852	0,023	8,692	0,009	62,311	0,055	1,296	39,812	1,215	915,940		
Prod Acc	1,000	0,996	0,964	1,000	0,310	1,000	0,987	1,000	0,380	1,000	1,000	0,999	0,995	1,000	0,950	1,000	0,993	0,594	0,000	0,966	0,603	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	Global Acc	0,966	

Overall accuracy = 96.6%

Zone 8: Poland- Confusion matrix CLC2012 - Blind and plausibility analysis

Table 34: Confusion matrix CLC 2012 for zone 8 (Poland) – BLIND ANALYSIS- In column the validation and line the production

CLC Class	111	112	121	122	123	124	131	132	133	141	142	211	222	231	242	243	311	312	313	321	322	324	331	332	333	411	412	511	512	521	523	Total	User Acc	
111	0,216	0,024																													0,240	0,900		
112		37,893								0,011	0,016	0,023			6,671	0,006																44,620	0,849	
121		0,185	3,535						0,185			0,021																				3,926	0,900	
122		0,038		0,852					0,122			0,006																				1,018	0,838	
123					0,078																											0,078	1,000	
124						0,541			0,011					0,083																		0,635	0,852	
131							2,078			0,075		0,023		0,011		0,011						0,022							0,075			2,296	0,905	
132		0,017					0,034	0,292																								0,343	0,850	
133		0,109	0,028	0,102					0,879			0,120			0,028				0,028													1,294	0,679	
141		0,026								0,440	0,026			0,026																		0,518	0,850	
142									0,179	1,518								0,089														1,786	0,850	
211							0,020					412,992		0,065																		413,077	1,000	
222												0,265	4,533		0,259																	5,058	0,896	
231												4,197		58,309	8,330	4,165						4,197				4,165						83,363	0,699	
242												2,546			20,384	2,546																25,475	0,800	
243												3,929		3,848	1,924	25,093		1,924														36,718	0,683	
311												0,010				4,678	37,369		2,328			2,388										46,773	0,799	
312												0,061				0,061		172,222	0,061			0,121										172,525	0,998	
313																0,099		0,049	75,240			0,296										75,684	0,994	
321													0,099							0,493	0,099	0,296										0,985	0,500	
322																					0,125											0,125	1,000	
324												0,882			0,026	6,647	0,086	4,227	1,047	0,024		6,357										19,297	0,329	
331																							0,141									0,141	1,000	
332																					0,015			0,015	0,030							0,061	0,250	
333																						0,025			0,226							0,251	0,900	
411													0,154									0,463				2,159	0,154		0,154			3,084	0,700	
412													0,018		0,018		0,018				0,018						0,185					0,258	0,714	
511																										0,228		2,053				2,282	0,900	
512							0,006			0,003		0,006																		11,948		11,964	0,999	
521																														0,017		0,017	1,000	
523																																0,034	0,034	1,000
Total	0,216	38,292	3,563	0,955	0,078	0,541	2,138	0,292	1,196	0,708	1,560	425,080	4,533	62,613	37,622	43,324	37,455	178,530	78,704	0,517	0,239	14,183	0,141	0,015	0,256	6,552	0,339	2,053	12,177	0,017	0,034	878,215		
Prod Acc	1,000	0,990	0,992	0,893	1,000	1,000	0,972	1,000	0,735	0,622	0,973	0,972	1,000	0,931	0,542	0,579	0,998	0,965	0,956	0,953	0,524	0,448	1,000	1,000	0,881	0,330	0,545	1,000	0,981	1,000	1,000			
																																	Glob Acc	0,921

Overall accuracy = 92.1%

Table 35: Confusion matrix CLC 2012 for zone 8 (Poland) – PLAUSIBILITY ANALYSIS- In column the validation and line the production

CLC Classe	111	112	121	122	123	124	131	132	133	141	142	211	222	231	242	243	311	312	313	321	322	324	331	332	333	411	412	511	512	521	523	Total	User Acc	
111	0,216	0,024																													0,240	0,900		
112		40,171								0,011	0,016	0,012			4,405	0,006																44,620	0,900	
121			3,736									0,190																				3,926	0,951	
122				1,001					0,011			0,006																				1,018	0,984	
123					0,078																											0,078	1,000	
124						0,593								0,042																		0,635	0,934	
131							2,089			0,075		0,023		0,011		0,011						0,011							0,075			2,296	0,910	
132		0,017						0,326																								0,343	0,950	
133		0,088	0,028	0,049			0,049		1,002			0,021			0,028				0,028													1,294	0,774	
141		0,026								0,466				0,026																		0,518	0,900	
142											1,696							0,089														1,786	0,950	
211												412,992		0,065		0,020																413,077	1,000	
222													4,780		0,278																	5,058	0,945	
231												4,185		62,474	8,330	4,165						0,044				4,165						83,363	0,749	
242												2,546			20,384	2,546																25,475	0,800	
243			0,020												1,924	30,925		1,924				1,924										36,718	0,842	
311													2,328			2,340	37,409		2,328			2,368										46,773	0,800	
312											0,061							172,222	0,121			0,121										172,525	0,998	
313																0,099			75,289			0,296										75,684	0,995	
321																		0,049		0,640	0,049	0,246										0,985	0,650	
322																					0,125											0,125	1,000	
324															5,088	0,036	1,245	0,999				11,928										19,297	0,618	
331																							0,141									0,141	1,000	
332																					0,015			0,046								0,061	0,750	
333																									0,251							0,251	1,000	
411																						0,154				2,930						3,084	0,950	
412															0,037							0,018					0,203					0,258	0,786	
511																										0,228		2,053				2,282	0,900	
512							0,006			0,003		0,003		0,003																11,948		11,964	0,999	
521																														0,017		0,017	1,000	
523																																0,034	0,034	1,000
Total	0,216	40,326	3,784	1,050	0,078	0,593	2,145	0,326	1,013	0,555	1,712	420,038	7,108	62,620	35,349	45,237	37,445	175,529	78,766	0,640	0,190	17,112	0,141	0,046	0,251	7,323	0,203	2,053	12,023	0,017	0,034	898,165		
Prod Acc	1,000	0,996	0,987	0,953	1,000	1,000	0,974	1,000	0,989	0,840	0,991	0,983	0,672	0,998	0,577	0,684	0,999	0,981	0,956	1,000	0,660	0,697	1,000	1,000	1,000	0,400	1,000	1,000	0,994	1,000	1,000			
																																Global Acc	0,942	

Overall accuracy = 94.2%

Table 36: Confusion matrix CLC 2012 for zone 9 (Italy) – BLIND ANALYSIS - In column the validation and line the production

Overall accuracy = 76.0%

[illegible]

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Zone 10: United Kingdom/Ireland confusion matrix CLC2012 - Blind and plausibility analysis

Table 38: Confusion matrix CLC 2012 for zone 10 (United Kingdom & Ireland) – BLIND ANALYSIS- In column the validation and line the production

CLC Classes	111	112	121	122	123	124	131	132	133	141	142	211	222	231	242	243	311	312	313	321	322	324	331	332	333	334	411	412	421	423	511	512	521	522	523	Total	User Acc	
111	0,867	0,144	0,217						0,144	0,072																										1,444	0,600	
112		54,661							0,047	1,930						1,930		0,016																		58,583	0,933	
121			0,439	8,063					0,947									0,016		0,031	0,016	0,039						0,010								9,559	0,843	
122				0,139	0,704									0,037																						0,880	0,800	
123						0,536				0,028																										0,564	0,950	
124							1,978							0,104																						2,082	0,950	
131		0,005						3,085		0,007		0,007		0,169								0,164									0,164					3,601	0,857	
132			0,018				0,018	0,232		0,018	0,036			0,018								0,018														0,357	0,650	
133			0,021	0,089	0,022		0,043		0,332			0,016		0,026							0,016															0,565	0,588	
141				0,141						2,285		0,141		0,308																						2,875	0,795	
142										0,620	11,280			0,655																						12,555	0,898	
211												269,374		6,216		5,720																				281,310	0,958	
213																						0,005														0,005	0,000	
222												0,052	0,232		0,077							0,026														0,387	0,600	
231		8,855										9,606		392,526	8,855	8,855		8,855				0,007														437,560	0,897	
242												0,563		0,375	2,064	0,188	0,188				0,188							0,188								3,753	0,550	
243												0,490		5,391		16,172					0,490	0,980						0,490								24,013	0,673	
311														1,147		2,295	17,212					2,295														22,950	0,750	
312														0,093				56,943	1,345				7,002													65,383	0,871	
313											0,278		0,278		0,557	1,949	0,835	10,026																		13,925	0,720	
321														3,021		3,011				45,169	3,011	0,020			3,011											57,241	0,789	
322		3,922						0,010												3,922	66,675	3,922														78,451	0,850	
324								0,010						0,360		0,030	0,512	3,192	0,512		1,588	33,161						0,565								39,931	0,830	
331																				0,097	0,097	0,097	1,556										0,097			1,946	0,800	
332																0,072				0,144	0,289			0,433	0,505											1,443	0,300	
333																				0,645	1,270				10,157											12,071	0,841	
334																					0,028					0,014										0,042	0,333	
411													0,217								0,069	0,069					0,966	0,069								1,390	0,695	
412													2,706		5,413		2,706		5,413	21,651								97,429								135,319	0,720	
421																													1,611								1,611	1,000
423																					0,058		0,117						0,175	0,757				0,058		1,165	0,650	
511												0,044																				0,400				0,445	0,900	
512																																	13,723				13,723	1,000
521																																	0,050	0,012			0,062	0,800
522																												0,018		0,018				0,268	0,054	0,357	0,750	
523																																			0,927		0,927	1,000
Total	0,867	68,047	8,666	0,725	0,536	1,978	3,166	0,232	1,477	4,953	11,322	280,521	0,232	413,693	10,996	44,242	19,862	72,563	11,883	55,421	95,445	47,806	1,673	0,433	13,672	0,014	0,966	98,751	1,804	0,757	0,418	13,888	0,050	0,378	1,039	1121,867		
Prod Acc	1,000	0,803	0,930	0,970	1,000	1,000	0,974	1,000	0,225	0,461	0,996	0,960	1,000	0,949	0,188	0,366	0,867	0,785	0,844	0,815	0,699	0,694	0,930	1,000	0,743	1,000	1,000	0,987	0,893	1,000	0,957	0,988	1,000	0,709	0,892			
																																					Global Acc	0,871

Overall accuracy = 87.1%

Table 39: Confusion matrix CLC 2012 for zone 10 (United Kingdom & Ireland) – PLAUSIBILITY ANALYSIS- In column the validation and line the production

CLC Class	111	112	121	122	123	124	131	132	133	141	142	211	222	231	242	243	311	312	313	321	322	324	331	332	333	334	411	412	421	423	511	512	521	522	523	Total	User Acc		
111	0,939	0,289	0,144							0,072																										1,444	0,650		
112		56,569							0,069		1,930							0,016																		58,583	0,966		
121			8,544						0,473									0,023		0,016	0,016	0,039						0,449								9,559	0,894		
122			0,139	0,704										0,037																						0,880	0,800		
123					0,536					0,028																										0,564	0,950		
124						2,082																														2,082	1,000		
131							3,095		0,007		0,007				0,164							0,164										0,164				3,601	0,859		
132							0,018	0,268			0,036				0,018							0,018														0,357	0,750		
133		0,021	0,063	0,022			0,022	0,016	0,347		0,016				0,042					0,016																0,565	0,614		
141			0,141							2,426					0,308																						2,875	0,844	
142										11,900				0,655																							12,555	0,948	
211											269,459			6,131		5,720																					281,310	0,958	
213																						0,005														0,005	0,000		
222												0,026	0,283		0,052							0,026															0,387	0,733	
231		8,855										9,451		419,254																							437,560	0,958	
242												0,563		0,375	2,064	0,188	0,188				0,188							0,188									3,753	0,550	
243										0,490	0,490			3,430		18,132						0,980						0,490									24,013	0,755	
311																2,295	18,360					2,295															22,950	0,800	
312														0,093				57,222	1,345			6,631															65,290	0,876	
313										0,278				0,278	0,278	1,114	0,278	11,418				0,278															13,925	0,820	
321														3,021		3,011					51,200																57,231	0,895	
322		3,922																		3,922	66,685	3,922															78,451	0,850	
324						0,010								0,324		0,020	0,007	2,987	1,011		0,529	34,479						0,565									39,931	0,863	
331																						0,097	1,848														1,946	0,950	
332																				0,072	0,216			1,010	0,144												1,443	0,700	
333																				0,010	0,635					11,426											12,071	0,947	
334																										0,042											0,042	1,000	
411														0,138													1,183	0,069									1,390	0,851	
412														2,706		2,706					5,413							124,493									135,319	0,920	
421																													1,611									1,611	1,000
423																				0,058									0,117	0,932						0,058	1,165	0,800	
511										0,044																						0,400					0,445	0,900	
512																																	13,723					13,723	1,000
521																																	0,050	0,012			0,062	0,800	
522																												0,018		0,018						0,286	0,036	0,357	0,800
523																																				0,927	0,927	1,000	
Total	0,939	69,656	9,031	0,725	0,536	2,082	3,144	0,284	0,896	2,526	14,379	280,267	0,283	437,021	2,116	32,350	19,669	60,526	13,774	55,220	73,756	48,934	1,848	1,010	11,571	0,042	1,183	126,254	1,746	0,932	0,418	13,888	0,050	0,298	1,021	1193,897			
Prod Acc	1,000	0,812	0,946	0,970	1,000	1,000	0,984	0,943	0,387	0,960	0,828	0,961	1,000	0,959	0,976	0,561	0,933	0,945	0,829	0,927	0,904	0,705	1,000	1,000	0,988	1,000	1,000	0,986	0,923	1,000	0,957	0,988	1,000	0,958	0,908				
																																					Global Acc	0,927	

Overall accuracy = 92.7%

Zone 11: Romania confusion matrix CLC2012 - Blind and plausibility analysis

Table 40: Confusion matrix CLC 2012 for zone 11 (Romania) – BLIND ANALYSIS -In column the validation and line the production

CLC Classes	112	121	122	123	124	131	132	133	141	142	211	213	221	222	231	242	243	311	312	313	321	322	324	331	332	333	411	421	423	511	512	521	523	Total	User Acc	
111	0,350																																	0,350	0,000	
112	23,540							0,042								11,730	6,703																	42,015	0,560	
121	0,005	3,897					0,206	0,048	0,008						0,005					0,005														4,174	0,934	
122	0,017	0,017	0,196	0,017				0,077																										0,326	0,602	
123				0,121																														0,121	1,000	
124					0,134						0,015																							0,149	0,900	
131		0,054	0,054			0,846		0,042							0,108								0,021						0,054				1,178	0,718		
132							0,140																											0,140	1,000	
133		0,011	0,032			0,011		0,074								0,032	0,011			0,021													0,192	0,387		
141									0,151								0,011																	0,162	0,933	
142	0,030							0,008		0,142						0,022								0,011							0,011			0,225	0,633	
211											292,216		6,639	6,639	6,650	13,278	6,639																	332,061	0,880	
212		0,033																																0,033	0,000	
213												1,184																						1,184	1,000	
221											3,604		3,003		0,601	1,802	2,403																	11,412	0,263	
222											1,393		0,011	5,581		2,785	3,482						0,739											13,990	0,399	
231	1,891	1,891						0,008			9,455				62,415		9,455						9,455											94,571	0,660	
242	1,509										1,509			1,509	1,509	21,129	1,509						1,509											30,184	0,700	
243														1,456	5,108	2,913	20,401						5,837											35,716	0,571	
311															0,009		3,758	176,689		0,018			7,627											188,102	0,939	
312										0,971									27,209	6,800			5,837											40,817	0,667	
313																				35,221	0,766		1,531											37,518	0,939	
321										1,131					1,131								12,444	3,394		4,525								22,625	0,000	
322																		0,141		0,141	1,980	0,141											2,404	0,824		
324											0,275			0,549		0,799	0,821	0,032	0,807	0,274			10,525											14,080	0,747	
331															0,025							0,025	0,356	0,025		0,025				0,051				0,509	0,700	
332																					0,016	0,047			0,031	0,047								0,140	0,222	
333																										0,132			0,019						0,151	0,875
411										0,203					0,609												10,207	0,600			0,600				12,220	0,835
421																										0,255									0,255	0,000
511																							0,347							6,601				6,949	0,950	
512																											0,606				5,502				6,107	0,901
521																																2,568			2,568	1,000
523																																	0,081		0,081	1,000
Total	27,343	5,903	0,282	0,139	0,134	0,857	0,346	0,301	0,160	1,114	309,526	1,184	9,653	15,461	78,718	53,691	55,170	177,510	27,382	42,872	1,181	1,995	55,737	4,108	0,056	4,704	11,093	0,600	0,019	6,706	6,113	2,568	0,081	712,274		
Prod Acc	0,861	0,660	0,695	0,874	1,000	0,988	0,404	0,246	0,947	0,128	0,944	1,000	0,311	0,361	0,793	0,394	0,370	0,995	0,994	0,822	0,000	0,992	0,189	0,087	0,549	0,028	0,920	0,000	0,000	0,984	0,900	1,000	1,000			
																																			Global Acc	0,789

Overall accuracy = 78.9%

Table 41: Confusion matrix CLC 2012 for zone 11 (Romania) – PLAUSIBILITY ANALYSIS -In column the validation and line the production

CLC Class	111	112	121	122	123	124	131	132	133	141	142	211	212	213	221	222	231	242	243	311	312	313	321	322	324	331	332	333	411	511	512	521	523	Total	User Acc		
111	0,175	0,175																																0,350	0,500		
112		37,826																3,351	0,838															42,015	0,900		
121		0,005	4,108					0,005	0,035									0,021																	4,174	0,984	
122				0,249					0,077																											0,326	0,763
123					0,121																															0,121	1,000
124						0,149																														0,149	1,000
131							1,028		0,021								0,054								0,075											1,178	0,873
132								0,140																												0,140	1,000
133		0,011	0,011	0,011			0,011		0,138									0,011																		0,192	0,721
141										0,151									0,011																	0,162	0,933
142		0,008									0,216																									0,225	0,962
211												318,772						0,011	13,278																332,061	0,960	
212			0,011									0,011	0,011																						0,033	0,333	
213														1,184																						1,184	1,000
221												1,201			4,204		1,802	3,003	1,201																	11,412	0,368
222														0,011	7,670	0,696	0,696	2,785	0,696						1,435										13,990	0,548	
231									1,900			5,673					85,107								1,891											94,571	0,900
242																	1,509	28,675																		30,184	0,950
243											0,011		0,728		0,728		4,369	2,913	24,771						2,195											35,716	0,694
311																			0,009	#####	0,009	0,018			3,841											188,102	0,979
312																					35,952	0,971			3,886											40,809	0,881
313																						36,752	0,766													37,518	0,980
321																				2,262			11,312		5,656	2,262				1,131						22,625	0,500
322																							0,141	2,262												2,404	0,941
324													0,008	0,532			0,532	0,821		0,274	0,008		0,008		11,903											14,080	0,845
331																							0,025		0,025	0,458										0,509	0,900
332																								0,016	0,047		0,047	0,031								0,140	0,333
333																												0,151								0,151	1,000
411											0,203						0,600													10,816		0,600				12,220	0,885
421																													0,255							0,255	0,000
511																									0,347					6,601						6,949	0,950
512																													0,606		5,502					6,107	0,901
521																																2,568				2,568	1,000
523																																		0,081	0,081		1,000
Total	0,175	38,025	4,130	0,259	0,121	0,149	1,039	0,144	2,171	0,151	0,216	325,871	0,011	1,913	4,215	8,407	94,681	51,949	30,148	#####	35,961	38,016	12,253	2,278	30,954	3,068	0,047	0,182	12,808	6,601	6,102	2,568	0,081	823,327			
Prod Acc	1,000	0,995	0,995	0,959	1,000	1,000	0,990	0,968	0,064	1,000	1,000	0,978	1,000	0,619	0,997	0,912	0,899	0,552	0,822	0,980	1,000	0,967	0,923	0,993	0,385	0,149	1,000	0,830	0,844	1,000	0,902	1,000	1,000		Global Acc 0,912		

Overall accuracy = 91.2%

Zone 12: Greece confusion matrix CLC2012 - Blind and plausibility analysis

Table 42: Confusion matrix CLC 2012 for zone 12 (Greece) –BLIND ANALYSIS - In column the validation and line the production

CLC Classes	111	112	121	122	123	124	131	132	133	141	142	211	212	213	221	222	223	231	241	242	243	311	312	313	321	322	323	324	331	332	333	334	411	421	422	511	512	521	522	523	Total	User Acc					
	111	1,832	0,131																																						1,963	0,933					
112		19,586															0,810																									20,396	0,960				
121			0,250	4,884	0,287				0,113																																		5,534	0,883			
122				1,809			0,106													0,106																							2,021	0,895			
123					0,103																																						0,103	1,000			
124						1,154																																					1,154	1,000			
131							2,457		0,174															0,174																			2,805	0,876			
132								0,073																																			0,073	1,000			
133			0,128	0,033			0,098		1,407																			0,098																1,764	0,798		
141										0,142																								0,036									0,178	0,800			
142		0,142									1,496									0,110																							1,748	0,856			
211												96,028						3,340		13,245							0,023	3,311															115,947	0,828			
212													62,071	0,120															0,017														62,209	0,998			
213												0,180		2,476																					0,180								2,836	0,873			
221															2,201		0,367			3,302	1,101	0,367																					7,338	0,300			
222												0,442				8,845				1,327	0,442																							11,056	0,800		
223																	41,212	1,873	1,873	7,493	1,873						1,873																	56,198	0,733		
231		0,310																5,290		0,929	0,619					0,648																		7,795	0,679		
241			0,175																1,053	0,526	0,526					0,175		0,175																2,632	0,400		
242			2,088								2,111					4,176		2,088	2,088	39,690	10,439																							62,679	0,633		
243																		2,861			88,734	2,861		2,861	2,861																		100,180	0,886			
311																				5,405	85,370		2,796					2,999																96,570	0,884		
312																					0,022	2,279	58,836	4,536																				67,929	0,866		
313																				1,386	0,036	0,036	36,701					2,808					1,386											42,353	0,867		
321																				2,374				47,514	4,749	7,123	19,399		2,374															83,533	0,569		
322																							1,119	2,239						0,373														3,731	0,600		
323																		4,761		9,523	0,203	4,761	4,761		162,535	4,964																		191,509	0,849		
324																		2,582		2,785	5,222	5,247	5,247	0,044	0,083	1,353	74,215		0,203	2,582	0,083														99,646	0,745	
331																		0,164												1,963					0,327									2,454	0,800		
332																					0,156			0,156		0,156			0,156	0,937	0,781													2,343	0,400		
333																						4,215	3,996	2,410	1,004						9,498	1,311												22,434	0,423		
334																		0,105						0,105								2,561												2,916	0,878		
411																																	1,547				0,119							1,785	0,867		
421												0,165																						2,309										2,474	0,933		
422																																			0,217									0,217	1,000		
511																											0,120																		1,798	0,933	
512																																													8,086	1,000	
521																																													0,654	1,000	
522																																													0,049	0,098	
523																																														2,812	1,000
Total	1,832	20,547	7,180	2,096	0,103	1,154	2,661	0,073	1,694	0,142	1,496	98,925	62,071	2,596	2,201	13,020	42,389	23,449	5,014	61,108	131,006	98,210	64,119	57,077	61,600	11,066	175,474	109,375	2,120	3,514	13,234	3,954	2,969	2,817	0,217	1,797	8,086	0,703	0,098	2,812	882,314						
Prod Acc	1,000	0,953	0,680	0,863	1,000	1,000	0,923	1,000	0,831	1,000	1,000	0,971	1,000	0,954	1,000	0,679	0,972	0,226	0,210	0,650	0,677	0,869	0,918	0,643	0,771	0,202	0,926	0,679	0,926	0,267	0,718	0,648	0,521	0,820	1,000	0,934	1,000	0,930	1,000	1,000					Global Acc	0,802	

Overall accuracy = 80.2%

Table 43: Confusion matrix CLC 2012 for zone 12 (Greece) –PLAUSIBILITY ANALYSIS - In column the validation and line the production

CLC Classe	111	112	121	122	123	124	131	132	133	141	142	211	212	213	221	222	223	231	241	242	243	311	312	313	321	322	323	324	331	332	333	334	411	421	422	511	512	521	522	523	Total	User Acc																																																																																																																																																																																																																																																																																																																																																					
111	1,832	0,131																																							1,963	0,933																																																																																																																																																																																																																																																																																																																																																					
112		20,396																																								20,396	1,000																																																																																																																																																																																																																																																																																																																																																				
121		0,250	4,932	0,287					0,066																																	5,534	0,891																																																																																																																																																																																																																																																																																																																																																				
122				1,809			0,106													0,106																							2,021	0,895																																																																																																																																																																																																																																																																																																																																																			
123					0,103																																						0,103	1,000																																																																																																																																																																																																																																																																																																																																																			
124						1,154																																					1,154	1,000																																																																																																																																																																																																																																																																																																																																																			
131			0,174				2,631																																				2,805	0,938																																																																																																																																																																																																																																																																																																																																																			
132								0,073																																			0,073	1,000																																																																																																																																																																																																																																																																																																																																																			
133		0,128					0,197		1,342																				0,098														1,764	0,760																																																																																																																																																																																																																																																																																																																																																			
141										0,142																								0,036									0,178	0,800																																																																																																																																																																																																																																																																																																																																																			
142		0,142									1,496									0,110																							1,748	0,856																																																																																																																																																																																																																																																																																																																																																			
211												96,028						3,340		13,245							0,023	3,311															115,947	0,828																																																																																																																																																																																																																																																																																																																																																			
212													62,174	0,035																													62,209	0,999																																																																																																																																																																																																																																																																																																																																																			
213														2,656																					0,180								2,836	0,937																																																																																																																																																																																																																																																																																																																																																			
221															4,769	0,367			1,468	0,367	0,367																						7,338	0,650																																																																																																																																																																																																																																																																																																																																																			
222											0,442					9,287			0,884	0,442																								11,056	0,840																																																																																																																																																																																																																																																																																																																																																		
223																	43,085		1,873	1,873	7,493	1,873																						56,198	0,767																																																																																																																																																																																																																																																																																																																																																		
231																		6,219		0,310	0,310				0,957																			7,795	0,798																																																																																																																																																																																																																																																																																																																																																		
241			0,175								0,351								1,053	0,702	0,175							0,175																2,632	0,400																																																																																																																																																																																																																																																																																																																																																		
242			2,088									2,111				4,176	2,088		2,088	43,866	6,263																							62,679	0,700																																																																																																																																																																																																																																																																																																																																																		
243																			2,861	88,705	2,890	2,861	2,861																					100,180	0,885																																																																																																																																																																																																																																																																																																																																																		
311																				5,405	90,775		0,093					0,296																96,570	0,940																																																																																																																																																																																																																																																																																																																																																		
312				0,022																		67,907																						67,929	1,000																																																																																																																																																																																																																																																																																																																																																		
313																							39,545					2,808																42,353	0,934																																																																																																																																																																																																																																																																																																																																																		
321																								57,011		9,497	17,025																		83,533	0,682																																																																																																																																																																																																																																																																																																																																																	
322																								0,933	2,798																			3,731	0,750																																																																																																																																																																																																																																																																																																																																																		
323																		4,761		4,761	0,203			4,761		176,819	0,203																		191,509	0,923																																																																																																																																																																																																																																																																																																																																																	
324																				2,785	0,057		2,665		0,083	1,156	90,236		0,083	2,582														99,646	0,906																																																																																																																																																																																																																																																																																																																																																		
331																		0,164											2,127					0,164										2,454	0,867																																																																																																																																																																																																																																																																																																																																																		
332			0,156																	0,156					0,156		0,156				1,250	0,469												2,343	0,533																																																																																																																																																																																																																																																																																																																																																		
333																										2,472	1,658				17,796	0,509												22,434	0,793																																																																																																																																																																																																																																																																																																																																																		
334																								0,045	0,211		0,056	0,105				2,500												2,916	0,857																																																																																																																																																																																																																																																																																																																																																		
411																											0,238								1,547									1,785	0,867																																																																																																																																																																																																																																																																																																																																																		
421											0,165																								2,309									2,474	0,933																																																																																																																																																																																																																																																																																																																																																		
422																																					0,217							0,217	1,000																																																																																																																																																																																																																																																																																																																																																		
511																																						1,798							1,798	1,000																																																																																																																																																																																																																																																																																																																																																	
512																																							8,086						8,086	1,000																																																																																																																																																																																																																																																																																																																																																	
521																																													0,654	1,000																																																																																																																																																																																																																																																																																																																																																	
522																																			0,049										0,148	0,667																																																																																																																																																																																																																																																																																																																																																	
523																																													2,812	2,812	1,000																																																																																																																																																																																																																																																																																																																																																
Total	1,832	21,047	7,525	2,118	0,103	1,154	2,934	0,073	1,407	0,142	1,496	99,096	62,174	2,690	4,769	13,462	45,540	14,483	5,014	65,425	#####	96,165	67,907	45,209	66,890	2,881	190,178	116,155	2,127	1,250	18,347	5,591	1,583	2,702	0,217	1,798	8,086	0,654	0,098	2,812	960,038																																																																																																																																																																																																																																																																																																																																																						
Prod Acc	1,000	0,969	0,655	0,854	1,000	1,000	0,897	1,000	0,953	1,000	1,000	0,969	1,000	0,987	1,000	0,690	0,946	0,429	0,210	0,670	0,759	0,944	1,000	0,875	0,852	0,971	0,930	0,777	1,000	1,000	0,970	0,447	0,978	0,855	1,000	1,000	1,000	1,000	1,000	1,000	1,000																																																																																																																																																																																																																																																																																																																																																						

Overall accuracy = 87.3%

Zone 13: Bulgaria confusion matrix CLC2012- Blind and plausibility analysis

Table 44: Confusion matrix CLC 2012 for zone 13 (Bulgaria) – BLIND ANALYSIS -In column the validation and line the production

CLC Classes	111	112	121	122	123	124	131	132	133	141	142	211	213	221	222	223	231	242	243	311	312	313	321	322	324	331	332	333	411	412	422	511	512	521	523	Total	User Acc	
111	0,063																																			0,063	1,000	
112		22,390	1,493															4,478	1,493																	29,853	0,750	
121			0,295	4,795					0,026			0,308		0,013					0,295															0,295		6,028	0,795	
122				0,067	0,266												0,013																			0,346	0,769	
123					0,059																															0,059	1,000	
124						0,441																														0,441	1,000	
131							1,606					0,013					0,672	0,134				0,134			0,143		0,134									2,837	0,566	
132								0,258																												0,258	1,000	
133				0,030			0,009		0,052																											0,091	0,573	
141										0,292									0,032																	0,325	0,900	
142		0,114	0,050								0,918														0,009											1,091	0,842	
211			5,930									273,831		0,013	6,008		5,995	6,059							0,026											297,862	0,919	
213													1,493																							1,493	1,000	
221												1,536		7,679	0,043				1,045																	10,303	0,745	
222												0,209		0,192	2,813		0,200	0,209	0,384																	4,007	0,702	
231		0,607	0,607			0,607						0,064				0,607	23,083		1,886					1,822	1,215											30,500	0,757	
242			0,998								0,998	1,062		1,995			0,998	3,991	9,976																	20,017	0,199	
243											4,057							64,917							8,115											77,089	0,842	
311																			170,164						3,675											173,839	0,979	
312																		0,833		35,833	3,333				1,667											41,667	0,860	
313										0,999												43,967			3,997												48,963	0,898
321											1,562					1,562							23,436		4,700												31,261	0,750
322																							0,412	1,647												2,059	0,800	
324				1,132						0,014		1,145		1,159	1,132		0,046		4,529	1,522	0,014	0,038	1,147		46,401											58,279	0,796	
331																	0,016								0,016	0,115								0,016		0,165	0,700	
332																				0,049		0,049					0,438	0,292								0,828	0,529	
333										0,151							0,301	0,151				0,301		0,151					1,808							2,862	0,632	
334																												0,034								0,034	0,000	
411					0,035							0,035					0,070												0,559							0,699	0,800	
412																														0,101						0,101	1,000	
422																															0,061					0,061	1,000	
511																		0,482							0,482							1,447			2,412	0,600		
512																									0,252		0,252							4,574		5,077	0,901	
521																											0,252		0,252						0,057		0,057	1,000
523																																			0,079		0,079	1,000
Total	0,063	23,406	13,940	1,428	0,094	0,441	2,222	0,258	0,078	0,307	3,066	283,823	1,493	11,052	9,996	0,607	32,940	8,694	92,218	171,686	35,896	47,472	25,344	3,469	70,849	0,115	0,438	2,520	0,559	0,101	0,061	1,447	4,869	0,057	0,095	739,635		
Prod Acc	1,000	0,957	0,344	0,186	0,628	1,000	0,723	1,000	0,662	0,953	0,300	0,965	1,000	0,695	0,281	0,000	0,701	0,459	0,704	0,991	0,998	0,926	0,925	0,475	0,655	1,000	1,000	0,717	1,000	1,000	1,000	0,939	1,000	0,827				
																																					Global Acc	0,869

Overall accuracy = 86.9%

Table 45: Confusion matrix CLC 2012 for zone 13 (Bulgaria) – PLAUSIBILITY ANALYSIS -In column the validation and line the production

CLC Class	111	112	121	122	123	124	131	132	133	141	142	211	213	221	222	231	242	243	311	312	313	321	322	324	331	332	333	411	412	422	511	512	521	523	Total	User Acc
111	0,063																																	0,063	1,000	
112		29,853																																29,853	1,000	
121		0,295	5,720									0,013																						6,028	0,949	
122			0,067	0,279																														0,346	0,808	
123					0,059																													0,059	1,000	
124						0,441																												0,441	1,000	
131							1,619									0,806	0,269							0,143										2,837	0,571	
132								0,258																										0,258	1,000	
133				0,030			0,009		0,052																									0,091	0,573	
141		0,032								0,292																								0,325	0,900	
142		0,114	0,050								0,918														0,009									1,091	0,842	
211												291,661		0,013	0,077	6,033		0,064						0,013										297,862	0,979	
213													1,493																					1,493	1,000	
221												1,003		8,234	0,021	0,555		0,491																10,303	0,799	
222												0,209			3,414		0,384																	4,007	0,852	
231												0,064				27,399							1,215	1,215										29,893	0,917	
242											0,998	1,062				1,995	8,979	6,983																20,017	0,449	
243																		73,032							4,057									77,089	0,947	
311																			173,789						0,050									173,839	1,000	
312																				40,833					0,833									41,667	0,980	
313																					45,965				2,998									48,963	0,939	
321																1,562	1,562					24,998		3,138										31,261	0,800	
322																						0,137	1,784											1,921	0,929	
324												0,013			1,132	1,132		2,264	1,389	0,014	0,019	0,014		52,291										58,270	0,897	
331												0,016														0,132							0,016	0,165	0,800	
332																				0,049						0,438	0,341							0,828	0,529	
333										0,151						0,151						0,452	0,151				1,958							2,862	0,684	
334																						0,034												0,034	0,000	
411					0,035							0,035																0,629						0,699	0,900	
412																													0,101					0,101	1,000	
422																														0,061				0,061	1,000	
511																															2,412			2,412	1,000	
512																							0,252			0,252						4,574		5,077	0,901	
521																																	0,057	0,057	1,000	
523																																	0,079	0,079	1,000	
Total	0,063	30,294	5,837	0,309	0,094	0,441	1,628	0,258	0,052	0,292	2,067	294,076	1,493	8,247	4,645	39,634	9,363	84,666	175,178	40,896	45,984	25,636	2,999	65,150	0,132	0,438	2,299	0,881	0,101	0,061	2,412	4,574	0,057	0,095	803,869	
Prod Acc	1,000	0,985	0,980	0,903	0,628	1,000	0,995	1,000	1,000	1,000	0,444	0,992	1,000	0,998	0,735	0,691	0,959	0,863	0,992	0,998	1,000	0,975	0,595	0,803	1,000	1,000	0,852	0,714	1,000	1,000	1,000	1,000	1,000	0,827		
																																			Global Acc	0,945

Overall accuracy = 94.5%

Zone 14: Island confusion matrix CLC2012 - Blind and plausibility analysis

Table 46: Confusion matrix CLC 2012 for zone 14 (Island) –BLIND ANALYSIS - In column the validation and line the production

CLC Classes	112	121	122	123	124	131	133	141	142	211	231	242	243	311	312	313	321	322	324	331	332	333	335	411	412	421	423	511	512	521	522	523	Total	User Acc
112	0,628																															0,628	1,000	
121		0,354		0,027													0,027					0,027										0,434	0,816	
122			0,017																													0,017	1,000	
123				0,043																												0,043	1,000	
124					0,171																											0,171	1,000	
131						0,072																										0,072	1,000	
133	0,037						0,036											0,037														0,111	0,323	
141								0,081										0,020														0,102	0,800	
142									0,889								0,018															0,907	0,980	
211										0,138																						0,138	1,000	
231										1,032	13,893		0,512				0,025	0,013							0,018							15,493	0,897	
242										0,016	0,020	0,253						0,032														0,320	0,789	
311														1,070				0,292	0,389													1,752	0,611	
312															0,094	0,056			0,037													0,187	0,500	
313																0,305	0,061	0,092														0,458	0,667	
321																	16,753	1,859					0,218									18,831	0,890	
322																		183,402				32,365										215,767	0,850	
324											0,113		0,095				0,204	0,773	0,853													2,037	0,419	
331																			18,965	2,253							0,318					21,536	0,881	
332																				117,863	14,448	14,648						0,018				146,977	0,802	
333																			2,844	10,941	68,615	0,109										82,509	0,832	
335																						64,568										64,568	1,000	
411																0,235			0,117	0,117				1,409	0,352			0,117				2,349	0,600	
412																	4,034	2,017						2,017	30,254							38,322	0,789	
421																								0,032		0,127						0,159	0,800	
423																									0,026	0,388			0,078	0,026		0,518	0,750	
511																				0,351								4,720				5,071	0,931	
512																												8,968				8,968	1,000	
521																													0,899			0,899	1,000	
522																											0,103			0,069		0,172	0,400	
523																															0,153	0,153	1,000	
Total	0,665	0,354	0,017	0,070	0,171	0,072	0,036	0,081	0,889	1,186	14,025	0,253	0,607	1,070	0,094	0,361	17,296	190,580	3,297	22,278	131,173	115,673	79,325	3,458	30,624	0,153	0,388	5,141	9,103	0,977	0,095	0,153	536,050	
Prod Acc	0,944	1,000	1,000	0,619	1,000	1,000	1,000	1,000	1,000	0,116	0,991	1,000	0,000	1,000	1,000	0,845	0,969	0,962	0,259	0,851	0,899	0,593	0,814	0,408	0,988	0,830	1,000	0,918	0,985	0,921	0,727	1,000		
																																	Global Acc	0,851

Overall accuracy = 85.1%

Table 47: Confusion matrix CLC 2012 for zone 14 (Island) –PLAUSIBILITY ANALYSIS - In column the validation and line the production

CLC Classe	112	121	122	123	124	131	133	141	142	211	231	242	243	311	312	313	321	322	323	324	331	332	333	335	411	412	421	423	511	512	521	522	523	Total	User Acc
112	0,628																																0,628	1,000	
121		0,381		0,027																			0,027										0,434	0,878	
122			0,017																														0,017	1,000	
123				0,043																													0,043	1,000	
124					0,171																												0,171	1,000	
131						0,072																											0,072	1,000	
133	0,037						0,036																										0,073	0,488	
141								0,102																									0,102	1,000	
142									0,889								0,018																0,907	0,980	
211										0,138																							0,138	1,000	
231										1,028	14,465																						15,493	0,934	
242											0,004	0,300																					0,304	0,986	
311													1,362					0,195		0,195													1,752	0,778	
312														0,131	0,037					0,019													0,187	0,700	
313															0,305	0,061	0,092																0,458	0,667	
321																17,901	0,930																18,831	0,951	
322																	204,979	10,788															215,767	0,950	
324											0,113		0,095				0,204	0,664		0,962													2,037	0,472	
331																					19,690	1,846											21,536	0,914	
332																						132,444		14,515					0,018				146,977	0,901	
333																					0,109	5,470	76,820	0,109									82,509	0,931	
335																							64,568										64,568	1,000	
411																0,235					0,117				1,762	0,117				0,117			2,349	0,750	
412																									2,017	34,288							36,305	0,944	
421																											0,159						0,159	1,000	
423																												0,414			0,078	0,026	0,518	0,800	
511																				0,351								4,720					5,071	0,931	
512																													8,968				8,968	1,000	
521																														0,899			0,899	1,000	
522																															0,172		0,172	1,000	
523																																0,153	0,153	1,000	
Total	0,665	0,381	0,017	0,070	0,171	0,072	0,036	0,102	0,889	1,166	14,582	0,300	0,095	1,362	0,131	0,343	18,419	206,858	10,788	1,176	20,267	139,760	76,847	79,192	3,778	34,405	0,159	0,414	4,720	9,103	0,977	0,198	0,153	587,938	
Prod Acc	0,944	1,000	1,000	0,619	1,000	1,000	1,000	1,000	1,000	0,118	0,992	1,000	0,000	1,000	1,000	0,891	0,972	0,991	0,000	0,819	0,972	0,948	1,000	0,815	0,466	0,997	1,000	1,000	0,985	0,921	0,870	1,000			
																																		Global Acc	0,937

Overall accuracy = 93.7%

Zone 15: Hungary confusion matrix CLC2012 - Blind and plausibility analysis

Table 48: Confusion matrix CLC 2012 for zone 15 (Hungary) – BLIND ANALYSIS -In column the validation and line the production

CLC Classes	111	112	121	122	123	124	131	132	133	141	142	211	213	221	222	231	242	243	311	312	313	321	324	333	411	412	511	512	Total	User Acc
111	0,228																												0,228	1,000
112		52,114	2,878		0,044												2,905												57,941	0,899
121			8,332	0,400				0,006	0,032	0,044		0,076				0,044		0,006											8,937	0,932
122			0,087	1,589					0,027			0,087											0,092						1,882	0,844
123					0,037																								0,037	1,000
124			0,183			0,733																							0,917	0,800
131							1,187					0,059		0,062		0,062			0,062			0,006	0,068					0,059	1,566	0,758
132				0,059				0,552				0,046				0,046								0,046					0,749	0,737
133		0,103	0,092	0,206			0,006		0,226			0,060					0,040	0,044					0,006						0,781	0,289
141										0,697																			0,697	1,000
142		0,226									4,167						0,226												4,619	0,902
211			0,023	12,582								594,943	0,154	12,766	1,187	1,710	1,326						12,603						637,294	0,934
213												0,540	0,540																1,080	0,500
221												0,766		13,453	0,017				0,731										14,967	0,899
222												1,219		0,089	7,841		0,609												9,759	0,804
231			3,774	0,171			0,012	0,025	0,171			11,993		0,482	0,482	76,104							2,031		0,012			1,825	97,084	0,784
242			3,757									0,059		1,878	0,079	0,482	28,296	3,757					0,039						38,348	0,738
243												0,965					18,388	2,163					1,082						22,598	0,814
311		0,119	3,989									3,989				0,334			158,390		3,871		29,139						199,832	0,793
312																			0,011	9,872	0,251		2,078						12,212	0,808
313												0,399							0,486	0,029	15,757		3,850						20,520	0,768
321																10,470						19,444							29,914	0,650
324			0,044									8,397		1,003	0,334	1,790		2,005	8,563	0,166	1,696		48,905	0,023				72,928	0,671	
333																								0,355					0,355	1,000
411																1,003						0,502	1,003		7,525			10,034	0,750	
412																0,247										0,987		1,234	0,800	
511																											6,067	6,067	1,000	
512												0,009				0,851												16,455	17,315	0,950
Total	0,228	52,562	23,160	15,007	0,081	0,733	1,205	0,582	0,456	0,740	4,167	623,607	0,694	29,735	9,941	93,143	33,402	24,199	170,407	10,067	21,574	19,951	100,897	0,424	7,538	0,987	6,067	18,339	1093,183	
Prod Acc	1,000	0,991	0,360	0,106	0,462	1,000	0,985	0,948	0,495	0,941	1,000	0,954	0,778	0,452	0,789	0,817	0,847	0,760	0,929	0,981	0,730	0,975	0,485	0,837	0,998	1,000	1,000	0,897		
																													Global Acc	0,861

Overall accuracy = 86.1%

Table 49: Confusion matrix CLC 2012 for zone 15 (Hungary) – PLAUSIBILITY ANALYSIS -In column the validation and line the production

CLC Classe	111	112	121	122	123	124	131	132	133	141	142	211	213	221	222	231	242	243	311	312	313	321	324	333	411	412	511	512	Total	User Acc
111	0,228																												0,228	1,000
112		55,063	2,878																										57,941	0,950
121			8,867					0,006		0,044		0,021																	8,937	0,992
122				1,642								0,087						0,061			0,006		0,087						1,882	0,872
123					0,037																								0,037	1,000
124						0,917																							0,917	1,000
131							1,371					0,059							0,062			0,006	0,068						1,566	0,875
132				0,059				0,644				0,046																	0,749	0,859
133		0,034	0,036	0,097					0,447			0,048				0,034		0,078	0,006										0,781	0,572
141										0,697																			0,697	1,000
142		0,452									4,167																		4,619	0,902
211				0,012								608,469	0,154	0,105	0,705	0,830	14,366	12,557					0,097						637,294	0,955
213													1,080																1,080	1,000
221												0,035		14,184	0,017			0,731											14,967	0,948
222												0,273		0,030	8,847		0,580												9,729	0,909
231			1,973	0,171					0,171			10,021	0,482	0,482	81,532					0,171	0,012	0,255		0,012			1,802	97,084	0,840	
242			3,757									0,039			0,020		30,755	3,757					0,020						38,348	0,802
243																		20,434	2,163										22,598	0,904
311			3,871															0,119	175,538		3,871		16,434						199,832	0,878
312																				11,641	0,251		0,320						12,212	0,953
313												0,399							0,486	0,058	17,098		2,480						20,520	0,833
321												1,496				4,487	1,496					20,939	1,496						29,914	0,700
324												4,272		1,337		0,089	1,002	3,062	1,044	0,270		61,851						72,928	0,848	
333																								0,355					0,355	1,000
411																1,505						0,502			8,027				10,034	0,800
412																0,247										0,987			1,234	0,800
511																										6,067			6,067	1,000
512												0,009											0,023					17,283	17,315	0,998
Total	0,228	55,549	21,381	1,982	0,037	0,917	1,371	0,649	0,618	0,740	4,167	625,273	1,234	16,137	10,071	88,724	45,700	40,236	181,317	12,743	21,666	20,958	83,631	0,355	8,040	0,987	6,067	19,085	1159,168	
Prod Acc	1,000	0,991	0,415	0,828	1,000	1,000	1,000	0,991	0,724	0,941	1,000	0,973	0,875	0,879	0,878	0,919	0,673	0,508	0,968	0,914	0,789	0,999	0,740	1,000	0,998	1,000	1,000	0,906		
																													Global Acc	0,913

Overall accuracy = 91.3%

Zone 16: Portugal confusion matrix CLC2012 - Blind and plausibility analysis

Table 50: Confusion matrix CLC 2012 for zone 16 (Portugal) – BLIND ANALYSIS - In column the validation and line the production

CLC Classes	111	112	121	122	123	124	131	132	133	141	142	211	212	213	221	222	223	231	241	242	243	244	311	312	313	321	322	323	324	331	332	333	334	411	412	421	422	423	511	512	521	522	523	Total	User Acc							
111	1,120	0,603																																											1,723	0,650						
112	1,710	30,956	1,710																																											34,376	0,901					
121			6,242								0,049										0,049																									6,340	0,984					
122				1,495																	0,071																									1,565	0,955					
123					0,246																																										0,246	1,000				
124						0,778											0,081																														0,858	0,906				
131				0,206			1,845																							0,206																		2,257	0,818			
132				0,021				0,193			0,021																																				0,235	0,823				
133				0,093	0,049				0,986		0,044										0,044																											1,215	0,811			
141		0,080	0,040							0,239	0,040																																				0,399	0,600				
142		0,139									2,205																																					2,344	0,941			
211												91,054	7,524					4,735		9,470	4,784									2,789																		120,357	0,757			
212												3,250	24,903					0,602		3,009																													31,764	0,784		
213														6,395				0,314																															6,709	0,953		
221												2,926			22,362					2,926	2,926																												31,140	0,718		
222																8,404			1,114		1,720																													11,238	0,748	
223												2,566					40,658		2,202	4,404	2,202																													52,031	0,781	
231												1,894						12,827			3,787						0,947																							19,454	0,659	
241		4,513																2,257	11,284	11,284	11,284	4,513																											45,135	0,250		
242												17,631		4,408				4,408		57,348	4,408																													88,202	0,650	
243		5,637																		5,637	90,661						5,637			5,637																				113,209	0,801	
244											4,300											81,741																													86,041	0,950
311															2,923							37,996	157,308																											214,312	0,734	
312																								68,902																	</											

Overall accuracy = 75.9%

[illegible]

Overall accuracy = 84.8%

Zone 17: Austria/Switzerland confusion matrix CLC2012 - Blind and plausibility analysis

Table 52: Confusion matrix CLC 2012 for zone 17 (Austria/Switzerland) – BLIND ANALYSIS- In column the validation and line the production

CLC Classes	111	112	121	122	123	124	131	133	141	142	211	221	222	231	242	243	311	312	313	321	322	324	331	332	333	335	411	412	511	512	Total	User Acc	
111	0,888	0,157																													1,045	0,850	
112		40,739													0,024	2,259		2,259													45,281	0,900	
121		0,187	3,618				0,004				0,015																				3,823	0,946	
122		0,048		0,519				0,016																							0,583	0,890	
123					0,031																										0,031	1,000	
124						0,447					0,049																				0,496	0,901	
131		0,044	0,044	0,016			0,757				0,093			0,088																	1,041	0,727	
132														0,035																	0,035	0,000	
133			0,022					0,011							0,022																0,056	0,200	
141		0,046							0,343						0,023		0,046														0,458	0,750	
142										1,591				0,008				0,495		0,134	0,016	0,016			0,235					2,496	0,637		
211											98,501	2,985		23,878	23,878																149,241	0,660	
221			0,296									5,335			0,296																5,927	0,900	
222		0,042										0,042	0,254		0,085																0,423	0,600	
231		3,937												35,429	11,810	11,810			7,873	3,937		0,031								74,826	0,473		
242											2,972				23,773	2,972															29,716	0,800	
243											2,148			1,074	6,445	10,741			1,074												21,482	0,500	
311																1,702	36,151		4,255			0,851									42,958	0,842	
312							4,227											152,209	12,682			38,122									207,240	0,734	
313																	4,444	2,222	102,772			2,222									111,661	0,920	
321															3,915			0,007		62,647	3,915	0,215			3,915						74,614	0,840	
322																		1,207			19,316	2,415									22,938	0,842	
324							0,007		0,137								0,478	1,927	0,248	0,186	0,093	5,011									8,088	0,620	
331																							0,030	0,119							0,148	0,200	
332																				2,364				40,408	4,727	0,097					47,596	0,849	
333									0,137					0,006						5,726	0,137				1,909	28,629					36,544	0,783	
335																								0,685		10,724					11,408	0,940	
411														0,088			0,088					0,088						1,496			1,760	0,850	
412														0,025			0,025											0,076			0,127	0,600	
511																													1,928		1,928	1,000	
512																											0,006				13,817	13,823	1,000
Total	0,888	45,200	3,981	0,535	0,031	0,447	4,988	0,034	0,343	1,866	103,777	8,362	0,254	60,631	70,271	29,483	41,120	160,439	128,904	74,993	23,478	48,971	0,030	43,120	37,506	10,821	1,503	0,076	1,928	13,817	698,193		
Prod Acc	1,000	0,901	0,909	0,970	1,000	1,000	0,152	0,326	1,000	0,853	0,949	0,638	1,000	0,584	0,338	0,364	0,879	0,949	0,797	0,835	0,823	0,102	1,000	0,937	0,763	0,991	0,996	1,000	1,000	1,000			
																																Global Acc	0,761

Overall accuracy = 76.1%

Table 53: Confusion matrix CLC 2012 for zone 17 (Austria / Switzerland) – PLAUSIBILITY ANALYSIS- In column the validation and line the production

CLC Class	111	112	121	122	123	124	131	132	133	141	142	211	221	222	231	241	242	243	311	312	313	321	322	324	331	332	333	335	411	412	511	512	Total	User Acc	
111	0,888	0,157																															1,045	0,850	
112		40,748															0,016	2,259		2,259													45,281	0,900	
121		0,187	3,618				0,004					0,015																					3,823	0,946	
122		0,048		0,535																													0,583	0,918	
123					0,031																												0,031	1,000	
124						0,496																											0,496	1,000	
131			0,044				0,877					0,016			0,044						0,044												1,025	0,856	
132								0,035																									0,035	1,000	
133			0,022						0,022								0,011																0,056	0,400	
141										0,389							0,023		0,046														0,458	0,850	
142											1,717				0,008					0,260		0,252		0,008			0,118						2,362	0,727	
211												104,470	2,985	20,893	20,893																		149,241	0,700	
221													5,927																				5,927	1,000	
222		0,042												0,296		0,042	0,042																0,423	0,700	
231															62,998			11,810						0,018									74,826	0,842	
242											2,972						26,744																	29,716	0,900
243														1,074			3,222	16,112			1,074												21,482	0,750	
311																		1,702	37,853		1,702			0,851									42,107	0,899	
312																		4,227		177,614	8,454			16,944									207,240	0,857	
313																					111,661												111,661	1,000	
321																				0,020		74,408		0,186									74,614	0,997	
322																				1,207			19,316	2,415									22,938	0,842	
324																			0,293	0,843	0,102			6,758									7,995	0,845	
331																									0,030	0,119							0,148	0,200	
332																						2,364				42,853	2,364	0,016					47,596	0,900	
333										0,137					0,006							3,817		0,137		1,909	30,537						36,544	0,836	
335																										0,228		11,180						11,408	0,980
411															0,088					0,088				0,088					1,496				1,760	0,850	
412															0,025					0,025										0,076			0,127	0,600	
511																															1,928		1,928	1,000	
512																													0,006				13,817	13,823	1,000
Total	0,888	41,182	3,684	0,535	0,031	0,496	0,881	0,035	0,022	0,389	1,854	107,473	8,912	0,296	85,137	0,042	50,952	36,110	38,191	182,316	123,037	80,840	19,316	27,406	0,030	45,108	33,018	11,196	1,503	0,076	1,928	13,817	795,431		
Prod Acc	1,000	0,989	0,982	1,000	1,000	1,000	0,996	1,000	1,000	1,000	0,926	0,972	0,665	1,000	0,740	0,000	0,525	0,446	0,991	0,974	0,908	0,920	1,000	0,247	1,000	0,950	0,925	0,999	0,996	1,000	1,000	1,000			
																																		Global Acc	0,868

Overall accuracy = 86.8%

Zone 18: Denmark/Netherland/Belgium /Luxembourg confusion matrix CLC2012 - Blind and plausibility analysis

Table 54: Confusion matrix CLC 2012 for zone 18 (Denmark, Netherland, Belgium, Luxembourg) – BLIND ANALYSIS - In column the validation and line the production

CLC Classes	111	112	121	122	123	124	131	132	133	141	142	211	221	222	231	242	243	311	312	313	321	322	324	331	411	412	421	423	511	512	521	522	523	Total	User Acc	
111	1,069	0,056																																1,125	0,950	
112	4,965	81,492	0,128	4,965					0,221			5,102						4,965																101,837	0,800	
121			14,546	0,788	0,035			0,788	0,930		0,160																							17,248	0,843	
122				2,375					0,011																									2,386	0,996	
123				0,111	2,217																													2,328	0,952	
124						1,864																												1,864	1,000	
131				0,076			1,429																0,076							0,152				1,733	0,825	
132				0,025				0,276	0,025						0,025								0,025											0,377	0,733	
133		0,035	0,037	0,037			0,020	0,032	2,651		0,084	0,130			0,130	0,072																		3,227	0,821	
141		0,140								2,701	0,140				0,140																			3,121	0,865	
142		0,640								0,732	13,241																							14,612	0,906	
211	15,690											353,346			0,020	0,020	7,845																	376,922	0,937	
221													0,141																					0,141	1,000	
222												0,199	0,090	1,225		0,269	0,090																	1,873	0,654	
231										6,791		0,060			115,453	13,583																		135,887	0,850	
241												0,008																						0,008	0,000	
242											17,969				13,477	80,883																		112,329	0,720	
243		3,113									3,113				12,451	3,113	40,487																	62,276	0,650	
311																		38,004					0,364											38,367	0,991	
312		1,773	0,887	0,119															35,584	4,433			1,892											44,688	0,796	
313		1,925										0,962						3,850	1,925	39,697														48,359	0,821	
321			0,040						0,040						1,440	0,383	0,343				4,842		0,423						0,119					7,630	0,635	
322															0,503						0,503	7,450	2,178										10,634	0,701		
324											0,080					1,020		0,618	0,632				9,217											11,567	0,797	
331																							1,894			0,105	0,105							2,105	0,900	
411								0,045															0,316	5,868					0,361	0,045				6,636	0,884	
412																	0,173		0,345	0,218	0,173	0,345		0,345	1,899								3,498	0,543		
421																				0,179				0,179		3,037					0,179			3,573	0,850	
423																										0,031	0,310					0,031	0,093	0,466	0,667	
511																												3,899	0,045				0,975	4,919	0,793	
512																												2,964	23,982				2,964	29,910	0,802	
521																														0,369				0,369	1,000	
522																																0,444		0,444	1,000	
523																																	0,748	0,748	1,000	
Total	6,033	104,864	15,739	8,395	2,251	1,864	1,449	1,096	3,923	3,433	20,417	380,971	0,230	1,225	143,641	98,322	49,784	46,991	38,127	45,107	5,742	7,622	14,836	1,894	6,392	1,899	3,174	0,416	7,224	24,344	0,548	4,414	0,842	892,642		
Prod Acc	0,177	0,777	0,924	0,283	0,985	1,000	0,986	0,252	0,676	0,787	0,649	0,927	0,611	1,000	0,804	0,823	0,813	0,809	0,933	0,880	0,843	0,977	0,621	1,000	0,918	1,000	0,957	0,747	0,540	0,985	0,674	0,101	0,889			
																																			Global Acc	0,848

Overall accuracy = 84.8%

Table 55: Confusion matrix CLC 2012 for zone 18 (Denmark, Netherland, Belgium, Luxembourg) –PLAUSIBILITY ANALYSIS - In column the validation and line the production

CLC Classe	111	112	121	122	123	124	131	132	133	141	142	211	221	222	231	242	243	311	312	313	321	322	324	331	411	412	421	423	511	512	521	522	523	Total	User Acc	
111	1,069	0,056																																1,125	0,950	
112	4,965	81,680	0,043	4,965					0,221			0,035			4,965				4,965															101,837	0,802	
121			16,173		0,035				0,880	0,160																								17,248	0,938	
122				2,375					0,011																									2,386	0,996	
123				0,111	2,217																													2,328	0,952	
124						1,864																												1,864	1,000	
131				0,076			1,429																0,076						0,152					1,733	0,825	
132				0,025				0,276	0,025						0,025								0,025											0,377	0,733	
133		0,014	0,068	0,037			0,020	0,032	2,803			0,093			0,093	0,014													0,032					3,207	0,874	
141		0,140								2,841	0,140																							3,121	0,910	
142										0,640	13,972																							14,612	0,956	
211		7,845								7,845		353,346			0,020	0,020	7,845																	376,922	0,937	
221													0,141																					0,141	1,000	
222												0,020		1,404		0,359	0,090																	1,873	0,750	
231										6,791	0,060				115,453	6,791	6,791																	135,887	0,850	
241																	0,008																	0,008	0,000	
242											11,231				11,231	89,867																		112,329	0,800	
243		3,113									3,113				3,113		52,938																	62,276	0,850	
311																		38,242					0,126											38,367	0,997	
312		1,773																	36,590	4,433			1,892											44,688	0,819	
313		0,962							0,962		0,962							0,962	0,962	40,660														45,472	0,894	
321									0,040						1,052	0,685					5,008		0,685						0,119					7,590	0,660	
322															0,503						0,503	8,960	0,668											10,634	0,843	
324									0,014		0,040						1,020		0,537	0,068			9,862											11,540	0,855	
331																							2,000			0,105								2,105	0,950	
411									0,045															6,184				0,361	0,045					6,636	0,932	
412																	0,173		0,173				0,218			2,935								3,498	0,839	
421																				0,179				0,179		3,037				0,179				3,573	0,850	
423																											0,403				0,031	0,031		0,466	0,867	
511																												3,899	0,045			0,975		4,919	0,793	
512																													26,946			2,964		29,910	0,901	
521																														0,369				0,369	1,000	
522																															0,444			0,444	1,000	
523																																	0,748	0,748	1,000	
Total	6,033	95,585	16,385	7,488	2,251	1,864	1,449	0,308	5,001	11,326	21,064	368,901	0,141	1,404	136,455	97,745	68,684	39,377	43,054	45,333	5,690	8,960	13,552	2,000	6,363	2,935	3,143	0,403	4,292	27,308	0,548	4,414	0,779	926,136		
Prod Acc	0,177	0,855	0,987	0,317	0,985	1,000	0,986	0,896	0,560	0,251	0,663	0,958	1,000	1,000	0,846	0,919	0,771	0,971	0,850	0,897	0,880	1,000	0,728	1,000	0,972	1,000	0,967	1,000	0,908	0,987	0,674	0,101	0,960			
																																			Global Acc	0,882

Overall accuracy = 88.2%

Zone 19: Albania/Serbia/Macedonia confusion matrix CLC2012 - Blind and plausibility analysis

Table 56: Confusion matrix CLC 2012 for zone 19 (Albania/Serbia/Macedonia) – BLIND ANALYSIS -In column the validation and line the production

CLC Classes	111	112	121	122	123	124	131	132	133	141	142	211	212	213	221	222	223	231	242	243	311	312	313	321	322	323	324	331	332	333	334	411	412	421	422	511	512	521	523	Total	User Acc	
111	0,082																																						0,082	1,000		
112		22,454																																					22,454	1,000		
121			2,576		0,005		0,099		0,081		0,020													0,005																2,787	0,924	
122			0,025	0,095					0,010																															0,130	0,727	
123					0,019																																			0,019	1,000	
124						0,270																																		0,270	1,000	
131							1,202	0,005										0,041		0,005	0,041										0,041					0,041				1,376	0,874	
132							0,015	0,247																																0,262	0,941	
133				0,034			0,020	0,013	0,400			0,044									0,005																			0,516	0,775	
141			0,022							0,219																														0,241	0,910	
142		0,018				0,036					0,244																													0,298	0,818	
211												104,413				6,110		0,051	25,580	6,060																				142,214	0,734	
212													1,238						0,147																					1,385	0,894	
213														0,416																										0,416	1,000	
221												0,163			2,105		0,137	0,163	0,137								0,163													2,869	0,734	
222												0,419			0,442	1,430					0,088						0,257													2,636	0,542	
223																1,606				1,285	0,100						0,321													3,312	0,485	
231												1,330						14,801	3,089		1,305			3,467	0,429		1,746													26,165	0,566	
241										0,046								0,006																						0,052	0,000	
242											0,051			6,748				4,217	83,564	6,748																				101,329	0,825	
243																		2,278	3,441	86,660	3,441		2,278				1,721													99,819	0,868	
244																				0,024																			0,024	0,000		
311										0,035									4,226	224,141			0,195				14,341													242,938	0,923	
312																					16,942	0,809					2,617													20,368	0,832	
313																					1,934	15,768	0,299				1,264													19,266	0,818	
321																		4,334						44,093	3,881	2,167	1,727			2,167										58,368	0,755	
322																								0,485	1,616		0,323													2,424	0,667	
323																			1,671	1,536				3,343		16,983	1,536													25,069	0,677	
324						0,026												2,490	1,731	3,369	12,577		1,678			2,470	80,561													104,901	0,768	
331																										0,055	1,580	0,055													1,689	0,935
332																								0,292				0,587	0,582											1,462	0,402	
333																		0,441						2,211	0,886		1,328			15,049									19,915	0,756		
334																					0,431		0,067				0,134				3,504								4,137	0,847		
411																	0,129										0,129						1,742			0,090	0,129			2,219	0,785	
412																																	0,003							0,003	1,000	
421																																		0,186						0,186	1,000	
422																																			0,123					0,123	1,000	
511																											1,078										2,899			3,977	0,729	
512																																					8,927			8,927	1,000	
521																																					0,434			0,434	1,000	
523																																						0,494			0,494	1,000
Total	0,082	22,472	2,623	0,128	0,024	0,307	1,363	0,265	0,491	0,219	0,264	106,454	1,284	0,416	9,295	7,540	1,606	28,482	118,157	110,185	243,665	18,876	20,796	54,194	6,811	21,621	108,223	2,658	0,642	17,798	3,504	1,783	0,003	0,186	0,123	2,989	9,097	0,434	0,494	759,672		
Prod Acc	1,000	0,999	0,982	0,737	0,777	0,882	0,882	0,931	0,814	1,000	0,924	0,981	0,965	1,000	0,226	0,190	1,000	0,520	0,707	0,786	0,920	0,898	0,758	0,814	0,237	0,786	0,744	0,594	0,915	0,846	1,000	0,977	1,000	1,000	0,970	0,981	1,000	1,000		Global Acc	0,821	

Overall accuracy = 82.1%

Table 57: Confusion matrix CLC 2012 for zone 19 (Albania/Serbia/Macedonia) – PLAUSIBILITY ANALYSIS In column the validation and line the production

CLC Classe	111	112	121	122	123	124	131	132	133	141	142	211	212	213	221	222	223	231	242	243	311	312	313	321	322	323	324	331	332	333	334	411	412	421	422	511	512	521	523	Total	User Acc		
111	0,082																																							0,082	1,000		
112		22,454																																						22,454	1,000		
121			2,701		0,005				0,081																															2,787	0,969		
122				0,130																																				0,130	1,000		
123					0,019																																			0,019	1,000		
124						0,270																																		0,270	1,000		
131							1,243	0,005											0,005	0,041									0,041											1,376	0,903		
132							0,015	0,247																																0,262	0,941		
133				0,010			0,010		0,472			0,024																												0,516	0,915		
141			0,022							0,219																														0,241	0,910		
142						0,018					0,280																													0,298	0,939		
211												136,078			0,025			0,051	6,060																					142,214	0,957		
212													1,238						0,147																					1,385	0,894		
213														0,416																										0,416	1,000		
221												0,163			2,405				0,137								0,163													2,869	0,838		
222												0,217				2,074					0,088						0,257													2,636	0,787		
223																	1,927				0,100						0,321													3,312	0,582		
231																		19,169	1,355	0,429					2,162			3,051												26,165	0,733		
241										0,046								0,006																						0,052	0,000		
242											0,051								98,746	2,532																				101,329	0,975		
243																		3,999	1,721	90,102				2,278																98,099	0,918		
244																				0,024																			0,024	0,000			
311																					238,527						4,411													242,938	0,982		
312																						19,163					1,205													20,368	0,941		
313																						0,967	17,703	0,299			0,297													19,266	0,919		
321																								52,308		2,167	1,727			2,167											58,368	0,896	
322																									1,939		0,485														2,424	0,800	
323																			1,536	1,536				0,271		20,190	1,536													25,069	0,805		
324							0,026											2,497	1,698	1,678	2,232		1,678			2,470	92,622													104,901	0,883		
331																										0,055	1,580	0,055													1,689	0,935	
332																								0,116		0,058		1,053	0,234												1,462	0,721	
333																								0,441			0,886			18,587										19,915	0,933		
334																											0,566														4,137	0,863	
411																																									2,219	1,000	
412																																									0,003	1,000	
421																																									0,186	1,000	
422																																									0,123	1,000	
511																												0,428														3,977	0,892
512																																									8,927	1,000	
521																																									0,434	1,000	
523																																									0,494	1,000	
Total	0,082	22,454	2,722	0,140	0,024	0,288	1,295	0,252	0,553	0,219	0,280	136,532	1,284	0,416	2,431	2,074	1,927	25,721	109,726	97,405	242,525	20,130	21,658	55,597	1,939	24,828	107,640	2,007	1,108	20,988	3,571	2,260	0,003	0,186	0,123	3,549	8,968	0,434	0,494	863,449			
Prod Acc	1,000	1,000	0,992	0,928	0,777	0,937	0,960	0,979	0,853	1,000	1,000	0,997	0,965	1,000	0,990	1,000	1,000	0,745	0,900	0,925	0,984	0,952	0,817	0,941	1,000	0,813	0,860	0,787	0,951	0,886	1,000	0,982	1,000	1,000	1,000	1,000	0,995	1,000	1,000	Global Acc	0,935		

Overall accuracy = 93.5%

Zone 20: Bosnia and Herzegovina/Croatia confusion matrix CLC2012 - Blind and plausibility analysis

Table 58: Confusion matrix CLC 2012 for zone 20 (Bosnia Herzegovina/Croatia) – BLIND ANALYSIS -In column the validation and line the production

CLC Classes	111	112	121	122	123	124	131	132	133	141	142	211	212	221	222	223	231	242	243	311	312	313	321	322	323	324	331	332	333	334	411	421	422	511	512	521	523	Total	User Acc	
111	0,012	0,037	0,012																																			0,061	0,200	
112		15,511	0,010																			0,815				0,015												16,350	0,949	
121			0,347	1,568																																		1,915	0,819	
122			0,053		0,846				0,044									0,159	0,053				0,053															1,208	0,700	
123					0,034																																	0,034	1,000	
124						0,295																																0,295	1,000	
131		0,047					0,834											0,010	0,010			0,047				0,047												0,994	0,839	
132			0,038				0,015	0,105																														0,157	0,667	
133			0,018				0,018		0,216		0,015												0,015															0,281	0,769	
141										0,149																												0,149	1,000	
142		0,181									0,422																											0,603	0,700	
211												39,726			0,013		0,066	4,417					0,010															44,233	0,898	
212													0,828																									0,828	1,000	
221												0,031		1,543				0,744	0,447						0,010	0,298												3,073	0,502	
222												0,165			0,686		0,055	0,123	0,123																			1,153	0,595	
223																0,366			0,855							0,214												1,435	0,255	
231													2,388				35,845		0,013	4,777			2,439		2,388	0,021												47,870	0,749	
241											0,005																											0,005	0,000	
242											19,378						0,013	90,429	12,918	6,459						0,010												129,208	0,700	
243											1,612						6,447	1,612	64,489	1,612		1,612			1,612	1,612												80,606	0,800	
311																		0,036	4,805	208,307		9,610			14,416	4,948												242,122	0,860	
312																			0,790	33,305	4,737					0,790												39,622	0,841	
313																	3,763			0,143	15,051	52,964				3,763												75,683	0,700	
321																			3,721	1,860	1,860		22,324		5,581	1,860												37,207	0,600	
322																				0,593				1,630		0,593			0,148									2,964	0,550	
323																			0,660			0,660			11,871													13,190	0,900	
324											0,026								1,404	7,219	0,015	1,404	4,212	5,616	4,212	47,895				0,051								72,053	0,665	
331																						0,027				0,027												0,054	0,000	
332																		0,090					0,090	0,181	0,090			1,085	0,271									1,808	0,600	
333																							2,908	0,831	0,702	0,415		0,415	3,323									8,595	0,387	
334																										0,287				0,642									0,928	0,691
335																									0,005													0,005	0,000	
411											0,086						0,171	0,086								0,086					1,325								1,753	0,756
412																															0,045								0,045	0,000
421																															0,047								0,047	0,000
422																																	0,064						0,064	1,000
423																																0,001							0,001	0,000
511																	0,278										0,020								2,498				2,796	0,894
512																																				3,158	0,166		3,324	0,950
521																																				0,003			0,003	0,000
523																																				0,277			0,277	1,000
Total	0,012	16,175	1,646	0,846	0,034	0,295	0,866	0,105	0,260	0,149	0,437	61,028	0,828	3,932	0,700	0,366	46,638	97,616	89,588	231,167	50,823	71,875	32,051	8,258	40,898	62,868	0,020	1,500	3,743	0,693	1,417	0,001	0,064	2,498	3,161	0,166	0,277	644,572		
Prod Acc	1,000	0,959	0,953	1,000	1,000	1,000	0,963	1,000	0,832	1,000	0,967	0,651	1,000	0,393	0,981	1,000	0,769	0,926	0,720	0,901	0,655	0,737	0,697	0,197	0,290	0,762	0,000	0,723	0,888	0,926	0,935	0,000	1,000	1,000	0,999	0,000	1,000			
Global Acc																																				0,774				

Overall accuracy = 77.4%

CLC Classe	111	112	121	122	123	124	131	132	133	141	142	211	212	221	222	223	231	241	242	243	311	312	313	321	322	323	324	331	332	333	334	411	412	421	422	511	512	521	523	Total	User Acc
111	0,037	0,012	0,012																																			0,061	0,600		
112		15,511							0,024														0,815																16,350	0,949	
121			0,173	1,742																																			1,915	0,909	
122				0,053	0,899				0,044										0,159				0,053																	1,208	0,744
123					0,034																																		0,034	1,000	
124						0,295																																	0,295	1,000	
131							0,937													0,010			0,047																0,994	0,943	
132							0,052	0,105																															0,157	0,667	
133				0,018			0,018		0,245																														0,281	0,873	
141										0,149																													0,149	1,000	
142		0,181									0,422																												0,603	0,700	
211												44,140			0,013	0,013	0,066																						44,233	0,998	
212													0,828																										0,828	1,000	
221														1,564		0,010			0,893	0,457							0,149												3,073	0,509	
222												0,165			0,700				0,288																				1,153	0,607	
223																0,651				0,427				0,142	0,071	0,142													1,435	0,454	
231							0,013										43,009				2,388			0,031	2,388	0,021													47,851	0,899	
241																		0,005																				0,005	1,000		
242												19,378					0,013		109,807							0,010</															

Overall accuracy = 89.3%

Zone 21: Czech Republic/Slovakia confusion matrix CLC2012 - Blind and plausibility analysis

Table 60: Confusion matrix CLC 2012 for zone 21 (Czech Republic/Slovakia) – BLIND ANALYSIS- In column the validation and line the production

CLC Classes	111	112	121	122	123	124	131	132	133	141	142	211	221	222	231	242	243	311	312	313	321	322	324	332	333	411	412	511	512	Total	User Acc
111	0,198																												0,198	1,000	
112	0,015	52,390	0,015						0,046			0,046			0,018	0,230														52,761	0,993
121			8,411									0,066			0,085					0,009										8,571	0,981
122				1,282								0,153																		1,436	0,893
123					0,028																									0,028	1,000
124						0,615																								0,615	1,000
131			0,088				1,606								0,088					0,009			0,088					0,097	1,977	0,813	
132			0,040				0,116	0,514							0,040								0,119					0,040	0,867	0,593	
133		0,051	0,089	0,098					0,205		0,025	0,038			0,022		0,077													0,605	0,339
141		0,034								0,469	0,101	0,034																0,034	0,670	0,700	
142											2,596	0,077					0,119		0,009				0,009							2,808	0,924
211		7,674	7,643					0,029			7,658	352,138		0,089	0,171				0,031				0,031							375,466	0,938
221												0,641	3,122																	3,763	0,830
222												0,796	0,234	2,211	0,171								0,234							3,646	0,607
231			0,015	0,015				0,029			0,155	14,057			78,590	5,348	3,720		0,155	0,155			0,059			0,155		0,015	102,468	0,767	
242																9,191														9,191	1,000
243															4,534		81,608													86,142	0,947
311																		100,059		0,048			7,367							107,474	0,931
312															0,021			0,069	107,984	0,207		0,069	71,325							179,676	0,601
313																		0,096	0,193	76,119			13,051							89,460	0,851
321															0,225					0,225	3,371		0,225							4,045	0,833
322																						1,148								1,148	1,000
324						0,021						0,107		0,021	0,122	0,594	0,637	1,405	1,068	0,060	0,030		31,959				0,015			36,039	0,887
332																								0,209	0,209					0,417	0,500
333																							0,044		0,265					0,309	0,857
334																							0,010							0,010	0,000
411												0,042			0,042			0,042					0,084			0,629				0,839	0,750
412																							0,242				0,161			0,403	0,400
511												0,130																1,166		1,295	0,900
512								0,015				0,015			0,316														6,114	6,460	0,946
Total	0,213	60,149	16,301	1,395	0,028	0,615	1,744	0,588	0,251	0,469	10,534	368,298	3,356	2,322	84,446	15,362	86,203	101,628	109,482	76,832	3,400	1,217	124,847	0,209	0,474	0,784	0,176	1,166	6,299	924,356	
Prod Acc	0,928	0,871	0,516	0,919	1,000	1,000	0,921	0,874	0,816	1,000	0,246	0,956	0,930	0,952	0,931	0,598	0,947	0,985	0,986	0,991	0,991	0,943	0,256	1,000	0,560	0,802	0,915	1,000	0,971		
																														Global Acc	0,857

Overall accuracy = 85.7%

Table 61: Confusion matrix CLC 2012 for zone 21 (Czech Republic/Slovakia) –PLAUSIBILITY ANALYSIS- In column the validation and line the production

CLC Classe	111	112	121	122	123	124	131	132	133	141	142	211	221	222	231	242	243	311	312	313	321	322	324	332	333	411	412	511	512	Total	User Acc	
111	0,198																													0,198	1,000	
112		52,530	0,015						0,015			0,046				0,153														52,761	0,996	
121			8,478									0,017			0,077															8,571	0,989	
122				1,387								0,048																		1,436	0,966	
123					0,028																									0,028	1,000	
124						0,615																								0,615	1,000	
131		0,088					1,703																0,088						0,097	1,977	0,862	
132								0,709															0,119						0,040	0,867	0,818	
133		0,051	0,064	0,025					0,379		0,025	0,038			0,022															0,605	0,627	
141										0,603	0,034	0,034																		0,670	0,900	
142											2,613	0,077					0,119													2,808	0,930	
211								0,015			0,015	375,176		0,059	0,109	0,030			0,031											375,435	0,999	
221												0,641	3,122																	3,763	0,830	
222												0,707	0,234	2,355	0,117								0,234							3,646	0,646	
231												7,633				85,538	5,348	3,735		0,155			0,059							102,468	0,835	
242																	9,191													9,191	1,000	
243															4,534		81,608													86,142	0,947	
311																		102,588		0,048			4,837							107,474	0,955	
312																			145,607	11,280			19,074							175,961	0,827	
313																				83,128			6,236							89,364	0,930	
321															0,225					0,225	3,371		0,225							4,045	0,833	
322																						1,148								1,148	1,000	
324							0,021					0,064				0,594	0,637	1,257	0,719	0,634	0,030		32,069				0,015			36,039	0,890	
332																						0,104		0,209	0,104					0,417	0,500	
333																							0,044		0,265					0,309	0,857	
334																							0,010							0,010	0,000	
411															0,042			0,042					0,084			0,629		0,042		0,839	0,750	
412																							0,202			0,202				0,403	0,500	
511																							0,130					1,166		1,295	0,900	
512												0,015																	6,445	6,460	0,998	
Total	0,198	52,670	8,557	1,413	0,028	0,615	1,725	0,724	0,395	0,603	2,687	384,495	3,356	2,415	90,662	15,315	86,099	103,845	146,400	95,469	3,400	1,252	63,410	0,209	0,369	0,629	0,216	1,166	6,624	1003,060		
Prod Acc	1,000	0,997	0,991	0,982	1,000	1,000	0,988	0,980	0,961	1,000	0,973	0,976	0,930	0,975	0,943	0,600	0,948	0,988	0,995	0,871	0,991	0,917	0,506	1,000	0,718	1,000	0,931	1,000	0,973			
																															Global Acc	0,933

Overall accuracy = 93.3%

Zone 22: Estonia/Latvia/Lithuania confusion matrix CLC2012 - Blind and plausibility analysis

Table 62: Confusion matrix CLC 2012 for zone 22 (Estonia/Latvia/Lithuania) – BLIND ANALYSIS -In column the validation and line the production

CLC Classes	111	112	121	122	123	124	131	132	133	141	142	211	222	231	242	243	311	312	313	321	322	324	331	411	412	421	511	512	521	523	Total	User Acc	
111	0,030	0,030																													0,060	0,500	
112		12,574							0,009			0,043		0,009								0,009									12,643	0,995	
121		0,169	3,137												0,169																3,475	0,903	
122				0,443					0,021																						0,464	0,955	
123					0,143																										0,143	1,000	
124						0,352																									0,352	1,000	
131							0,857		0,036			0,016					0,036					0,016									0,961	0,892	
132							0,028	0,160										0,020													0,208	0,768	
133		0,029	0,043						0,108						0,010																0,190	0,568	
141										0,716						0,038															0,754	0,950	
142			0,033						0,016		0,549							0,033	0,033												0,663	0,828	
211												180,002		2,205								0,017									182,225	0,988	
222								0,029	0,404				0,144																		0,578	0,700	
231												20,321		32,766	6,467	6,553															66,107	0,496	
242							0,017					7,011		3,505	59,592																70,126	0,850	
243															2,582	46,935								2,582							52,099	0,901	
311																0,030	51,282		3,091			6,212									60,615	0,846	
312																	0,048	107,746	0,048			0,381									108,222	0,996	
313																	0,209	0,104	111,205			12,646									124,164	0,896	
321															0,102			0,017		1,117		0,153		0,609							1,998	0,559	
322																		0,089			0,387	0,089		0,030							0,595	0,650	
324								0,185		0,019		0,125	7,721	4,115	1,211						71,543										84,919	0,842	
331																						0,022	0,312								0,334	0,933	
333																						0,045	0,045	0,090							0,181	0,000	
411																								4,856	0,540						5,396	0,900	
412																	0,017	0,720		0,720	0,737				12,300						14,493	0,849	
421																										0,015					0,015	1,000	
511																											1,700				1,700	1,000	
512																												19,359			19,359	1,000	
521																												0,018			0,018	1,000	
523																													0,193		0,193	1,000	
Total	0,030	12,802	3,213	0,443	0,143	0,352	0,902	0,160	0,189	0,716	0,549	207,608	0,404	38,505	68,964	53,782	59,296	112,141	116,306	1,117	1,152	91,870	0,402	8,077	12,840	0,015	1,700	19,359	0,018	0,193	720,802		
Prod Acc	1,000	0,982	0,976	1,000	1,000	1,000	0,950	1,000	0,568	1,000	1,000	0,867	1,000	0,851	0,864	0,873	0,865	0,961	0,956	1,000	0,336	0,779	0,775	0,601	0,958	1,000	1,000	1,000	1,000	1,000			
																																Global Acc	0,886

Overall accuracy = 88.6%

Table 63: Confusion matrix CLC 2012 for zone 22 (Estonia/Latvia/Lithuania) – PLAUSIBILITY ANALYSIS -In column the validation and line the production

CLC Classe	111	112	121	122	123	124	131	132	133	141	142	211	222	231	242	243	311	312	313	321	322	324	331	411	412	421	511	512	521	523	Total	User Acc			
111	0,030	0,030																												0,060	0,500				
112		12,614									0,014	0,016																			12,643	0,998			
121		0,169	3,137													0,169																3,475	0,903		
122			0,021	0,443																												0,464	0,955		
123					0,143																											0,143	1,000		
124						0,352																										0,352	1,000		
131							0,893					0,016										0,052										0,961	0,929		
132							0,020	0,188																									0,208	0,904	
133		0,029	0,029									0,010																					0,190	0,640	
141											0,754																						0,754	1,000	
142											0,565							0,033	0,033			0,033											0,663	0,852	
211												180,375		1,833								0,017										182,225	0,990		
222												0,029	0,462		0,087																		0,578	0,800	
231												19,954		39,531		6,553						0,069											66,107	0,598	
242							0,017					7,011		3,505	59,592																		70,126	0,850	
243																49,517								2,582									52,099	0,950	
311																0,030	51,372		3,121			6,092											60,615	0,848	
312																	0,048	108,031	0,048			0,095											108,222	0,998	
313																	0,104		117,841			6,219											124,164	0,949	
321																		0,034		1,422		0,034		0,508									1,998	0,712	
322																		0,060			0,387	0,119		0,030									0,595	0,650	
324											0,011		0,017		0,095	3,875	3,668	0,258				76,477											84,402	0,906	
331																						0,022	0,312										0,334	0,933	
333																					0,045	0,090	0,045										0,181	0,000	
411																							4,856	0,540									5,396	0,900	
412																	0,017								14,476									14,493	0,999
421																										0,015								0,015	1,000
511																											1,700							1,700	1,000
512																												19,359						19,359	1,000
521																													0,018					0,018	1,000
523																															0,193		0,193	1,000	
Total	0,030	12,842	3,187	0,443	0,143	0,352	0,930	0,188	0,121	0,754	0,579	207,421	0,462	44,887	59,679	56,364	55,400	111,843	121,301	1,422	0,432	89,319	0,357	7,976	15,015	0,015	1,700	19,359	0,018	0,193	745,177				
Prod Acc	1,000	0,982	0,984	1,000	1,000	1,000	0,960	1,000	1,000	1,000	0,976	0,870	1,000	0,881	0,999	0,879	0,927	0,966	0,971	1,000	0,895	0,856	0,874	0,609	0,964	1,000	1,000	1,000	1,000	1,000					
																																		Global Acc 0,917	

Overall accuracy = 91.7%

Zone 23: Overseas departments confusion matrix CLC2012 - Blind and plausibility analysis

Table 64: Confusion matrix CLC 2012 for zone 23 (Guyana/Guadeloupe/La reunion/Martinique/Mayotte) – BLIND ANALYSIS- In column the validation and line the production

CLC Classes	111	112	121	122	123	124	131	132	133	141	142	211	213	222	231	242	243	311	312	321	322	323	324	331	332	333	411	423	511	512	521	522	523	Total	User Acc
111	0,060																																	0,060	1,000
112		8,661							0,023							0,017	0,017																	8,718	0,993
121		0,046	0,541						0,017																									0,604	0,896
122		0,023		0,250																														0,273	0,916
123					0,077																													0,077	1,000
124						0,216																												0,216	1,000
131							0,151																											0,151	1,000
132								0,011																										0,011	1,000
133									0,017																									0,017	1,000
141										0,064																								0,064	1,000
142											0,104																							0,104	1,000
211												8,448				0,444																		8,892	0,950
213													0,953																					0,953	1,000
222														1,674		0,239	0,478																	2,391	0,700
231		0,371												0,023	2,620	0,023	0,371						0,383											3,791	0,691
242																4,894	0,544																	5,438	0,900
243		1,135							0,012							1,135	8,638	0,012				0,012	0,590											11,532	0,749
244																							0,309											0,309	0,000
311																		113,040														0,045	113,085	1,000	
312																			0,312															0,312	1,000
313																			0,772															0,772	0,000
321													0,015							3,727														3,741	0,996
322																					2,023		0,276											2,299	0,880
323																						1,885	1,256											3,141	0,600
324																1,156	1,696				0,555		7,564		0,555									11,526	0,656
331																								0,039										0,039	1,000
332																									1,043									1,043	1,000
333																		0,157		0,315						1,102								1,574	0,700
411																		0,544									15,119							15,664	0,965
421																											0,106							0,106	0,000
423																												0,010						0,010	1,000
511																								0,153					1,222		0,153			1,527	0,800
512																														0,062				0,062	1,000
521																															0,027			0,027	1,000
522																													0,386			3,108	0,408	3,902	0,796
523																																	209,235	209,235	1,000
Total	0,060	10,236	0,541	0,250	0,077	0,216	0,151	0,011	0,069	0,064	0,104	8,448	0,953	1,696	2,635	6,752	11,203	116,221	0,312	3,727	2,893	1,896	10,378	0,191	1,043	1,657	15,225	0,010	1,607	0,062	0,027	3,260	209,688	396,896	
Prod Acc	1,000	0,846	1,000	1,000	1,000	1,000	1,000	1,000	0,249	1,000	1,000	1,000	1,000	0,987	0,994	0,725	0,771	0,973	1,000	1,000	0,699	0,994	0,729	0,202	1,000	0,665	0,993	1,000	0,760	1,000	0,953	0,998			
																																		Global Acc	0,964

Overall accuracy = 96.4%

Table 65: Confusion matrix CLC 2012 for zone 23 (Guyana/Guadeloupe/La reunion/Martinique/Mayotte) – PLAUSIBILITY ANALYSIS- In column the validation and line the production

CLC Classe	111	112	121	122	123	124	131	132	133	141	142	211	213	222	231	242	243	311	312	313	321	322	323	324	331	332	333	411	423	511	512	521	522	523	Total	User Acc
111	0,060																																	0,060	1,000	
112		8,661							0,023							0,017	0,017																	8,718	0,993	
121			0,046	0,541																				0,017										0,604	0,896	
122				0,023		0,250																												0,273	0,916	
123					0,077																													0,077	1,000	
124						0,216																												0,216	1,000	
131							0,151																											0,151	1,000	
132								0,011																										0,011	1,000	
133									0,017																									0,017	1,000	
141										0,064																								0,064	1,000	
142											0,104																							0,104	1,000	
211												8,448				0,444																		8,892	0,950	
213													0,953																					0,953	1,000	
222														1,674		0,239	0,478																	2,391	0,700	
231														0,023	2,991	0,023	0,371							0,383										3,791	0,789	
242																	4,894	0,544																5,438	0,900	
243		1,135							0,012							1,135	8,661	0,012					0,012	0,567										11,532	0,751	
244																								0,309										0,309	0,000	
311																		113,040															0,045	113,085	1,000	
312																			0,312															0,312	1,000	
313																				0,772														0,772	1,000	
321													0,015								3,727													3,741	0,996	
322																						2,276		0,023										2,299	0,990	
323																							2,827	0,314										3,141	0,900	
324																1,156	1,696							8,674										11,526	0,753	
331																									0,039									0,039	1,000	
332																										1,043								1,043	1,000	
333																		0,157			0,315						1,102							1,574	0,700	
411																		0,023										15,641							15,664	0,999
421																												0,106							0,106	0,000
423																													0,010						0,010	1,000
511																									0,153					1,374					1,527	0,900
512																															0,062				0,062	1,000
521																																0,027			0,027	1,000
522																																	3,902		3,902	1,000
523																																		209,235	209,235	1,000
Total	0,060	9,865	0,541	0,250	0,077	0,216	0,151	0,011	0,051	0,064	0,104	8,448	0,953	1,696	3,006	6,752	11,226	114,927	0,312	0,772	3,727	2,591	2,838	10,287	0,191	1,043	1,102	15,747	0,010	1,374	0,062	0,027	3,902	209,279	401,836	
Prod Acc	1,000	0,878	1,000	1,000	1,000	1,000	1,000	1,000	0,331	1,000	1,000	1,000	1,000	0,987	0,995	0,725	0,771	0,984	1,000	1,000	1,000	0,878	0,996	0,843	0,202	1,000	1,000	0,993	1,000	1,000	1,000	1,000	1,000	1,000	Global Acc 0,976	

Overall accuracy = 97.6%

Annex 4. Producer and user accuracy per bio-geographical region – Blind and plausibility analysis

Table 66: Producer, user and overall accuracy for Blind and plausibility analysis per bio-geographical region (6 regions from ALP to BOR region)

	ALP				ANA				ARC				ATL				BLS				BOR			
	Alpine				Anatolian				Arctic				Atlantic				BlackSea				Boreal			
	Blind Analysis		Plausibility analysis		Blind Analysis		Plausibility analysis		Blind Analysis		Plausibility analysis		Blind Analysis		Plausibility analysis		Blind Analysis		Plausibility analysis		Blind Analysis		Plausibility analysis	
	Producer	User	Producer	User	Producer	User	Producer	User	Producer	User	Producer	User	Producer	User	Producer	User	Producer	User	Producer	User	Producer	User	Producer	User
111	0,550	1,000	0,550	1,000	0,245	1,000	0,285	1,000					0,671	0,459	0,725	0,481	1,000	1,000	1,000	1,000	0,769	1,000	0,769	1,000
112	0,820	0,905	0,904	0,966	0,970	0,929	0,994	0,933	1,000	0,944	1,000	0,944	0,931	0,871	0,941	0,892	0,990	0,915	1,000	0,986	0,934	0,984	0,983	0,986
121	0,866	0,769	0,924	0,992	0,942	0,942	0,956	0,974	0,816	1,000	0,878	1,000	0,848	0,966	0,877	0,973	0,846	0,991	0,985	0,992	0,890	0,845	0,891	0,910
122	0,947	0,935	0,926	0,975					1,000	1,000	1,000	1,000	0,867	0,574	0,867	0,613	0,946	0,988	0,988	0,989	0,960	0,984	0,957	0,991
123	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	0,619	1,000	0,619	0,930	0,947	0,930	0,945	1,000	0,854	1,000	1,000	0,993	0,993	0,993	0,993
124	0,939	1,000	0,939	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	0,956	0,943	0,981	0,944	1,000	1,000	1,000	1,000	1,000	0,835	1,000	0,835
131	0,795	0,937	0,823	0,939	0,967	0,982	1,000	1,000	1,000	1,000	1,000	1,000	0,860	0,983	0,863	0,990	0,310	1,000	0,310	1,000	0,875	0,913	0,896	0,948
132	0,807	0,955	0,807	1,000	0,511	1,000	0,511	1,000					0,728	0,976	0,773	0,953	1,000	0,065	1,000	1,000	0,688	0,752	0,850	0,789
133	0,530	0,849	0,741	0,981	0,877	1,000	0,947	0,975	0,323	1,000	0,488	1,000	0,836	0,451	0,859	0,503	0,733	0,108	0,868	0,500	0,788	0,329	0,830	0,867
141	0,736	1,000	1,000	1,000	1,000	1,000	1,000	1,000	0,800	1,000	1,000	1,000	0,837	0,634	0,870	0,584	1,000	0,761	1,000	1,000	0,907	0,985	0,943	1,000
142	0,757	0,646	0,832	0,964	0,667	1,000	0,667	1,000	0,980	1,000	0,980	1,000	0,852	0,863	0,916	0,793	0,850	0,763	0,889	0,771	0,947	0,996	0,965	0,993
211	0,912	0,988	0,910	0,991	0,893	0,750	0,946	0,828	1,000	0,116	1,000	0,118	0,925	0,961	0,936	0,972	0,819	0,982	0,838	0,998	0,874	0,852	0,894	0,865
212	1,000	1,000	1,000	1,000	0,944	0,960	1,000	0,962					0,500	0,827	0,500	0,827	0,500	0,989	1,000	0,994				
213					1,000	0,966	1,000	0,966									0,750	1,000	0,750	1,000				
221	0,852	0,492	0,903	0,981	0,178	1,000	0,342	1,000					0,945	0,760	1,000	0,772	0,552	1,000	0,552	1,000				
222	0,639	0,716	0,670	0,726	0,838	0,894	0,838	0,905					0,538	0,987	0,571	0,988	0,588	0,635	0,588	0,635	0,680	1,000	0,770	1,000
223					0,030	1,000	0,030	1,000																
231	0,642	0,830	0,873	0,858	0,739	0,607	0,834	0,711	0,897	0,991	0,934	0,992	0,857	0,926	0,897	0,936	0,927	0,318	0,993	0,333	0,491	0,519	0,597	0,615
241															0,007	1,000								
242	0,768	0,602	0,945	0,816	0,652	0,725	0,652	0,838	0,789	1,000	0,986	1,000	0,869	0,731	0,901	0,809	0,574	0,466	0,574	0,630	0,800	0,758	0,813	0,872
243	0,691	0,680	0,767	0,793	0,691	0,796	0,761	0,905					0,798	0,473	0,877	0,550	1,000	0,290	1,000	0,292	0,791	0,822	0,836	0,835
244																								
311	0,885	0,944	0,934	0,986	0,692	1,000	0,902	1,000	0,611	1,000	0,778	1,000	0,907	0,971	0,913	0,979	0,864	0,943	0,864	1,000	0,841	0,882	0,859	0,962
312	0,794	0,936	0,919	0,984	1,000	0,895	1,000	1,000	0,500	1,000	0,700	1,000	0,898	0,896	0,933	0,969	1,000	0,888	1,000	0,888	0,831	0,948	0,866	0,984
313	0,846	0,834	0,954	0,904	0,913	1,000	0,913	1,000	0,667	0,845	0,667	0,891	0,854	0,737	0,936	0,756	0,762	0,751	0,903	0,782	0,859	0,688	0,947	0,750
321	0,813	0,780	0,909	0,896	0,811	0,562	0,945	0,693	0,890	0,969	0,951	0,972	0,770	0,716	0,867	0,831	1,000	0,736	1,000	0,736	0,459	0,987	0,584	0,990
322	0,890	0,891	0,952	0,949					0,850	0,962	0,950	0,991	0,806	0,811	0,838	0,946					0,986	0,584	0,986	0,714
323	0,174	0,721	0,336	1,000	0,667	1,000	0,667	1,000																
324	0,778	0,350	0,891	0,562	0,852	0,837	0,926	0,961	0,419	0,259	0,472	0,819	0,819	0,756	0,862	0,810	0,816	0,822	0,817	0,823	0,791	0,645	0,912	0,739
331	0,789	0,735	0,837	0,747	0,690	0,821	0,766	0,718	0,881	0,851	0,914	0,972	0,832	0,947	0,931	1,000	0,926	0,975	0,982	1,000	0,847	0,883	0,847	0,883
332	0,847	0,953	0,941	0,985	0,698	0,911	0,721	0,913	0,811	0,903	0,905	0,950	0,843	1,000	0,934	1,000	1,000	1,000	1,000	1,000	0,616	1,000	0,616	1,000
333	0,908	0,922	0,936	0,973	0,591	0,823	0,773	0,933	0,832	0,593	0,931	1,000	0,969	0,937	0,992	0,998	0,332	0,992	0,332	1,000	0,970	0,953	0,970	0,998
334	0,474	0,912	0,519	1,000									0,956	0,927	1,000	1,000					0,500	0,008	0,750	1,000
335	0,957	0,822	0,976	1,000	1,000	1,000	1,000	1,000	1,000	0,814	1,000	0,815												
411	0,680	0,779	0,725	0,995	0,917	0,901	0,958	0,901	0,600	0,408	0,750	0,466	0,868	0,998	0,919	0,999	0,861	0,930	1,000	0,944	0,896	0,715	0,904	0,723
412	0,891	1,000	0,982	1,000					0,789	0,988	0,944	0,997	0,745	0,988	0,930	0,988					0,940	0,841	0,990	0,909
421									0,800	0,830	1,000	1,000	0,917	0,898	0,917	0,922					0,706	0,581	0,868	0,521
422					1,000	1,000	1,000	1,000					0,800	1,000	1,000	1,000	1,000	1,000	1,000	1,000				
423	1,000	1,000	1,000	1,000					0,753	1,000	0,803	1,000	0,715	0,944	0,854	0,976					1,000	1,000	1,000	1,000
511	0,967	0,977	0,967	1,000	0,832	1,000	0,845	1,000	0,931	0,918	0,931	1,000	0,925	0,772	0,925	0,924	0,847	1,000	0,847	1,000	0,945	1,000	0,945	1,000
512	1,000	0,977	1,000	0,979	1,000	0,960	1,000	0,960	1,000	0,985	1,000	0,985	0,933	0,981	0,966	0,982	0,954	0,977	0,954	0,977	0,989	0,999	0,989	0,999
521					1,000	1,000	1,000	1,000	1,000	0,921	1,000	0,921	0,869	1,000	0,869	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000
522									0,400	0,727	1,000	0,870	0,906	0,516	0,914	0,531					0,667	1,000	0,667	1,000
523	1,000	1,000	1,000	1,000					1,000	1,000	1,000	1,000	0,929	0,936	0,929	0,954	0,644	0,974	1,000	0,983	0,869	1,000	0,895	1,000
Global Acc	0,843		0,925		0,771		0,864		0,853		0,937		0,877		0,913		0,817		0,844		0,843		0,891	

Table 67: Producer, user and overall accuracy for Blind and plausibility analysis per bio-geographical region (6 regions from CON to STE region)

	Continental				Oversea department				Macaronesia				Mediterranean				Pannonian				Steppic			
	Blind Analysis		Plausibility analysis		Blind Analysis		Plausibility analysis		Blind Analysis		Plausibility analysis		Blind Analysis		Plausibility analysis		Blind Analysis		Plausibility analysis		Blind Analysis		Plausibility analysis	
	Producer	User	Producer	User	Producer	User	Producer	User	Producer	User	Producer	User	Producer	User	Producer	User	Producer	User	Producer	User	Producer	User	Producer	User
111	0,753	0,526	0,853	0,558	1,000	1,000	1,000	1,000	0,025	1,000	0,025	1,000	0,856	0,966	0,860	0,969	1,000	1,000	1,000	1,000				
112	0,865	0,925	0,940	0,981	0,993	0,846	0,993	0,878	1,000	1,000	1,000	1,000	0,968	0,799	0,973	0,807	0,964	0,994	0,965	0,994	0,667	0,495	1,000	1,000
121	0,903	0,708	0,973	0,850	0,896	1,000	0,896	1,000	1,000	0,651	1,000	0,651	0,921	0,692	0,945	0,746	0,943	0,424	0,991	0,478	1,000	0,390	1,000	1,000
122	0,850	0,813	0,921	0,971	0,916	1,000	0,916	1,000	1,000	1,000	1,000	1,000	0,936	0,960	0,954	0,967	0,870	0,126	0,893	0,846	0,487	1,000	0,609	1,000
123	1,000	0,973	1,000	0,973	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	0,944	0,657	0,944	0,657	1,000	0,543	1,000	0,831	1,000	1,000	1,000	1,000
124	0,962	0,996	0,986	0,996	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	0,952	0,995	0,960	1,000	0,844	1,000	1,000	1,000	0,750	1,000	1,000	1,000
131	0,819	0,765	0,873	0,967	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	0,847	0,702	0,858	0,983	0,727	0,986	0,830	1,000				
132	0,758	0,747	0,873	0,979	1,000	1,000	1,000	1,000					0,966	0,951	0,972	0,915	0,757	0,930	0,870	0,971	1,000	1,000	1,000	1,000
133	0,653	0,791	0,751	0,582	1,000	0,249	1,000	0,331	0,964	1,000	0,964	1,000	0,786	0,783	0,750	0,786	0,339	0,888	0,630	1,000				
141	0,888	0,548	0,913	0,664	1,000	1,000	1,000	1,000					0,775	1,000	0,783	1,000	0,948	0,956	0,948	0,956	1,000	1,000	1,000	1,000
142	0,927	0,557	0,978	0,589	1,000	1,000	1,000	1,000	1,000	0,906	1,000	0,906	0,914	0,895	0,916	0,900	0,870	0,278	0,879	0,995	1,000	1,000	1,000	1,000
211	0,909	0,945	0,950	0,960	0,950	1,000	0,950	1,000					0,860	0,934	0,908	0,946	0,918	0,964	0,972	0,983	1,000	0,948	1,000	0,992
212			0,500	1,000					1,000	1,000	1,000	1,000	0,958	0,982	0,962	0,997								
213	0,950	0,881	0,997	1,000	1,000	1,000	1,000	1,000					0,974	0,996	0,989	0,999	0,500	0,778	1,000	0,875	1,000	1,000	1,000	1,000
221	0,651	0,605	0,747	0,925									0,661	0,980	0,750	0,982	0,821	0,411	0,868	0,905	0,167	1,000	0,500	1,000
222	0,544	0,564	0,656	0,844	0,700	0,987	0,700	0,987					0,703	0,873	0,766	0,938	0,725	0,340	0,839	0,897				
223													0,858	0,887	0,902	0,925								
231	0,733	0,806	0,814	0,914	0,691	0,994	0,789	0,995	1,000	0,900	1,000	1,000	0,790	0,443	0,832	0,551	0,773	0,821	0,856	0,926	0,500	0,954	0,833	0,973
241	0,992	0,729	0,996	0,715									0,350	0,190	0,473	0,264								
242	0,642	0,507	0,778	0,649	0,900	0,725	0,900	0,725					0,702	0,649	0,757	0,733	0,695	0,725	0,890	0,782	0,333	0,512	1,000	0,790
243	0,802	0,620	0,894	0,791	0,749	0,771	0,751	0,771	0,741	1,000	0,741	1,000	0,823	0,695	0,848	0,771	0,828	0,842	0,881	0,633	1,000	0,503	1,000	1,000
244													0,928	0,765	0,931	0,804								
311	0,879	0,949	0,926	0,978	1,000	0,973	1,000	0,984	1,000	1,000	1,000	1,000	0,860	0,934	0,913	0,960	0,828	0,944	0,893	0,974	0,667	0,966	1,000	0,977
312	0,860	0,973	0,945	0,986	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	0,899	0,937	0,953	0,939	0,819	0,977	0,945	0,898				
313	0,916	0,773	0,979	0,821			1,000	1,000					0,785	0,794	0,883	0,918	0,768	0,736	0,833	0,786				
321	0,380	0,830	0,617	0,947	0,996	1,000	0,996	1,000	0,500	0,090	1,000	0,668	0,538	0,750	0,642	0,892	0,649	0,975	0,699	0,999				
322	0,579	0,834	0,579	0,777	0,880	0,699	0,990	0,878	1,000	0,121	1,000	0,121	0,783	0,504	0,846	0,771								
323	0,857	1,000	1,000	0,378	0,600	0,994	0,900	0,996	1,000	0,835	1,000	0,835	0,790	0,799	0,886	0,861								
324	0,614	0,419	0,800	0,641	0,656	0,729	0,753	0,843	1,000	0,669	1,000	0,669	0,792	0,665	0,880	0,736	0,681	0,475	0,854	0,704	0,238	0,081	0,619	1,000
331	0,644	0,189	0,844	0,322	1,000	0,202	1,000	0,202	0,667	1,000	1,000	1,000	0,917	0,946	0,976	0,968					0,800	1,000	1,000	1,000
332	0,695	0,832	0,695	1,000	1,000	1,000	1,000	1,000	0,754	1,000	0,754	1,000	0,492	0,875	0,630	0,957								
333	0,458	0,249	0,618	1,000	0,700	0,665	0,700	1,000	0,548	0,982	0,877	1,000	0,532	0,714	0,642	0,856	1,000	0,837	1,000	1,000				
334	0,458	1,000	0,458	1,000					0,888	1,000	0,888	1,000	0,833	0,841	0,807	0,741								
335																								
411	0,778	0,564	0,882	0,565	0,965	0,993	0,999	0,993					0,962	0,866	0,962	0,997	0,781	0,988	0,873	0,990	0,838	0,933	0,840	0,933
412	0,564	0,846	0,848	0,996					1,000	1,000	1,000	1,000					0,800	1,000	0,800	1,000				
421	0,668	1,000	0,668	0,961									0,886	0,928	0,908	0,955								
422	0,349	0,483	1,000	0,729									0,856	0,941	0,907	0,953								
423	1,000	0,139	1,000	0,335	1,000	1,000	1,000	1,000					0,794	1,000	0,992	1,000								
511	0,882	0,900	0,942	1,000	0,800	0,760	0,900	1,000					0,824	0,935	0,868	0,956	1,000	1,000	1,000	1,000	1,000	0,982	1,000	1,000
512	0,965	0,986	0,978	0,994	1,000	1,000	1,000	1,000					0,984	0,975	0,992	0,975	0,958	0,911	0,998	0,915	0,891	0,804	0,891	0,804
521	0,873	0,890	0,897	0,892	1,000	1,000	1,000	1,000					0,969	0,974	0,982	1,000								
522	1,000	1,000	1,000	1,000	0,796	0,953	1,000	1,000					0,925	0,929	0,925	0,929								
523	0,936	0,141	0,936	1,000	1,000	0,998	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000								
Global Acc	0,839		0,907		0,964		0,976		0,803		0,903		0,807		0,865		0,869		0,935		0,870		0,964	

Annex 5. Confusion matrix CLC2012 per Bio-geographical region

Bio-geographical region ALP - Alpine - confusion matrix CLC2012 - Blind and plausibility analysis

Table 68: Confusion matrix CLC 2012 for Alpine region –BLIND ANALYSIS- In column the validation and line the production

CLC Classes	111	112	121	122	123	124	131	132	133	141	142	211	212	221	222	223	231	242	243	311	312	313	321	322	323	324	331	332	333	334	335	411	412	423	511	512	523	Total	User Acc		
111	0,236	0,186	0,008																																			0,430	0,550		
112		34,598							0,008									2,766	3,562		1,278																	42,212	0,820		
121		0,216	2,471																		0,004														0,160		2,852	0,866			
122		0,027		0,649					0,009																													0,685	0,947		
123					0,003																																	0,003	1,000		
124		0,020				0,306																																0,326	0,939		
131		0,025					0,980										0,034		0,073		0,073					0,044									0,004		1,233	0,795			
132							0,054	0,227																														0,281	0,807		
133		0,013	0,025	0,045			0,012	0,011	0,173			0,042					0,006																					0,326	0,530		
141		0,017								0,092	0,017																											0,125	0,736		
142		0,010							0,014		2,261						0,005	0,024		0,017	0,308		0,076	0,028		0,113			0,133									2,988	0,757		
211												53,107			0,007		5,110				0,005					0,012												58,242	0,912		
212													1,034																									1,034	1,000		
221			0,168									0,316		2,780																								3,264	0,852		
222		0,024									0,217		0,082	2,033				0,762								0,063												3,181	0,639		
231		2,228										1,547		0,330	78,778	6,736	16,374				4,455	5,960				6,235												122,643	0,642		
242			0,542									1,244				0,797	1,652	30,845	1,682	4,184																		40,148	0,768		
243																	6,720	8,685	69,599	3,221		1,652				9,992												100,665	0,691		
244																			0,016																			0,016	0,000		
311																			4,011	423,111		9,407				41,541												478,070	0,885		
312										1,063							1,081		0,452	1,081	331,728	18,598		0,034		63,857													417,894	0,794	
313																			7,323	13,838	192,032		0,985			12,838													227,015	0,846	
321																	1,064	2,215	1,205	1,205	1,209	1,143	143,728	15,191		6,305			3,592										176,858	0,813	
322																			0,830			1,222		9,612	296,295		14,563		3,937	6,344								332,803	0,890		
323																						1,735		1,622	5,965													9,322	0,174		
324		0,872								0,081							0,332		3,692	7,917	4,801	2,749	2,722	2,090		89,222	0,206			0,033								114,716	0,778		
331																					0,017		0,034		0,017	0,572	0,067	0,017										0,725	0,789		
332																		0,059		0,046		3,220	1,335	0,628	0,348		142,180	15,354		4,666								167,835	0,847		
333										0,078							0,003					16,313	10,332		1,905		2,310	304,324											335,265	0,908	
334																			0,120		0,056					0,186			0,018	0,344								0,724	0,474		
335																												0,744	0,224		21,514								22,482	0,957	
411																	0,050									0,050						0,667		0,044	0,169				0,980	0,680	
412																	0,040				0,014			7,125		1,779					0,186	89,281			1,779				100,203	0,891	
423																																0,014							0,014	1,000	
511											0,064																								1,885				1,949	0,967	
512																														0,003						89,779				89,782	1,000
523																																							0,365	0,365	1,000
Total	0,236	38,235	3,214	0,694	0,003	0,306	1,046	0,238	0,204	0,092	3,500	53,746	1,034	5,652	2,837	0,330	94,875	51,270	102,316	448,178	354,449	230,186	184,350	332,464	2,250	255,036	0,778	149,238	330,006	0,377	26,180	0,856	89,281	0,014	1,930	91,892	0,365	2408,836			
Prod Acc	1,000	0,905	0,769	0,935	1,000	1,000	0,937	0,955	0,849	1,000	0,646	0,988	1,000	0,492	0,716	0,000	0,830	0,602	0,680	0,944	0,936	0,834	0,780	0,891	0,721	0,350	0,735	0,953	0,922	0,912	0,822	0,779	1,000	1,000	0,977	0,977	1,000		Global Acc 0,843		

Overall accuracy = 84.3%

Table 69: Confusion matrix CLC 2012 for Alpine region –PLAUSIBILITY ANALYSIS- In column the validation and line the production

CLC Classe	111	112	121	122	123	124	131	132	133	141	142	211	212	213	221	222	231	242	243	311	312	313	321	322	323	324	331	332	333	334	335	411	412	423	511	512	523	Total	User Acc	
111	0,236	0,186	0,008																																			0,430	0,550	
112		38,173																1,844	0,917		1,278																	42,212	0,904	
121		0,216	2,636																0,146			0,025					0,044								0,004			1,233	0,823	
122		0,027		0,634								0,024																										0,685	0,926	
123					0,003																																	0,003	1,000	
124		0,020				0,306																																0,326	0,939	
131							1,014												0,146			0,025					0,044								0,004			1,233	0,823	
132							0,054	0,227																														0,281	0,807	
133		0,013	0,013	0,016			0,012			0,242		0,013	0,013				0,006																						0,326	0,741
141										0,125																												0,125	1,000	
142		0,005							0,005	2,424							0,005			0,006	0,152		0,142			0,109			0,067									2,912	0,832	
211												53,028			0,007		5,124	0,078			0,005																	58,242	0,910	
212												1,034																											1,034	1,000
221												0,316																											3,264	0,903
222		0,024										0,144			0,058	2,131			0,762								0,063												3,181	0,670
231																	107,032		13,407				1,096			1,107													122,643	0,873
242																	2,194	37,954																					40,148	0,945
243														0,797	0,797		9,201	6,640	77,225	1,784		0,608				3,613													100,665	0,767
244																		0,016																				0,016	0,000	
311																	0,023	0,482	446,042	0,010	4,742					26,290													477,589	0,934
312																		2,392			382,174	14,037				17,447													416,050	0,919
313																					3,245	216,249	0,985			6,126													226,605	0,954
321																			1,205	0,079	1,143	160,776	6,191		6,221			1,243											176,858	0,909
322																				0,971		3,163	316,654		10,478			1,462											332,728	0,952
323																					0,226			3,131	5,965														9,322	0,336
324		0,872															1,163	1,999	3,442	0,614	2,362	1,789			102,107	0,206													114,553	0,891
331																					0,017	0,034					0,607	0,067											0,725	0,837
332																					3,154	0,386			0,708		157,925	5,652		0,009									167,835	0,941
333											0,078						0,003					8,000	10,254		1,000		2,041	313,889											335,265	0,936
334																						0,018				0,330				0,376									0,724	0,519
335																												0,323	0,224		21,936								22,482	0,976
411																	0,050								0,050							0,711			0,169				0,980	0,725
412																	0,014	0,025		0,014													98,370		1,779				100,203	0,982
423																																		0,014					0,014	1,000
511																									0,064										1,885				1,949	0,967
512																													0,003							89,779			89,782	1,000
523																																					0,365		0,365	1,000
Total	0,236	39,534	2,657	0,651	0,003	0,306	1,080	0,227	0,246	0,125	2,514	53,523	1,034	0,797	3,006	2,935	124,815	46,515	97,371	452,478	388,543	239,165	179,366	333,519	3,131	181,722	0,813	160,356	322,536	0,376	21,945	0,714	98,370	0,014	1,885	91,731	0,365	2640,361		
Prod Acc	1,000	0,966	0,992	0,975	1,000	1,000	0,939	1,000	0,981	1,000	0,964	0,991	1,000	0,000	0,981	0,726	0,858	0,816	0,793	0,986	0,984	0,904	0,896	0,949	1,000	0,562	0,747	0,985	0,973	1,000	1,000	0,995	1,000	1,000	1,000	0,979	1,000		Global Acc 0,925	

Overall accuracy = 92.5%

Bio-geographical region ANA - Anatolian- confusion matrix CLC2012 - Blind and plausibility analysis

Table 70: Confusion matrix CLC 2012 for Anatolian region – BLIND ANALYSIS- In column the validation and line the production

CLC Classes	111	112	121	123	124	131	132	133	141	142	211	212	213	221	222	223	231	242	243	311	312	313	321	322	323	324	331	332	333	335	411	422	511	512	521	Total	User Acc	
111	0,274	0,843																																		1,116	0,245	
112		17,875				0,038												0,451					0,070													18,434	0,970	
121			0,099	3,198													0,049						0,049													3,395	0,942	
122				0,050																																0,050	0,000	
123					0,002																															0,002	1,000	
124						1,027																														1,027	1,000	
131				0,038			2,068																							0,032						2,138	0,967	
132							0,079				0,075																									0,154	0,511	
133			0,033	0,111				1,721																						0,098						1,963	0,877	
141									0,090																											0,090	1,000	
142										0,030														0,015												0,045	0,667	
211											129,300	0,028					7,732	7,749																		144,808	0,893	
212											6,440	184,332					4,522																			195,294	0,944	
213													1,190																							1,190	1,000	
221			0,384											0,610	0,994			0,948	0,497																3,434	0,178		
222															16,746			1,617	1,617																	19,980	0,838	
223															0,964	0,030																				0,994	0,030	
231											1,676	1,576					51,290						14,855													69,397	0,739	
241																			0,009																	0,009	0,000	
242											5,941	6,022	0,042		0,030		2,940	44,854	8,952																	68,783	0,652	
243											24,054						12,554	6,277	109,875					0,017						6,277						159,054	0,691	
311																				29,274	2,672		2,069			8,308										42,324	0,692	
312																					22,733															22,733	1,000	
313																						14,919				1,428										16,347	0,913	
321																	4,902						205,457	4,902		9,804		4,902	23,314						253,280	0,811		
322																										0,008										0,008	0,000	
323																			1,790						3,579										5,369	0,667		
324																			5,570				10,962			126,568			5,426						148,527	0,852		
331																	0,288									0,095	0,851								1,234	0,690		
332																							19,917				49,902	1,678							71,497	0,698		
333											4,839								9,695					111,576		4,856		188,967							319,933	0,591		
335																													0,025							0,025	1,000	
411												0,025					0,284									0,030					0,025	7,086		0,302		7,727	0,917	
421																								0,390					3,897		0,779			1,559		6,625	0,000	
422																															0,341					0,341	1,000	
511																							0,030			0,186	0,186							1,991		2,393	0,832	
512																																		44,496		44,496	1,000	
521																																			0,012	0,012		1,000
Total	0,274	19,234	3,396	0,002	1,027	2,106	0,079	1,721	0,090	0,030	172,325	191,983	1,233	0,610	18,734	0,030	84,562	61,896	138,005	29,274	25,405	14,919	365,407	4,902	3,579	151,283	1,037	54,803	229,690	0,025	7,865	0,341	1,991	46,357	0,012	1260,789		
Prod Acc	1,000	0,929	0,942	1,000	1,000	0,982	1,000	1,000	1,000	1,000	0,750	0,960	0,966	1,000	0,894	1,000	0,607	0,725	0,796	1,000	0,895	1,000	0,562	0,000	1,000	0,837	0,821	0,911	0,823	1,000	0,901	1,000	1,000	0,960	1,000			
																																					Global Acc	0,771

Overall accuracy = 77.1%

Table 71: Confusion matrix CLC 2012 for Anatolian region – PLAUSIBILITY ANALYSIS- In column the validation and line the production

CLC Clas	111	112	121	123	124	131	132	133	141	142	211	212	213	221	222	223	231	242	243	311	312	313	321	323	324	331	332	333	335	411	422	511	512	521	Total	User Acc	
111	0,319	0,798																																	1,116	0,285	
112		18,317						0,048															0,070												18,434	0,994	
121		0,099	3,247																				0,049												3,395	0,956	
122			0,050																																0,050	0,000	
123				0,002																															0,002	1,000	
124					1,027																														1,027	1,000	
131						2,138																													2,138	1,000	
132							0,079				0,075																								0,154	0,511	
133		0,033	0,038					1,858															0,009					0,025							1,963	0,947	
141									0,090																										0,090	1,000	
142										0,030													0,015												0,045	0,667	
211											137,031	0,028					7,732	0,017																	144,808	0,946	
212											0,030	195,236					0,028																		195,294	1,000	
213													1,190																						1,190	1,000	
221		0,384												1,173	0,769			0,723	0,384																3,434	0,342	
222															16,746			1,617	1,617																19,980	0,838	
223															0,964	0,030																			0,994	0,030	
231											3,251	1,576					57,857						6,683							0,030					69,397	0,834	
241																		0,009																	0,009	0,000	
242											5,941	6,022	0,042		0,030		2,969	44,826	8,952																68,783	0,652	
243											19,243						12,554	6,277	120,963				0,017												159,054	0,761	
311																				38,165					4,159										42,324	0,902	
312																					22,733														22,733	1,000	
313																						14,919			1,428										16,347	0,913	
321																							239,372				4,902	9,007							253,280	0,945	
322																									0,008										0,008	0,000	
323																			1,790					3,579										5,369	0,667		
324																							10,962		137,564										148,527	0,926	
331																	0,288									0,946									1,234	0,766	
332																								14,956			51,580	4,961							71,497	0,721	
333																								72,690				247,243							319,933	0,773	
335																													0,025						0,025	1,000	
411											0,025																								7,727	0,958	
421																							0,390					3,897	0,779					1,559	6,625	0,000	
422																															0,341				0,341	1,000	
511																										0,372						2,021			2,393	0,845	
512																																		44,496		44,496	1,000
521																																			0,012	0,012	1,000
Total	0,319	19,630	3,335	0,002	1,027	2,138	0,079	1,906	0,090	0,030	165,572	202,887	1,233	1,173	18,509	0,030	81,428	53,468	133,707	38,165	22,733	14,919	345,212	3,579	143,160	1,318	56,481	265,134	0,025	8,209	0,341	2,021	46,357	0,012	1412,552		
Prod Acc	1,000	0,933	0,974	1,000	1,000	1,000	1,000	0,975	1,000	1,000	0,828	0,962	0,966	1,000	0,905	1,000	0,711	0,838	0,905	1,000	1,000	1,000	0,693	1,000	0,961	0,718	0,913	0,933	1,000	0,901	1,000	1,000	0,960	1,000		Global Acc 0,864	

Overall accuracy = 86.4%

Bio-geographical region ARC - Arctic - confusion matrix CLC2012 - Blind and plausibility analysis

Table 72: Confusion matrix CLC 2012 for Arctic region – BLIND ANALYSIS-In column the validation and line the production

CLC Classes	112	121	122	123	124	131	133	141	142	211	231	242	243	311	312	313	321	322	324	331	332	333	335	411	412	421	423	511	512	521	522	523	Total	User Acc
112	0,431																															0,431	1,000	
121		0,243		0,018														0,018				0,018										0,298	0,816	
122			0,012																													0,012	1,000	
123				0,030																												0,030	1,000	
124					0,118																											0,118	1,000	
131						0,049																										0,049	1,000	
133	0,026						0,025											0,026														0,076	0,323	
141								0,056										0,014														0,070	0,800	
142									0,611								0,012															0,623	0,980	
211										0,095																						0,095	1,000	
231										0,709	9,544		0,352				0,017	0,009							0,012							10,644	0,897	
242										0,011	0,014	0,174						0,022														0,220	0,789	
311														0,735				0,201	0,267													1,203	0,611	
312															0,064	0,039			0,026													0,129	0,500	
313																0,210	0,042	0,063														0,315	0,667	
321																	11,510	1,277				0,150										12,937	0,890	
322																		126,001				22,235										148,236	0,850	
324										0,077		0,065					0,140	0,531	0,586													1,400	0,419	
331																				13,029	1,548						0,218					14,795	0,881	
332																					85,585	9,926	10,063				0,012					105,587	0,811	
333																				1,954	7,516	47,140	0,075									56,685	0,832	
335																						44,360										44,360	1,000	
411																0,161			0,081	0,081			0,968	0,242			0,081				1,614	0,600		
412																	2,771	1,386					1,386	20,785							26,328	0,789		
421																					0,022				0,087							0,109	0,800	
423																									0,018	0,272			0,053	0,018		0,360	0,753	
511																			0,241								3,243					3,484	0,931	
512																												6,161				6,161	1,000	
521																													0,618			0,618	1,000	
522																										0,071				0,047		0,118	0,400	
523																															0,105	0,105	1,000	
Total	0,457	0,243	0,012	0,048	0,118	0,049	0,025	0,056	0,611	0,815	9,636	0,174	0,417	0,735	0,064	0,248	11,883	130,932	2,265	15,305	94,730	79,470	54,498	2,376	21,039	0,105	0,272	3,532	6,254	0,671	0,065	0,105	372,892	
Prod Acc	0,944	1,000	1,000	0,619	1,000	1,000	1,000	1,000	1,000	0,116	0,991	1,000	0,000	1,000	1,000	0,845	0,969	0,962	0,259	0,851	0,903	0,593	0,814	0,408	0,988	0,830	1,000	0,918	0,985	0,921	0,727	1,000		
	Global Acc																															0,853		

Overall accuracy = 85.3%

Table 73: Confusion matrix CLC 2012 for Arctic region – PLAUSIBILITY ANALYSIS-In column the validation and line the production

CLC Classe	112	121	122	123	124	131	133	141	142	211	231	242	243	311	312	313	321	322	323	324	331	332	333	335	411	412	421	423	511	512	521	522	523	Total	User Acc
112	0,431																																0,431	1,000	
121		0,261		0,018																			0,018										0,298	0,878	
122			0,012																														0,012	1,000	
123				0,030																													0,030	1,000	
124					0,118																												0,118	1,000	
131						0,049																											0,049	1,000	
133	0,026						0,025																										0,050	0,488	
141								0,070																									0,070	1,000	
142									0,611								0,012																0,623	0,980	
211										0,095																							0,095	1,000	
231										0,706	9,938																						10,644	0,934	
242											0,003	0,206																					0,209	0,986	
311													0,936				0,134		0,134													1,203	0,778		
312														0,090	0,026					0,013												0,129	0,700		
313															0,210	0,042	0,063															0,315	0,667		
321																12,298	0,639															12,937	0,951		
322																	140,824	7,412														148,236	0,950		
324										0,077		0,065					0,140	0,456			0,661											1,400	0,472		
331																					13,532	1,268										14,800	0,914		
332																						95,602		9,972				0,012				105,587	0,905		
333																					0,075	3,758	52,777	0,075								56,685	0,931		
335																							44,360									44,360	1,000		
411																0,161				0,081				1,210	0,081			0,081				1,614	0,750		
412																								1,386	23,556							24,942	0,944		
421																										0,109							0,109	1,000	
423																											0,289			0,053	0,018		0,360	0,803	
511																				0,241								3,243				3,484	0,931		
512																													6,161				6,161	1,000	
521																														0,618			0,618	1,000	
522																															0,118		0,118	1,000	
523																																0,105	0,105	1,000	
Total	0,457	0,261	0,012	0,048	0,118	0,049	0,025	0,070	0,611	0,801	10,018	0,206	0,065	0,936	0,090	0,235	12,654	142,115	7,412	0,808	13,929	100,629	52,795	54,406	2,596	23,637	0,109	0,289	3,243	6,254	0,671	0,136	0,105	408,545	
Prod Acc	0,944	1,000	1,000	0,619	1,000	1,000	1,000	1,000	1,000	0,118	0,992	1,000	0,000	1,000	1,000	0,891	0,972	0,991	0,000	0,819	0,972	0,950	1,000	0,815	0,466	0,997	1,000	1,000	1,000	0,985	0,921	0,870	1,000		
																																		Global Acc	0,937

Overall accuracy = 93.7%

Bio-geographical region ATL - Atlantic - confusion matrix CLC2012 - Blind and plausibility analysis

Table 74: Confusion matrix CLC 2012 for Atlantic region – BLIND ANALYSIS- In column the validation and line the production

CLC Classes	111	112	121	122	123	124	131	132	133	141	142	211	212	221	222	231	242	243	311	312	313	321	322	323	324	331	332	333	334	411	412	421	422	423	511	512	521	522	523	Total	User Acc		
111	1,919	0,496	0,222						0,148	0,074																															2,859	0,671	
112	2,238	175,548	0,064	2,238					2,220	1,980		2,309						1,980		0,016					0,050																188,644	0,931	
121		1,169	26,192	0,355	0,104				1,447		0,072	0,021				0,691	0,014			0,016		0,032	0,016		0,731					0,010											30,871	0,848	
122			0,142	3,684					0,005							0,202			0,216																						4,249	0,867	
123				0,050	1,954					0,029						0,034		0,034																							2,101	0,930	
124			0,082			4,064										0,107																									4,254	0,956	
131		0,005	0,034				6,563		0,007		0,007					0,173									0,203			0,019			0,016					0,607					7,634	0,860	
132			0,036				0,018	0,598	0,066	0,018	0,037					0,030									0,018																0,821	0,728	
133		0,079	0,165	0,037	0,006		0,044	0,014	3,255		0,038	0,115				0,086	0,032		0,006				0,017																		3,894	0,836	
141			0,145							5,317	0,063	0,145				0,530	0,151																								6,350	0,837	
142		0,629				0,247			0,966	20,664	0,587					1,165																									24,259	0,852	
211		7,074					0,009					913,729	0,038			19,735	36,812	10,609							0,005						0,011										988,023	0,925	
212												0,183				0,183																										0,367	0,500
213																									0,005																	0,005	0,000
221													15,580				0,915																									16,495	0,945
222											0,465		0,040	2,150			1,316								0,026																	3,998	0,538
231		9,086							0,009		3,062	10,008				593,006	27,785	39,616		9,086		0,052	0,052		0,008			0,052	0,019												691,840	0,857	
241		0,688									0,028						1,461	2,064																								4,270	0,000
242												19,699	4,878			3,351	190,659	0,193	0,193				0,193									0,193									219,358	0,869	
243		1,403									3,070					9,741	1,403	75,349	1,284		0,186		0,503		1,006						0,503										94,447	0,798	
311																1,177		9,751	283,616	0,005	14,857				3,163						0,018										312,589	0,907	
312		0,800		0,054												0,095		4,831		215,948	2,979				15,683											0,030					240,419	0,898	
313		0,434									0,286					0,286	2,225	3,278	2,970	58,635					0,526																68,640	0,854	
321			0,018						0,018							6,051	0,173	5,470				60,996	3,089		0,211		3,089								0,054						79,168	0,770	
322		4,024					0,010									0,227		0,830		2,889		17,582	144,666		9,157																179,386	0,806	
324							0,020					0,331				0,370	0,009	0,573	3,524	7,296	2,805	0,026	5,464		96,587				0,030	0,009	0,580			0,313							117,937	0,819	
331	0,027																					0,100	0,100		0,100	2,912						0,047		0,095				0,120			3,501	0,832	
332																		0,074				0,148	0,315				5,668	0,518													6,724	0,843	
333									0,009												0,662	1,669	0,183	0,036				81,260													83,820	0,969	
334																						0,029						0,626													0,655	0,956	
411									0,020							0,232							0,071		0,213					6,211	0,071	0,156		0,163	0,020						7,157	0,868	
412																2,795		5,554		2,777	0,078	5,631	22,214				1,784				119,018										159,850	0,745	
421																0,368																	4,053								4,420	0,917	
422																	0,053																0,212								0,265	0,800	
423									0,005		0,005					0,044							0,060		0,005	0,164		0,005			0,005	0,237		1,618			0,014	0,102			2,263	0,715	
511																0,046																									6,717	0,925	
512			0,009				0,014																																		40,034	0,933	
521																																				0,084	0,013				0,097	0,869	
522		0,121																														0,018		0,018				2,048	0,055		2,261	0,906	
523																	0,122							0,052																	2,289	2,463	0,929
Total	4,184	201,556	27,111	6,419	2,064	4,311	6,678	0,612	7,209	8,384	23,943	950,798	0,222	20,499	2,178	640,725	260,783	159,274	292,117	241,002	79,539	85,229	178,457	0,236	127,734	3,075	5,668	86,727	0,676	6,220	120,424	4,512	0,212	1,713	8,042	38,070	0,084	3,971	2,446	3170,410			
Prod Acc	0,459	0,871	0,966	0,574	0,947	0,943	0,983	0,976	0,451	0,634	0,863	0,961	0,827	0,760	0,987	0,926	0,731	0,473	0,971	0,896	0,737	0,716	0,811	0,000	0,756	0,947	1,000	0,937	0,927	0,998	0,988	0,898	1,000	0,944	0,772	0,981	1,000	0,516	0,936				
Global Acc																																						0,877					

Overall accuracy = 87.7%

Table 75: Confusion matrix CLC 2012 for Atlantic region – PLAUSIBILITY ANALYSIS- In column the validation and line the production

CLC Classe	111	112	121	122	123	124	131	132	133	141	142	211	212	221	222	231	241	242	243	311	312	313	321	322	323	324	331	332	333	334	411	412	421	422	423	511	512	521	522	523	Total	User Acc							
111	2,072	0,565	0,148							0,074																															2,859	0,725							
112	2,238	177,567	0,025	2,238					2,243		1,980	0,048				2,238					0,016					0,050																188,644	0,941						
121		0,691	27,086		0,104				0,923		0,072	0,043				0,691		0,014			0,024		0,016	0,016		0,731						0,461										30,871	0,877						
122			0,142	3,684					0,005							0,202				0,216																						4,249	0,867						
123			0,034	0,050	1,954					0,029									0,034																								2,101	0,930					
124			0,082			4,171																																					4,254	0,981					
131			0,034				6,591		0,007		0,007					0,168										0,203						0,016					0,607					7,634	0,863						
132			0,018				0,018	0,634	0,066	0,037						0,030										0,018																0,821	0,773						
133		0,070	0,103	0,037	0,006		0,022	0,031	3,338		0,067	0,082				0,085		0,006		0,006				0,017											0,014								3,885	0,859					
141			0,145							5,525	0,063					0,467	0,151																										6,350	0,870					
142						0,247				0,288	22,218	0,587				0,918																												24,259	0,916				
211		3,537					0,009			3,537		925,176	0,038			19,368	25,743	10,614																									988,023	0,936					
212													0,183						0,183																									0,367	0,500				
213																										0,005																		0,005	0,000				
221													16,495																															16,495	1,000				
222											0,398			2,284			1,290									0,026																		3,998	0,571				
231		9,086									3,062	9,848				620,491	15,637	33,593					0,052	0,052		0,019																		691,840	0,897				
241		0,688									0,028			0,028			3,497																											4,270	0,007				
242											12,795	4,878				3,342		197,563	0,193	0,193		0,009		0,193								0,193												219,358	0,901				
243		1,403								0,503	1,786					4,923			82,854	1,284		0,186				1,006						0,503												94,447	0,877				
311																			9,751	285,481		14,792				2,565																			312,589	0,913			
312		0,800														0,095			4,831		224,324	2,979				7,296																			240,324	0,933			
313		0,434									0,286					0,286		0,286	1,577	0,745	64,215					0,812																			68,640	0,936			
321									0,018							5,241	0,309	4,438				68,608				0,472											0,054								79,140	0,867			
322		4,024														0,430		0,830	0,830		12,489	150,281			10,503																				179,386	0,838			
324							0,020				0,385					0,332	0,009	0,250	2,841	5,587	2,794	1,282	1,839		101,696				0,009	0,580					0,313									117,937	0,862				
331		0,027																							0,100	3,259													0,020						3,501	0,931			
332																							0,074	0,222				6,279	0,148				0,047		0,048									6,724	0,934				
333																							0,010	0,651		0,036																			83,820	0,992			
334																															0,655														0,655	1,000			
411								0,020								0,151															6,576	0,071	0,156			0,163	0,020							7,157	0,919				
412																2,795		2,777						5,554		0,078						148,647													159,850	0,930			
421																0,368																		4,053											4,420	0,917			
422																																				0,265									0,265	1,000			
423					0,005											0,049								0,065		0,005								0,120											0,014	0,074	2,263	0,854	
511																0,046																														6,717	0,925		
512			0,009							0,014																																				40,034	0,966		
521																																														0,097	0,869		
522			0,121																																0,018			0,018								2,067	0,037	2,261	0,914
523																		0,122							0,052																						2,289	2,463	0,929
Total	4,310	199,014	27,827	6,010	2,068	4,418	6,660	0,665	6,633	9,453	28,009	951,464	0,222	21,374	2,312	662,716	0,028	244,219	150,756	291,596	231,525	84,974	82,531	158,889	0,052	125,621	3,259	6,279	83,270	0,655	6,585	150,470	4,394	0,265	1,980	6,720	39,377	0,084	3,889	2,399	3298,632								
Prod Acc	0,481	0,892	0,973	0,613	0,945	0,944	0,990	0,953	0,503	0,584	0,793	0,972	0,827	0,772	0,988	0,936	1,000	0,809	0,550	0,979	0,969	0,756	0,831	0,946	0,000	0,810	1,000	1,000	0,998	1,000	0,999	0,988	0,922	1,000	0,976	0,924	0,982	1,000	0,531	0,954			Global Acc	0,913					

Overall accuracy = 91.3%

Bio-geographical region BLS – Black sea - confusion matrix CLC2012 - Blind and plausibility analysis

Table 76: Confusion matrix CLC 2012 for Black sea region – BLIND ANALYSIS -In column the validation and line the production

CLC Classes	111	112	121	122	123	124	131	132	133	141	142	211	212	213	221	222	231	242	243	244	311	312	313	321	324	331	332	333	411	421	422	511	512	521	523	Total	User Acc
111	0,255																																		0,255	1,000	
112		2,360							0,023																										2,383	0,990	
121			1,345					0,221	0,014	0,009																									1,589	0,846	
122				0,012	0,988	0,019											0,025																		1,044	0,946	
123					0,112																														0,112	1,000	
124						0,064																													0,064	1,000	
131							0,132		0,220																0,073										0,425	0,310	
132								0,015																											0,015	1,000	
133				0,012					0,205																			0,038					0,025		0,280	0,733	
141										0,029																									0,029	1,000	
142		0,034									0,264															0,012									0,310	0,850	
211												287,358					9,913	13,387	40,196																350,855	0,819	
212												4,522	4,522																						9,044	0,500	
213														0,870				0,290																	1,159	0,750	
221												0,658			0,811																				1,468	0,552	
222																5,244		0,104	3,574																8,922	0,588	
231												0,035					4,635		0,330																5,000	0,927	
242																3,011		12,044	2,930	3,011															20,997	0,574	
243																			19,243																19,243	1,000	
311																				51,487		8,017		0,075											59,578	0,864	
312																					24,399														24,399	1,000	
313									1,428												3,068	3,068	24,195												31,761	0,762	
321																							28,614												28,614	1,000	
324																				0,040				5,426	24,214										29,680	0,816	
331																										0,463			0,028						0,009	0,500	0,926
332																											4,961								4,961	1,000	
333										0,082														4,839	4,839			4,856								14,616	0,332
411																													4,058	0,657						4,715	0,861
421																													0,279							0,279	0,000
422																															0,033					0,033	1,000
511																								0,245									1,360		1,605	0,847	
512												0,052																					1,078		1,130	0,954	
521																																	2,902		2,902	1,000	
523		0,185																																	0,334	0,519	0,644
Total	0,255	2,580	1,357	1,000	0,131	0,064	0,132	0,236	1,891	0,039	0,346	292,573	4,574	0,870	0,811	8,255	14,573	25,826	66,273	3,011	54,595	27,467	32,212	38,880	29,446	0,475	4,961	4,894	4,365	0,657	0,033	1,360	1,104	2,902	0,343	513,452	
Prod Acc	1,000	0,915	0,991	0,988	0,854	1,000	1,000	0,065	0,108	0,761	0,763	0,982	0,989	1,000	1,000	0,635	0,318	0,466	0,290	0,000	0,943	0,888	0,751	0,736	0,822	0,975	1,000	0,992	0,930	0,000	1,000	1,000	0,977	1,000	0,974		
																																				Global Acc	0,817

Overall accuracy = 81.7%

Table 77: Confusion matrix CLC 2012 for Black sea region – PLAUSIBILITY ANALYSIS -In column the validation and line the production

CLC Class	111	112	121	122	123	124	131	132	133	141	142	211	212	213	221	222	231	242	243	244	311	312	313	321	324	331	332	333	411	422	511	512	521	523	Total	User Acc	
111	0,255																																		0,255	1,000	
112		2,383																																	2,383	1,000	
121			1,566						0,024																										1,589	0,985	
122			0,012	1,032																															1,044	0,988	
123				0,112																															0,112	1,000	
124					0,064																														0,064	1,000	
131						0,132		0,220																	0,073										0,425	0,310	
132							0,015																												0,015	1,000	
133			0,012					0,243																								0,025			0,280	0,868	
141									0,029																										0,029	1,000	
142		0,034								0,276																										0,310	0,889
211										#####							9,948	6,694	40,162																350,855	0,838	
212											9,044																									9,044	1,000
213												0,870						0,290																		1,159	0,750
221								0,658				0,811																								1,468	0,552
222													5,244				0,104	3,574																	8,922	0,588	
231								0,035									4,965																			5,000	0,993
242													3,011				12,044	2,930	3,011																	20,997	0,574
243																		19,243																		19,243	1,000
311																				51,487		8,017		0,075												59,578	0,864
312																					24,399															24,399	1,000
313																					3,068	28,692														31,761	0,903
321																							28,614													28,614	1,000
324																							5,426	24,254												29,680	0,817
331																									0,491								0,009	0,500	0,982		
332																										4,961										4,961	1,000
333						0,082																		4,839	4,839			4,856								14,616	0,332
411																													4,715							4,715	1,000
421																													0,279							0,279	0,000
422																														0,033						0,033	1,000
511																								0,245								1,360				1,605	0,847
512												0,052																					1,078			1,130	0,954
521																																	2,902			2,902	1,000
523																																		0,519	0,519	1,000	
Total	0,255	2,418	1,578	1,044	0,112	0,064	0,132	0,015	0,487	0,029	0,358	#####	9,096	0,870	0,811	8,255	14,913	19,132	65,909	3,011	51,487	27,467	36,709	38,880	29,486	0,491	4,961	4,856	4,994	0,033	1,360	1,104	2,902	0,528	530,743		
Prod Acc	1,000	0,986	0,992	0,989	1,000	1,000	1,000	1,000	0,500	1,000	0,771	0,998	0,994	1,000	1,000	0,635	0,333	0,630	0,292	0,000	1,000	0,888	0,782	0,736	0,823	1,000	1,000	1,000	0,944	1,000	1,000	0,977	1,000	0,983			
																																Global Acc		0,844			

Overall accuracy = 84.4%

Bio-geographical region BOR - Boreal - confusion matrix CLC2012 - Blind and plausibility analysis

Table 78: Confusion matrix CLC 2012 for Boreal region – BLIND ANALYSIS- In column the validation and line the production

CLC Classes	111	112	121	122	123	124	131	132	133	141	142	211	212	222	231	242	243	311	312	313	321	322	324	331	332	333	334	411	412	421	423	511	512	521	522	523	Total	User Acc
111	0,196	0,059																																			0,255	0,769
112		40,279	1,376						0,720			0,050			0,008		0,683		0,012				0,008														43,135	0,934
121		0,431	8,184				0,010	0,194	0,002			0,018				0,157			0,002				0,194														9,191	0,890
122				1,336					0,036										0,019																		1,391	0,960
123			0,002		0,338																																0,340	0,993
124						1,269																															1,269	1,000
131		0,049	0,018				2,509		0,035			0,015				0,062		0,034	0,127				0,018														2,866	0,875
132							0,205	0,586		0,024									0,037																		0,852	0,688
133		0,041	0,044		0,002		0,002		0,401							0,009			0,004										0,005								0,510	0,788
141		0,056	0,032							1,594	0,014	0,013					0,035						0,014														1,757	0,907
142		0,014	0,030						0,015		3,815								0,044	0,030			0,030		0,019							0,030				4,027	0,947	
211											293,427				24,625	8,812	7,610		0,040	1,304			0,079					0,012								335,909	0,874	
213																							0,002														0,002	0,000
222												0,027		0,405		0,134			0,015				0,015														0,595	0,680
231												18,826			34,321	5,990	7,740	0,004	0,592	0,569	0,014		1,805		0,014		0,027					0,002				69,904	0,491	
242						0,250	0,016					6,493			4,747	58,357	2,830			0,250			0,005			0,014		0,027								72,948	0,800	
243												13,116			1,904	3,513	102,292	0,002	3,363	1,907			0,783				2,391									129,271	0,791	
311																	0,028	93,382		9,908			7,743					0,003								111,064	0,841	
312				0,022								11,833					0,005	0,044	1565,133	146,106			133,492			2,812		23,662								1883,108	0,831	
313																		5,264	20,481	377,111			36,149					0,006								439,011	0,859	
321														0,403		0,094		0,016		1,035			0,141				0,564									2,253	0,459	
322																					7,906		0,083				0,028									8,016	0,986	
324						0,005		0,004				0,308	2,724		0,075		2,905	7,170	60,008	9,809		2,724	344,322					5,476								435,529	0,791	
331																						0,057	0,315													0,371	0,847	
332																						0,314	0,651	2,054	0,314											3,333	0,616	
333																						0,132	0,042	0,042		7,072										7,288	0,970	
334																			0,012				0,012				0,023									0,046	0,500	
411																												7,576	0,571			0,312					8,459	0,896
412								0,007											0,031	0,674		2,445	7,813					173,332								184,302	0,940	
421																0,024	0,012	0,012			0,012	0,012					0,012		0,204							0,290	0,706	
423																																0,005					0,005	1,000
511																0,127		0,248	0,127													8,695				9,197	0,945	
512																													3,031				267,517				270,547	0,989
521																																	0,129				0,129	1,000
522																							0,027											0,053			0,080	0,667
523												0,074							0,074				0,074						0,147							2,440	2,808	0,869
Total	0,196	40,928	9,687	1,358	0,340	1,519	2,747	0,780	1,221	1,618	3,829	344,198	2,724	0,405	66,083	77,032	124,374	105,913	1650,195	547,868	1,048	13,534	533,569	0,356	2,054	7,418	2,836	10,598	206,099	0,352	0,005	8,695	267,861	0,129	0,053	2,440	3407,611	
Prod Acc	1,000	0,984	0,845	0,984	0,993	0,835	0,913	0,752	0,329	0,985	0,996	0,852	0,000	1,000	0,519	0,758	0,822	0,882	0,948	0,688	0,987	0,584	0,645	0,883	1,000	0,953	0,008	0,715	0,841	0,581	1,000	1,000	0,999	1,000	1,000	1,000	Global Acc	0,843

Overall accuracy = 84.3%

Table 79: Confusion matrix CLC 2012 for Boreal region – PLAUSIBILITY ANALYSIS- In column the validation and line the production

CLC Classe	111	112	121	122	123	124	131	132	133	141	142	211	222	231	242	243	311	312	313	321	322	324	331	332	333	334	411	412	421	423	511	512	521	522	523	Total	User Acc	
111	0,20	0,06																																		0,255	0,769	
112		42,40	0,69						0,01		0,01	0,01						0,00																		43,135	0,983	
121			0,43	8,19			0,01	0,19	0,02							0,16							0,19													9,191	0,891	
122				0,02	1,33				0,02							0,00							0,02													1,391	0,957	
123				0,00		0,34																														0,340	0,993	
124						1,27																														1,269	1,000	
131		0,05	0,02				2,57					0,01				0,06		0,10					0,05													2,866	0,896	
132							0,11	0,72	0,02																											0,852	0,850	
133		0,03	0,03		0,00		0,00		0,42			0,01						0,00										0,01								0,510	0,830	
141		0,01	0,05							1,66	0,01	0,01											0,01													1,757	0,943	
142		0,00									3,85							0,04	0,03				0,03									0,03				3,992	0,965	
211												300,24		19,26	7,52	8,81		0,01					0,06					0,00								335,896	0,894	
213																							0,00													0,002	0,000	
222							0,03	0,46				0,03	0,46		0,08			0,02					0,02													0,595	0,770	
231												18,48		41,70		7,18		0,04	0,56	0,01			1,88		0,01		0,03									69,904	0,597	
242						0,25	0,02					6,49		4,50	59,27	2,17			0,25				0,01													72,948	0,813	
243												9,75		1,90	1,12	108,05	0,00	3,36	1,91				0,78				2,39									129,271	0,836	
311																0,03	95,32		8,08				7,60													111,028	0,859	
312				0,01								11,83				0,00	0,04	1621,24	122,42				103,69					11,84								1871,082	0,866	
313																0,10			410,78				23,06					0,00								433,940	0,947	
321														0,40				0,03		1,32			0,03					0,47								2,253	0,584	
322																					7,91						0,03									8,016	0,986	
324							0,01					0,02		0,02		2,83	3,61	22,88	3,45		2,72	396,72						2,73								434,989	0,912	
331																						0,06	0,31													0,371	0,847	
332																					0,31	0,97			2,05											3,333	0,616	
333																				0,13	0,04	0,04			7,07											7,288	0,970	
334																		0,01								0,03										0,046	0,750	
411																											7,65	0,57				0,24					8,459	0,904
412																		0,02				1,78						182,50								184,302	0,990	
421												0,01	0,01					0,01									0,01		0,24							0,277	0,868	
423																														0,00							0,005	1,000
511														0,13		0,25	0,13														8,69					9,197	0,945	
512																												3,03				267,52					270,547	0,989
521																																0,13					0,129	1,000
522																						0,03											0,05			0,080	0,667	
523																						0,07							0,22					2,51		2,808	0,895	
Total	0,20	42,99	9,00	1,34	0,34	1,52	2,71	0,92	0,49	1,66	3,88	346,90	0,46	67,78	67,99	129,43	99,09	1648,02	547,61	1,33	11,08	537,19	0,36	2,05	7,09	0,03	10,58	200,68	0,46	0,00	8,69	267,79	0,13	0,05	2,51	3584,720		
Prod Acc	1,000	0,986	0,910	0,991	0,993	0,835	0,948	0,789	0,867	1,000	0,993	0,865	1,000	0,615	0,872	0,835	0,962	0,984	0,750	0,990	0,714	0,739	0,883	1,000	0,998	1,000	0,723	0,909	0,521	1,000	1,000	0,999	1,000	1,000	1,000			
																																					Global Acc	0,891

Overall accuracy = 89.1%

Bio-geographical region CON - Continental - confusion matrix CLC2012 - Blind and plausibility analysis

Table 80: Confusion matrix CLC 2012 for Continental region – BLIND ANALYSIS - In column the validation and line the production

CLC Classes	111	112	121	122	123	124	131	132	133	141	142	211	212	213	221	222	231	241	242	243	311	312	313	321	322	323	324	331	332	333	334	411	412	421	422	423	511	512	521	522	523	Total	User Acc				
111	1,912	0,574	0,053																																							2,539	0,753				
112	1,723	244,485	2,562														0,013		21,838	4,955	2,238		0,528				0,009																282,667	0,865			
121		1,170	43,612				0,085	0,355	0,349	2,085	0,022	2,160			0,007		0,766		0,016	0,160		0,691	0,696																				48,282	0,903			
122		0,087	0,181	3,883					0,191			0,083					0,007		0,103	0,034																							4,571	0,850			
123					0,680																																						0,680	1,000			
124						3,932			0,015			0,028					0,115																										4,089	0,962			
131		0,182	0,127	0,068			10,521	0,111	0,046	0,103	0,107	0,243					0,609		0,006	0,030	0,034	0,010	0,035				0,196		0,073						0,059	0,288						12,848	0,819				
132		0,024	0,119				0,105	1,567									0,102										0,131												0,020				2,066	0,758			
133		0,530	0,335	0,181			0,112	0,027	2,909		0,018	0,146							0,075	0,059	0,004		0,062																				4,458	0,653			
141		0,125	0,018							4,222	0,219	0,024					0,036		0,037	0,029	0,026																0,017					4,752	0,888				
142		0,100	0,027			0,015				0,246	11,549	0,038										0,123					0,004					0,341					0,012					12,455	0,927				
211		3,815	13,751				0,028	0,015			6,755	1890,619			8,964	10,596	25,669		96,942	15,603	0,121	0,015					6,893																2079,787	0,909			
212			0,024																																								0,024	0,000			
213												0,099		13,527								0,617																						14,243	0,950		
221												2,696								4,750	2,654																							30,864	0,651		
222												2,835			0,485	17,871	0,216	0,317	6,243	3,659							1,241																	32,867	0,544		
231		5,731	0,337	0,027			0,330	0,015	0,102		0,973	47,038					507,007		58,741	43,297	9,584	0,095	0,082	0,033	0,989			11,862					5,825					0,007					692,075	0,733			
241												0,007						0,853																										0,860	0,992		
242		3,795								0,542	27,705		1,829	3,048	1,788	46,278		204,467	27,482								1,652																	318,585	0,642		
243		1,784									13,253				0,797	17,479		7,249	229,037	6,420	2,653						5,246																	285,703	0,802		
311											5,856	0,013					2,920	0,023	11,074	704,410		52,295				24,650											1,784							801,242	0,879		
312			0,400				2,392					0,084					9,922		7,850	0,034	504,865	19,004				37,275																	4,956	586,781	0,860		
313		1,650								0,542	1,650						3,653		1,216	1,352	4,780	2,723	263,095				6,406																		287,069	0,916	
321										1,032	2,087						9,828			4,472							16,828	3,715		4,954															69,443	0,380	
322																	0,830			1,033		0,083				6,350	2,472	0,203																	10,970	0,579	
323																				0,113						0,679																			0,792	0,857	
324		0,313		0,615			0,032	0,004	0,008		2,795			0,629	0,635	1,388		1,774	15,946	13,371	7,440	4,295	3,605			84,169						0,007													137,026	0,614	
331							0,096	0,014									0,064		0,009						0,048	0,289	1,200	0,028	0,057							0,059									1,863	0,644	
332																					0,026				0,017		0,017		0,138															0,198	0,695		
333							0,036										0,164			0,082				1,673			0,035	0,042		1,731						0,021									3,783	0,458	
334																						0,120																							0,222	0,458	
411					0,019						0,113						1,049					0,071					0,875																			12,006	0,778
412																				0,025	0,078	0,025	0,078	0,020	0,078		0,301					0,329	1,211						0,320						2,146	0,564	
421																					0,011			0,081								0,081			0,685	0,044									1,026	0,668	
422			0,077																																	0,041										0,119	0,349
423																																					0,028									0,028	1,000
511																	0,180		0,262								0,262	1,183				0,315			0,098	17,186									19,486	0,882	
512							0,022	0,007		0,004		0,016					0,156		0,456	0,456							0,137			0,137		0,345														49,196	0,965
521																	0,055										0,09,																				

Overall accuracy = 83.9%

Table 81: Confusion matrix CLC 2012 for Continental region – PLAUSIBILITY ANALYSIS - In column the validation and line the production

CLC Class	111	112	121	122	123	124	131	132	133	141	142	211	212	213	221	222	223	231	241	242	243	311	312	313	321	322	323	324	331	332	333	334	411	412	421	422	423	511	512	521	522	523	Total	User Acc			
111	2,16	0,32	0,05																																								2,539	0,853			
112	1,72	265,83	0,89	0,00					0,03	2,08	0,02	0,05								7,98	1,29		2,24	0,53																				282,667	0,940		
121		0,17	46,97	0,00			0,00		0,02			0,36						0,04		0,04				0,69																				48,282	0,973		
122		0,03	0,14	4,21					0,04			0,01								0,10				0,03																					4,571	0,921	
123					0,68																																							0,680	1,000		
124						4,03												0,06																										4,089	0,986		
131		0,20	0,02				11,21	0,00	0,03	0,10	0,11	0,04						0,54			0,03	0,03		0,03				0,21											0,29				12,839	0,873			
132		0,02	0,06				0,02	1,80																				0,13											0,02				2,066	0,873			
133		0,21	0,29	0,10			0,17	0,03	3,35			0,21								0,05	0,01			0,04																				4,458	0,751		
141		0,12	0,02							4,34	0,17							0,06		0,01	0,01	0,03																						4,752	0,913		
142		0,08	0,03			0,02					12,18	0,04											0,12																						12,455	0,978	
211			6,76					0,01			6,76	1974,96			1,72	0,12		6,94	75,44	0,08	0,10	0,02					6,88																	2079,772	0,950		
212											0,01	0,01																																0,024	0,500		
213												0,05		14,19																															14,243	0,997	
221												0,54			23,04	0,01		2,27		3,30	1,70																								30,864	0,747	
222												0,57			0,01	21,55		0,82	0,34	4,17	2,64	0,76						2,00																	32,867	0,656	
231				0,02			0,01		0,08		0,90	46,00						563,25		36,93	27,15	6,95		0,08		0,66		3,97						5,75										691,732	0,814		
241																			0,86	0,00																								0,860	0,996		
242		1,97									0,54	26,69			0,14			24,13		247,92	12,32			4,88																				318,585	0,778		
243		1,78	0,03									4,24					1,78	8,92		4,62	254,14	1,23	2,65					4,87																	284,266	0,894	
311									1,68			2,96			3,21					3,75	741,94		38,98					8,72																	801,242	0,926	
312												0,08								2,81		554,49	11,27				18,12																		586,781	0,945	
313									0,43			0,43								0,14	0,03	0,46	279,68					4,55																	285,719	0,979	
321																		5,72		3,55	2,48			42,88	1,10	1,30	8,39	2,48					1,56												69,443	0,617	
322																		0,83		1,03		0,06		0,20	6,35		2,30	0,20																	10,970	0,579	
323																											0,79																		0,792	1,000	
324		0,31					0,03		0,09			0,06			0,62			1,85		1,71	10,77	4,60	2,27	4,19	0,90		109,59						0,01												137,009	0,800	
331							0,10					0,01						0,04						0,03	0,05		0,07	1,57																1,863	0,844		
332																							0,03			0,02		0,02		1,57		0,14												0,198	0,695		
333							0,04											0,08							1,20			0,12				2,34													3,783	0,618	
334																												0,12					0,10												0,222	0,458	
411					0,02							0,02						0,80				0,06	0,07					0,45					10,59											12,006	0,882		
412																					0,03	0,08		0,08				0,15						1,82									2,146	0,848			
421																						0,01			0,08								0,08		0,68	0,04			0,12					1,026	0,668		
422																																					0,12								0,119	1,000	
423																																					0,03								0,028	1,000	
511																		0,18											0,63					0,31				18,36						19,486	0,942		
512							0,01			0,00		0,01						0,00			0,46							0,14					0,47						48,11						49,196	0,978	
521																											0,09								0,03						1,03					1,151	0,897
522																																										0,05				0,053	1,000
523																																								0,06					0,81	0,870	0,936
Total	3,88	271,04	55,26	4,34	0,70	4,05	11,59	1,84	5,75	6,53	20,67	2057,35	0,01	14,19	24,91	25,51	1,78	616,53	1,20	382,27	321,43	758,75	562,41	340,48	45,29	8,17	2,09	170,88	4,88	0,14	2,34	0,10	18,76	1,83	0,71	0,16	0,08	18,36	48,41	1,16	0,05	0,81	5278,173				
Prod Acc	0,																																														

Overall accuracy = 90.7%

Bio-geographical region - Macaronesia - confusion matrix CLC2012 - Blind and plausibility analysis

Table 82: Confusion matrix CLC 2012 for Macaronesia region – BLIND ANALYSIS - In column the validation and line the production

CLC Classes	111	112	121	122	123	124	131	133	142	212	231	242	243	311	312	313	321	322	323	324	331	332	333	334	412	523	Total	User Acc
111	0,026																0,509		0,509								1,044	0,025
112		1,027																									1,027	1,000
121			0,025																								0,025	1,000
122				0,008																							0,008	1,000
123					0,090																						0,090	1,000
124						0,074																					0,074	1,000
131							0,063																				0,063	1,000
133			0,013					0,356																			0,369	0,964
141									0,012																		0,012	0,000
142									0,117																		0,117	1,000
212										2,563																	2,563	1,000
231											2,309																2,309	1,000
243												1,719	4,922														6,640	0,741
311														2,673													2,673	1,000
312															5,659												5,659	1,000
321										0,256						0,256	0,512										1,023	0,500
322																		0,025									0,025	1,000
323																			10,421								10,421	1,000
324																				1,256							1,256	1,000
331																					0,285				0,142		0,427	0,667
332																						1,820					2,414	0,754
333																	4,633	0,183	1,544							7,721	14,081	0,548
334																				0,027					0,217		0,244	0,888
412																									0,225		0,225	1,000
523																										0,052	0,052	1,000
Total	0,026	1,027	0,038	0,008	0,090	0,074	0,063	0,356	0,129	2,563	2,565	1,719	4,922	2,673	5,659	0,256	5,653	0,209	12,475	1,878	0,285	1,820	7,863	0,217	0,225	0,052	42,445	
Prod Acc	1,000	1,000	0,651	1,000	1,000	1,000	1,000	1,000	0,906	1,000	0,900	0,000	1,000	1,000	1,000	0,000	0,090	0,121	0,835	0,669	1,000	1,000	0,982	1,000	1,000	1,000		
																											Global Acc	0,803

Overall accuracy = 80.3%

Table 83: Confusion matrix CLC 2012 for Macaronesia region – PLAUSIBILITY ANALYSIS - In column the validation and line the production

CLC Classe	111,00	112,00	121,00	122,00	123,00	124,00	131,00	133,00	142,00	212,00	231,00	242,00	243,00	311,00	312,00	321,00	322,00	323,00	324,00	331,00	332,00	333,00	334,00	412,00	523,00	Total	User Acc	
111	0,03															0,51		0,51								1,044	0,025	
112		1,03																									1,027	1,000
121			0,02																								0,025	1,000
122				0,01																							0,008	1,000
123					0,09																						0,090	1,000
124						0,07																					0,074	1,000
131							0,06																				0,063	1,000
133			0,01					0,36																			0,369	0,964
141									0,01																		0,012	0,000
142									0,12																		0,117	1,000
212										2,56																	2,563	1,000
231											2,31																2,309	1,000
243												1,72	4,92														6,640	0,741
311														2,67													2,673	1,000
312															5,66												5,659	1,000
321																1,02											1,023	1,000
322																	0,03										0,025	1,000
323																		10,42									10,421	1,000
324																			1,26								1,256	1,000
331																				0,43							0,427	1,000
332																			0,59		1,82						2,414	0,754
333																	0,18	1,54				12,35					14,081	0,877
334																			0,03				0,22				0,244	0,888
412																								0,22			0,225	1,000
523																									0,05		0,052	1,000
Total	0,026	1,027	0,038	0,008	0,090	0,074	0,063	0,356	0,129	2,563	2,309	1,719	4,922	2,673	5,659	1,532	0,209	12,475	1,878	0,427	1,820	12,354	0,217	0,225	0,052	47,732		
Prod Acc	1,000	1,000	0,651	1,000	1,000	1,000	1,000	1,000	0,906	1,000	1,000	0,000	1,000	1,000	1,000	0,668	0,121	0,835	0,669	1,000	1,000	1,000	1,000	1,000	1,000			
																											Global Acc	0,903

Overall accuracy = 90.3%

Table 84: Confusion matrix CLC 2012 for Mediterranean region – BLIND ANALYSIS - In column the validation and line the production

Overall accuracy = 80.7%

Table 85: Confusion matrix CLC 2012 for Mediterranean region – PLAUSIBILITY ANALYSIS - In column the validation and line the production

CLC Classe	111	112	121	122	123	124	131	132	133	141	142	211	212	213	221	222	223	231	241	242	243	244	311	312	313	321	322	323	324	331	332	333	334	411	421	422	423	511	512	521	522	523	Total	User Acc			
111	17,94	2,11	0,29						0,51																																	20,858	0,860				
112	0,56	83,70	0,63						0,01								0,01			0,14																							86,061	0,973			
121		0,18	29,63	0,16	0,37				0,66		0,02									0,37																								31,371	0,945		
122				5,00			0,06		0,03			0,03								0,12																								5,237	0,954		
123	0,02	0,02			0,70																																							0,744	0,944		
124						3,12					0,08															0,05																		3,248	0,960		
131			0,13				10,46	0,05	0,37			0,01								0,22						0,11			0,37			0,11							0,37				12,185	0,858			
132			0,02				0,01	0,97																																			0,994	0,972			
133		0,53	0,54				0,11	0,04	6,42		0,51	0,36																	0,01	0,05														8,568	0,750		
141		0,16	0,10							1,26	0,07																							0,02										1,609	0,783		
142		0,50									6,10									0,06																								6,656	0,916		
211												641,89	0,13			0,71	0,06	17,60	6,76	15,46	7,48	6,76							0,04	1,83														707,084	0,908		
212		4,52										0,34	223,78	0,02		0,22				3,82																								232,711	0,962		
213												0,04	0,03	14,92																						0,10								15,093	0,989		
221												2,34	0,04		81,78	0,81	1,34		1,13	19,10	1,70		0,20			0,38																		108,999	0,750		
222		1,20										2,77				78,27	0,03		6,41	10,20	3,08																							102,194	0,766		
223		0,05										2,57				0,20	194,26		2,66	6,52	4,99		1,11	0,05	0,05	0,09		0,05	2,77															215,357	0,902		
231							0,01		0,04								0,29	60,61		1,08	0,67						8,29		1,55	0,31															72,844	0,832	
241		0,69	0,10								0,62	0,02						0,43	9,45	5,01	1,90	1,54																							19,973	0,473	
242		3,03	2,97						0,03		18,96			1,46	2,29	7,81	6,79	3,09	220,27	22,26									0,01	1,83															290,789	0,757	
243		1,72									0,04						1,48	5,05	6,30	18,11	285,46	1,48	3,37			2,69	3,37		1,04	6,41															336,525	0,848	
244											1,31									2,96	127,49								2,59	2,59															136,946	0,931	
311															0,89					3,88	6,24	507,82	0,89	2,72				14,87	18,91																556,217	0,913	
312				0,01																4,83		0,51	412,29	4,33				4,83	5,99																432,790	0,953	
313																				1,90		5,37	6,14	163,16																						184,872	0,883
321		4,12							0,07			4,12					4,50	16,74		7,01	12,90							203,43	6,82	21,03	36,04															316,788	0,642
322																												0,51	35,41		5,94															41,863	0,846
323			5,21									0,85					0,03	2,61		12,95		0,87					3,46		374,60	22,31															422,890	0,886	
324		1,26							0,04		2,18	0,07			0,07			0,04		8,77	2,19	9,77	11,42	4,81	0,87	2,60	6,65	381,84				0,05	1,48												434,098	0,880	
331																		0,09												9,19	0,05					0,09									9,413	0,976	
332			0,09																	0,09							0,09	0,88	0,68	0,05		21,23	10,58												33,670	0,630	
333																											7,12	0,18	6,70	20,87		0,90	64,62	0,28											100,685	0,642	
334			0,01													0,10				0,03		0,13		0,02	0,16	0,03	0,26	0,43			0,02	5,03													6,235	0,807	
335																												0,00																		0,003	0,000
411																			0,06															5,72				0,04							5,940	0,962	
421										0,09	0,39															0,14										6,51					0,04				7,163	0,908	
422																																	0,11				0,08	1,87							2,056	0,907	
423																																				0,00		0,10									

Overall accuracy = 86.5%

Bio-geographical region PAN – Pannonian - confusion matrix CLC2012 - Blind and plausibility analysis

Table 86: Confusion matrix CLC 2012 for Pannonian region – BLIND ANALYSIS- In column the validation and line the production

CLC Classes	111	112	121	122	123	124	131	132	133	141	142	211	213	221	222	231	242	243	311	312	313	321	324	333	411	412	511	512	Total	User Acc
111	0,083																												0,083	1,000
112		26,803	0,920		0,014							0,008					0,052												27,796	0,964
121			3,493	0,128	0,004			0,007	0,010	0,014	0,008	0,024				0,014		0,002											3,705	0,943
122			0,028	0,626					0,009			0,028											0,029						0,720	0,870
123					0,022																								0,022	1,000
124			0,059			0,318																							0,376	0,844
131							0,413					0,019		0,020		0,020			0,020			0,002	0,022		0,034		0,019	0,569	0,727	
132				0,019				0,196				0,015				0,015								0,015					0,259	0,757
133		0,039	0,029	0,113			0,002		0,150		0,013	0,052				0,004	0,024	0,014					0,002						0,443	0,339
141										0,302		0,017																	0,319	0,948
142		0,087									1,463						0,072	0,059											1,682	0,870
211			0,007	4,023							3,784	307,515	0,049	4,081	5,462	0,589	5,485						4,029						335,024	0,918
212			0,012																										0,012	0,000
213												0,173	0,173																0,345	0,500
221												0,984		6,128	0,006	0,114			0,234										7,465	0,821
222												0,759		0,086	2,966	0,013	0,195		0,074										4,093	0,725
231			1,207	0,055			0,004	0,008			4,924		0,154	0,154	26,323								0,649		0,004		0,583	34,065	0,773	
242			1,201								0,019		4,122	0,025	0,154	15,355	1,201						0,013						22,091	0,695
243											0,308						10,347	0,692					1,143						12,490	0,828
311		0,038	1,275								1,304					0,107			69,606		1,261		10,508						84,100	0,828
312																			0,004	2,696	0,004		0,588						3,291	0,819
313											0,128								0,155	0,009	5,037		1,231						6,560	0,768
321															3,347							6,216	0,011						9,574	0,649
324			0,014								2,685		0,321	0,107	0,572		0,641	2,913	0,053	0,542		16,763	0,007					24,619	0,681	
333																								0,113					0,113	1,000
411															0,429	0,021						0,160	0,289		3,212				4,111	0,781
412															0,079											0,316			0,395	0,800
511																										2,892			2,892	1,000
512												0,003				0,272												6,196	6,471	0,958
Total	0,083	26,967	8,246	4,963	0,040	0,318	0,419	0,211	0,169	0,316	5,268	318,962	0,222	14,913	8,719	32,052	21,184	12,284	73,697	2,759	6,845	6,379	35,278	0,136	3,250	0,316	2,892	6,799	515,725	
Prod Acc	1,000	0,994	0,424	0,126	0,543	1,000	0,986	0,930	0,888	0,956	0,278	0,964	0,778	0,411	0,340	0,821	0,725	0,842	0,944	0,977	0,736	0,975	0,475	0,837	0,988	1,000	0,911			
																													Global Acc	0,869

Overall accuracy = 86.9%

Table 87: Confusion matrix CLC 2012 for Pannonian region – PLAUSIBILITY ANALYSIS- In column the validation and line the production

CLC Classe	111	112	121	122	123	124	131	132	133	141	142	211	213	221	222	231	242	243	311	312	313	321	324	333	411	412	511	512	Total	User Acc
111	0,08																												0,083	1,000
112		26,83	0,92									0,01					0,04												27,796	0,965
121			3,67		0,00			0,01		0,01		0,01																	3,705	0,991
122				0,64								0,03						0,02			0,00		0,03						0,720	0,893
123					0,02																								0,022	1,000
124						0,38																							0,376	1,000
131							0,47					0,02							0,02			0,00	0,02		0,03				0,569	0,830
132				0,02				0,23				0,01																	0,259	0,870
133		0,02	0,01	0,04					0,28			0,04				0,02	0,01	0,02	0,00										0,443	0,630
141										0,30		0,02																	0,319	0,948
142		0,14									1,48							0,06											1,682	0,879
211				0,00							0,01	325,76	0,05	0,03	0,23	0,31	4,59	4,01					0,03						335,024	0,972
212			0,01																										0,012	0,000
213													0,35																0,345	1,000
221												0,09		6,48	0,01		0,66	0,23											7,465	0,868
222												0,33		0,07	3,43		0,19		0,07										4,083	0,839
231			0,63	0,05								3,20		0,15	0,15	29,15					0,05	0,00	0,08		0,00			0,58	34,065	0,856
242			1,20									0,01			0,01		19,66	1,20					0,01						22,091	0,890
243																		11,00	0,69				0,80						12,490	0,881
311			1,24															0,04	75,11		1,26		6,46						84,100	0,893
312																				3,11	0,08		0,10						3,291	0,945
313												0,13							0,16	0,02	5,47		0,79						6,560	0,833
321												0,48				1,43	0,48					6,69	0,49						9,574	0,699
324												1,37		0,43		0,03	0,32	1,03	0,33	0,09		21,02							24,619	0,854
333																							0,11						0,113	1,000
411																0,48						0,02		3,59			0,02		4,111	0,873
412																0,08										0,32			0,395	0,800
511																										2,89			2,892	1,000
512												0,00											0,01					6,46	6,471	0,998
Total	0,08	26,99	7,69	0,76	0,03	0,38	0,47	0,23	0,28	0,32	1,49	331,51	0,39	7,16	3,82	31,49	25,15	17,39	77,09	3,46	6,95	6,70	29,86	0,11	3,63	0,32	2,89	7,06	554,973	
Prod Acc	1,000	0,994	0,478	0,846	0,831	1,000	1,000	0,971	1,000	0,956	0,995	0,983	0,875	0,905	0,897	0,926	0,782	0,633	0,974	0,898	0,786	0,999	0,704	1,000	0,990	1,000	1,000	0,915		
																													Global Acc	0,935

Overall accuracy = 93.5%

Bio-geographical region STE - Steppic - confusion matrix CLC2012 - Blind and plausibility analysis

Table 88: Confusion matrix CLC 2012 for steppic region – BLIND ANALYSS- In column the validation and line the production

CLC Classes	112	121	122	123	124	132	133	141	142	211	213	221	222	231	242	243	311	321	324	331	411	511	512	Total	User Acc
112	3,669														0,917	0,917								5,503	0,667
121		1,325																						1,325	1,000
122	0,019		0,077				0,061																	0,157	0,487
123				0,027																				0,027	1,000
124					0,049					0,016														0,065	0,750
132						0,015																		0,015	1,000
141								0,012																0,012	1,000
142									0,012															0,012	1,000
211										109,058														109,058	1,000
213											0,778													0,778	1,000
221										1,973		0,658			0,658	0,658								3,945	0,167
231	2,070	2,070								2,070				6,211										12,421	0,500
242	1,652									1,652					1,652									4,957	0,333
243																1,594								1,594	1,000
311																	8,229		4,115					12,344	0,667
324												0,291	0,291				0,291	0,291	0,365					1,530	0,238
331																				0,111		0,028		0,139	0,800
411									0,222					0,009							4,601		0,657	5,490	0,838
511																						1,521		1,521	1,000
512																					0,332		2,699	3,030	0,891
Total	7,411	3,396	0,077	0,027	0,049	0,015	0,061	0,012	0,012	114,991	0,778	0,658	0,291	6,511	3,227	3,169	8,520	0,291	4,479	0,111	4,933	1,549	3,356	142,663	
Prod Acc	0,495	0,390	1,000	1,000	1,000	1,000	0,000	1,000	1,000	0,948	1,000	1,000	0,000	0,954	0,512	0,503	0,966	0,000	0,081	1,000	0,933	0,982	0,804		
																								Global Acc	0,870

Overall accuracy = 87.0%

Table 89: Confusion matrix CLC 2012 for steppic region – PLAUSIBILITY ANALYSIS In column the validation and line the production

CLC Classe	112	121	122	123	124	132	133	141	142	211	213	221	231	242	243	311	324	331	411	511	512	Total	User Acc
112	5,503																					5,503	1,000
121		1,325																				1,325	1,000
122			0,096				0,061															0,157	0,609
123				0,027																		0,027	1,000
124					0,065																	0,065	1,000
132						0,015																0,015	1,000
141								0,012														0,012	1,000
142									0,012													0,012	1,000
211										109,058												109,058	1,000
213											0,778											0,778	1,000
221										0,658		1,973		1,315								3,945	0,500
231							2,070						10,351									12,421	0,833
242														4,957								4,957	1,000
243															1,594							1,594	1,000
311																12,344						12,344	1,000
324													0,291				0,291	0,948				1,530	0,619
331																		0,139				0,139	1,000
411									0,222										4,611		0,657	5,490	0,840
511																				1,521		1,521	1,000
512																			0,332		2,699	3,030	0,891
Total	5,503	1,325	0,096	0,027	0,065	0,015	2,132	0,012	0,012	109,937	0,778	1,973	10,642	6,272	1,594	12,635	0,948	0,139	4,942	1,521	3,356	158,027	
Prod Acc	1,000	1,000	1,000	1,000	1,000	1,000	0,000	1,000	1,000	0,992	1,000	1,000	0,973	0,790	1,000	0,977	1,000	1,000	0,933	1,000	0,804		
																					Global Acc		0,964

Overall accuracy = 96.4%

Bio-geographical region DOM - Overseas department (tropical region) - confusion matrix CLC2012 - Blind and plausibility analysis

Table 90: Confusion matrix CLC 2012 for Tropical region – BLIND ANALYSIS- In column the validation and line the production

CLC Classes	111	112	121	122	123	124	131	132	133	141	142	211	213	222	231	242	243	311	312	321	322	323	324	331	332	333	411	423	511	512	521	522	523	Total	User Acc
111	0,018																																	0,018	1,000
112		2,565							0,007							0,005	0,005																	2,582	0,993
121		0,014	0,160						0,005																									0,179	0,896
122		0,007		0,074																														0,081	0,916
123					0,023																													0,023	1,000
124						0,064																												0,064	1,000
131							0,045																											0,045	1,000
132								0,003																										0,003	1,000
133									0,005																									0,005	1,000
141										0,019																								0,019	1,000
142											0,031																							0,031	1,000
211												2,502				0,131																		2,633	0,950
213													0,282																					0,282	1,000
222														0,496		0,071	0,142																0,708	0,700	
231		0,110												0,007	0,776	0,007	0,110						0,113										1,123	0,691	
242																1,449	0,161																	1,610	0,900
243		0,336							0,003							0,336	2,558	0,003				0,003	0,175										3,415	0,749	
244																						0,091											0,091	0,000	
311																		33,476														0,013	33,489	1,000	
312																			0,092														0,092	1,000	
313																	0,229																0,229	0,000	
321													0,004							1,104													1,108	0,996	
322																					0,599		0,082										0,681	0,880	
323																						0,558	0,372										0,930	0,600	
324																0,342	0,502			0,164			2,240		0,164								3,413	0,656	
331																							0,011										0,011	1,000	
332																								0,309									0,309	1,000	
333																	0,047		0,093						0,326								0,466	0,700	
411																	0,161									4,477							4,639	0,965	
421																										0,031							0,031	0,000	
423																											0,003						0,003	1,000	
511																							0,045						0,362		0,045		0,452	0,800	
512																														0,018			0,018	1,000	
521																															0,008		0,008	1,000	
522																												0,114				0,920	0,121	1,155	0,796
523																																	61,963	61,963	1,000
Total	0,018	3,031	0,160	0,074	0,023	0,064	0,045	0,003	0,020	0,019	0,031	2,502	0,282	0,502	0,780	2,000	3,318	34,418	0,092	1,104	0,857	0,562	3,073	0,057	0,309	0,491	4,509	0,003	0,476	0,018	0,008	0,965	62,097	117,537	
Prod Acc	1,000	0,846	1,000	1,000	1,000	1,000	1,000	1,000	0,249	1,000	1,000	1,000	1,000	0,987	0,994	0,725	0,771	0,973	1,000	1,000	0,699	0,994	0,729	0,202	1,000	0,665	0,993	1,000	0,760	1,000	1,000	0,953	0,998		
																																		Global Acc	0,964

Overall accuracy = 96.4%

Table 91: Confusion matrix CLC 2012 for Tropical region – PLAUSIBILITY ANALYSIS- In column the validation and line the production

CLC Classe	111	112	121	122	123	124	131	132	133	141	142	211	213	222	231	242	243	311	312	313	321	322	323	324	331	332	333	411	423	511	512	521	522	523	Total	User Acc	
111	0,018																																	0,018	1,000		
112		2,565							0,007							0,005	0,005																		2,582	0,993	
121			0,014	0,160																				0,005												0,179	0,896
122			0,007		0,074																															0,081	0,916
123					0,023																															0,023	1,000
124						0,064																														0,064	1,000
131							0,045																													0,045	1,000
132								0,003																												0,003	1,000
133									0,005																											0,005	1,000
141										0,019																										0,019	1,000
142											0,031																									0,031	1,000
211												2,502				0,131																				2,633	0,950
213													0,282																							0,282	1,000
222														0,496		0,071	0,142																		0,708	0,700	
231														0,007	0,886	0,007	0,110							0,113											1,123	0,789	
242																1,449	0,161																			1,610	0,900
243		0,336							0,003							0,336	2,565	0,003						0,003	0,168											3,415	0,751
244																								0,091												0,091	0,000
311																		33,476															0,013	33,489	1,000		
312																			0,092																	0,092	1,000
313																				0,229																0,229	1,000
321															0,004						1,104															1,108	0,996
322																						0,674		0,007												0,681	0,990
323																							0,837	0,093												0,930	0,900
324																	0,342	0,502						2,569												3,413	0,753
331																									0,011											0,011	1,000
332																										0,309										0,309	1,000
333																		0,047				0,093					0,326								0,466	0,700	
411																		0,007										4,632								4,639	0,999
421																												0,031								0,031	0,000
423																													0,003							0,003	1,000
511																									0,045					0,407						0,452	0,900
512																															0,018					0,018	1,000
521																																0,008				0,008	1,000
522																																	1,155			1,155	1,000
523																																		61,963	61,963	1,000	
Total	0,018	2,921	0,160	0,074	0,023	0,064	0,045	0,003	0,015	0,019	0,031	2,502	0,282	0,502	0,890	2,000	3,325	34,035	0,092	0,229	1,104	0,767	0,841	3,046	0,057	0,309	0,326	4,663	0,003	0,407	0,018	0,008	1,155	61,976	119,000		
Prod Acc	1,000	0,878	1,000	1,000	1,000	1,000	1,000	1,000	0,331	1,000	1,000	1,000	1,000	0,987	0,995	0,725	0,771	0,984	1,000	1,000	1,000	0,878	0,996	0,843	0,202	1,000	1,000	0,993	1,000	1,000	1,000	1,000	1,000	1,000			
																																				Global Acc	0,976

Overall accuracy = 97.6%

Annex 6. Confusion matrix of CLCCH2006-12 – European level – Blind and plausibility analysis

Table 92: Confusion matrix of change CLCCH2006-2012 for blind analysis (at level 2 of CLC nomenclature) between the validation and the production

0000	0001	0002	0003	0004	0005	0006	0007	0008	0009	0010	0011	0012	0013	0014	0015	0016	0017	0018	0019	0020	0021	0022	0023	0024	0025	0026	0027	0028	0029	0030	0031	0032	0033	0034	0035	0036	0037	0038	0039	0040	0041	0042	0043	0044	0045	0046	0047	0048	0049	0050	0051	0052	0053	0054	0055	0056	0057	0058	0059	0060	0061	0062	0063	0064	0065	0066	0067	0068	0069	0070	0071	0072	0073	0074	0075	0076	0077	0078	0079	0080	0081	0082	0083	0084	0085	0086	0087	0088	0089	0090	0091	0092	0093	0094	0095	0096	0097	0098	0099	0100	0101	0102	0103	0104	0105	0106	0107	0108	0109	0110	0111	0112	0113	0114	0115	0116	0117	0118	0119	0120	0121	0122	0123	0124	0125	0126	0127	0128	0129	0130	0131	0132	0133	0134	0135	0136	0137	0138	0139	0140	0141	0142	0143	0144	0145	0146	0147	0148	0149	0150	0151	0152	0153	0154	0155	0156	0157	0158	0159	0160	0161	0162	0163	0164	0165	0166	0167	0168	0169	0170	0171	0172	0173	0174	0175	0176	0177	0178	0179	0180	0181	0182	0183	0184	0185	0186	0187	0188	0189	0190	0191	0192	0193	0194	0195	0196	0197	0198	0199	0200	0201	0202	0203	0204	0205	0206	0207	0208	0209	0210	0211	0212	0213	0214	0215	0216	0217	0218	0219	0220	0221	0222	0223	0224	0225	0226	0227	0228	0229	0230	0231	0232	0233	0234	0235	0236	0237	0238	0239	0240	0241	0242	0243	0244	0245	0246	0247	0248	0249	0250	0251	0252	0253	0254	0255	0256	0257	0258	0259	0260	0261	0262	0263	0264	0265	0266	0267	0268	0269	0270	0271	0272	0273	0274	0275	0276	0277	0278	0279	0280	0281	0282	0283	0284	0285	0286	0287	0288	0289	0290	0291	0292	0293	0294	0295	0296	0297	0298	0299	0300	0301	0302	0303	0304	0305	0306	0307	0308	0309	0310	0311	0312	0313	0314	0315	0316	0317	0318	0319	0320	0321	0322	0323	0324	0325	0326	0327	0328	0329	0330	0331	0332	0333	0334	0335	0336	0337	0338	0339	0340	0341	0342	0343	0344	0345	0346	0347	0348	0349	0350	0351	0352	0353	0354	0355	0356	0357	0358	0359	0360	0361	0362	0363	0364	0365	0366	0367	0368	0369	0370	0371	0372	0373	0374	0375	0376	0377	0378	0379	0380	0381	0382	0383	0384	0385	0386	0387	0388	0389	0390	0391	0392	0393	0394	0395	0396	0397	0398	0399	0400	0401	0402	0403	0404	0405	0406	0407	0408	0409	0410	0411	0412	0413	0414	0415	0416	0417	0418	0419	0420	0421	0422	0423	0424	0425	0426	0427	0428	0429	0430	0431	0432	0433	0434	0435	0436	0437	0438	0439	0440	0441	0442	0443	0444	0445	0446	0447	0448	0449	0450	0451	0452	0453	0454	0455	0456	0457	0458	0459	0460	0461	0462	0463	0464	0465	0466	0467	0468	0469	0470	0471	0472	0473	0474	0475	0476	0477	0478	0479	0480	0481	0482	0483	0484	0485	0486	0487	0488	0489	0490	0491	0492	0493	0494	0495	0496	0497	0498	0499	0500	0501	0502	0503	0504	0505	0506	0507	0508	0509	0510	0511	0512	0513	0514	0515	0516	0517	0518	0519	0520	0521	0522	0523	0524	0525	0526	0527	0528	0529	0530	0531	0532	0533	0534	0535	0536	0537	0538	0539	0540	0541	0542	0543	0544	0545	0546	0547	0548	0549	0550	0551	0552	0553	0554	0555	0556	0557	0558	0559	0560	0561	0562	0563	0564	0565	0566	0567	0568	0569	0570	0571	0572	0573	0574	0575	0576	0577	0578	0579	0580	0581	0582	0583	0584	0585	0586	0587	0588	0589	0590	0591	0592	0593	0594	0595	0596	0597	0598	0599	0600	0601	0602	0603	0604	0605	0606	0607	0608	0609	0610	0611	0612	0613	0614	0615	0616	0617	0618	0619	0620	0621	0622	0623	0624	0625	0626	0627	0628	0629	0630	0631	0632	0633	0634	0635	0636	0637	0638	0639	0640	0641	0642	0643	0644	0645	0646	0647	0648	0649	0650	0651	0652	0653	0654	0655	0656	0657	0658	0659	0660	0661	0662	0663	0664	0665	0666	0667	0668	0669	0670	0671	0672	0673	0674	0675	0676	0677	0678	0679	0680	0681	0682	0683	0684	0685	0686	0687	0688	0689	0690	0691	0692	0693	0694	0695	0696	0697	0698	0699	0700	0701	0702	0703	0704	0705	0706	0707	0708	0709	0710	0711	0712	0713	0714	0715	0716	0717	0718	0719	0720	0721	0722	0723	0724	0725	0726	0727	0728	0729	0730	0731	0732	0733	0734	0735	0736	0737	0738	0739	0740	0741	0742	0743	0744	0745	0746	0747	0748	0749	0750	0751	0752	0753	0754	0755	0756	0757	0758	0759	0760	0761	0762	0763	0764	0765	0766	0767	0768	0769	0770	0771	0772	0773	0774	0775	0776	0777	0778	0779	0780	0781	0782	0783	0784	0785	0786	0787	0788	0789	0790	0791	0792	0793	0794	0795	0796	0797	0798	0799	0800	0801	0802	0803	0804	0805	0806	0807	0808	0809	0810	0811	0812	0813	0814	0815	0816	0817	0818	0819	0820	0821	0822	0823	0824	0825	0826	0827	0828	0829	0830	0831	0832	0833	0834	0835	0836	0837	0838	0839	0840	0841	0842	0843	0844	0845	0846	0847	0848	0849	0850	0851	0852	0853	0854	0855	0856	0857	0858	0859	0860	0861	0862	0863	0864	0865	0866	0867	0868	0869	0870	0871	0872	0873	0874	0875	0876	0877	0878	0879	0880	0881	0882	0883	0884	0885	0886	0887	0888	0889	0890	0891	0892	0893	0894	0895	0896	0897	0898	0899	0900	0901	0902	0903	0904	0905	0906	0907	0908	0909	0910	0911	0912	0913	0914	0915	0916	0917	0918	0919	0920	0921	0922	0923	0924	0925	0926	0927	0928	0929	0930	0931	0932	0933	0934	0935	0936	0937	0938	0939	0940	0941	0942	0943	0944	0945	0946	0947	0948	0949	0950	0951	0952	0953	0954	0955	0956	0957	0958	0959	0960	0961	0962	0963	0964	0965	0966	0967	0968	0969	0970	0971	0972	0973	0974	0975	0976	0977	0978	0979	0980	0981	0982	0983	0984	0985	0986	0987	0988	0989	0990	0991	0992	0993	0994	0995	0996	0997	0998	0999	1000	1001	1002	1003	1004	1005	1006	1007	1008	1009	1010	1011	1012	1013	1014	1015	1016	1017	1018	1019	1020	1021	1022	1023	1024	1025	1026	1027	1028	1029	1030	1031	1032	1033	1034	1035	1036	1037	1038	1039	1040	1041	1042	1043	1044	1045	1046	1047	1048	1049	1050	1051	1052	1053	1054	1055	1056
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[illegible]

For the plausibility analysis, the overall accuracy of CLCCH2006-2012 at the third level is 88.6%. The overall accuracy of CLCCH2006-2012 at the second level is 91.2%.

Zone 1: Turkey - Confusion matrix of CLCCH2006-2012 - Blind and plausibility analysis

Table 94: Confusion matrix CLCCH2006-2012 level 2 for zone 1 (Turkey) – BLIND ANALYSIS - In column the validation and line the production

Overall accuracy at the second level = 83.2%
Overall accuracy at the third level = 77.3%

[illegible]

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Table 96: Confusion matrix CLCCH2006-2012 for zone 2 (France) –BLIND ANALYSIS- In column the validation and line the production

Overall accuracy at the second level = 88.4%
Overall accuracy at the third level = 84.8%

[illegible]

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Table 98: Confusion matrix CLCCH2006-2012 for zone 3 (Spain) – BLIND ANALYSIS -In column the validation and line the production

Overall accuracy at the second level = 89.6%
Overall accuracy at the third level = 84.2%

[illegible]

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Table 100: Confusion matrix CLCCH2006-2012 for zone 4 (Sweden) – BLIND ANALYSIS- In column the validation and line the production

Overall accuracy at the second level = 82.2%
Overall accuracy at the third level = 81.4%

[illegible]

Overall accuracy at the second level = 90.0%
Overall accuracy at the third level = 89.5%

Table 102: Confusion matrix CLCCH2006-2012 for zone 5 (Germany) – BLIND ANALYSIS- In column the validation and line the production

[illegible]

Overall accuracy at the third level = 80.6%

[illegible]

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Table 104: Confusion matrix CLCCH2006-2012 for zone 6 (Finland) – BLIND ANALYSIS- In column the validation and line the production

Overall accuracy at the second level = 85.9%
Overall accuracy at the third level = 72.9%

[illegible]

Overall accuracy at the third level = 80.7%

Zone 7: Norway - Confusion matrix of CLCCH2006-2012 - Blind and plausibility analysis

Table 106: Confusion matrix CLCCH2006-2012 for zone 7 (Norway) – BLIND ANALYSIS- In column the validation and line the production

CLC Classes	11-11	12-12	13-11	13-12	13-13	14-13	14-14	21-11	21-12	21-13	21-21	23-12	23-23	24-11	24-12	24-13	24-24	31-11	31-12	31-13	31-14	31-21	31-31	31-32	32-11	32-12	32-13	32-14	32-31	32-32	33-13	33-33	41-11	41-13	41-41	42-13	42-42	51-51	52-52	Total	User Acc
11-11	5,504																																						5,504	1,000	
12-12	0,014	0,873													0,003				0,017	0,010																			0,918	0,951	
13-12				0,004																																			0,004	1,000	
13-13		0,011		0,004	0,209														0,005	0,020			0,004											0,011					0,264	0,791	
13-14							0,004																																0,004	0,000	
13-32				0,004																					0,004														0,004	0,000	
14-11																									0,004														0,004	0,000	
14-12						0,004																																	0,004	0,000	
14-14							1,523					0,009																			0,071								1,604	0,950	
21-11			0,002					0,002	0,002	0,002	0,002																												0,011	0,200	
21-12									0,002	0,002		0,002																												0,006	0,333
21-13										0,002																														0,002	1,000
21-21											15,442							1,716																						17,158	0,900
23-23													0,599				0,037														0,075		0,037						0,748	0,800	
24-11														0,002	0,002																								0,004	0,500	
24-12															0,002																								0,002	1,000	
24-13																0,002												0,002												0,004	0,500
24-14						0,002																																	0,002	0,000	
24-24																32,084															1,439								33,523	0,957	
31-11			0,002															0,010	0,002	0,007				0,005															0,026	0,400	
31-12																			0,013		0,007			0,013															0,027	0,500	
31-13																0,013			0,040					0,027															0,080	0,500	
31-14							0,003											0,010	0,003	0,051				0,025															0,092	0,550	
31-21																				0,004			0,008	0,004						0,004									0,019	0,200	
31-31																							285,899							25,345									311,244	0,919	
31-32										0,078													0,567	2,860				0,003		0,083										3,591	0,796
32-11																									0,013														0,013	1,000	
32-12															0,013												0,013												0,027	0,500	
32-14																							0,013				0,013		0,013		0,013								0,040	0,333	
32-21																							0,004															0,004	0,000		
32-31																							0,454						0,204	0,354									1,013	0,202	
32-32																							1,565	0,391							149,489		5,612						157,058	0,952	
33-13																																0,013	0,013						0,027	0,500	
33-14																																	0,013						0,013	0,000	
33-33																														4,871		297,478							302,349	0,984	
41-13																																	0,004	0,004					0,008	0,500	
41-14																												0,004						0,004					0,008	0,000	
41-21																																			0,008				0,008	0,000	
41-41																														2,543		1,271		58,491					62,305	0,939	
42-42										0,003																					0,003		0,003		0,003	0,003	0,052			0,069	0,750
51-51																																					41,108			41,108	1,000
52-52																	0,087																						1,215	1,302	0,933
Total	5,518	0,885	0,004	0,008	0,213	0,004	1,533	0,002	0,004	0,006	15,535	0,002	0,599	0,002	0,008	0,029	33,924	0,020	0,037	0,080	0,051	0,004	288,586	3,255	0,017	0,013	0,002	0,020	0,204	184,290	0,013	304,429	0,004	0,004	58,517	0,003	0,052	41,108	1,215	893,222	
Prod Acc	0,997	0,987	0,000	0,492	0,982	0,000	0,994	1,000	0,500	0,333	0,994	0,000	1,000	1,000	0,281	0,074	0,946	0,517	0,358	0,501	1,000	1,000	0,991	0,879	0,774	1,000	0,000	0,671	1,000	0,811	1,000	0,977	0,000	1,000	1,000	0,000	1,000	1,000	1,000	Global Acc	0,950

Overall accuracy at the second level = 95.0%

Overall accuracy at the third level = 93.1%

Table 107: Confusion matrix CLCCH2006-2012 for zone 7 (Norway) – PLAUSIBILITY ANALYSIS- In column the validation and line the production

CLC Classe	11-11	12-12	13-11	13-12	13-13	13-32	14-14	21-11	21-12	21-13	21-21	23-12	23-23	24-11	24-12	24-13	24-24	31-11	31-12	31-13	31-14	31-21	31-31	31-32	32-11	32-12	32-13	32-14	32-21	32-31	32-32	32-33	33-13	33-14	33-33	41-13	41-14	41-21	41-24	41-41	42-42	51-51	52-52	Total	User Acc						
11-11	5,504																																											5,504	1,000						
12-12	0,014	0,873															0,003		0,017	0,010																									0,918	0,951					
13-12					0,004																																								0,004	1,000					
13-13		0,011			0,004	0,213													0,005	0,016				0,004																	0,011				0,264	0,806					
13-14							0,004																																						0,004	0,000					
13-32					0,004																																								0,004	0,000					
14-11																									0,004																				0,004	0,000					
14-12		0,004																																											0,004	0,000					
14-14							1,523					0,009																																		1,604	0,950				
21-11			0,002					0,004	0,002	0,002																																			0,011	0,400					
21-12									0,002	0,002		0,002																																		0,006	0,333				
21-13										0,002																																			0,002	1,000					
21-21											16,300								0,858																											17,158	0,950				
23-23														0,673																																0,748	0,900				
24-11															0,002	0,002																														0,004	0,500				
24-12																0,002																													0,002	1,000					
24-13																	0,002												0,002																	0,004	0,500				
24-14							0,002																																						0,002	0,000					
24-24																	32,084																												33,523	0,957					
31-11			0,002															0,012	0,002	0,007			0,003																							0,026	0,467				
31-12						0,013													0,013																											0,027	0,500				
31-13																				0,080																										0,080	1,000				
31-14																		0,007	0,003	0,065			0,017																							0,092	0,708				
31-21																					0,003	0,065		0,017						0,004																0,019	0,400				
31-31																						0,008			0,008								13,050												311,244	0,958					
31-32																							298,19											0,005													3,591	0,910			
32-11																							0,318	3,268			0,013																				0,013	1,000			
32-12														0,013														0,013																				0,027	0,500		
32-14																													0,040																	0,040	1,000				
32-21																														0,004																0,004	1,000				
32-31																							0,250									0,655	0,108													1,013	0,646				
32-32																							0,783	0,391									155,88														157,058	0,993			
33-13																																		0,027													0,027	1,000			
33-14																																			0,013												0,013	1,000			
33-33																																			4,848	0,023											299,062	0,984			
41-13																																					0,004										0,008	0,500			
41-14																																						0,008									0,008	1,000			
41-21																																						0,004	0,004							0,008	0,500				
41-41																																															61,033	1,000			
42-42		0,003											0,003																																			0,069	0,800		
51-51																																																	41,108	1,000	
52-52																																																	1,215	1,302	0,933
Total	5,518	0,892	0,004	0,008	0,217	0,013	1,529	0,004	0,004	0,006	16,309	0,002	0,677	0,002	0,004	0,015	33,032	0,019	0,037	0,116	0,065	0,008	299,58	3,659	0,017	0,013	0,002	0,040	0,008	0,655	175,49	0,023	0,027	0,013	294,19	0,004	0,008	0,004	0,004	61,049	0,055	41,108	1,215	913,096							
Prod Acc	0,997	0,979	0,000	0,492	0,982	0,000	0,996	1,000	0,500	0,333	0,999	0,000	0,995	1,000	0,500	0,138	0,971	0,652	0,358	0,691	1,000	1,000	0,995	0,893	0,774	1,000	0,000	1,000	0,500	1,000	0,888	0,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	Global Acc	0,976					

Overall accuracy at the second level = 97.6%
Overall accuracy at the third level = 96.0%

[illegible]

Overall accuracy at the second level = 94.8%
Overall accuracy at the third level = 93.8%

Table 110: Confusion matrix CLCCH2006-2012 for zone 9 (Italy) – BLIND ANALYSIS- In column the validation and line the production

Overall accuracy at the second level = 80.2%
Overall accuracy at the third level = 75.2%

[illegible]

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Table 112: Confusion matrix CLCCH2006-2012 for zone 10 (United Kingdom, Ireland) – BLIND ANALYSIS - In column the validation and line the production

Overall accuracy at the second level = 87.1%
Overall accuracy at the third level = 85.6%

[illegible]

Overall accuracy at the third level = 91.7%

Zone 11: Romania - Confusion matrix of CLCCH2006-2012 - Blind and plausibility analysis

Table 114: Confusion matrix CLCCH2006-2012 for zone 11 (Romania) – BLIND ANALYSIS- In column the validation and line the production

CLC Classes	11-11	12-12	13-11	13-12	13-13	13-23	13-51	14-14	14-21	21-11	21-12	21-13	21-21	21-22	21-23	21-24	21-31	21-32	21-51	22-21	22-22	23-11	23-12	23-13	23-21	23-23	23-31	24-24	31-12	31-13	31-14	31-31	31-32	32-13	32-22	32-31	32-32	33-21	33-23	33-32	33-33	33-51	41-32	41-33	41-41	41-51	42-42	51-13	51-51	52-52	Total	User Acc							
11-11	23,810																											18,433																						42,243	0,564								
12-12	0,017	4,281			0,202							0,035	0,015										0,017																												4,568	0,937							
13-11			0,017																																															0,017	1,000								
13-12								0,008			0,008													0,008																												0,025	0,000						
13-13		0,118		0,011	0,893	0,108					0,011	0,011				0,011								0,011				0,032	0,011																						0,054			1,268	0,704				
13-14	0,008																																																			0,008	0,000						
13-51							0,008																																												0,008	1,000							
14-14	0,022							0,294																				0,033																								0,011			0,370	0,793			
21-11	0,042		0,021									0,042																																									0,106	0,000					
21-12		0,014		0,005						0,005	0,053	0,065			0,005								0,005				0,005																											0,155	0,343				
21-13												0,148																																											0,169	0,875			
21-21		0,033											286,666		6,639						6,639	13,278						19,917																											333,173	0,860			
21-22														0,011				0,042				0,011																																0,063	0,167				
21-23																									0,011																														0,011	0,000			
21-41												0,203																																											0,203	0,000			
22-13																							0,021																																0,021	0,000			
22-21												0,021													0,011																																0,032	0,000	
22-22									0,696			3,699								0,601	8,574					0,601	10,471											0,696																			25,339	0,338	
22-24																									0,011		0,011											0,011																	0,032	0,000			
23-13																																							0,021																0,021	0,000			
23-21												0,032												0,042																																0,074	0,571		
23-23		1,891										9,455										1,891				62,404		9,455										9,455																			94,552	0,660	
23-32														0,008																																										0,008	0,000		
23-41																									0,008																														0,008	0,000			
23-51																																																							0,008			0,008	0,000
24-12											0,021																																														0,021	0,000	
24-13																																																										0,021	0,000
24-24	1,509												1,509								2,966					6,607	45,942												7,335																		65,868	0,697	
31-13																															0,008																										0,008	1,000	
31-31																												3,758					0,971	241,341	6,261				9,381																		261,713	0,922	
31-32																									0,008																																	1,530	0,928
32-14					0,008																																																				0,008	0,000	
32-31																										0,009												0,009	0,110																				

Table 115: Confusion matrix CLCCH2006-2012 for zone 11 (Romania) – PLAUSIBILITY ANALYSIS- In column the validation and line the production

CLC Classe	11-11	12-12	13-11	13-12	13-13	13-23	13-32	13-51	14-13	14-14	21-11	21-12	21-13	21-21	21-22	21-32	22-21	22-22	23-12	23-13	23-21	23-23	23-41	24-13	24-24	31-13	31-31	31-32	32-31	32-32	32-33	32-51	33-32	33-33	41-32	41-41	41-51	51-13	51-41	51-51	52-52	Total	User Acc		
11-11	38,053																								4,189																	42,243	0,901		
12-12		4,515											0,035						0,017																						4,568	0,989			
13-11			0,017																																							0,017	1,000		
13-12									0,008			0,008								0,008																						0,025	0,000		
13-13			0,011	0,011	0,011	1,054	0,054	0,054					0,021							0,021					0,011	0,011	0,011															1,268	0,831		
13-14	0,008																																									0,008	0,000		
13-51								0,008																																		0,008	1,000		
14-14										0,359															0,011																		0,370	0,971	
21-11	0,021		0,021								0,063																															0,106	0,600		
21-12		0,005		0,005							0,005	0,067	0,065						0,009																								0,155	0,432	
21-13													0,169																													0,169	1,000		
21-21		0,011												319,883											13,278																	333,173	0,960		
21-22															0,011	0,042		0,011																								0,063	0,167		
21-23														0,203									0,011																				0,011	0,000	
21-41																																											0,203	0,000	
22-13																				0,021																						0,021	0,000		
22-21																	0,021					0,011																					0,032	0,667	
22-22													0,601			0,601	11,864					2,498			7,686			0,696	1,393														25,339	0,468	
22-24													0,011											0,011																			0,032	0,000	
23-13																																											0,021	0,000	
23-21													0,032								0,042																						0,074	0,571	
23-23													5,673							1,891			85,096							1,891													94,552	0,900	
23-32																		0,008																								0,008	0,000		
23-41																							0,008																			0,008	1,000		
23-51																																						0,008				0,008	0,000		
24-12																									0,021																	0,021	0,000		
24-13																			0,021																							0,021	0,000		
24-24																		0,728			0,728	5,879			56,348						2,185													65,868	0,855
31-13																										0,008																0,008	1,000		
31-31																											253,303	3,758		4,651														261,713	0,968
31-32																												0,030	1,500														1,530	0,980	
32-14										0,008																																0,008	0,000		
32-31																									0,009	0,054		0,037	0,082														0,182	0,201	
32-32																								0,532	0,532	3,061	0,532	0,266	27,832	1,131					1,131	1,131	1,131						37,282	0,747	
32-51																																											0,008	1,000	
33-33																																											0,800	0,859	
33-51								0,008																																		0,008	0,000		
41-41																						0,600																				12,009	0,900		
42-42																																											0,255	0,000	
51-23																																											0,008	0,000	
51-51																																											0,303	12,069	
52-52																																											2,649	2,649	
Total	38,083	4,542	0,049	0,015	1,054	0,054	0,054	0,017	0,008	0,368	0,068	0,075	0,291	326,402	0,011	0,042	0,622	12,611	0,027	1,963	0,771	94,627	0,008	0,011	82,096	0,019	256,449	5,791	0,999	38,154	1,131	0,008	0,025	2,165	1,131	12,497	0,609	0,008	0,303	12,069	2,649	826,483			
Prod Acc	0,999	0,994	0,347	0,000	1,000	0,000	0,000	0,500	0,000	0,977	0,933	0,888	0,582	0,980	1,000	0,000	0,034	0,941	0,000	0,000	0,055	0,899	1,000	0,000	0,686	0,442	0,988	0,259	0,037	0,729	0,000	1,000	0,000	0,317	0,000	0,865	0,000	0,000	0,000	1,000	1,000	Global Acc	0,920		

Overall accuracy at the second level = 92.0%
Overall accuracy at the third level = 90.9%

Table 116: Confusion matrix CLCCH2006-2012 for zone 12 (Greece) – BLIND ANALYSIS- In column the validation and line the production

Overall accuracy at the second level = 85.1%
Overall accuracy at the third level = 78.6%

Table 117: Confusion matrix CLCCH2006-2012 for zone 12 (Greece) – PLAUSIBILITY ANALYSIS- In column the validation and line the production

CLC Class	11-11	12-12	13-12	13-13	13-14	13-23	13-31	13-32	13-51	14-14	21-11	21-12	21-13	21-21	21-41	22-13	22-22	23-12	23-13	23-23	23-51	24-12	24-13	24-24	24-32	31-13	31-31	31-32	31-33	32-12	32-13	32-14	32-23	32-31	32-32	32-33	33-12	33-32	33-33	41-41	42-42	51-51	52-52	Total	User Acc					
11-11	22,212																																											22,212	1,000					
12-12	0,250	6,955	0,212	0,106																		0,106		0,106						0,106																7,840	0,887			
13-11	0,147																																														0,147	0,000		
13-12			0,569																																												0,569	1,000		
13-13			0,174	2,929													0,197							0,073							0,197																	3,668	0,799	
13-14					0,037																																											0,037	1,000	
13-23						0,029		0,029																																								0,057	0,500	
13-31							0,029																																									0,029	1,000	
13-32								0,171																																								0,171	1,000	
13-51									0,057																																							0,057	1,000	
14-14	0,110									1,569														0,110																									1,824	0,860
21-12												0,095																																				0,095	1,000	
21-13											0,048		0,190																																				0,238	0,800
21-21														160,892									3,311			13,245											3,311												180,940	0,889
21-24															0,023																																	0,023	0,000	
22-13																	0,143																															0,143	1,000	
22-22															0,442			57,508																															74,591	0,771
22-24																																																	0,023	0,000
23-12																			0,048					0,048																									0,095	0,500
23-13																				0,095																												0,095	1,000	
23-21																																																	0,023	0,000
23-23																				6,190						0,619																							7,738	0,800
23-32																																																	0,029	0,000
23-51									0,030														0,542																										0,572	0,947
24-12																								0,048																								0,048	1,000	
24-13	0,048																							0,190																								0,238	0,800	
24-24		2,263												2,439											145,691			5,723																					165,416	0,881
24-32																										0,029																						0,029	1,000	
31-13																											0,098																					0,098	1,000	
31-31																									5,405			195,718										2,773											203,896	0,960
31-32																												0,140	4,470									1,589									6,281	0,712		
31-33																												0,045	0,017	1,235							0,017	0,045										1,359	0,909	
32-12																																																	0,164	0,600
32-13																																																	0,131	1,000
32-14	0,033																																																0,066	0,500
32-21																																																	0,029	0,000

Overall accuracy at the second level = 90.6

Zone 13: Bulgaria- Confusion matrix of CLCCH2006-2012 - Blind and plausibility analysis

Table 118: Confusion matrix CLCCH2006-2012 for zone 13 (Bulgaria) – BLIND ANALYSIS In column the validation and line the production

CLC Classes	11-11	12-12	13-12	13-13	13-14	13-23	13-51	14-14	21-13	21-21	21-22	22-13	22-21	22-22	22-23	22-24	22-32	23-12	23-13	23-21	23-22	23-23	23-32	24-13	24-21	24-24	31-12	31-13	31-14	31-31	31-32	32-12	32-13	32-31	32-32	33-23	33-33	41-12	41-21	41-23	41-41	42-42	51-51	52-52	Total	User Acc				
11-11	22,453	1,493																									5,971																		29,916	0,751				
12-12	0,295	5,553								0,295																	0,295																		0,295		6,733	0,825		
13-13			0,030	1,488			0,403												0,134				0,134				0,134		0,134																		2,996	0,497		
13-14					0,053																									0,013																	0,066	0,800		
13-51							0,026																																							0,026	1,000			
14-14	0,100							1,095																			0,032	0,050		0,050																	1,328	0,824		
21-12			0,013						0,013	0,013				0,013				0,026				0,013																								0,093	0,000			
21-13									0,013	0,013																																					0,026	0,500		
21-21		5,930								274,295					5,930							5,930				5,930																					298,017	0,920		
21-22										0,081	0,194				0,293							0,008					0,016																					0,592	0,328	
22-12												0,013																																			0,013	0,000		
22-13																								0,013																							0,013	0,000		
22-21										0,513			0,192	0,064											0,064	0,064																						0,898	0,214	
22-22										1,472	0,192		0,192	10,048									0,192				1,558																						13,654	0,736
22-23										0,064							0,064																																0,128	0,000
22-24										0,064																																					0,064	0,000		
22-32										0,013				0,026				0,013																0,013														0,066	0,200	
23-12																																																0,013	1,000	
23-13				0,013																																											0,013	0,000		
23-21										0,194	0,013										0,039																											0,259	0,150	
23-23	0,607	0,607													0,607				0,607		0,607	22,475	1,215			1,822																							29,765	0,755
23-32																													0,013																			0,013	0,000	
24-12																																																0,013	0,000	
24-13				0,026															0,026																0,013													0,066	0,000	
24-14	0,013																																															0,013	0,000	
24-21																						0,064				0,064																						0,128	0,000	
24-22																										0,064																						0,064	0,000	
24-24		0,998						0,998		5,055				1,995								0,998			78,884																								97,042	0,813
24-32																																																0,013	0,000	
31-13																													0,035																			0,035	1,000	
31-31								0,999															0,833						253,096	4,540																		264,134	0,958	
31-32								0,014														0,046							0,419	1,050																		1,580	0,664	
31-51							0,013																																								0,013	0,000		
32-12																																																0,009	1,000	
32-13				0,009																																0,009												0,035	0,250	
32-14																																																0,009	0,000	
32-21										0,013				0,013																																		0,040	0,000	
32-31																													0,118	0,017																			0,336	0,250
32-32		1,132																																																

Overall accuracy at the second level = 89.0%

Overall accuracy at the third level = 85.5%

Table 119: Confusion matrix CLCCH2006-2012 for zone 13 (Bulgaria) – PLAUSIBILITY ANALYSIS In column the validation and line the production

CLC Classe	11-11	12-12	13-11	13-12	13-13	13-14	13-23	13-51	14-14	21-12	21-13	21-21	21-22	22-12	22-21	22-22	22-23	22-24	22-32	23-12	23-13	23-21	23-22	23-23	23-32	24-13	24-21	24-24	24-32	31-13	31-14	31-31	31-32	31-51	32-12	32-13	32-31	32-32	33-32	33-33	41-12	41-21	41-41	42-42	51-51	52-52	Total	User Acc					
11-11	29,916																																														29,916	1,000					
12-12	0,295	6,438																																															6,733	0,956			
13-13				0,030	1,488			0,269													0,134				0,537			0,269		0,134																				2,996	0,497		
13-14						0,066																																										0,066	1,000				
13-51								0,026																																								0,026	1,000				
14-14	0,133	0,050							1,095																						0,050																		1,328	0,824			
21-12				0,013						0,026		0,013									0,040																												0,093	0,286			
21-13											0,026																																					0,026	1,000				
21-21												292,087													5,930																								298,017	0,980			
21-22											0,038	0,358				0,187							0,008																										0,592	0,605			
22-12													0,013																																				0,013	1,000			
22-13																											0,013																							0,013	0,000		
22-21											0,192			0,449	0,064							0,064			0,064		0,064		0,064																						0,898	0,500	
22-22											0,981	0,192		0,192	10,923		0,192							0,491					0,683																						13,654	0,800	
22-23											0,064						0,064																																	0,128	0,500		
22-24														0,064																																				0,064	0,000		
22-32														0,013					0,040																															0,066	0,600		
23-12																				0,013																														0,013	1,000		
23-13																					0,013																													0,013	1,000		
23-21											0,104											0,104	0,013	0,039																											0,259	0,400	
23-23																								27,335	0,607																									29,765	0,918		
23-32																																		0,013																0,013	0,000		
24-12																																																				0,013	0,000
24-13					0,013																						0,053																							0,066	0,800		
24-14			0,013																																															0,013	0,000		
24-21																											0,064	0,064																						0,128	0,500		
24-22																								0,064																										0,064	0,000		
24-24									0,998			0,998												1,995				88,994																							97,042	0,917	
24-32																													0,013																						0,013	1,000	
31-13																															0,035																				0,035	1,000	
31-31																																	260,303	0,999															264,134	0,985			
31-32																																	0,290	1,220																1,580	0,772		
31-51																																		0,013																0,013	1,000		
32-12																																				0,009														0,009	1,000		
32-13					0,009																													0,009																	0,035	0,500	
32-14																																																				0,009	0,000
32-21											0,013																																										

Overall accuracy at the second level = 95.6%

Overall accuracy at the third level = 94.4%

Zone 14: Island- Confusion matrix of CLCCH2006-2012 - Blind and plausibility analysis

Table 120: Confusion matrix CLCCH2006-2012 for zone 14 (Island) – BLIND ANALYSIS- In column the validation and line the production

CLC Classes	11-11	12-12	13-11	13-12	13-13	13-14	13-33	13-51	14-14	21-21	23-21	23-23	24-24	31-31	32-13	32-21	32-23	32-32	32-33	32-51	33-33	33-51	33-52	41-13	41-41	42-42	51-33	51-51	52-12	52-33	52-52	Total	User Acc
11-11	0,565		0,063																													0,628	0,900
12-12		0,533		0,017														0,027			0,027											0,603	0,883
12-32																		0,018														0,018	0,000
12-33																					0,018											0,018	0,000
13-12				0,018																												0,018	1,000
13-13			0,037		0,072																											0,109	0,657
13-14					0,018																											0,018	1,000
13-33						0,018	0,018																									0,036	0,500
14-14								0,934										0,020														0,955	0,979
21-21									0,138																							0,138	1,000
23-14																		0,018														0,018	0,000
23-23										0,512	0,512	13,311	0,512																			14,847	0,897
23-32												0,018																				0,018	0,000
24-24												0,016	0,253					0,032														0,300	0,842
31-31														1,525				0,871														2,397	0,636
32-13															0,018																	0,018	1,000
32-14								0,018																								0,018	0,000
32-23												0,013				0,008	0,021	0,038														0,080	0,263
32-24												0,004																				0,004	0,000
32-32												0,095	0,095					193,038				32,365										225,593	0,856
32-51																				0,619												0,619	1,000
33-32																					0,218											0,218	0,000
33-33																					311,969						0,407					312,376	0,999
33-51																					0,083	0,920					0,028					1,030	0,893
41-13																								0,018								0,018	1,000
41-23											0,036															0,018						0,054	0,000
41-41																		4,269	0,117		0,117				32,015		0,117					36,636	0,874
41-51																					0,018											0,018	0,000
42-42																								0,032	0,515						0,104	0,650	0,792
51-33																					0,254						0,417	0,318				0,989	0,422
51-51																					0,223	0,395					11,359					11,977	0,948
52-12																												0,018				0,018	1,000
52-33																													0,018			0,018	1,000
52-52																						0,060					0,103			1,061	1,225	0,867	
Total	0,565	0,533	0,100	0,035	0,072	0,018	0,018	0,018	0,952	0,650	0,512	13,492	0,859	1,525	0,018	0,008	0,021	198,330	0,117	0,619	345,292	1,315	0,060	0,018	32,065	0,515	0,852	11,898	0,018	0,018	1,165	569,392	
Prod Acc	1,000	1,000	0,000	0,508	1,000	1,000	1,000	0,000	0,981	0,212	0,000	0,987	0,294	1,000	1,000	0,000	1,000	0,973	0,000	1,000	0,903	0,700	0,000	1,000	0,998	1,000	0,490	0,955	1,000	1,000	0,911		
																																Global Acc	0,931

Overall accuracy at the second level = 93.1%

Overall accuracy at the third level = 82.9%

Table 121: Confusion matrix CLCCH2006-2012 for zone 14 (Island) – PLAUSIBILITY ANALYSIS- In column the validation and line the production

CLC Classe	11-11	12-12	13-11	13-12	13-13	13-14	13-33	13-51	14-14	21-21	23-21	23-23	24-24	31-31	32-13	32-21	32-23	32-32	32-33	32-51	33-32	33-33	33-51	41-13	41-23	41-41	42-42	51-33	51-51	52-12	52-33	52-52	Total	User Acc
11-11	0,597		0,031																													0,628	0,950	
12-12		0,559		0,017																		0,027										0,603	0,927	
12-32																		0,018					0,018									0,018	0,000	
12-33																						0,018										0,018	0,000	
13-12				0,018																												0,018	1,000	
13-13			0,037		0,072																											0,109	0,657	
13-14					0,018																											0,018	1,000	
13-33						0,018	0,018																									0,036	0,500	
14-14								0,955																								0,955	1,000	
21-21										0,138																						0,138	1,000	
23-14																		0,018														0,018	0,000	
23-23										0,512	0,512	13,823																				14,847	0,931	
23-32												0,018																				0,018	0,000	
24-24													0,300																			0,300	1,000	
31-31														1,836				0,561														2,397	0,766	
32-13															0,018																	0,018	1,000	
32-14									0,018																							0,018	0,000	
32-23												0,021				0,004	0,055															0,080	0,684	
32-24																	0,004															0,004	0,000	
32-32												0,095	0,095					225,403														225,593	0,999	
32-51																			0,619													0,619	1,000	
33-32																				0,218												0,218	1,000	
33-33																					312,376											312,376	1,000	
33-51																					0,110	0,920										1,030	0,893	
41-13																							0,018									0,018	1,000	
41-23											0,036													0,018		0,018						0,054	0,333	
41-41																		0,235	0,117							36,167			0,117			36,636	0,987	
41-51																						0,018										0,018	0,000	
42-42																										0,547						0,104	0,650	0,841
51-33																						0,318						0,671				0,989	0,679	
51-51																						0,223							11,754			11,977	0,981	
52-12																													0,018			0,018	1,000	
52-33																														0,018		0,018	1,000	
52-52																															1,225	1,225	1,000	
Total	0,597	0,559	0,069	0,035	0,072	0,018	0,018	0,018	0,973	0,650	0,512	13,993	0,395	1,836	0,018	0,004	0,059	226,234	0,117	0,619	0,218	313,089	0,920	0,018	0,018	36,167	0,547	0,671	11,872	0,018	0,018	1,328	608,378	
Prod Acc	1,000	1,000	0,000	0,508	1,000	1,000	1,000	0,000	0,982	0,212	0,000	0,988	0,760	1,000	1,000	0,000	0,929	0,996	0,000	1,000	1,000	0,998	1,000	1,000	1,000	1,000	1,000	0,990	1,000	1,000	0,922			
																																	Global Acc	0,995

Overall accuracy at the second level = 99.5%

Overall accuracy at the third level = 93.3%

Table 122: Confusion matrix CLCCH2006-2012 for zone 15 (Hungary) – BLIND ANALYSIS- In column the validation and line the production

Overall accuracy at the second level = 84.7%
Overall accuracy at the third level = 84.0%

[illegible]

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Table 124: Confusion matrix CLCCH2006-2012 for zone 16 (Portugal) –BLIND ANALYSIS- In column the validation and line the production

Overall accuracy at the second level = 78.9%
Overall accuracy at the third level = 68.3%

[illegible]

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Zone 17: Austria / Swiss - Confusion matrix of CLCCH2006-2012 - Blind and plausibility analysis

Table 126: Confusion matrix CLCCH2006-2012 for zone 17 (Austria / Swiss) – BLIND ANALYSIS - In column the validation and line the production

CLC Classes	11-11	12-12	13-12	13-13	13-23	13-32	13-51	14-14	21-11	21-12	21-13	21-14	21-21	22-12	22-22	23-11	23-14	23-23	24-24	31-12	31-13	31-14	31-24	31-31	31-32	31-51	32-14	32-31	32-32	32-33	33-32	33-33	41-41	51-51	Total	User Acc
11-11	41,703																		2,259					2,259											46,221	0,902
12-12	0,235	4,413	0,048										0,049																						4,745	0,930
13-12		0,006	0,037																																0,043	0,857
13-13	0,044	0,011	0,055	0,670	0,122								0,044						0,022																0,969	0,691
13-14								0,006																											0,006	0,000
13-21													0,006																						0,006	0,000
13-23						0,006																													0,006	0,000
13-32																													0,006						0,006	0,000
13-51							0,012																												0,012	1,000
14-14	0,046							1,871											0,023					0,398					0,118			0,235		2,691	0,695	
21-11	0,049								0,016																										0,065	0,250
21-12		0,011	0,020							0,069	0,020		0,015							0,004															0,138	0,500
21-13			0,016								0,049		0,016																						0,081	0,600
21-14												0,016					0,016																		0,032	0,500
21-21													98,495		2,985			23,878	23,878																149,235	0,660
22-22	0,042													0,296	5,631				0,381															6,350	0,887	
23-11																0,016																			0,016	1,000
23-23																	3,937		35,429	23,620					7,873				3,937						74,796	0,474
23-33																		0,006																	0,006	0,000
24-11																			0,016																0,016	0,000
24-13										0,016			0,016																						0,032	0,000
24-24													5,120					1,074	42,444					1,074											49,712	0,854
31-11																							0,008												0,008	0,000
31-12		0,008																																	0,008	0,000
31-13																					0,033					0,016	0,008								0,033	1,000
31-14																						0,016		0,016	0,008										0,041	0,400
31-23																									0,018				0,006						0,025	0,000
31-31				4,227															1,702					307,816	19,131			21,987							354,864	0,867
31-32																				0,007	0,137			1,130	1,737				0,329					3,339	0,520	
31-33																													0,137						0,137	0,000
31-51																										0,006									0,006	1,000
32-14																		0,008									0,008		0,033						0,049	0,167
32-31																																				

Overall accuracy at the second level = 80.5%

Overall accuracy at the third level = 75.3%

Table 127: Confusion matrix CLCCH2006-2012 for zone 17 (Austria / Swiss) – PLAUSIBILITY ANALYSIS - In column the validation and line the production

CLC Classe	11-11	12-12	13-12	13-13	13-21	13-23	13-32	13-51	14-14	21-11	21-12	21-13	21-14	21-21	22-22	23-11	23-14	23-23	24-13	24-21	24-24	31-11	31-13	31-14	31-23	31-31	31-32	31-51	32-14	32-23	32-31	32-32	33-14	33-33	41-41	51-51	Total	User Acc	
11-11	41,703																				2,259					2,259											46,221	0,902	
12-12	0,235	4,486	0,024																																		4,745	0,945	
13-12		0,006	0,037																																		0,043	0,857	
13-13		0,011	0,055	0,749		0,044						0,044							0,011		0,011					0,044											0,969	0,772	
13-14									0,006																												0,006	0,000	
13-21					0,006																																0,006	1,000	
13-23						0,006																															0,006	0,000	
13-32																																	0,006				0,006	0,000	
13-51								0,012																													0,012	1,000	
14-14									1,917												0,023					0,281							0,235	0,118	0,118			2,691	0,712
21-11	0,049									0,016																											0,065	0,250	
21-12		0,011	0,020								0,089	0,004		0,015																							0,138	0,644	
21-13												0,081																									0,081	1,000	
21-14													0,016				0,016																				0,032	0,500	
21-21														104,464	2,985			20,893			20,893																149,235	0,700	
22-22	0,042															6,223					0,085																6,350	0,980	
23-11																	0,016																				0,016	1,000	
23-23																		62,986			11,810																74,796	0,842	
23-33																		0,006																			0,006	0,000	
24-11																					0,016																0,016	0,000	
24-13				0,016																0,016																	0,032	0,000	
24-24													2,972					1,074			44,593					1,074											49,712	0,897	
31-11																						0,008															0,008	1,000	
31-12		0,008																																			0,008	0,000	
31-13																							0,033														0,033	1,000	
31-14																								0,016		0,016	0,008										0,041	0,400	
31-23																									0,012		0,006					0,006					0,025	0,500	
31-31																					5,929					331,175	4,227					13,533					354,864	0,933	
31-32																										0,700	2,639										3,339	0,790	
31-33																											0,137										0,137	0,000	
31-51																												0,006									0,006	1,000	
32-14																										0,008			0,016	0,008		0,016					0,049	0,333	
32-31																										0,275	0,007			0,757	0,028						1,066	0,710	
32-32																										1,765	0,650					99,502					101,917	0,976	
32-33																													0,137								0,137	0,000	
33-33																																	6,181	81,585			87,766	0,930	
41-41																	0,113									0,113						0,088		1,573			1,887	0,833	
41-51																																		0,006			0,006	0,000	
51-51																																			15,726	15,726	1,000		
Total	42,029	4,523	0,136	0,765	0,006	0,044	0,006	0,012	1,923	0,016	0,089	0,129	0,016	107,450	9,208	0,016	0,016	85,072	0,011	0,016	85,618	0,008	0,033	0,016	0,012	337,710	7,675	0,006	0,154	0,008	0,757	119,595	0,118	81,702	1,579	15,726	800,444		
Prod Acc	0,992	0,992	0,271	0,979	1,000	0,000	0,000	1,000	0,997	1,000	1,000	0,630	1,000	0,972	0,676	1,000	0,000	0,740	0,000	0,000	0,521	1,000	1,000	1,000	1,000	0,981	0,344	1,000	0,107	0,000	1,000	0,832	0,000	0,999	0,996	1,000			
																																						Global Acc	0,887

Overall accuracy at the second level = 88.7%

Overall accuracy at the third level = 86.4%

Table 128: Confusion matrix CLCCH2006-2012 for zone 18 (Denmark/Netherland/Belgium/Luxembourg) – BLIND ANALYSIS- In column the validation and line the production

Overall accuracy at the second level = 86.8%
Overall accuracy at the third level = 84.4%

[illegible]

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Table 130: Confusion matrix CLCCH2006-2012 for zone 19 (Albania/Serbia/Macedonia) – BLIND ANALYSIS - in column the validation and line the production

Overall accuracy at the second level = 84.2%
Overall accuracy at the third level = 80.9%

Table 131: Confusion matrix CLCCH2006-2012 for zone 19 (Albania / Serbia / Macedonia) – PLAUSIBILITY ANALYSIS - in column the validation and line the production

[illegible]

Overall accuracy at the second level = 94.2%

Overall accuracy at the third level = 92.6%

Table 132: Confusion matrix CLCCH2006-2012 for zone 20 (Bosnia and Herzegovina / Croatia) –BLIND ANALYSIS- In column the validation and line the production

Overall accuracy at the second level = 84.8%
Overall accuracy at the third level = 76.9%

[illegible]

Overall accuracy at the second level = 90.8%
Overall accuracy at the third level = 88.9%

Table 134: Confusion matrix CLCCH2006-2012 for zone 21 (Czech Republic / Slovakia) – BLIND ANALYSIS - In column the validation and line the production

Overall accuracy at the second level = 84.2%
Overall accuracy at the third level = 84.1%

[illegible]

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Table 136: Confusion matrix CLCCH2006-2012 for zone 22 (Estonia / Latvia / Lithuania) – BLIND ANALYSIS- In column the validation and line the production

Overall accuracy at the second level = 87.8%
Overall accuracy at the third level = 87.0%

Table 137: Confusion matrix CLCCH2006-2012 for zone 22 (Estonia / Latvia / Lithuania) – PLAUSIBILITY ANALYSIS- In column the validation and line the production

[illegible]

Overall accuracy at the second level = 90.7%

Overall accuracy at the third level = 90.2%

Table 139: Confusion matrix CLCCH2006-2012 for zone 23 (Guadeloupe /Guyana / La reunion/ Martinique/ Mayotte) – PLAUSIBILITY ANALYSIS- In column the validation and line the production

CLC Class	11-11	12-12	13-11	13-12	13-13	14-14	21-11	21-13	21-21	21-52	22-22	23-23	23-32	24-11	24-24	24-32	31-12	31-13	31-23	31-24	31-31	31-32	31-52	32-13	32-21	32-24	32-31	32-32	33-33	41-41	42-42	51-51	52-31	52-52	Total	User Acc	
11-11	8,613																																		8,613	1,000	
12-12		1,039																																	1,039	1,000	
13-12	0,023		0,046	0,023																															0,091	0,250	
13-13					0,138																														0,138	1,000	
14-14						0,169																													0,169	1,000	
21-11							0,034																												0,034	1,000	
21-13								0,017																											0,017	1,000	
21-21									9,366						0,444																				9,810	0,955	
21-23												0,023																							0,023	0,000	
21-52										0,069																									0,069	1,000	
22-22											1,674				0,717																				2,391	0,700	
22-23												0,023																							0,023	0,000	
23-23												2,969			0,371							0,371													3,711	0,800	
23-32													0,023																						0,023	1,000	
24-11							0,017							0,034	0,034																				0,085	0,400	
24-12																0,017																			0,017	0,000	
24-23															0,023																				0,023	0,000	
24-24	1,135														15,083													0,876							17,094	0,882	
31-11	0,023																																		0,023	0,000	
31-12																	0,023																		0,023	1,000	
31-13																		0,023																	0,023	1,000	
31-23																						0,012													0,012	0,000	
31-24																				0,069	0,012					0,012									0,092	0,750	
31-31																					113,374														113,374	1,000	
31-32																					0,015	0,045	0,030	0,375											0,465	0,807	
31-41																						0,023													0,023	0,000	
31-52																							0,367												0,367	1,000	
32-11																	0,023																		0,023	0,000	
32-21																									0,035										0,035	1,000	
32-24															0,012					0,023				0,012		0,035		0,012							0,092	0,375	
32-31																					0,029					0,073									0,102	0,714	
32-32															0,555					0,555	1,666						17,443								20,219	0,863	
33-33																					0,157						0,315	2,184							2,656	0,822	
41-41																														15,641						15,641	1,000
42-42																														0,106	0,010					0,116	0,088
51-51																													0,153			1,437				1,589	0,904
52-31																							0,022										0,649	0,022	0,694	0,936	
52-52																																		212,727	212,727	1,000	
Total	9,794	1,039	0,046	0,023	0,138	0,169	0,051	0,017	9,366	0,069	1,696	2,991	0,023	0,034	17,240	0,017	0,023	0,046	0,015	0,693	115,290	0,758	0,390	0,012	0,035	0,046	0,073	18,645	2,336	15,747	0,010	1,437	0,649	212,749	403,714		
Prod Acc	0,879	1,000	0,000	1,000	1,000	1,000	0,667	1,000	1,000	1,000	0,987	0,992	1,000	1,000	0,875	0,000	1,000	0,500	0,000	0,100	0,983	0,495	0,943	0,000	1,000	0,750	1,000	0,936	0,935	0,993	1,000	1,000	1,000			Global Acc 0,981	

Overall accuracy at the second level = 98.1%

Overall accuracy at the third level = 97.6%

Bio-geographical region ALP - Alpine - Confusion matrix of CLCCH2006-2012 - Blind and plausibility analysis

Table 140: Confusion matrix CLCCH2006-2012 for Alpine region –BLIND ANALYSIS- In column the validation and line the production

Overall accuracy at the second level = 88.6%
Overall accuracy at the third level = 83.5%

[illegible]

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Table 142: Confusion matrix CLCCH2006-2012 for Anatolian region– BLIND ANALYSIS- In column the validation and line the production

Overall accuracy at the second level = 79.3%
Overall accuracy at the third level = 75.6%

[illegible]

Overall accuracy at the third level = 85.9%

Bio-geographical region ARC - Arctic - Confusion matrix of CLCCH2006-2012 - Blind and plausibility analysis

Table 144: Confusion matrix CLCCH2006-2012 for Arctic region –BLIND ANALYSIS- In column the validation and line the production

CLC Classes	11-11	12-12	13-11	13-12	13-13	13-14	13-33	13-51	14-14	21-21	23-21	23-23	24-24	31-31	32-13	32-21	32-23	32-32	32-33	32-51	33-33	33-51	33-52	41-13	41-41	42-42	51-33	51-51	52-12	52-33	52-52	Total	User Acc	
11-11	0,388		0,043																													0,431	0,900	
12-12		0,366		0,012														0,018			0,018											0,414	0,883	
12-32																		0,012														0,012	0,000	
12-33																					0,012											0,012	0,000	
13-12				0,012																												0,012	1,000	
13-13			0,026		0,049																											0,075	0,657	
13-14					0,012																											0,012	1,000	
13-33						0,012	0,012																									0,025	0,500	
14-14									0,642									0,014														0,656	0,979	
21-21										0,095																						0,095	1,000	
23-14																		0,012														0,012	0,000	
23-23										0,352	0,352	9,145	0,352																			10,200	0,897	
23-32												0,012																				0,012	0,000	
24-24												0,011	0,174					0,022														0,206	0,842	
31-31														1,048				0,599														1,647	0,636	
32-13															0,012																	0,012	1,000	
32-14									0,012																							0,012	0,000	
32-23												0,009				0,006	0,015	0,026														0,055	0,263	
32-24												0,003																				0,003	0,000	
32-32												0,065	0,065					132,621				22,235										154,986	0,856	
32-51																				0,425												0,425	1,000	
33-32																						0,150										0,150	0,000	
33-33																					218,940							0,280				219,219	0,999	
33-51																					0,057	0,632						0,019				0,708	0,893	
41-13																								0,012								0,012	1,000	
41-23											0,025															0,012						0,037	0,000	
41-41																		2,933	0,081		0,081				21,995		0,081					25,170	0,874	
41-51																				0,012												0,012	0,000	
42-42																									0,022	0,359						0,071	0,452	0,794
51-33																					0,175							0,287	0,218			0,680	0,422	
51-51																				0,153	0,271							7,804			8,229	0,948		
52-12																													0,012			0,012	1,000	
52-33																														0,012		0,012	1,000	
52-52																							0,041				0,071				0,729	0,842	0,867	
Total	0,388	0,366	0,069	0,024	0,049	0,012	0,012	0,012	0,654	0,446	0,352	9,269	0,590	1,048	0,012	0,006	0,015	136,257	0,081	0,425	241,833	0,903	0,041	0,012	22,029	0,359	0,585	8,174	0,012	0,012	0,800	395,798		
Prod Acc	1,000	1,000	0,000	0,508	1,000	1,000	1,000	0,000	0,981	0,212	0,000	0,987	0,294	1,000	1,000	0,000	1,000	0,973	0,000	1,000	0,905	0,700	0,000	1,000	0,998	1,000	0,490	0,955	1,000	1,000	0,911			
	Global Acc																											0,932						

Overall accuracy at the second level = 93.2%

Overall accuracy at the third level = 83.1%

Table 145: Confusion matrix CLCCH2006-2012 for Arctic region –PLAUSIBILITY ANALYSIS- In column the validation and line the production

CLC Class	11-11	12-12	13-11	13-12	13-13	13-14	13-33	13-51	14-14	21-21	23-21	23-23	24-24	31-31	32-13	32-21	32-23	32-32	32-33	32-51	33-32	33-33	33-51	41-13	41-23	41-41	42-42	51-33	51-51	52-12	52-33	52-52	Total	User Acc
11-11	0,410		0,022																													0,431	0,950	
12-12		0,384		0,012																		0,018										0,414	0,927	
12-32																		0,012					0,018									0,012	0,000	
12-33																						0,012										0,012	0,000	
13-12				0,012																												0,012	1,000	
13-13			0,026		0,049																											0,075	0,657	
13-14					0,012																											0,012	1,000	
13-33						0,012	0,012																									0,025	0,500	
14-14									0,656																							0,656	1,000	
21-21										0,095																						0,095	1,000	
23-14																		0,012														0,012	0,000	
23-23										0,352	0,352	9,497																				10,200	0,931	
23-32												0,012																				0,012	0,000	
24-24													0,206																			0,206	1,000	
31-31														1,261					0,385													1,647	0,766	
32-13															0,012																	0,012	1,000	
32-14									0,012																							0,012	0,000	
32-23												0,015				0,003	0,038															0,055	0,684	
32-24																	0,003															0,003	0,000	
32-32												0,065	0,065					154,856														154,986	0,999	
32-51																				0,425												0,425	1,000	
33-32																					0,150											0,150	1,000	
33-33																						219,219										219,219	1,000	
33-51																						0,076	0,632									0,708	0,893	
41-13																							0,012									0,012	1,000	
41-23											0,025													0,012		0,012						0,037	0,333	
41-41																		0,161	0,081							24,847			0,081			25,170	0,987	
41-51																						0,012										0,012	0,000	
42-42																											0,380					0,071	0,452	0,842
51-33																							0,218					0,461				0,680	0,679	
51-51																							0,153						8,076			8,229	0,981	
52-12																													0,012			0,012	1,000	
52-33																															0,012	0,012	1,000	
52-52																															0,842	0,842	1,000	
Total	0,410	0,384	0,047	0,024	0,049	0,012	0,012	0,012	0,668	0,446	0,352	9,613	0,271	1,261	0,012	0,003	0,041	155,427	0,081	0,425	0,150	219,709	0,632	0,012	0,012	24,847	0,380	0,461	8,156	0,012	0,012	0,913	422,582	
Prod Acc	1,000	1,000	0,000	0,508	1,000	1,000	1,000	0,000	0,982	0,212	0,000	0,988	0,760	1,000	1,000	0,000	0,929	0,996	0,000	1,000	1,000	0,998	1,000	1,000	1,000	1,000	1,000	0,990	1,000	1,000	0,922			
																																	Global Acc	0,995

Overall accuracy at the second level = 99.5%

Overall accuracy at the third level = 93.4%

Table 146: Confusion matrix CLCCH2006-2012 for Atlantic region –BLIND ANALYSIS- In column the validation and line the production

Overall accuracy at the second level = 88.0%
Overall accuracy at the third level = 85.9%

[illegible]

Overall accuracy at the third level = 89.8%

Bio-geographical region BLS – Black sea - Confusion matrix of CLCCH2006-2012 - Blind and plausibility analysis

Table 148: Confusion matrix CLCCH2006-2012 for black sea region –BLIND ANALYSIS - In column the validation and line the production

CLC Classes	11-11	12-12	13-12	13-13	13-14	13-23	13-51	14-14	21-13	21-21	21-22	21-51	22-11	22-21	22-22	22-33	23-13	23-21	23-23	23-51	24-13	24-24	24-51	31-12	31-13	31-14	31-31	31-32	32-13	32-31	32-32	32-51	33-33	41-41	42-42	51-13	51-51	52-52	Total	User Acc		
11-11	2,546																								0,266														2,546	1,000		
12-12		1,705	0,366	0,221		0,025																																	2,583	0,660		
13-12		0,059	0,118					0,009									0,009																						0,196	0,603		
13-13		0,012		0,147																						0,220					0,073								0,452	0,326		
13-14					0,021																					0,007													0,029	0,750		
13-51							0,018																																0,018	1,000		
14-14	0,027							0,237																		0,027								0,012					0,304	0,781		
21-11									0,023																														0,023	0,000		
21-12									0,005																														0,005	0,000		
21-13									0,038																														0,038	1,000		
21-21										297,231										9,913		53,839																	360,983	0,823		
21-22											0,009				0,012																								0,021	0,434		
21-51										0,052		0,042																											0,093	0,448		
22-11													0,070																										0,070	1,000		
22-13																0,038																							0,038	0,000		
22-21										0,035													0,035																0,070	0,000		
22-22													0,658	6,034								3,678																	10,369	0,582		
22-23										0,035																													0,035	0,000		
23-21																		0,007																					0,007	1,000		
23-23																			4,635			0,330																	4,965	0,934		
24-13																					0,025		0,025																0,050	0,500		
24-14	0,007																																						0,007	0,000		
24-24											3,011											37,229																	40,240	0,925		
24-31																														0,022										0,022	0,000	
31-12																								0,025						0,022										0,025	1,000	
31-13																									0,055														0,055	1,000		
31-31																									1,428															114,705	0,988	
31-32																										0,040	0,348													0,389	0,896	
32-13																											0,040	0,348												0,055	1,000	
32-31																											0,075				0,861	0,075								1,010	0,852	
32-32																																57,877								57,877	1,000	
32-51																	0,063																0,063							0,126	0,500	
33-32																																0,029								0,029	0,000	
33-33								0,082																								0,029								20,078	0,512	
33-51																																9,679	10,280	0,028			0,014			0,014	0,000	
41-41																																			4,058	0,657				4,715	0,861	
42-42																																		0,279	0,033					0,312	0,106	
51-13																									0,011										0,021					0,032	0,667	
51-51																																	0,245							2,238	2,483	0,901
52-52	0,185																																							3,236	3,421	0,946
Total	2,765	1,776	0,484	0,368	0,021	0,025	0,018	0,329	0,066	297,352	0,009	0,042	0,070	0,658	9,057	0,038	0,009	0,007	14,549	0,063	0,025	95,110	0,025	0,292	1,714	0,034	113,392	0,348	0,055	0,883	67,977	0,063	10,292	4,365	0,690	0,021	2,252	3,245	542,541			
Prod Acc	0,921	0,960	0,244	0,400	1,000	0,000	1,000	0,723	0,572	1,000	1,000	1,000	1,000	0,000	0,666	0,000	0,000	1,000	0,319	0,000	1,000	0,391	0,000	0,086	0,032	0,000	0,999	1,000	1,000	0,976	0,851	1,000	0,999	0,930	0,048	1,000	0,994	0,997		Global Acc	0,863	

Overall accuracy at the second level = 86.3%

Overall accuracy at the third level = 81.5%

Table 149: Confusion matrix CLCCH2006-2012 for black sea region –PLAUSIBILITY ANALYSIS - In column the validation and line the production

CLC Clas	11-11	12-12	13-11	13-12	13-13	13-14	13-51	14-13	14-14	21-11	21-13	21-21	21-22	21-51	22-11	22-13	22-21	22-22	23-13	23-21	23-23	23-51	24-13	24-24	24-51	31-12	31-13	31-14	31-31	31-32	32-13	32-31	32-32	32-51	33-33	41-41	42-42	51-13	51-51	52-52	Total	User Acc		
11-11	2,546																																								2,546	1,000		
12-12		1,951		0,366																		0,266																			2,583	0,755		
13-12			0,059	0,118				0,009											0,009								0,220						0,073								0,196	0,603		
13-13			0,012		0,147																								0,220					0,073							0,452	0,326		
13-14						0,029																																			0,029	1,000		
13-51							0,018																																		0,018	1,000		
14-14	0,027								0,249																				0,027													0,304	0,821	
21-11										0,023																																0,023	1,000	
21-12											0,005																															0,005	0,000	
21-13											0,038																															0,038	1,000	
21-21												303,9										9,913			####																		360,983	0,842
21-22													0,009					0,012																								0,021	0,434	
21-51														0,052			0,042																									0,093	0,448	
22-11															0,070																											0,070	1,000	
22-13																0,038																										0,038	1,000	
22-21																	0,035					0,035																				0,070	0,500	
22-22																	0,658	6,034						3,678																		10,369	0,582	
22-23												0,035																														0,035	0,000	
23-21																				0,007																						0,007	1,000	
23-23																					4,965																					4,965	1,000	
24-13																							0,025			0,025																0,050	0,500	
24-14			0,007																																							0,007	0,000	
24-24																		3,011						####																			40,240	0,925
24-31																																										0,022	0,000	
31-12																										0,025																0,025	1,000	
31-13																											0,055															0,055	1,000	
31-31																												114,7														114,705	1,000	
31-32																													0,389													0,389	1,000	
32-13																																0,055										0,055	1,000	
32-31																																	0,936	0,075							1,010	0,926		
32-32																																		57,88							57,877	1,000		
32-51																			0,063																0,063							0,126	0,500	
33-32																																		0,029								0,029	0,000	
33-33								0,082																										9,679	10,31				0,009	20,078	0,513			
33-51																							0,014																			0,014	0,000	
41-41																																										4,715	1,000	
42-42																																			0,279	0,033						0,312	0,106	
51-13																																										0,032	1,000	
51-51																																	0,245								2,238	2,483	0,901	
52-52																																										3,421	3,421	1,000
Total	2,573	2,022	0,007	0,484	0,147	0,029	0,018	0,009	0,331	0,023	0,043	304,0	0,009	0,042	0,070	0,038	0,692	9,057	0,009	0,007	14,9	0,063	0,025	88,1	0,039	0,292	0,275	0,027	114,7	0,389	0,055	0,957	67,98	0,063	10,31	4,994	0,033	0,032	2,238	3,430	552,349			
Prod Acc	0,989	0,965	0,000	0,244	1,000	1,000	1,000	0,000	0,753	1,000	0,883	1,000	1,000	1,000	1,000	1,000	0,050	0,666	0,000	1,000	0,333	0,000	1,000	0,423	0,000	0,086	0,200	0,000	1,000	1,000	1,000	0,977	0,851	1,000	1,000	0,944	1,000	1,000	1,000	0,997				
																																											Global Acc	0,879

Overall accuracy at the second level = 87.9%

Overall accuracy at the third level = 84.3%

Table 150: Confusion matrix CLCCH2006-2012 for Boreal region – BLIND ANALYSIS- In column the validation and line the production

Overall accuracy at the second level = 84.8%
Overall accuracy at the third level = 80.2%

[illegible]

Overall accuracy at the second level = 90.1%
Overall accuracy at the third level = 86.7%

Table 152: Confusion matrix CLCCH2006-2012 for Continental region – BLIND ANALYSIS- In column the validation and line the production

Overall accuracy at the second level = 85.3%
Overall accuracy at the third level = 83.0%

[illegible]

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Bio-geographical region MAC– Macaronesia- Confusion matrix of CLCCH2006-2012 - Blind and plausibility analysis

Table 154: Confusion matrix CLCCH2006-2012 for Macaronesia region – IBLIND ANALYSIS- In column the validation and line the production

CLC Classes	11-11	12-12	13-12	13-13	13-14	14-14	21-21	23-23	24-11	24-12	24-24	31-31	32-14	32-31	32-32	32-33	33-33	41-41	52-52	Total	User Acc
11-11	1,045														1,018					2,063	0,507
12-12		0,164																		0,164	1,000
13-12			0,025																	0,025	1,000
13-13			0,013	0,419																0,432	0,969
13-14					0,012															0,012	1,000
14-14						0,102														0,102	1,000
21-11									0,008											0,008	0,000
21-21							2,563													2,563	1,000
23-23								2,309												2,309	1,000
24-12										0,008										0,008	1,000
24-24											6,640									6,640	1,000
31-31												8,332								8,332	1,000
32-14													0,015							0,015	1,000
32-32								0,256						0,256	12,189					12,701	0,960
32-33																0,052				0,052	1,000
33-32															0,025					0,025	0,000
33-33															6,981	0,164	9,968			17,114	0,582
41-41																		0,225		0,225	1,000
52-52																			0,052	0,052	1,000
Total	1,045	0,164	0,038	0,419	0,012	0,102	2,563	2,565	0,008	0,008	6,640	8,332	0,015	0,256	20,214	0,217	9,968	0,225	0,052	44,121	
Prod Acc	1,000	1,000	0,651	1,000	1,000	1,000	1,000	0,900	0,000	1,000	1,000	1,000	1,000	0,000	0,603	0,242	1,000	1,000	1,000		
																				Global Acc	0,835

Overall accuracy at the second level = 83.5%

Overall accuracy at the third level = 79.9%

Table 155: Confusion matrix CLCCH2006-2012 for Macaronesia region – PLAUSIBILITY ANALYSIS- In column the validation and line the production

CLC Class	11-11	12-12	13-12	13-13	13-14	14-14	21-21	23-23	24-11	24-12	24-24	31-31	32-14	32-32	32-33	33-33	41-41	52-52	Total	User Acc
11-11	1,045													1,018					2,063	0,507
12-12		0,164																	0,164	1,000
13-12			0,025																0,025	1,000
13-13			0,013	0,419															0,432	0,969
13-14					0,012														0,012	1,000
14-14						0,102													0,102	1,000
21-11									0,008										0,008	0,000
21-21							2,563												2,563	1,000
23-23								2,309											2,309	1,000
24-12										0,008									0,008	1,000
24-24											6,640								6,640	1,000
31-31												8,332							8,332	1,000
32-14													0,015						0,015	1,000
32-32														12,701					12,701	1,000
32-33															0,052				0,052	1,000
33-32														0,025					0,025	0,000
33-33														2,349	0,164	14,601			17,114	0,853
41-41																	0,225		0,225	1,000
52-52																		0,052	0,052	1,000
Total	1,045	0,164	0,038	0,419	0,012	0,102	2,563	2,309	0,008	0,008	6,640	8,332	0,015	16,093	0,217	14,601	0,225	0,052	49,265	
Prod Acc	1,000	1,000	0,651	1,000	1,000	1,000	1,000	1,000	0,000	1,000	1,000	1,000	1,000	0,789	0,242	1,000	1,000	1,000		
																			Global Acc	0,932

Overall accuracy at the second level = 93.2%

Overall accuracy at the third level = 79.9%

Table 156: Confusion matrix CLCCH2006-2012 for Mediterranean region – BLIND ANALYSIS- In column the validation and line the production

Overall accuracy at the second level = 85.8%
Overall accuracy at the third level = 79.1%

[illegible]

Overall accuracy at the third level = 85.5%

Table 158: Confusion matrix CLCCH2006-2012 for Pannonian region – BLIND ANALYSIS- In column the validation and line the production

Overall accuracy at the second level = 85.8%
Overall accuracy at the third level = 85.3%

[illegible]

Overall accuracy at the third level = 92.3%

Bio-geographical region STE – Steppic - Confusion matrix of CLCCH2006-2012 - Blind and plausibility analysis

Table 160: Confusion matrix CLCCH2006-2012 for Steppic region – BLIND ANALYSIS- In column the validation and line the production

CLC Classes	11-11	12-12	13-13	13-51	14-14	21-13	21-21	21-51	22-22	23-11	23-23	24-24	31-31	31-32	32-22	32-32	33-33	33-51	41-32	41-41	41-51	51-51	Total	User Acc
11-11	3,669											1,834											5,503	0,667
12-12	0,019	1,477				0,038	0,016																1,551	0,952
13-13			0,015																				0,015	1,000
14-14					0,024																		0,024	1,000
21-12						0,023																	0,023	0,000
21-21							109,801																109,801	1,000
21-41							0,222																0,222	0,000
22-21							0,023																0,023	0,000
22-22							1,973		0,658			1,315											3,945	0,167
23-21							0,012																0,012	0,000
23-23		2,070					2,070			2,070	6,211												12,421	0,500
23-41											0,009												0,009	0,000
23-51																				0,009			0,009	0,000
24-24	1,652						1,652					3,247											6,551	0,496
31-31													8,229			4,115							12,344	0,667
31-32														0,073									0,073	1,000
32-32											0,291	0,291			0,291	0,291			0,291				1,457	0,200
32-51																						0,009	0,009	0,000
33-33																	0,111	0,028					0,139	0,800
33-51				0,009																			0,009	0,000
41-41								0,657												4,601			5,259	0,875
51-51																				0,332		4,192	4,524	0,927
Total	5,340	3,548	0,015	0,009	0,024	0,061	115,769	0,657	0,658	2,070	6,511	6,396	8,520	0,073	0,291	4,406	0,111	0,028	0,291	4,933	0,009	4,201	142,600	
Prod Acc	0,687	0,416	1,000	0,000	1,000	0,000	0,948	0,000	1,000	0,000	0,954	0,508	0,966	1,000	0,000	0,066	1,000	0,000	0,000	0,933	0,000	0,998		
																							Global Acc	0,870

Overall accuracy at the second level = 87.0%

Overall accuracy at the third level = 87.0%

Table 161: Confusion matrix CLCCH2006-2012 for Steppic region – PLAUSIBILITY ANALYSIS- In column the validation and line the production

CLC Class	11-11	12-12	13-13	13-51	14-14	21-13	21-21	22-21	22-22	23-13	23-23	23-41	24-24	31-31	31-32	32-31	32-32	32-51	33-33	41-41	41-51	51-41	51-51	Total	User Acc
11-11	5,503																							5,503	1,000
12-12		1,513				0,038																		1,551	0,975
13-13			0,015																					0,015	1,000
14-14					0,024																			0,024	1,000
21-12						0,023																		0,023	0,000
21-21							109,801																	109,801	1,000
21-41							0,222																	0,222	0,000
22-21								0,023																0,023	1,000
22-22							0,658		1,973				1,315											3,945	0,500
23-21							0,012																	0,012	0,000
23-23										2,070	10,351													12,421	0,833
23-41												0,009												0,009	1,000
23-51																					0,009			0,009	0,000
24-24													6,551											6,551	1,000
31-31														12,344										12,344	1,000
31-32															0,073									0,073	1,000
32-32											0,291					0,291	0,874							1,457	0,600
32-51																		0,009						0,009	1,000
33-33																			0,139					0,139	1,000
33-51				0,009																				0,009	0,000
41-41																				4,601	0,657			5,259	0,875
51-51																						0,332	4,192	4,524	0,927
Total	5,503	1,513	0,015	0,009	0,024	0,061	110,692	0,023	1,973	2,070	10,642	0,009	7,866	12,344	0,073	0,291	0,874	0,009	0,139	4,601	0,667	0,332	4,192	157,997	
Prod Acc	1,000	1,000	1,000	0,000	1,000	0,000	0,992	1,000	1,000	0,000	0,973	1,000	0,833	1,000	1,000	0,000	1,000	1,000	1,000	1,000	0,000	0,000	1,000		
																							Global Acc		0,964

Overall accuracy at the second level = 96.4%

Overall accuracy at the third level = 96.4%

Bio-geographical region DOM – Oversea departments - Confusion matrix of CLCCH2006-2012 - Blind and plausibility analysis

Table 162: Confusion matrix CLCCH2006-2012 for Oversea departments (Guadeloupe, French Guyana, Martinique, La Reunion, Mayotte) – BLIND ANALYSIS- In column the validation and line the production

CLC Classes	11-11	12-12	13-11	13-12	13-13	13-32	14-14	21-11	21-13	21-21	21-52	22-22	23-11	23-23	23-32	24-11	24-13	24-24	31-12	31-13	31-23	31-24	31-31	31-32	31-52	32-13	32-21	32-24	32-31	32-32	33-33	41-41	42-42	51-51	52-31	52-52	Total	User Acc		
11-11	2,551																																				2,551	1,000		
12-12		0,308																																				0,308	1,000	
13-12	0,007		0,014	0,007																																		0,027	0,250	
13-13				0,041																																		0,041	1,000	
14-14					0,050																																	0,050	1,000	
21-11						0,050																																0,010	1,000	
21-13							0,010																															0,005	1,000	
21-21								0,005																														2,774		
21-23																			0,131																			2,905	0,955	
21-52																																							0,007	0,000
22-22																																							0,020	1,000
22-23																																							0,496	0,700
23-23																																							0,007	0,000
23-32																																							0,110	0,700
24-11																																							0,769	0,700
24-12																																							0,007	1,000
24-13																																							0,015	0,600
24-23																																							0,010	0,000
24-24																																							0,005	0,000
31-11																																							0,007	0,000
31-12																																							0,007	0,000
31-13																																							0,007	1,000
31-23																																							0,007	1,000
31-24																																							0,007	0,000
31-31																																							0,003	0,000
31-32																																							0,003	0,000
31-52																																							0,003	0,000
32-11																																							0,003	0,000
32-21																																							0,003	0,000
32-24																																							0,003	0,000
32-31																																							0,003	0,000
32-32																																							0,003	0,000
33-33																																							0,003	0,000
41-41																																							0,003	0,000
42-42																																							0,003	0,000
51-51																																							0,003	0,000
52-31																																							0,003	0,000
52-52																																							0,003	0,000
Total	2,900	0,308	0,014	0,007	0,041	0,003	0,050	0,010	0,005	2,774	0,020	0,502	0,110	0,776	0,007	0,015	0,005	5,116	0,007	0,014	0,004	0,195	34,297	0,231	0,115	0,003	0,010	0,007	0,021	5,354	0,856	4,509	0,003	0,494	0,192	62,935	118,966			
Prod Acc	0,879	1,000	0,000	1,000	1,000	0,000	1,000	1,000	1,000	1,000	1,000	0,987	0,000	0,991	1,000	1,000	0,000	0,873	1,000	0,500	0,000	0,088	0,979	0,481	0,943	0,000	1,000	1,000	1,000	0,934	0,755	0,993	1,000	0,769	1,000	0,999				
	Global Acc																																							0,976

Overall accuracy at the second level = 97.6%

Overall accuracy at the third level = 96.4%

Table 163: Confusion matrix CLCCH2006-2012 for Oversea departments (Guadeloupe, French Guyana, Martinique, La Reunion, Mayotte) – PLAUSIBILITY ANALYSIS- In column the validation and line the production

CLC Class	11-11	12-12	13-11	13-12	13-13	14-14	21-11	21-13	21-21	21-52	22-22	23-23	23-32	24-11	24-24	24-32	31-12	31-13	31-23	31-24	31-31	31-32	31-52	32-13	32-21	32-24	32-31	32-32	33-33	41-41	42-42	51-51	52-31	52-52	Total	User Acc
11-11	2,551																																	2,551	1,000	
12-12		0,308																																0,308	1,000	
13-12	0,007		0,014	0,007																														0,027	0,250	
13-13				0,041																														0,041	1,000	
14-14					0,050																													0,050	1,000	
21-11						0,010																												0,010	1,000	
21-13							0,005																											0,005	1,000	
21-21								2,774							0,131																			2,905	0,955	
21-23												0,007																						0,007	0,000	
21-52										0,020																								0,020	1,000	
22-22											0,496				0,212																			0,708	0,700	
22-23											0,007																							0,007	0,000	
23-23												0,879			0,110							0,110												1,099	0,800	
23-32													0,007																					0,007	1,000	
24-11						0,005								0,010	0,010																			0,025	0,400	
24-12																0,005																		0,005	0,000	
24-23															0,007																			0,007	0,000	
24-24	0,336														4,467													0,259						5,062	0,882	
31-11	0,007																																	0,007	0,000	
31-12																	0,007																	0,007	1,000	
31-13																		0,007																0,007	1,000	
31-23																				0,021	0,003					0,003								0,027	0,750	
31-24																					0,003					0,003									0,027	0,750
31-31																					33,575													33,575	1,000	
31-32																		0,004	0,013	0,009	0,111													0,138	0,807	
31-41																				0,007														0,007	0,000	
31-52																						0,109												0,109	1,000	
32-11																		0,007																0,007	0,000	
32-21																								0,010										0,010	1,000	
32-24															0,003				0,007					0,003		0,010		0,003						0,027	0,375	
32-31																				0,009							0,021							0,030	0,714	
32-32															0,164				0,164	0,493								5,166						5,988	0,863	
33-33																				0,047								0,093	0,647					0,787	0,822	
41-41																															4,632			4,632	1,000	
42-42																															0,031	0,003		0,034	0,088	
51-51																													0,045				0,425	0,471	0,904	
52-31																							0,007										0,192	0,007	0,205	0,936
52-52																																		62,997	62,997	1,000
Total	2,900	0,308	0,014	0,007	0,041	0,050	0,015	0,005	2,774	0,020	0,502	0,886	0,007	0,010	5,105	0,005	0,007	0,014	0,004	0,205	34,142	0,224	0,115	0,003	0,010	0,014	0,021	5,522	0,692	4,663	0,003	0,425	0,192	63,004	119,556	
Prod Acc	0,879	1,000	0,000	1,000	1,000	1,000	0,667	1,000	1,000	1,000	0,987	0,992	1,000	1,000	0,875	0,000	1,000	0,500	0,000	0,100	0,983	0,495	0,943	0,000	1,000	0,750	1,000	0,936	0,935	0,993	1,000	1,000	1,000		Global Acc	0,981

Overall accuracy at the second level = 98.1%

Overall accuracy at the third level = 97.6%

Annex 9. Observations for the main problematic countries showing land cover changes

Turkey

Table 164: Commission and omission of land cover changes and main problematic CLC 2012 classes for Turkey (zone 1)

CLCCHN3P	User accuracy	CLCCHN3V	Producer accuracy	Main confusion with CLCCHN3V	CLCCHN3P	User accuracy	CLCCHN3V	Producer accuracy	Main confusion with CLCCHN3V
111-111	0,648	111-111	0,991	confusion with 112-112	211-223	1,000	211-223	1,000	
112-111	0,400	112-111	0,547	confusion with 111-111; 112-112	211-231	0,200	211-231	0,440	confusion 211-211; 231-231
112-112	0,969	112-112	0,754		211-242	0,467	211-242	1,000	
121-121	0,818	121-121	0,916		211-243	0,000			
121-321		121-321	0,000		211-321	1,000	211-321	1,000	
122-122	0,752	122-122	0,947		211-324	0,500	211-324	1,000	
123-123	1,000	123-123	0,209	Confusion with 121-121	211-512	1,000	211-512	0,799	
124-124	1,000	124-124	1,000		212-112	1,000	212-112	1,000	
131-121	1,000	131-121	1,000		212-121	1,000	212-121	1,000	
131-131	0,617	131-131	0,976	overestimation of 131 in 2006 - XXX - 131	212-131	1,000	212-131	0,241	
131-211		131-211	0,000		212-133	0,786	212-133	0,543	
131-231	0,000				212-212	0,930	212-212	0,971	
131-242	0,000				212-213	0,400	212-213	0,096	confusion with 212 -212; 213-213
131-321	0,000	131-321	0,000		212-222	0,000			
131-324	1,000	131-324	0,210	omission change 131-324 instead to 131-131	212-231	0,000			
131-512	0,000				212-242	0,625	212-242	0,059	confusion with 242-242
132-132	0,487	132-132	1,000	confusion with 211-211	212-411	0,000			
133-112	0,800	133-112	0,755		212-512	0,800	212-512	1,000	
133-121	0,633	133-121	0,386	omission change 133-121 instead to 121-121	213-213	0,864	213-213	0,994	
133-122	0,750	133-122	1,000		213-512	1,000	213-512	1,000	
133-133	0,326	133-133	0,976	Confusion with 121-121 or XXX (2006) - 133	221-221	0,414	221-221	1,000	confusion with 242-242
133-142		133-142	0,000		221-512	0,000			
133-231	0,000	133-231	0,000		222-112	1,000	222-112	1,000	
133-242	0,000				222-133	0,000			
133-321	0,000	133-321	0,000		222-222	0,675	222-222	0,857	confusion with 242 or 243
133-324		133-324	0,000		222-333		222-333	0,000	
133-333		133-333	0,000		222-512	0,000			
133-511	0,000				223-223	0,850	223-223	0,678	omission from 242; 222
133-512	1,000	133-512	0,689	other changes 411-512	231-112	0,000	231-112	0,000	
141-141	0,802	141-141	1,000		231-121	0,000	231-121	0,000	
142-142	0,602	142-142	1,000	confusion with 112 and other change 133	231-131	0,308	231-131	1,000	
211-112	0,432	211-112	1,000	confusion with 242-242	231-133	0,683	231-133	0,785	
211-121	1,000	211-121	0,365	omission from 133-133	231-211	0,425	231-211	0,864	
211-131	1,000	211-131	0,488	omission from 131-131	231-212	0,429	231-212	1,000	
211-133	0,794	211-133	0,403	omission from 133-133	231-213	1,000	231-213	1,000	
211-211	0,841	211-211	0,882		231-231	0,764	231-231	0,584	Omission from 211,212, 242..
211-212	0,200	211-212	1,000	confusion 211-211; 212 -212	231-242	0,717	231-242	0,032	
211-213	0,000	211-213	0,000		231-512	1,000	231-512	0,182	other change 321-512
211-222		211-222	0,000						

CLCCHN3P	User accuracy	CLCCHN3V	Producer accuracy	Main confusion with CLCCHN3V	CLCCHN3P	User accuracy	CLCCHN3V	Producer accuracy	Main confusion with CLCCHN3V
241-241	0,000	241-241	0,000		321-231	0,000			
242-112	0,694	242-112	0,301		321-243	1,000	321-243	1,000	
242-121	1,000	242-121	1,000		321-321	0,774	321-321	0,610	confusion with 333; 231 and 332
242-212	0,400	242-212	1,000		321-324	0,497	321-324	0,061	
242-222	1,000	242-222	0,026		321-333		321-333	0,000	
242-223	0,400	242-223	1,000		321-512	0,740	321-512	1,000	
242-231	0,000				322-131		322-131	0,000	
242-242	0,629	242-242	0,704		322-322	0,000	322-322	0,000	
242-324		242-324	0,000		322-332		322-332	0,000	
242-411		242-411	0,000		323-112	1,000	323-112	0,441	
242-512	0,428	242-512	0,633		323-133		323-133	0,000	
243-112	0,500	243-112	1,000		323-142		323-142	0,000	
243-133	0,800	243-133	0,420		323-321	0,000			
243-211	0,000	243-211	0,000		323-323	0,700	323-323	0,638	Confusion with 333; 334 and 321
243-212	1,000	243-212	1,000		323-324		323-324	0,000	
243-222	0,500	243-222	1,000		323-334		323-334	0,000	
243-243	0,769	243-243	0,716		323-512	1,000	323-512	1,000	
243-311	0,000				324-131	0,824	324-131	0,472	
243-324	0,000				324-133	0,500	324-133	0,667	
243-512	0,747	243-512	0,630		324-142	0,000			
244-244		244-244	0,000		324-211	0,000			
311-121	0,344	311-121	1,000		324-223	1,000	324-223	1,000	
311-131	1,000	311-131	1,000		324-231		324-231	0,000	
311-311	0,806	311-311	0,915		324-243	0,000			
311-324	0,739	311-324	0,946		324-311	0,624	324-311	0,272	
311-512	1,000	311-512	1,000		324-312	0,700	324-312	1,000	
312-121	1,000	312-121	1,000		324-313	0,400	324-313	0,043	
312-122		312-122	0,000		324-321	0,660	324-321	0,660	
312-131	0,000	312-131	0,000		324-324	0,835	324-324	0,781	
312-133	1,000	312-133	0,939		324-333	0,000			
312-223	1,000	312-223	1,000		324-334		324-334	0,000	
312-312	0,906	312-312	0,909		324-512	0,828	324-512	0,827	
312-321	0,000				331-121	1,000	331-121	1,000	
312-324	0,933	312-324	0,411		331-142	1,000	331-142	1,000	
312-512		312-512	0,000		331-324		331-324	0,000	
313-133		313-133	0,000		331-331	0,900	331-331	0,944	
313-313	0,715	313-313	0,766	confusion with 311 and 312	331-411	0,000			
313-324	0,467	313-324	0,863		331-511	1,000	331-511	1,000	
313-334		313-334	0,000		331-512	0,000			
321-112		321-112	0,000		332-133	1,000	332-133	1,000	
321-121	1,000	321-121	0,429		332-332	0,669	332-332	1,000	
321-131	0,800	321-131	0,456		332-333		332-333	0,000	
321-132		321-132	0,000		333-112	0,000			
321-133	0,508	321-133	0,332		333-131	0,661	333-131	0,805	
321-211	0,765	321-211	0,649		333-133	0,291	333-133	0,117	
321-212		321-212	0,000		333-211	0,000			
321-223	0,000				333-232		333-232	0,000	
321-223		321-223	0,000		333-242	0,000			

CLCCHN3P	User accuracy	CLCCHN3V	Producer accuracy	Main confusion with CLCCHN3V
333-243	1,000	333-243	1,000	
333-324	0,000			
333-333	0,546	333-333	0,790	confusion with 321 and 324
334-321	1,000	334-321	1,000	
334-324	0,318	334-324	0,811	confusion with 324
334-333	0,000			
334-334	0,000	334-334		
335-335	1,000	335-335	1,000	
411-211		411-211	0,000	
411-212	0,500	411-212	1,000	
411-213	1,000	411-213	1,000	
411-231	0,500	411-231	1,000	confusion with 231-231
411-411	0,897	411-411	0,915	
411-512	0,587	411-512	0,218	omission from 411-411
421-421	0,101	421-421	1,000	333-333 (Halophyte steppe)
422-422	0,800	422-422	1,000	
511-133	0,667	511-133	1,000	other change 312-133
511-511	0,800	511-511	1,000	
511-512	0,952	511-512	0,813	
512-131	0,500	512-131	1,000	commission from 131-131
512-211	1,000	512-211	1,000	
512-512	1,000	512-512	0,967	
521-521	1,000	521-521	1,000	
522-522	1,000	522-522	1,000	
523-523	0,910	523-523	1,000	

Finland

Table 165: Commission and omission of land cover changes and main problematic CLC 2012 classes for Finland (Zone 6)

CLCCHN3P	User accuracy	CLCCHN3V	Producer accuracy	Main confusion with CLCCHN3V	CLCCHN3P	User accuracy	CLCCHN3V	Producer accuracy	Main confusion with CLCCHN3V
112-112	0,789	112-112	0,964				231-512	0,000	
121-121	0,850	121-121	0,622	confusion 112-112			242-242	0,000	
121-123	0,000				243-112	0,000			
122-122	1,000	122-122	1,000		243-142	0,000			
123-123	0,800	123-123	0,860		243-243	0,646	243-243	0,849	confusion with 211-211
123-133	0,500	123-133	0,210	other change	243-324	0,083	243-324	1,000	
124-124	1,000	124-124	1,000		243-332	0,000			
131-131	0,800	131-131	0,887		243-412	0,000			
131-132	0,000						311-112	0,000	
132-132	0,500	132-132	0,984	confusion with 131-131	311-311	0,605	311-311	0,747	Confusion with 313-313/ 324-324
132-141		132-141	0,000		311-324	0,273	311-324	0,170	
132-313	0,000				312-112	0,722	312-112	0,730	
132-324	0,000				312-121	0,000	312-121	0,000	
133-112	1,000	133-112	0,246	omission with 211-112	312-131	0,353	312-131	0,214	
133-121	1,000	133-121	0,017		312-132	0,000			
133-122	0,500	133-122	1,000	confusion with 211 -122	312-133	0,450	312-133	0,533	
133-123	1,000	133-123	0,066		312-142	0,000			
133-131	0,000						312-211	0,000	
133-133	0,000						312-212	0,000	
133-142	1,000	133-142	0,073		312-312	0,681	312-312	0,877	confusion with 313-313
141-141	0,611	141-141	1,000	comission 112-112	312-324	0,727	312-324	0,636	
142-142	0,842	142-142	0,924		312-412	0,350	312-412	1,000	
211-112	0,300	211-112	1,000		313-112	1,000	313-112	0,604	
211-121	1,000	211-121	0,732				313-121	0,000	
211-122		211-122	0,000		313-131	1,000	313-131	0,147	
211-133	0,667	211-133	0,170				313-132	0,000	
211-142	0,500	211-142	1,000				313-133	0,000	
211-211	0,918	211-211	0,706	omission with 243-243 and 312-312			313-142	0,000	
211-311		211-311	0,000		313-313	0,818	313-313	0,555	omission from 312-312
211-324	0,063	211-324	1,000		313-324	0,680	313-324	0,067	
211-412	0,167	211-412	1,000	overestimation of change (412-412)	313-412	0,000	313-412	0,000	
		231-112	0,000		321-321	0,533	321-321	0,195	
		231-133	0,000		322-322	0,500	322-322	0,574	
		231-211	0,000		324-112	0,000	324-112	0,000	
231-231	0,263	231-231	0,057	confusion with 243-243 / 324 -324	324-121	0,158	324-121	0,792	
231-324	0,063	231-324	0,307		324-123	0,000			
231-332	0,000				324-131	0,333	324-131	1,000	
231-412	0,000				324-133	0,667	324-133	1,000	

CLCCHN3P	User accuracy	CLCCHN3V	Producer accuracy	Main confusion with CLCCHN3V
324-142	1,000	324-142	1,000	
324-211	0,176	324-211	0,929	
324-231	0,000	324-231	0,000	
324-311	0,091	324-311	1,000	
324-312	0,235	324-312	0,993	
324-313	0,262	324-313	0,171	
324-324	0,511	324-324	0,527	
324-412	0,364	324-412	0,670	
331-131	0,000			
331-331	0,000			
332-332	0,368	332-332	1,000	
333-333	0,950	333-333	0,639	
411-231	0,000			
411-243	0,000			
411-411	0,850	411-411	0,969	
412-121	0,000	412-121	0,000	
		412-131	0,000	
		412-133	0,000	
412-211	0,250	412-211	1,000	
412-243	0,000			
412-311	0,000			
412-312	0,000			
412-313	0,000			
412-324	0,000	412-324	0,000	
412-412	0,804	412-412	0,691	omission from 312-312 and 324-324
421-231	0,000			
421-324	0,000			
421-421	0,647	421-421	0,476	commission from other land cover and omission from 523
511-511	0,895	511-511	1,000	
512-512	1,000	512-512	0,985	
523-523	0,667	523-523	1,000	

Problematic classes and changes concern essentially the natural classes 311, 312, 313, 324, 322, 321

Italy

Table 166: Commission and omission of land cover changes and main problematic CLC 2012 classes for Italy (zone 9)

CLCCHN3P	User accuracy	CLCCHN3V	Producer accuracy	Main confusion with CLCCHN3V	CLCCHN3P	User accuracy	CLCCHN3V	Producer accuracy	Main confusion with CLCCHN3V
111-111	0,900	111-111	0,753	omission from 112-112	212-212	0,800	212-212	0,991	
112-112	0,900	112-112	0,873		213-213	0,950	213-213	0,859	
		112-121	0,000		221-221	0,000			
112-133	0,000				221-211	1,000	221-211	0,696	
121-121	0,950	121-121	0,656	omission from 112-112 and 242-242	221-221	0,450	221-221	0,855	242-242 or 211-211
121-133	0,500	121-133	1,000		222-221	0,500	222-221	1,000	
122-122	0,700	122-122	0,986		222-133	1,000	222-133	0,883	
123-123	0,850	123-123	1,000		222-142	1,000	222-142	1,000	
124-124	0,950	124-124	1,000		222-211	0,153	222-211	1,000	
131-131	0,700	131-131	0,985		222-222	0,750	222-222	0,765	commission with 242, 243, 211 and omission from 223
131-133	1,000	131-133	1,000		222-242	0,000			
131-231	0,000				223-112	0,000			
131-324	1,000	131-324	1,000		223-121	0,000			
		132-112	0,000		223-133	1,000	223-133	1,000	
		132-121	0,000		223-223	0,750	223-223	0,912	
		132-122	0,000		223-323	0,000			
132-132	0,600	132-132	0,403	Omission from 131-131			231-121	0,000	
132-133	1,000	132-133	1,000		231-133	0,000	231-133	0,000	
133-111	0,000				231-231	0,700	231-231	0,243	
133-112	0,709	133-112	0,957				231-243	0,000	
133-121	0,600	133-121	0,606				231-324	0,000	
133-122	0,888	133-122	0,757		231-512	0,500	231-512	1,000	
133-123	1,000	133-123	0,611		241-131	0,000			
133-133	0,200	133-133	1,000	commission with changes XXX-133	241-211	0,000			
133-142	1,000	133-142	0,843		241-241	0,350	241-241	0,257	commission with 211; 231; 242; 243; and omission from 211 and 242
		133-231	0,000		242-112	0,500	242-112	0,036	
		133-512	0,000		242-121	0,450	242-121	0,156	
141-141	0,850	141-141	1,000				242-131	0,000	
142-142	1,000	142-142	0,134	omission from 211-211	242-133	0,750	242-133	0,862	
142-324	1,000	142-324	1,000		242-211	0,468	242-211	1,000	
211-112	0,300	211-112	1,000	commission with other change or 112-112			242-223	0,000	
211-121	0,825	211-121	0,063		242-242	0,540	242-242	0,495	commission with 211; 231; 223; 243; and omission from 211; 221; 243
211-122	1,000	211-122	1,000				242-243	0,000	
211-131	0,750	211-131	1,000				243-112	0,000	
		211-132	0,000				243-121	0,000	
211-133	1,000	211-133	0,707				243-131	0,000	
211-142	1,000	211-142	1,000				243-133	0,000	
211-211	0,740	211-211	0,904	commission with other agriculture classes 242, 231, 243,244			243-211	0,000	
211-213	0,000				243-243	0,540	243-243	0,595	commission with 231; 242; 311; 324 and omission from 211; 311
211-221	0,700	211-221	1,000		243-323	0,500	243-323	1,000	
211-222	1,000	211-222	1,000		243-324	0,000			
		211-243	0,000		243-331	0,000			
211-311	0,265	211-311	1,000	311-311 or other changes			243-334	0,000	
211-324	0,000						243-511	0,000	
211-334	0,000				244-244	0,850	244-244	0,429	omission with 211-211
211-411	0,333	211-411	1,000	no change (231-231) or (411-411)	311-121	1,000	311-121	1,000	
		211-511	0,000		311-122	1,000	311-122	1,000	
211-512	0,750	211-512	1,000		311-131	1,000	311-131	1,000	

CLCCHN3P	User accuracy	CLCCHN3V	Producer accuracy	Main confusion with CLCCHN3V	CLCCHN3P	User accuracy	CLCCHN3V	Producer accuracy	Main confusion with CLCCHN3V
311-142	1,000	311-133	0,000				324-131	1,000	
311-211	0,604	311-142	1,000				324-133	0,000	
311-213	1,000	311-211	0,303				324-211	1,000	
		311-213	1,000				324-231	0,000	
		311-244	0,000				324-311	0,880	
311-311	0,920	311-311	0,897		324-313	1,000	324-313	1,000	
311-324	0,724				324-324	0,420	324-324	0,403	commission with 243, 311,321,323 and omission from 211, 243, 323, 311
311-334	0,500	311-324	0,957		324-331	0,750	324-331	0,632	
311-511	0,000	311-334	0,342				324-334	0,000	
311-512	1,000						324-423	0,000	
312-142	1,000	311-512	1,000				324-511	0,000	
312-312	0,939	312-142	1,000				331-324	0,571	
312-324	0,850	312-312	1,000		331-324	0,500	331-324	0,571	
313-131	1,000	312-324	0,980		331-331	0,850	331-331	0,931	
		313-131	1,000		331-511	0,000	331-511	0,000	
		313-132	0,000		331-523	1,000	331-523	1,000	
313-313	0,800	313-132	0,000		332-332	0,900	332-332	0,990	
313-324	0,700	313-313	0,866		333-323	0,000			
321-121	1,000	313-324	1,000		333-333	0,750	333-333	0,854	
321-131	1,000	321-121	1,000				334-311	0,000	
		321-131	1,000		334-313	0,000			
321-312	0,000	321-311	0,000		334-323	1,000	334-323	1,000	
321-321	0,750	321-312	0,000				334-324	0,000	
321-323	0,000	321-321	0,752		334-334	0,050	334-334	0,733	confusion with other land cover type
		321-323	0,000				335-332	0,000	
		321-324	0,000		335-335	0,850	335-335	1,000	
322-322	0,800	321-334	0,000		411-411	0,900	411-411	0,964	
323-112	0,500	322-322	0,578	omission from 321	411-512	1,000	411-512	1,000	
323-121	1,000	323-112	1,000		412-412	0,000			
323-133	1,000	323-121	1,000		421-421	0,750	421-421	0,971	
323-211	0,000	323-133	0,500		422-422	0,800	422-422	0,695	
					511-331	0,000			
323-243	0,000	323-231	0,000		511-511	0,700	511-511	0,433	omission from 243
323-323	0,800	323-243	0,000	omission from 311, 321,324	512-512	1,000	512-512	0,980	
		323-323	0,715		521-421	0,000			
		323-324	0,000		521-521	0,900	521-521	0,974	
323-334	0,333	323-334	0,316		522-522	0,000			
324-121	1,000				523-523	1,000	523-523	1,000	
324-131	1,000								
324-133	0,000								
324-211	0,333	324-131	0,614						
		324-121	1,000						

Greece

Table 167: Commission and omission of land cover changes and main problematic CLC 2012 classes for Greece (zone 12)

CLCCHN3P	User accuracy	CLCCHN3V	Producer accuracy	Main confusion with CLCCHN3V	CLCCHN3P	User accuracy	CLCCHN3V	Producer accuracy	Main confusion with CLCCHN3V
111-111	0,933	111-111	1,000		223-223	0,733	223-223	0,972	
112-112	0,960	112-112	0,952		231-121	0,500	231-121	1,000	
121-121	0,850	121-121	0,649		231-131	1,000	231-131	1,000	
122-122	0,533	122-122	0,772	commission with 133-122 and omission from 121	231-133	1,000	231-133	1,000	
123-123	1,000	123-123	1,000		231-142		231-142	0,000	
124-124	1,000	124-124	1,000		231-211	0,000			
131-131	0,867	131-131	0,917		231-231	0,640	231-231	0,272	commission with 242,243,321 and omission from 211, 242,243,312,324
131-231	0,000				231-323	0,000			
131-311	1,000	131-311	1,000		231-512	0,200	231-512	1,000	
131-321	1,000	131-321	0,500		241-241	0,400	241-241	0,210	commission with 242,243 and omission from 223,24
131-323	1,000	131-323	1,000		242-121	1,000	242-121	1,000	
131-324	0,750	131-324	1,000		242-133	0,500	242-133	0,250	
131-512	1,000	131-512	1,000		242-242	0,633	242-242	0,649	commission with 242,243,321 and omission from 211, 242,243,312,324
132-132	0,000				243-122		243-122	0,000	
133-112	0,000	133-112	0,000		243-131	1,000	243-131	1,000	
133-121	0,600	133-121	1,000		243-132		243-132	0,000	
133-122	1,000	133-122	0,521		243-133	1,000	243-133	0,492	
133-133	0,400	133-133	0,693	commission with XXX-133-and omission from 131	243-243	0,886	243-243	0,677	Omission from 223, 242,311,323
133-142	1,000	133-142	1,000		243-324	1,000	243-324	1,000	
133-231	1,000	133-231	1,000		311-131	1,000	311-131	1,000	
133-512	1,000	133-512	1,000		311-133	1,000	311-133	1,000	
141-141	0,800	141-141	1,000		311-311	0,886	311-311	0,859	
142-142	0,867	142-142	1,000		311-324	0,531	311-324	1,000	
211-112		211-112	0,000		311-334		311-334	0,000	
211-121	1,000	211-121	1,000		312-312	0,867	312-312	0,918	
211-122	1,000	211-122	1,000		312-313		312-313	0,000	
211-133	0,400	211-133	1,000		312-321	0,000			
211-211	0,829	211-211	0,971		312-322		312-322	0,000	
211-242	0,000				312-324	0,633	312-324	0,302	
212-121		212-121	0,000		312-333	0,000			
212-212	1,000	212-212	0,997		312-334	1,000	312-334	0,371	
212-213	0,167	212-213	1,000		313-313	0,800	313-313	0,647	omission from 243, 311, 312, 323, 32
212-411		212-411	0,000		313-324	0,764	313-324	0,910	
213-212	0,000				313-334	0,800	313-334	0,360	
213-213	0,867	213-213	0,909		321-121	0,000	321-121	0,000	
221-221	0,300	221-221	1,000	commission with 242, 243	321-122		321-122	0,000	
221-242	0,000				321-131	0,000			
222-133		222-133	0,000		321-133		321-133	0,000	
222-222	0,800	222-222	0,679	omission from 242	321-142	0,000			
223-133	1,000	223-133	0,592		321-231		321-231	0,000	

CLCCHN3P	User accuracy	CLCCHN3V	Producer accuracy	Main confusion with CLCCHN3V	CLCCHN3P	User accuracy	CLCCHN3V	Producer accuracy	Main confusion with CLCCHN3V
321-321	0,559	321-321	0,766	commission with 243, 322, 323,324 and omission from 243, 323,333	411-411	0,800	411-411	0,501	omission from 313
		321-323	0,000		421-421	0,929	421-421	0,809	
		321-334	0,000		422-422	1,000	422-422	1,000	
		322-131	0,000		511-511	0,933	511-511	1,000	
322-322	0,600	322-322	0,203	commission with 321 and omission from 321 , 333	512-512	1,000	512-512	0,942	
		322-332	0,000		521-521	1,000	521-521	0,930	
323-121	0,750	323-121	1,000		522-522	0,667	522-522	1,000	
323-131	1,000	323-131	1,000		523-523	1,000	523-523	1,000	
323-133	1,000	323-133	0,200						
323-142	0,000								
		323-231	0,000						
323-211	0,000								
323-321	0,000	323-321	0,000						
323-323	0,850	323-323	0,925						
323-324	0,000	323-324	0,000						
323-333	0,200	323-333	1,000						
323-334	0,900	323-334	0,303						
		324-112	0,000						
324-133	0,000								
324-243	0,000								
324-311	0,200	324-311	0,634						
324-312	0,500	324-312	0,480						
324-313	0,360	324-313	0,064						
		324-321	0,000						
324-323	0,000	324-323	0,000						
324-324	0,686	324-324	0,628	commission with 243,311, 312,313 and omission from 211,311,313,321,323					
324-333	0,000	324-334	0,000						
		324-511	0,000						
331-331	0,800	331-331	0,926						
332-332	0,400	332-332	1,000	commission with 333					
333-133	0,000								
333-333	0,440	333-333	0,666	commission with 321,322,323 and omission from 324					
334-323	0,563	334-323	1,000						
334-324	0,376	334-324	0,712						
334-334	0,000								

Portugal

Table 168: Commission and omission of land cover changes and main problematic CLC 2012 classes for Portugal (zone 16)

CLCCHN3P	User accuracy	CLCCHN3V	Producer accuracy	Main confusion with CLCCHN3V	CLCCHN3P	User accuracy	CLCCHN3V	Producer accuracy	Main confusion with CLCCHN3V
111-111	0,650	111-111	0,384	commission with 112 and omission from 112	213-213	0,850	213-213	1,000	
112-112	0,900	112-112	0,665	omission from 241, 243	213-231	0,000			
121-121	0,800	121-121	0,688	omission from 112	221-121	0,000			
		121-133	0,000				221-133	0,000	
122-122	0,650	122-122	1,000		221-211	0,000	221-211	0,000	
123-123	0,900	123-123	1,000		221-221	0,700	221-221	0,823	
124-124	0,800	124-124	1,000		222-212	0,000			
131-131	0,700	131-131	1,000		222-222	0,700	222-222	0,727	omission from 311
		131-321	0,000				223-121	1,000	
131-324	0,000	131-324	0,000		223-121	1,000			
131-333	0,000						223-142	0,000	
132-132	0,500	132-132	1,000		223-211	0,500	223-211	0,308	
132-321	1,000	132-321	1,000		223-223	0,650	223-223	0,980	commission with 242,241
		132-324	0,000				223-243	0,000	
133-112	1,000	133-112	1,000		223-323	0,000			
133-121	1,000	133-121	0,380		223-324	1,000	223-324	1,000	
133-122	1,000	133-122	0,223		223-512	0,667	223-512	1,000	
		133-123	0,000				231-132	0,000	
		133-132	0,000		231-133	0,600	231-133	1,000	
133-133	0,133	133-133	1,000	commission with XXX-133	231-142	0,000			
133-142	0,750	133-142	0,383		231-221	0,500	231-221	1,000	
133-323	0,000				231-223	0,000			
133-324	0,000				231-231	0,650	231-231	0,644	commission with 243,321 and omission from 241, 211
133-512	1,000	133-512	1,000				231-243	0,000	
141-141	0,600	141-141	1,000	commission with 112, 121	231-321	0,000			
141-142	1,000	141-142	1,000		231-323	0,500	231-323	0,500	
142-142	0,750	142-142	0,960	commission with 112 and change XXX-142			231-324	0,500	
		142-211	0,000				241-112	1,000	
211-112	0,000				231-324	1,000			
211-121	1,000	211-121	0,226		241-112	1,000			
		211-124	0,000				241-211	0,000	
211-132	0,000				241-241	0,250	241-241	0,773	commission with 242,243 and omission from 222,223
		211-142	0,000		241-512	1,000	241-512	0,500	
211-211	0,740	211-211	0,764	commission with 212,242,243			242-112	0,000	
211-212	0,750	211-212	0,509				242-121	0,000	
211-221	1,000	211-221	0,667		242-122	1,000	242-122	1,000	
211-223	0,873	211-223	0,927				242-133	0,000	
211-243	0,000	211-243	0,000				242-231	0,000	
211-323	1,000	211-323	1,000		242-242	0,650	242-242	0,609	commission with 211,221,243 and omission from 211,212, 223, 241
211-324	0,889	211-324	0,215				242-512	0,000	
211-512	0,750	211-512	0,750		243-133	0,667	243-133	0,552	
212-211	0,000						243-211	0,000	
212-212	0,780	212-212	0,752	commission with 211,242 and omission from 211	243-212	0,000			
212-213	1,000	212-213	0,401		243-243	0,800	243-243	0,704	omission from 211,221, 231,241,242,324
		212-222	0,000				243-321	0,000	
212-223	1,000	212-223	0,222		243-323	0,500	243-323	0,500	

CLCCHN3P	User accuracy	CLCCHN3V	Producer accuracy	Main confusion with CLCCHN3V	CLCCHN3P	User accuracy	CLCCHN3V	Producer accuracy	Main confusion with CLCCHN3V
243-324	1,000	243-324	0,032		323-512	0,500	323-512	1,000	
243-512	0,500	243-512	1,000		324-121	1,000	324-121	0,142	
		244-142	0,000				324-122	0,000	
244-244	0,950	244-244	0,656	omission from 311, 241	324-124	1,000	324-124	1,000	
244-324	0,200	244-324	1,000		324-131	1,000	324-131	0,490	
311-221	0,000	311-221	0,000				324-132	0,000	
311-223	1,000	311-223	1,000		324-133	0,800	324-133	0,523	
311-231	0,000				324-142	1,000	324-142	0,500	
311-244	0,000				324-211	0,750	324-211	0,257	
311-311	0,520	311-311	0,741	commission with 244, 324 and omission from 313 and change 324-311, 334-311	324-221	0,500	324-221	0,500	
311-324	0,840	311-324	0,491		324-222	0,500	324-222	1,000	
312-121	1,000	312-121	1,000		324-223	1,000	324-223	1,000	
312-131	1,000	312-131	1,000		324-231	1,000	324-231	0,500	
312-133	0,000	312-133	0,000		324-242	1,000	324-242	1,000	
312-142	0,000						324-243	0,000	
312-312	0,900	312-312	0,921				324-244	0,000	
312-324	0,760	312-324	0,328		324-311	0,747	324-311	0,762	
		312-334	0,000		324-312	0,850	324-312	0,232	
313-121	0,500	313-121	1,000		324-313	0,850	324-313	0,809	
313-313	0,820	313-313	0,987				324-321	0,000	
		313-322	0,000				324-322	0,000	
		313-323	0,000		324-324	0,440	324-324	0,804	commission with 322, 312-324, 322-324
313-324	0,740	313-324	0,636		324-334	0,850	324-334	0,576	
		313-334	0,000		331-331	0,867	331-331	1,000	
		321-131	0,000		332-332	0,750	332-332	1,000	
		321-133	0,000				333-323	0,000	
321-142	1,000	321-142	0,500						
		321-211	0,000		333-333	0,650	333-333	1,000	
321-321	0,600	321-321	0,387	commission with 231, 243, 322, 323 and omission from 322, 324	333-512	0,000			
		321-322	0,000		334-311	0,000			
		321-323	0,000		334-322	0,400	334-322	1,000	
		321-324	0,000		334-324	0,700	334-324	0,353	
321-512	1,000	321-512	0,333		334-334	0,000	334-334	0,000	
322-133	0,000	322-133	0,000		411-411	1,000	411-411	1,000	
322-211	1,000	322-211	1,000		412-412	1,000	412-412	1,000	
322-322	0,800	322-322	0,471	omission from 324	421-421	0,950	421-421	0,846	
323-121	0,000				422-422	0,900	422-422	0,927	
		322-324	0,000		423-423	0,800	423-423	1,000	
		322-334	0,000		511-511	1,000	511-511	0,752	omission from 512
323-131	1,000	323-131	1,000				512-132	0,000	
		323-142	0,000		512-512	0,850	512-512	1,000	
323-221	1,000	323-221	1,000		521-521	0,800	521-521	1,000	
323-243	0,000				522-522	0,500	522-522	0,774	omission from 421
323-323	0,900	323-323	0,970		523-523	1,000	523-523	1,000	
323-324	1,000	323-324	0,965						

Annex 10. Illustrations of some confusion or difficulties of interpretation met during the validation process.

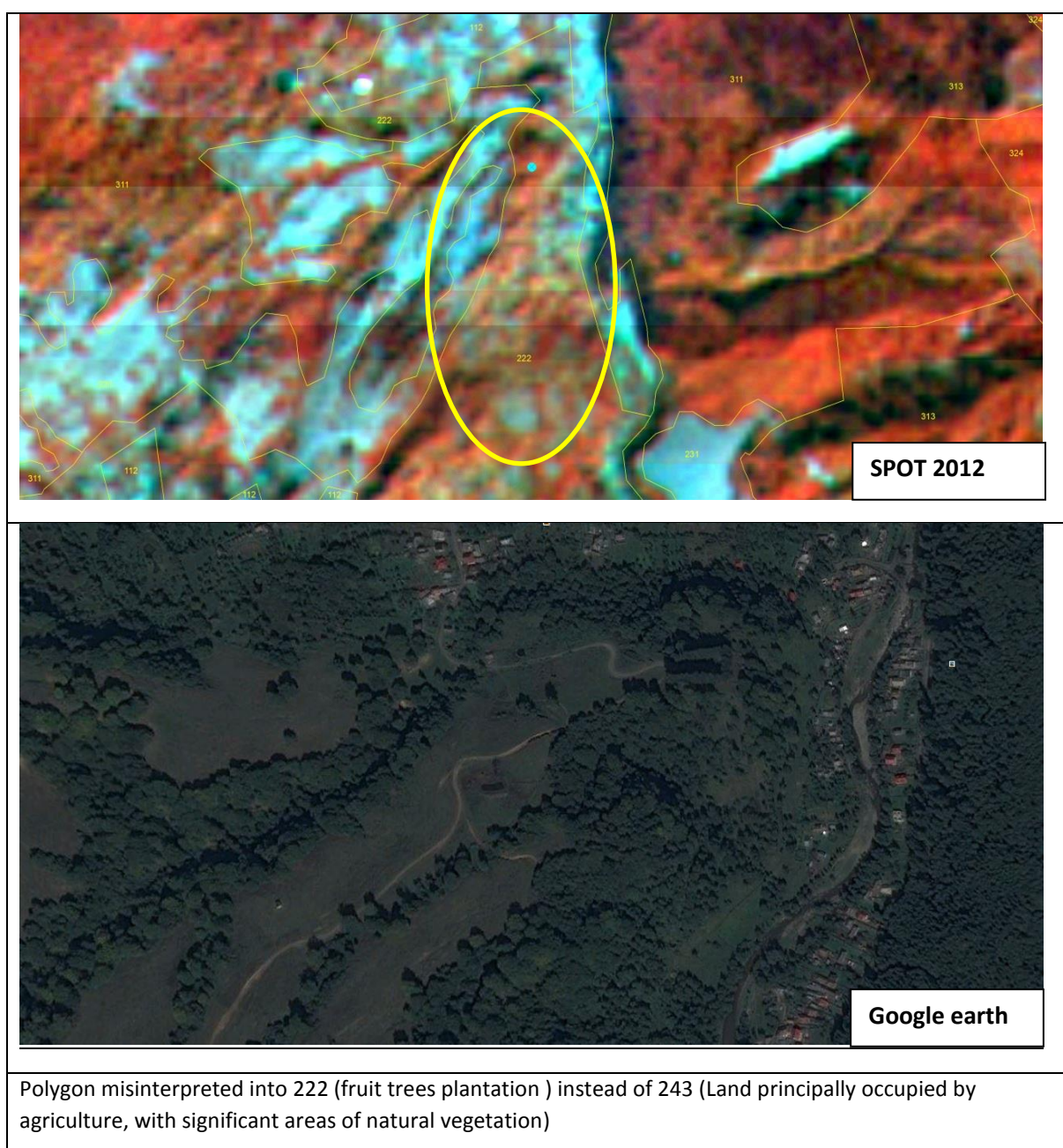
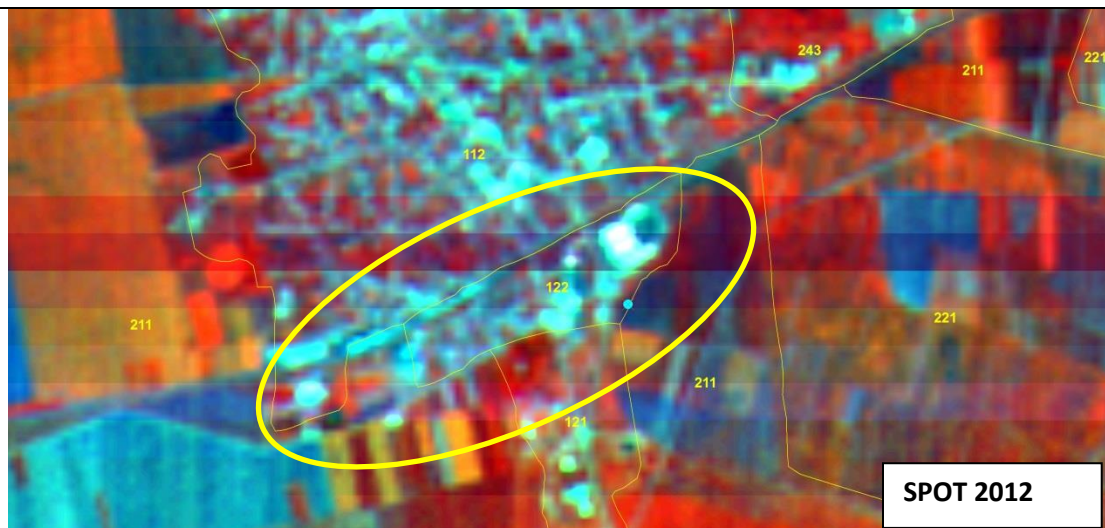
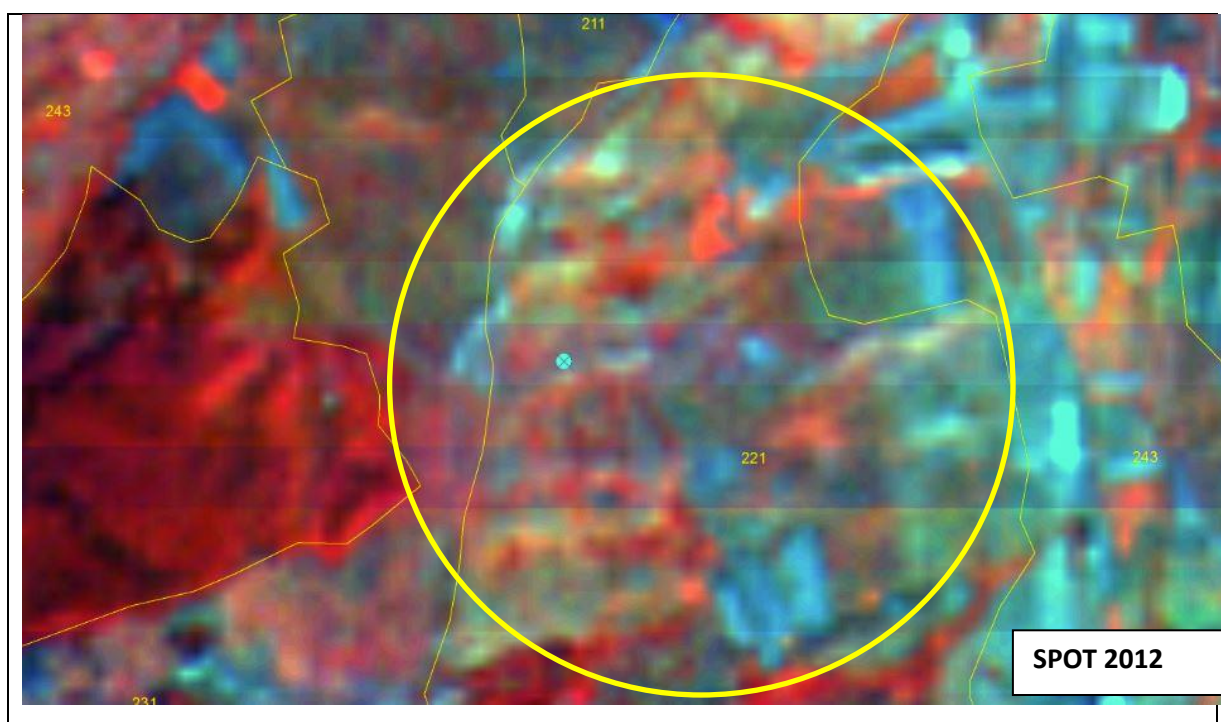


Figure 6: illustration of wrong polygon in CLC2012 – Wrong 222



Polygon misclassified into 122 (road and rail networks) instead of 121 (industrial and commercial units)

Figure 7: illustration of wrong polygon in CLC2012 – Wrong 122



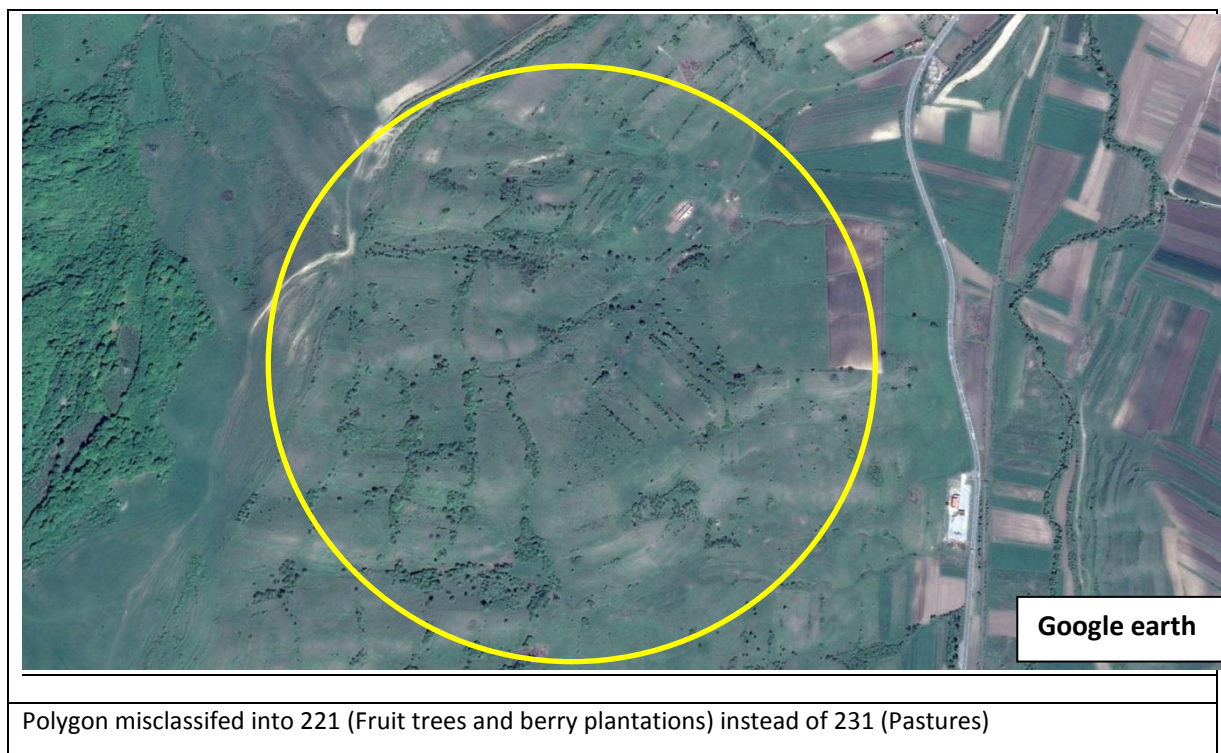
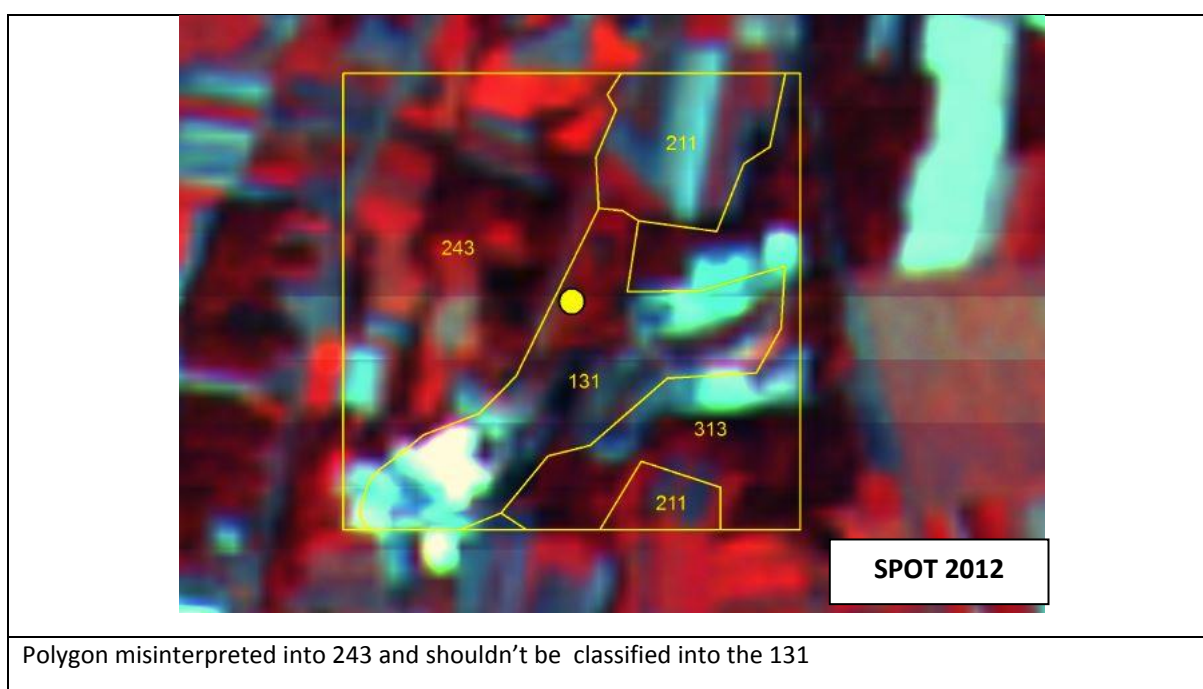


Figure 8: illustration of wrong polygon in CLC2012 – Wrong 221

Example of some bad vector polygon drawing of CLC2012



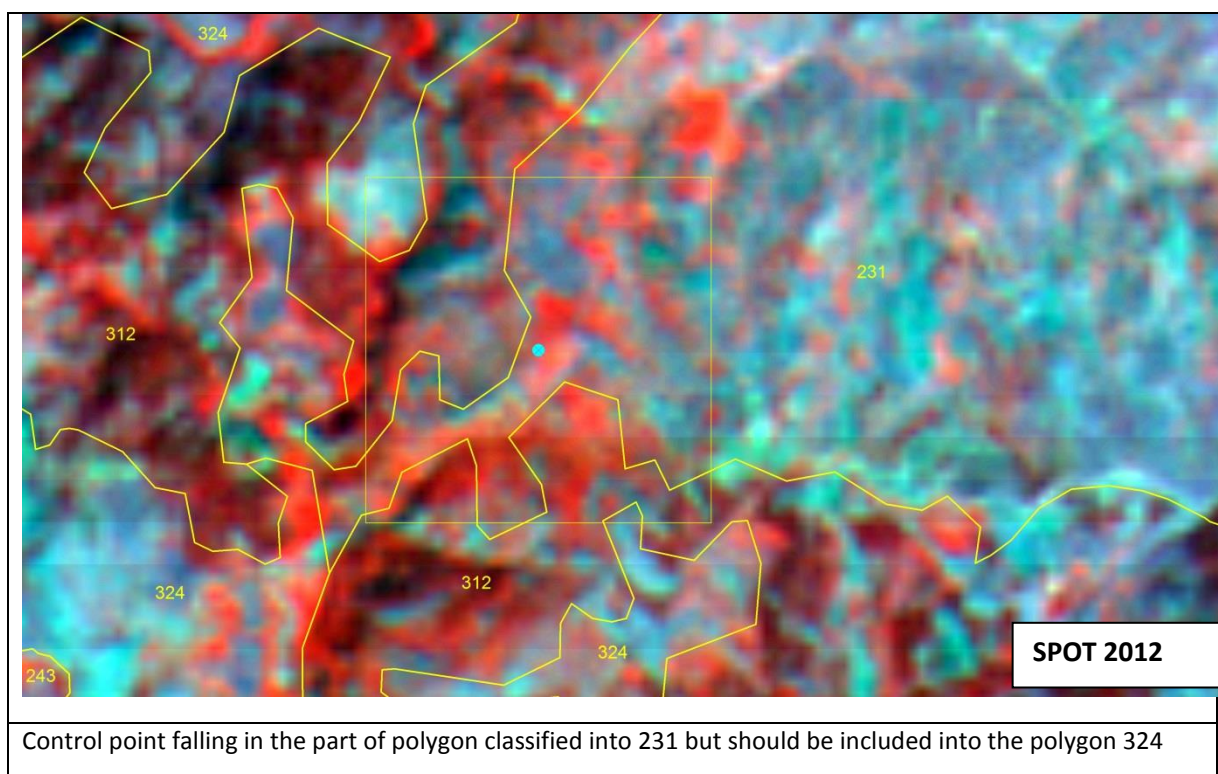


Figure 9: illustration of bad polygon drawing in CLC2012

Interpretation difficulties due to the location of the control point

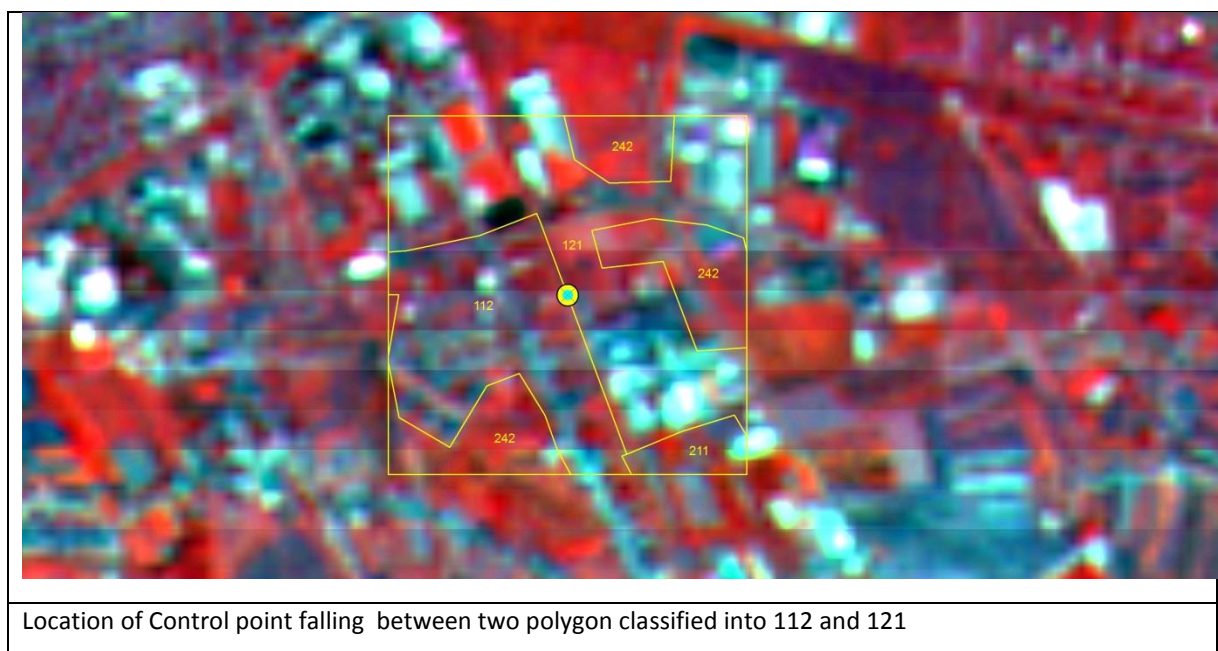


Figure 10: illustration of interpretation difficulties due to the location of the control point